

Draft



**PROBLEMS OF WATER AND FOOD INSECURITY IN THE INDIAN
PART OF THE KOSI RIVER BASIN: A STUDY ON AGRICULTURE**

TEAM

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Preface

This research was directed at understanding the realities of the Kosi region in Bihar which is well known for natural hazards and specifically for recurrent and devastating floods. Periodic droughts and the threats of climate change are also drawing attention these days. There is a fairly long tradition of research interest being focused on the area in context of the floods and their causes and solutions and also, though less commonly, on their effects. However, in-depth study of socio-economic conditions and livelihood in particular is rare. The study, based on both secondary sub-state data and primary household data backed by field insights seeks to understand what factors keep the region underdeveloped and migration prone. The agronomic endowments, risks and climatic vulnerabilities and the food insecurity concerns are some of the factors that describe the character of the region.

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1. Introduction

Development is not growth and growth can translate into development only if the nuances of regions and sections of people are factored into the economic policies, transcending the core focus on economic growth. Every region and its sub-regions have their own subtle but distinctive specificities and people of different economic histories, cultural backgrounds and social hierarchies need differentiated treatment for development. Also any economic policy cannot be expected to have a uniform impact on all regions and sections. Special attention is to needed for regions and sections of people who are likely to be deprived from the fruits of mainstream economic policies or who have special needs due to natural or historical disadvantages. In particular, a principle of justice demands that greatest benefit flows to the least-advantaged members of society (Rawls, 1971).

Bihar, located towards the east of India and once associated historically with flourishing Magadh kingdom and its legendary ruler Ashoka, is today one of least developed states of India. Bihar ranks 14th out of the 28 states in its income measured by its gross state domestic product (GSDP) but it has the lowest per capita GSDP and a high rate of poverty. However, there are pockets of high affluence in the state which makes the income inequality even astounding. The state of the economy has a lot to do with agricultural performance in this principally agrarian state where the iniquity is made worse by the lopsided distribution of land and the failure of land reforms. Despite many natural endowments, Bihar and its agriculture are severely disadvantaged in being a frequent victim of natural calamities, flood being a leading one. Addressing the economy of this state and its agriculture therefore demands priority in national economic policies.

After the state bifurcation, Bihar has become one of the least developed states of India. As districts that were relatively endowed by nature (mostly in mineral, industries and forests) moved out to form Jharkhand in 2000, people began to lack resources and alternative livelihood opportunities and agriculture increasingly became the mainstay of the populace. Yet pressure of population on limited land resources, dismal income potential from farm production and the loss of appeal of the occupation beleaguered the state's agriculture. Recurrent natural calamities and shortcomings of governance intensified the suffering. Civil tensions and criminalization too resulted as a consequence. Economic migration, a legacy of the past, remains a redeeming feature and a conspicuous economic phenomenon of the state.

Bihar is located on the plains of river Ganga and its tributaries which include the Kosi. On the positive side, the abundance of water both on surface and in sub-soil aquifers and the silts brought by the rivers create considerable potential for agricultural growth. Bihar has a high irrigation intensity by Indian standards although the performance of agriculture fails to prove the efficiency of its use. Bihar is endowed with many rivers. Although associated historically with civilization of an economy, the presence of rivers is a mixed blessing for Bihar. Frequent floods, persistent water-logging and occasional droughts create water management a challenge for the state. North Bihar constituted of the sub-Himalayan Region and South Bihar embodying the Gangetic Plains display distinct characteristics.

Kosi river, a major tributary of river Ganga, which arises in Tibet and traverses Nepal before entering Bihar, is especially linked with Bihar's struggles with nature. The trans-boundary dimension of the river and its diverse upper course in the Himalayan terrain marked by sharp slopes in terrain, climate and vegetation before it descends on the plains creates more discharges that can be controlled effectively with existent capacity. Management and efficient

utilization of the abundant water resources flowing into Bihar is difficult requiring cooperative planning on the part of the state government, the union government of India, the government of Nepal and concerned international development agencies. As shown by past many years of experiences and engineering efforts, solutions do not lie in structural methods of flood control alone. Kosi basin in Bihar is the subject of this research. Bihar is also drained and occasionally inundated by River Ganga which comes from Himalayas via many other Indian states receiving water from tributaries.

Possible socio-economic responses are also important for development planning of the river basin. It is important to take account of the profile and behavior of the target population and understand their constraints, problems and prospects to benefit. The largest section of people in Bihar dwell in rural areas and engage in agriculture for livelihood and food security. Propensity to migrate to cities for earning make women important stake-holders as well as victims of the underdevelopment. Their decisions to grow specific crops, use farm inputs and indulge in allied agricultural activities such as dairy and fishery are deeply associated with the conditions they live in and building up their means to make investment. In turn their decisions also influence the direction and intensity of the development outcomes. On the other hand, welfare efforts of the state like distribution of subsidized food and public works employment, also temper the decisions of economic agents. State actions in information and knowledge transmission via early warning of floods, extension and market intelligence, support and relief during droughts and floods and creation of infrastructure, especially to improve marketing of farm products.

1.2. Objective and Methodology

In this Report an attempt is made to understand the socio-economic profile of the area, i.e., the Kosi basin region of Bihar and its people with special focus on their livelihood mostly as farmers. The report investigates how far

agriculture can serve as a viable answer to livelihood and food security needs of the people of the study area. Analyses of both secondary and primary data collected through multi-stage random sampling are used in the methodology.

1.3. Organization

The Introduction (this chapter) outlines the motivation and the objectives of the present study and reviews the literature as a background. Chapter 2 discusses the geographical specificities of the region under study i.e., the Kosi basin in Bihar, with emphasis on natural disasters both floods and droughts through reviews of literature and media reports. Before initiating the substantial analyses, Chapter 3 deliberates on the theoretical and historical background of the issues that will emerge in the course of the study such as the changing place of agriculture in contemporary economy, food security, natural disasters, monsoon, climate change and gender. Chapter 4 presents the details of the methods used as well as all the definitions, specifications and notations used in the chapters to follow. The substantive part of the study begins from Chapter 5.

Chapter 5, 6 and 7 present the analyses of secondary data pertaining to the entire study region using district level data on Bihar gathered from official sources in public domain. Chapter 5 describes the region by its general environment, land, water, power, infrastructure and industrialization, socio-economic aspects-occupation, poverty, agrarian structure and gender status both at the average level and differentiated by sub-regions while Chapter 6 does a similar portrayal focusing only on agriculture. Base-level database (district data for the region) is built up during the analysis and presented in the appendices. The analysis involves simple statistical methods, tabulation and graphical representation. In Chapter 7, a model is estimated using the secondary data and making use of the variability that is observed over a recent time span and across the districts to explain yield rates of three major crops of

the region namely Rice, Wheat and Maize. The model is econometric which takes account of natural, infrastructural and economic factors in shaping crop yield rates. The highlight of the model is the incorporation of spatial inter-linkages such as among proximate agro-ecology and practices and that between upstream and downstream regions of the basin via effect of rainfall, slope and river not overlooking the trans-region area in Nepal.

In Chapter 8 a detail discussion is made of findings from analysis of primary data collected from selected study regions in two districts based on multi-stage stratified random sampling of households where the regions are selected purposively based on their proximity to the river and the strata are decided to give representation to different classes of farm households including the land less subject to availability constraints. The data covers various socio-economic attributes so that the residents of the regions are studied with respect to their sociology, demography, migration and livelihood as also hazards and land loss due to natural and man-made factors. Performance of agriculture measured by cropping pattern, crop diversification, crop specific yield rates, incomes and input use are covered.

The method used in Chapter 8 is as in Chapter 5 and 6. Special emphasis is placed on commercial maize farming in a situation where accessing food rice and wheat is important. Moreover as in Chapter 7 model is built to explain the determination of yield of the crops rice, wheat and maize using cross section household data and econometric model correcting for sample selection bias as not all households grow all the crops. The model is estimated using various relevant locational, social and economic variables but the identification and role of the economic incentive is of special interest in context of the peculiarity of the situation where food insecurity is a major concern, commercial farming is gaining popularity and food is availed from both the open market and the subsidized state distribution outlets.

Chapter 9 shifts the focus from crops to the farms and enquires about farm level input use, practices and incomes as well as food security status of households. In the process the marketing practices and channels are examined as also the sources of food, role of subsistence and market oriented farming and the surplus production of food. Similar analysis is also made of the allied agricultural activity of animal husbandry to learn about animal wealth, milk production, income from milk production and market orientation. In both cases of crops and milk incomes from all production with imputed value and cash incomes from selling milk are assessed.

The appendices to chapter 8 and 9 present detailed data by farm classes and also the institutions for food, agriculture and water under current policy transition. Chapter 10, which is more descriptive lays down the insights gained during field visits to the study regions in the Kosi basin as well as other areas in Bihar outside the basin and field discussions with authorities and farmers, traders and processors as also details on the villages that are explored in the primary investigations. Chapter 11 gives some concluding remarks and list out the mandated delivery of research results.

1.4. Literature Review

Socio-economic development indicators are found poor in this region. Even though, a temporal analysis of above indicators does not suggest any significant difference either between flood and non-flood years or before and after the major flood of 2008, there is a general perception that the frequency of flooding is the major reason of the under-development of this region. While this vulnerability, as any other risk, is likely to be a deterrent to investment and productivity regardless of the actual occurrence of a disaster, the role of flood in the area's dismal performance on the economic front is not universally accepted. The accountability of floods for agricultural backwardness is

challenged by a recent study (Somanathan, 2013) which found no difference among frequently flooded, unexpectedly flooded, and non-flooded villages.

Notwithstanding, some improvements in public distribution system have been noted in the region. Public distribution system is the major support system which provides rice and wheat at very low price to poor households across the country. Recently, Government of India has made significant change in the policy which increases its coverage. Apart from the above, Government of Bihar government initiated a system of coupons in year 2007 to administer the public distribution system (PDS). The system is supposed to operate as follows. Every beneficiary household is provided with 12 coupons annually for wheat, rice and kerosene, which specify their entitlements and the price they have to pay for each commodity. Coupons are distributed through village camps, organised by the local gram panchayat under the monitoring and surveillance of officials from the block development office (the process of coupon distribution is video-recorded). Every month, beneficiary households redeem one coupon each against each of the specified PDS commodities at a local fair price shop (FPS) (beneficiaries can ostensibly choose the FPS they use). Then, the FPS owner forwards coupons to the block/district level authorities in order to get next month's supplies.

Studies (Mooij, 2001, Thorat and Lee, 2003, Dreze and Khera, 2015 and many more) conducted in Bihar focusing on PDS have highlighted two major problems – One, Caste discrimination and Two, Leakages.

Mooij (2001) analysed the way PDS is implemented in Bihar and reported several problems - large-scale misappropriation of foodgrains at all levels; the distribution of cards to BPL families is unsatisfactory; the Bihar State Food and Civil Supplies Corporation is financially not able to perform its task and high level corruption and involvement of political influence. He suggested that the

improving of PDS is political process in which the well-willing stakeholders come together to act strategically.

Thorat and Lee (2005) studied two major government programmes namely Mid-Day Meal Scheme (MDM) and Public Distribution System in five states of the country (Rajasthan, Uttar Pradesh, Bihar, Andhra Pradesh and Tamil Nadu). This study reported high level of discrimination in both the above programme based on caste and religion. However, MDM was not evaluated in Uttar Pradesh and Bihar in this study because it was not implement in both the states during the study. They observed most of villages without PDS shop are in Uttar Pradesh and Bihar. In Bihar, 16 per cent villages were lacking of PDS shops while this figure was 39 per cent for Uttar Pradesh. The study finally suggested to important policies in order to tackle the above exclusion/discrimination problem - First, the government can relocate or newly locate MDM centres and PDS shops in dalit colonies or other accessible caste-neutral localities. Second, the government can seek partnerships with dalit women's groups and community to jointly implement and monitor the programme.

Dreze and Khera (2015) studied PDS leakages using National Sample Survey data from 1999 – 2000 to 2009 – 10 and estimated leakages in the range of 75% – 90% throughout the 2000s. These studies suggested that Bihar's PDS was so bad that it had virtually no impact on rural poverty. Dreze and Khera (2015) noted improvement in PDS and reported a sharp decline in PDS leakages observed in Bihar for recent years between 2004 -05 and 2011 – 12. Despite decline in PDS leakages, high level of corruptions and malpractices still persist in the state, as is reported by Choithani and Pritchard (2015) and Dreze et al. (2015).

Choithani and Pritchard (2015) assessed Bihar's coupons based system of PDS delivery using a survey of 350 households in 10 strategically selected villages, spread across nine administrative blocks in a single district of western Bihar.

Findings from the study provided a somewhat confounding and ambiguous reading of the operation of the coupon system. The study found that high rates of foodgrains leakage persist in the context of coupons. Maladministrative practices lead to many households receiving expired coupons, and FPS owners devise a series of strategies to extract profits from beneficiaries – either by way of inducing artificial shortages by side-selling PDS grains on the open market, or by requiring beneficiaries to procure a month's ration using two or three coupons.

Dreze et al. (2015) pointed that Bihar still had a long way to go in ensuring minimum standards of transparency and accountability at the ration shops. Several incidents of irregularity were reported by this study. For instance, many of the ration shops had little to show by way of an information board. The system of “doorstep delivery,” officially operational at that time, was equally notional in many places. Fixed distribution schedules, another helpful safeguard, were either lacking or ignored. Making fake entries in ration cards was an accepted practice. The dealer's attitude was often casual in the extreme, with some dealers even admitting openly that they cheated and seeking to justify it on the grounds that everyone else was cheating too.

Considering flood as major problem in the region, the government of India has approved a project comprised on embankment, barrage and canals to tame floods in the Kosi river and utilize water resources for irrigation purposes. This project yielded very mixed results – a study conducted by Government of India (1979) through its planning commission has favoured the above project, while several independent studies (Mishra, 2008, Sinha et al. 2008, Somanathan and Sonanathan 2009; Somanathan 2013) conducted by eminent scientists and social scientists have raised a question on embankment strategy for taming floods in the Kosi river. These studies along with planning commission studies are briefly discussed below.

Government of India (1979) conducted an evaluation study of the Kosi Project in 1977-78 through Planning Commission to find out whether there had been any increase in the income and employment opportunities in the Kosi basin following the construction of the embankments. It also evaluated impact on promoting health and sanitation and on the overall development processes. To accomplish the above objective, the study area was divided into five major zones - (1) Villages trapped between the embankments; (2) Villages outside the eastern Kosi embankment but within a distance of 5 km from it; (3) Villages located at the very end of the embankment; (4) Villages that were hit by the floods of 1948 but not included in (3) above, and (5) Villages trapped in the floods of 1954 but not included in any of (1) to (4) above. The evaluation study concluded that a sum of Rs. 40.32cr had been spent on the embankments till 1978 and they had protected an area of 160,000ha against recurring floods and because of them an annual loss of property worth Rs. 6-10cr had been prevented. The study had pointed out several positive impacts of the Kosi embankments - a marked improvement in fisheries, communication had improved, there had been plantations on the canal banks, cottage industry had come up, the general health of the people occupying the basin had improved and there had been an increase in land prices.

The study, however, noted undesirable impacts as far as agriculture was concerned. It said that even 20 years after the construction of the embankments, the anticipated agricultural development had not taken place. Interestingly, according to the report, the cultivated area within the embankments nearly doubled but the cultivated area in the protected countryside in zone (2) had reduced considerably because of waterlogging. There was an almost uniform decline in the per hectare yield of paddy. The area under paddy increased between the embankments but the yield suffered because of sand casting there. There was minimal use of chemical fertilizers and use of organic fertilizer was almost non-existent. The team was generally satisfied with the flood control part of the project.

Mishra (2008) and Somnathan (2013) have arrived at observations contrary to the findings of the above evaluation study. Mishra (2008) claimed that Kosi project has failed to achieve its both flood control and irrigation objective. He pointed that after construction of embankments all major and devastating floods such as 2008 etc. were caused by embankment breaches. Embankments causes to waterlogging and sedimentation problems that harms agriculture production and livelihood in the region. Poor maintenance of embankments, high level corruptions, poor design of embankments, frequent change in course of embankments due to political influences and lack of people's participation were majorly caused in failure of achieving objectives of Kosi project.

Somnathan (2013) evaluated Kosi embankments to answer a question 'whether embankments should be used to control floods' using data from 504 households in 28 villages, that were divided into three groups – unexpectedly flooded villages, regularly flooded villages, and villages that neither regularly flooded by river overflow, nor unexpectedly flooded by embankment breaches. This study suggested to three major findings – One, the gross value of crop output in the regularly flooded villages is higher than that in third category of villages. Two, mean wages of agricultural and casual workers are not lower in regularly flooded villages compared to the third category of villages. Three, these villages do not worse on measures of schooling, health, wealth and household amenities. These findings nullify one of presumption of protecting embankments that these constructions help to increase crop production and livelihood.

Sinha et al. (2008) revisited the flooding problem in the Kosi river basin and presented an in-depth analysis of flood hydrology using hydrological analysis integrated with a GIS-based flood risk mapping in parts of the basin. The study noted hydrological characteristics of the Kosi river that include very high discharge variability, and high sediment flux from an uplifting hinterland. It

further pointed out that these natures of the river have been almost ignored in the embankment strategy of taming the floods in the Kosi river and hence this strategy should be reviewed. They suggested that the basis for selection of an appropriate method has to be based on a careful analysis of hydrological data to understand the processes involved and then to extrapolate the changes in the hydraulic regime after applying the flood control measures.

Summing-up, though hunger problem in the Kosi basin has been tackling by improving condition of public distribution system coupled with initiatives taken by both state and center governments such as coupon system and National Food Security Bill etc, socio-economic condition of the people living the region is very bleak. Flood has been considered as major problem and embankment strategy has been adopted to face this challenge. But, this strategy seems to be failed and hence the embankment strategy should be reviewed. Besides, people participation and nature of rivers should also be considered in the future plan. Future plan needs to be made immediately since climate change is likely to increase flooding events in the region. Sharma et al. (2000) analysed climatic and hydrologic information to assess the impacts of existing and potential future land-use and climatic changes over the basin. Results of the study showed that the population grew at a compound growth rate of about one percent per annum over the basin during the last four decades. Despites, the study did not find noticeable trends in land-use change. Climatic data analysed for the period from 1947 to 1993 showed using parametric and nonparametric statistics indicated some increasing tendency for temperature and precipitation, however hydrological data indicated an overall decrease of discharge along mainstream Kosi River and its major tributaries. The decreasing trends of streamflow were found more significant during low-flow months. Further, the study analysed the hydrological sensitivity of the basin to possible land-use and climatic changes using water balance, statistical correlation, and distributed deterministic modeling approaches. The results indicated a stronger influence of basin characteristics compared to climatic

characteristics on flow regime. Among the climatic variables, hydrologic response was found much more sensitive to changes in precipitation, and the response was found more significant in the drier areas of the basin.

Chapter 2

The Kosi Flood Plains

The Kosi river is associated with the woes of underdevelopment in the state of Bihar where large sections of people live in poverty, risk and lack opportunities of agriculture and industrialization. People are compelled to migrate for a reasonable living creating societal pain and burden on women. Fear of flood is a part of their being, guiding them in their day to day decisions on economic and social lives.

Controlling Kosi floods is a challenge comparable to only a handful of rivers of the world. Moreover, the region is drained by many other rivers including the Ganga and the tributaries of both Ganga and the Kosi. The entire river basin is linked by the pressure created by precipitation in various geographically associated regions. The economic status and prospects of the people cannot also be seen in isolation from those of people in proximate regions outside the Kosi basin. Development of the region therefore has much to do with understanding the river, its linkages and its effects.

Because the Kosi drains a large densely populated plain area in Bihar, the understanding of the river can be effectively utilized for development only with a strong socio-economic perspective. Solutions lie in designing development policies that are based on knowledge, evidences, information exchange, federal cooperation and dialogues among nations that share the basin in a holistic and integrated manner. This is a background Chapter providing information gathered from official documents and websites, literature and media reports that describes the river and the river basin in terms of the past and present history of the river's course, geography and impacts.

2.2. The Kosi river

Kosi River¹ is a major tributary of river Ganga River System (Map A2.1). A confluence of three rivers from Tibet in China and Nepal, the river rises in the Himalayas 7000 m above sea level and drains the foot hills to the east of Kathmandu. Traversing the Nepal Himalayas, where it encounters a confluence of seven tributaries at Chatra george, it enters India near Bheem Nagar. In India, the river flows through the plains of North Bihar, forming an alluvial 'fan' and joins River Ganga near Kursela. It is a long river with a considerable history in its up-stream stages.

Kosi is a trans-boundary river among three countries; China, India, and Nepal (see Map 4.1) with a vast upper catchment encompassing six geological and climatic belts lying both in the Tibetan plateau and the Himalayan ranges and accommodating eight peaks, 36 glaciers and 296 glacier lakes. The total length of river Kosi is around 729 kilometres and its total drainage area is around 74, 500 sq kilometres, out of which only 11,000 sq kilometres lies in India.

Fed by snow in the upper reaches and by rainfall in the slopes, Kosi is one of the heaviest sediment carrying rivers in the world owing to the unconsolidated terrain that it traverses. Of the major tributaries, the Arun brings the greatest amount of coarse silt in proportion to its total sediment load. The river encounters sharp changes in gradients in its large basin in higher mountainous altitudes. About 50% of the entire Kosi basin lies above 4000m above sea level (ASL) and only 16% of the Kosi basin lies below 120m ASL which does not generate enough space to accommodate the enormous runoff and discharge created at the upstream levels. The compression is especially aggravated when precipitation increases and creates not only devastating floods but also numerous interlacing channels that shift laterally over the fan from time to time allowing the floods to spread out very widely. Controlling

¹ Kosi (also Koshi) is one of the ancient rivers of India described in mythology as Kaushiki. It is formed when Sun-Kosi, Arun and Tamur join near Barakshetra.

Kosi flood long been a testing undertaking before engineers and planners. Providing flood relief is a demanding public responsibility.

2.3. Vulnerability to Flood and the apprehension of Climate change

The Kosi basin in Bihar has been known for its floods, waterlogging and untamed environment since historical times as recorded by revenue administrators of Mughal rulers and even earlier emperors. Over time the problem, far from being solved, was aggravated with the avulsions and channeling of the river especially in recent decades. Numerous efforts at flood control could arguably have made the effects worse. The association of climate change with the floods and the droughts is tenuous and shrouded somewhere between the geographical reality and administrative lapses, the former being even termed as 'a safe refuge for hiding inefficiency on the water front' (Mishra, 2009). Presenting a more disturbing future, global warming, with its predicted effects on the snows and glaciers, provides a valuable imperative for management of disasters and their adversities.

Glaciers are additional sources of discharge in the Kosi river and its upstream tributaries. In the Himalayas the melting of glaciers and now their retreat due to climate change produce lakes that are insecurely dammed by ice or moraines. When these dams break owing to heavy rain, higher temperature or other stimulus, a Glacial Lake Outburst Flood (GLOF) can occur resulting in voluminous flows that ravage mountain landscape, residents and tourists without sparing the societies, infrastructure and environment downstream.

2.4. Bihar and its Kosi basin

Bihar, a state in the eastern portion of India, is bounded by Himalayas in the north and Chota Nagpur Plateau in the south. Until 2000 Chota Nagpur Plateau was included in the state of Bihar but now that part has become a separate state named Jharkhand. The state of Bihar is divided into two unequal parts by river Ganga which flows through the middle from west to

east. Sharing a common border with the country Nepal, and the states West Bengal, Jharkhand and Uttar Pradesh, the state presents itself mainly as a flat alluvial plain. Bihar has hills that are part of Himalayas in the northern part of the state and in the south they are extension of the Vindhya Range.

The general slope of the state is from west to east. The entire state is drained mainly by the Ganga, Kosi, Gandak, Son and their tributaries apart from other seasonal streams. The northern part of the state is host to 8 major rivers Ghaghra, Gandak, Budhi Gandak, Bagmati, Kamala, Bhutahi Balan, Kosi, Mahananda that are all tributaries of the Ganges. Due to the presence of large rivers (Map A2.1 (c)), most parts of this state are flooded in the rainy season. Land quality is very fertile due to the deposition of alluvium. Being far from the sea, the state enjoys tropical monsoon type of climate. The southwest monsoon brings rainfall from June to September. Rainfall also occurs in October and November due to Northeasterly winds. Mildly cold in winter Bihar is known for its very hot summers.

Kosi river in India flows only in the northern part of Bihar where it is typically braided in structure with a migratory behaviour due to intermittent sediment deposition. Not surprisingly, the region is sometimes described as an inland delta. The river flows towards south west but takes a short turn towards south east to flow almost parallel to Ganga in a lower most reach before finally meeting Ganga. The dynamics of the river in the region is generally described by its 'Avulsive' shifts which make the zone of impact of a flood uncertain.

The vast plain in general has a north to south slope but is flatter in the south. The distribution of the basin area in Bihar is skewed towards lower levels that form a fan whose flat surface is dotted with water-logged depressions called chauras and is dissected by numerous small channels called dhars. The alluvial plain created by Kosi is fertile, water-rich and is extensively cultivated.

Although, owing to the frequent and devastating floods, the Kosi River is known as the "Sorrow of Bihar", in practice, it delivers both benefits and

adversities. The history of the river's course and the floods that it causes, make Kosi's plains fertile with abundant ground-water, necessary for agriculture. The implications are rather tortuous. Because of the benefits of Kosi the fan is extensively cultivated and is densely populated but being flood-prone, heavy loss of life and property results from the floods. As more than one country is involved in the Kosi basin, international initiatives have been imperative for meeting the problem. The governments of India and Nepal signed the Kosi treaty to address this common and inter-related problem.

2.5. Kosi flood plains within the Ganga basin

Below the Siwaliks, the river builds up a mega-fan, some 15,000 km² in extent, with an international dimension, breaking into more than twelve distinct channels, all of which present laterally shifting courses due to the frequent flooding. The vast alluvial fan formed by the deposition which can be fertility enhancing in many cases, extends some 180 km from its apex where it leaves the foothills and the at border on to the Ganga. The Kosi alluvial fan is one of the largest in the world.

The solid load carried by the river reflects the chemistry of the mountains at various points in its basin as well as the character of human activities such as land use, agriculture, constructions, forests and garbage disposal. The river erodes its channels and transports the sediments down the steep gradients and narrow gorges in the mountains and foothills but on the plains beyond Chatra, the gradient falls as the river approaches the Ganga and deposits the sediment load.

Besides the gradient and the topography changes, another distinctive feature of the Kosi basin is that it consistently adjoins other major basins. In the far north, ridges separate it from the Yarlung Tsangpo River. In Bihar specifically, it is surrounded by ridges separating it from the Gandaki in the west and the Mahananda in the east. The hydrological linkages cannot be dissociated from the devastation and casualty of floods. The vast tract of very fertile land

in North Bihar lying between river Kamla and Bagmati is known as the Adhwara Sub-Basin which is traversed by a network of 'Adhwara' group of rivers that originate from the foot hills of Nepal and join into two distinct drainage channels and further into a combined channel that flows southwards to fall into river Bagmati. Kosi is a tributary of river Ganga and shares its basin with adjoining basins of numerous other diverse rivers that confluence with Ganga and Kosi (Appendix 2.3 (a)).

2.5.1. Recent history of Floods and Droughts

In 2008, more than three million people were rendered homeless in Bihar when the Kosi river breached its banks upstream in Nepal and changed course. It was said to be the worst flood in the state in over 50 years. Preceding 2008 floods were reported in 1998, 1999, 2000, 2001, 2002, 2003, 2004, 2007 having their toll on human lives and property. The basin is flood prone not just because of Kosi but the spate arising in the integrated river system of the basin and not just due to rainfall in Bihar but rainfall upstream.

The years 2009 and 2010 were relatively uneventful but Mahanada river breached an embankment in 2009 washing out 70 villages displacing sixty thousand people and exposing people of Katihar in North Bihar to misery. As Rice and Maize crops stood on fields of water, residents reportedly lost confidence in the repair work even while 2009 was a year of serious drought in India, and also in Bihar (see below). In 2011, an unprecedented flood was experienced in river Son as water level rose in Ganga near Bhagalpur although the monsoon in India was normal. In 2012 too rivers Gandak and Ganga exerted pressure on embankments causing a few flash floods. All these rivers being connected by channels, the entire basin seemed to be in pressure in 2011 and 2012. Meanwhile, India's disaster management, monitored by legal institutions, civil societies and the media, was becoming stronger. Flood did not spare the region in 2013 too, a year again more marked for a drought arising from insufficient rainfall and a long summer.

The threat of flood resurfaced in 2014 when landslides struck the Himalayas. In early August of 2014, the Nepalese government removed debris in Bhote Kosi, a tributary of Kosi, using low intensity blasts which in turn led to a rise of water level and heavy discharge of water in Indian Kosi. Bihar government invoked forcible evacuation in nearly 9 districts mostly of people living in the space between the river and the embankment. The National Disaster Response Force (NDRF), Army and State Disaster Response Force (SDRF) all were deployed for saving human lives in districts Supaul, Saharsa, Madhepura, Khagaria, Araria, Madhubani, Bhagalpur, Purnia and Darbhanga by evacuation and relief (PTI, 2014a). Supaul is the usual headquarters.

The governments negotiated over the means of clearing the debris. People were allowed to return home but later in the same month, the rising water levels of the Ganga and other rivers at many places in Bihar again made the Water Resources Department (WRD) watchful (PTI, 2014b). Bagmati, Burhi Gandak, Kamala Balan and Adhwara group of rivers also swelled above the danger levels as breaches in the embankments of Kamala Balan river in Darbhanga and Gandak river in West Champaran had to be restored.

Table 2.1: Flood affected districts of Bihar in 2014 and 2016

Year	2014 Flood		2016 Flood
Districts affected	Baghalpur, Madhubani, Supaul, Saharsa, Khagaria, Darbhanga and Madhepura.	Gopalganj, Sitamarhi,	Araria, Supaul, Darbhanga, Purnea, Kishanganj, Katihar, Madhepura, Bahgalpur, Saharsa, Gopalganj, East Champaran and Muzaffarpur and later, West Champaran

Source: Media reports (Various)

The following year the state had to deal with another kind of natural calamity. The earth quake of 2015 happily brought the Union and the State government together in cooperation. However, 2015 was also a year of poor rainfall in much of India and Bihar but questionable information protocols and political compulsions would determine official notification of a drought.

In the current year of 2016, monsoon was, as predicted by IMD normal in India, but as many as 9 states (Assam, Arunachal Pradesh, J&K, Gujarat, Bihar, Madhya Pradesh, Maharashtra, Meghalaya, Odisha and Uttar Pradesh) faces floods. In end July, water levels in the Kosi river network had started rising, threatening hundreds of villages in over half a dozen districts. The rain-swollen Kosi river first flooded about 50 villages in Supaul where thatched houses were washed away forcing people to abandon their homes. In the first week of August the fear of flood was again gripping villages in Supaul as Kosi and other rivers kept rising because of heavy rains in Bihar and catchment areas of neighbouring Nepal. The state government alerted engineers to keep vigil and be ready with necessary equipment and boulders to 'face any situation' and to protect the embankments. The eastern Kosi embankment that had breached in 2008 had been repaired before the season and was declared safe but swirling flood waters continued to wreak havoc in the state as two more districts of East Champaran and Muzaffarpur were declared as flood-hit. Flood hit districts are listed in Table 2.1.

Nearly 100 deaths were reported in 2016, 26.19 lakh people residing in 2,034 villages in 60 flood-affected blocks of 12 districts were rendered homeless. Thirteen districts were affected. Initial estimates put around 1.86 lakh hectares of land to be affected by the floods caused due to rise in the water levels of rivers Mahananda, Bakhra, Kankai, Parmar and Kosi but later reports said that crops spread over 3.39 lakh hectares were affected but admittedly that the quantum of loss was an under-assessment. Around 4.81 lakh people have been displaced while 2.47 lakh people were sheltered in the 412 relief camps being run by the government. The state government deployed 112 medical teams to treat sick people in the flood-affected areas dispatched to the flood-affected areas after an aerial survey. News on the misery of people was disseminated by media. There were reports of poor Dalit families living on rats for food. By August 4th the alert was withdrawn in six districts.

The situation eased with no district, blocks or Panchayts remaining under water but the Ganga and few other rivers kept flowing above the danger mark

in Patna and Bhagalpur districts. On 18th August at the time of drafting this report incessant rain in north Bihar brought back the threat in some districts even though for past three days officially, no district was under flood. River Ganga flowed above danger level at Patna and Bhagalpur, Punpun river in Sripalpur, Budhi Gandak in Khagaria and Kosi in Baltara in Khagaria district. Water entered in riverine areas of Patna, Buxar, Bhojpur, Munger, Saran and Vaishali districts. Relief camp started functioning.

2.5.2. The Causes of floods

The recurrent incidences of Kosi floods have been traced to natural and geological changes in the Himalayas. The large upper basin in southern Tibet and eastern Nepal lies on mountainous terrain which is being elevated by about 1 cm a year by tectonic forces expected to be accompanied by erosion leading to loads of sediments being carried downstream. Though the origin of the floods is not fully understood, empirical measurements of the river's sediment load confirm that the area is rising. Bhote Kosi site, associated with the flood in 2014 as also 2016, was intensely affected by 2015 earthquake.

River gradient ranges from more than 10 meters/km for major upper tributaries in the mountains to as little as 6 cm/km as the Kosi nears the Ganges. The gradient decline slows down current and the turbulence that hold in suspension the sediments which settle out on the riverbed but eventually that raises the channel above the surrounding terrain. In times of heavier flows, the river breaks out to seek lower terrain. It spreads out over the surface of a cone-shaped alluvial fan that it creates, which it again proceeds to be elevated by deposition. The cone-shaped Kosi alluvial fan, one of the largest in the world, extends from outermost foothills to the Ganga valley. Flow rates over 25,000 m³/s have been measured where the Kosi exits the Himalayan foothills.

The history of the floods also shows that rainfall during different periods of the monsoon season has been responsible for floods, but excess rainfall in the

beginning of the season, specifically in late in July in 1998, 2000, 2004, 2007, 2008 and 2012 initiated the surge. While rainfall happened to be intense in Bihar in the concerned years, both Nepal and India appeared as affected in 1999, 2001, 2008 when high rainfall in Nepal was also reported. The flood of 2014 and 2016 are directly related to rainfall and river water management in Nepal (Table A2.2 (a)). Kosi floods thus have a strong international dimension. More information is required in the part of Kosi basin north of Nepal and in Tibetan plateau of China. Meteorological data available further upstream found opposing trends that could be on account of climate change that affects mountain ecosystems and also human activities in gentle slopes leading to contraction of grasslands. In a study area at the border of China and Nepal in middle Himalayas at the site of rivers Arun, Sun and Tamur with Mt Everest at the centre weather data (Wang et al., 2012), both temperature and rainfall showed steep gradients. In the northern, middle and southern slopes, precipitation is 300 mm, 600mm and 1000-2500 mm. The vertical zonation is also reflected in the sharp transitions from glaciers to grassland to shrub land and forests, vegetation ranging from alpine to semi-arid and varying vegetative productivity. More satellite derived dataset and growing number of weathers stations will gradually help in understanding the spatio-temporal patterns in the Chinese parts of the Kosi river basin.

2.5.3. Kosi Flood structural management

After the disastrous floods of 1954, a National Flood Control Policy planned to control floods through a series of structures and river training works. Based on 10 years investigation, the Kosi project was conceptualized holistically to include a barrage to provide irrigation and power benefits to Nepal and India, to build embankments and to create a high multipurpose dam within Nepal at Barakshetra to provide a substantial flood cushion (see appendix A2.3 (b)). A Kosi Agreement between Nepal and India signed on 25 April 1954 and revised on 19 December 1966 to address Nepal's concerns are important joint initiatives.

2.5.4. *The Kosi Flood of 2008*

Despite 5 decades of flood management, Kosi continued to cause extensive flooding. The damage caused by the Kosi flood of 2008 is the highest in five decades of flood history in Bihar (Kale, 2008) and according to a Nepalese senior officer from the Ministry of Home Affairs and member of the Central Disaster Relief Committee, the worst in the entire flood history of Nepal.

On 18th August 2008, when a breach occurred at the eastern afflux bund at Kusha in Nepal, upstream of Kosi barrage, the river burst through the embankment and shifted by about 120 km eastward in contrast to the dominant westward shift of the last 200 years. Picking up an old channel abandoned over 100 years ago near the border, 95% of the river flowed through the new course submerging several districts of Nepal and Bihar. Five districts out of the 38 districts of Bihar, namely Supaul, Saharsha, Madhepura, Araria, Purnia were worst hit. Parts of Khagaria and Bhagalpur were affected. The region lacked preparedness because no significant flooding was experienced since the completion of the barrage.

The devastating 2008 flood inundated nearly 1,000 villages in these five districts. Over 500 lives were lost and 2500 went missing. More than 2.20 lakh houses were damaged, over 1,000 acres of agricultural land submerged and 19323 cattle heads were lost. For relief and rehabilitation, the State government got hundreds of crores of rupees. Several non-government organizations too came forward to support the government's programmes.

The economic and political fall-out was also far reaching. Despite the relief expenses, the people of the area have been struggling all through these years after the deluge, thousands of residents having migrated to cities in search of livelihood. Several were pushed to penury. Despite the government's assurances and constitution of committees to estimate the loss, a report on the event is still awaited after 6 years while the opposition and the flood-hit continue agitation demanding early rehabilitation.

2.5.5. Revisiting Kosi Flood control after 2008

The limitation of structural methods was exposed by the 2008 floods. After the construction of the barrage all major floods were caused by embankment breaches. Even before 2008 similar events were associated with the floods of the years 1963, 1968, 1971, 1980, 1984, 1987 and 1991 though downstream of the barrage in Bihar in India². Debates and blame-games followed the 2008 flood. Despite prior indications of an imminent embankment breach (Shrestha et al., 2010) and joint discussions of strengthening it by Nepal and India in 2001, the hierarchical communication mechanism prevented action. Though, heavy rain during the monsoon season is a main cause of floods, many have held the poor maintenance of embankments more responsible for the breaches (Mishra, 2008, Dixit, 2009, Shrestha et al., 2010) that also magnified the repercussion of the floods. The hypothesis is evidently validated by the river discharge data³.

A high level international workshop on 'River Dynamics and Flood Hazard assessment with special reference to the Koshi river' organized on 23rd March, 2011 in Patna, Bihar emphasized the problem of sediments raising a doubt on hard engineering solutions. Inappropriate technology for a high-sediment carrying river, poor maintenance of embankment infrastructure, lack of warning mechanism, institutional dysfunction and government deficit and poor capacity of respond to the humanitarian crisis made up the weaknesses of the system.

2.5.6. Recent approach at Flood management

Early warning, weather forecasts, evacuation preparedness, and quick relief to manage disasters have become more important in context of Kosi floods in

² Detailed information in http://en.wikipedia.org/wiki/Kosi_embankment

³ According to data collected from Department of Hydrology and Meteorology in Nepal, the river discharge at the time of the embankment breach in 2008 at Chatara, was only slightly higher than the long-term average of August (1985-2010) but much less than the maximum discharge that was recorded at the Chatara station between the years 1985 and 2010, which was 9,610 m³/sec on 11th August 1987.

the last few years, a fight in which media and civil societies play a significant role. The advancements in space science is highlight in this process (See chapter 3). Mapping based on remote sensing studies (Bapalu & Sinha) help to identify the vulnerable zones. Developments in meteorological, agronomic and hydrological modeling is concomitant in forecasting weather amidst climate change, understanding of the river hydraulics and hydrology and planning agriculture. Flooding in less valuable zones is becoming accepted as a measure with differentiated safety standards according to needs and ‘zoning’ off of development.

2.5.7. Structural measures not abandoned

Experiences showed that structural intervention of engineering nature too helped flood control by moderating the intensity or reducing the frequency along with other spill-over beneficial effects of off-season water supply, irrigation and electricity. However, the approach has become even more holistic. A World Bank funded project currently in process supports flood control as well reconstruction and rehabilitation. The Phase I, under implementation since March 2011, was designed to help address the short-to medium-term recovery needs of the flood-affected population through building of houses, bridges and roads to restore and enhance connectivity.

On December 8, 2015 the World Bank approved US\$ 250 Million for Bihar Kosi Basin Development Project, India to address the long-term challenges for flood management capacity and agricultural production building on the ongoing reconstruction work that started with the 2008 Kosi river floods. Under this phase the state will enhance its resilience to floods and increase agricultural production and productivity in the targeted districts. The project will also help improve the Government of Bihar’s capacity to respond promptly and effectively to future disasters. The project will help improve the overall productivity in the Kosi River Basin by unlocking the agricultural potential of the area. It will help farmers adopt innovative and

climate-resilient technologies and extend their access to markets through better roads

Departing towards structural solutions the Bihar government is again contemplating high dam construction in the Himalayas which needs the collaboration of Nepal (A2.3(c)). Containing floods in the Sapt and Sun Kosi area, which also affects the Himalayan country, it can have added benefits for Nepal to chain water for its own use, meet power concerns and sell surplus power to India while improving quality of life of its own people. India's power needs are burgeoning and Nepal can earn from feeding India by harnessing the potential in a process similar to Bhutan, another Himalayan country which used this option to assure the best quality of life to its citizens. Bihar government too feels that Bihar will largely gain from hydel projects in Nepal. However, there are dissenting voices in India, with experts saying that a high dam in a highly unstable seismic area could be a huge risk to India.

2.5.8. Managing droughts in Bihar

More recently droughts too are arresting attention. Various regions of India are vulnerable to monsoon deficiency (A2.2 (b)). High cost of irrigation makes agriculture in Bihar particularly vulnerable to droughts. It is believed that droughts would have less impact on farm economy and the farmers if irrigation becomes more affordable. Under current technology adapting to droughts requires policy recognition of drought as a natural calamity.

In India drought relief funds and disaster responses are deployed when a district is notified as drought affected. Generally speaking, declaring a district as drought hit depends on average rainfall compared with the long term average along with statistics on crop sown area. Data on vegetative and moisture index provided by the Centre are additional dimensions.

Certain civil society experts find the declaration of drought arbitrary, based on political calculation and patronage and also claim that the states tend to

be in denial. Resistance by states towards declaring drought can be attributed to avoiding legally-binding commitments. In case of Bihar the average rainfall appears higher also due to higher moisture and even floods in certain region despite clear drought-like situations in others. In general northern Bihar is prone to floods and southern part to droughts but both do get affected by the two calamities.

Table 2.2: Drought affected districts in Bihar

Year	District Number	District Name
2009	26	Arwal, Aurangabad, Banka, Begusarai, Bhagalpur, Bhojpur, Buxar, Gaya, Jamui, Jehanabad, Kaimur, Katihar, Kishanganj, Lakhisaria, Madhepura, Munger, Muzaffarpur, Nalanda, Nawadha, Patna, Rohtas, Saran, Shekhpura, Sitamarhi, Siwan, Vaishali
2010	38	Araria, Arwal, Aurangabad, Banka, Begusarai, Bhagalpur, Bhojpur, Buxar, Darbhanga, Gaya, Gopalganj, Jamui, Jehanabad, Kaimur, Katihar, Khagaria, Kishanganj, Lakhisaria, Madhepura, Madhubani, Munger, Muzaffarpur, Nalanda, Nawadha, Pashchim Champaran, Patna, Purba Champaran, Purnea, Rohtas, Saharsa, Samastipur, Saran, Shekhpura, Sheohar, Sitamarhi, Siwan, Supaul, Vaishali
2013	33	Aurangabad, Begusarai, Bhagalpur, Bhojpur, Buxar, Darbhanga, Gaya, Gopalganj, Jamui, Jehanabad, Kaimur, Katihar, Khagaria, Lakhisaria, Madhepura, Madhubani, Munger, Muzaffarpur, Nalanda, Nawadha, Pashchim Champaran, Patna, Purba Champaran, Purnea, Saharsa, Samastipur, Saran, Shekhpura, Sheohar, Sitamarhi, Siwan, Supaul, Vaishali

Source: Farmer Portal (Website)

Droughts affected the state of Bihar in 2009, 2010 and 2013 in the recent past. The drought in 2009 is said to have led to an increase in the number of poor people in the state from 2004-2005 to 2009-2010, in spite of rapid growth of gross domestic product in this period. In 2013 Bihar declared 33 districts as drought hit (Table 2.2). At 668.6 mm between 1st June 2013 and 11th September 2013 the average rainfall received in Bihar was 25% less than the long term annual average of 892.2 mm of that period. The State Government of Bihar on 18 September 2013 approved the proposal of Disaster Management department for declaring the 33 districts of Bihar as drought-hit. The rainfall scarcity slowed down the sowing of paddy, adversely affected

the paddy cultivation and cattle rearing, depleted the underground water table and undermined employment. The governments mobilized funds for the State Disaster Response Force (SDRF) and the National Disaster Response Force (NDRF) to fight drought in Bihar. The State Government of Bihar put on hold the collection of cess and land rent, electricity bills, irrigation charge as well as cooperative loans from the farmers in affected districts for 2013-14. Apart from this, the Bihar Government decided to take up relief measures for salvaging the standing crops.

India witnessed back to back scanty rainfall conditions in 2014 and 2015, affecting crop production severely and bringing misery to the farming classes. Drinking water also became scarce and the Supreme Court of India ordered rescheduling of a cricket match while news of trains transporting water was reported in media. Not less than nine states were identified as drought hit though causing serious contention.

Bihar too recorded only 122 mm of rain in the month of June in year 2015 against the normal 169 mm and generating a deficit of 28 percent. IMD reported deficient or scant rains in 23 of the 38 districts and in its forecasts it had ruled out the possibility of good rainfalls till July 17th (Kishore and Pandey, 2015). Moreover, the year was preceded by another one marked by erratic rainfall. The resistance of states declaring drought was critiqued by social activists. The Honourable Supreme Court also expressed concern over the drought or the drought like situation in the state in 2015 even as the state government, going by drought declaration manual, refused to declare it. This was the sixth year of irregular monsoon and drought-like conditions in Bihar in the last seven year and concerns arose about droughts too becoming frequent in the state and over the impact of climate change. Lack of steady progress in irrigation development, even decline of irrigated area in some periods and excessive dependence on private bore-wells in a surface water-rich region has raised questions on the accountability of the state irrigation department.

On the policy front, at the national level expanding irrigation is receiving high priority (Pradhan Mantri Krishi Sinchayee Yojana (PMKSY)). In the largest conditional cash transfer scheme in agriculture sector in the developing world, the state government provides cash subsidy of Rs. 25/liter on diesel to farmers in drought affected blocks to make irrigation more affordable (See Appendix 9). Between 2008 and 2014, despite large allocations the well-intended scheme has been poorly implemented (Kishore and Pandey, 2015). On the other hand, a year-long study of 160 farmers with access to cheap irrigation from solar powered pump-sets in Bihar showed that these farmers grew paddy in all their land in Kharif in 2013, in spite of low rainfall and reaped nearly 20 percent higher yields compared to their neighbors suggesting that affordable groundwater irrigation is essential for effective drought proofing in Bihar.

2.6 Wetland ecology in Kosi basin

Area estimates of various wetland categories for Bihar have been carried out using GIS layers of wetland boundary, water-spread, aquatic vegetation and turbidity (National Wetland Atlas: Bihar, 2010). Total wetland area estimated is around 4.4 per cent of the geographic area of the state. The major wetland type is River/Stream in Bihar (Figure 2.1). Natural waterlogged, Lakes/Ponds, and Ox-bow lakes/ Cut-off meanders are others. The inland natural dominate the wetlands in Bihar which comprise about 92 per cent of the total wetland extent while the man-made accounts for about 3.5 per cent.

Most of the wetlands in Bihar state show various types of floral and faunal diversity. The common fauna in the water are various type of fishes, amphibians and reptiles. Few fishes such as rohu, catla, mrigal, tangra, lata, etc. are widely cultured in these wetlands. A wide variety of migratory and resident birds, blue-bull, fox, deer and wild pigs are seen around the wetlands.

District-wise distribution of wetlands showed that Begusarai followed by Katihar and Saharsa, all in Kosi basin (Table A2.3 (C)), have the highest extent of wetlands about 10 per cent of the geographical area of the respective districts. The least extents (less than 1 per cent) of wetlands has been shown in Kaimur (Bhabua), Nalanda, Sheikhpura and Sitamarhi. Seasonal changes in the extent of open water is highest in Gaya district and least in Madhubani district. Supaul district even shows a slight increase in the extent of open water from post-monsoon to pre-monsoon which is due to non-seasonal rainfall. Analysis of data on turbidity reveals that Bhagalpur has the largest extent in post-monsoon.

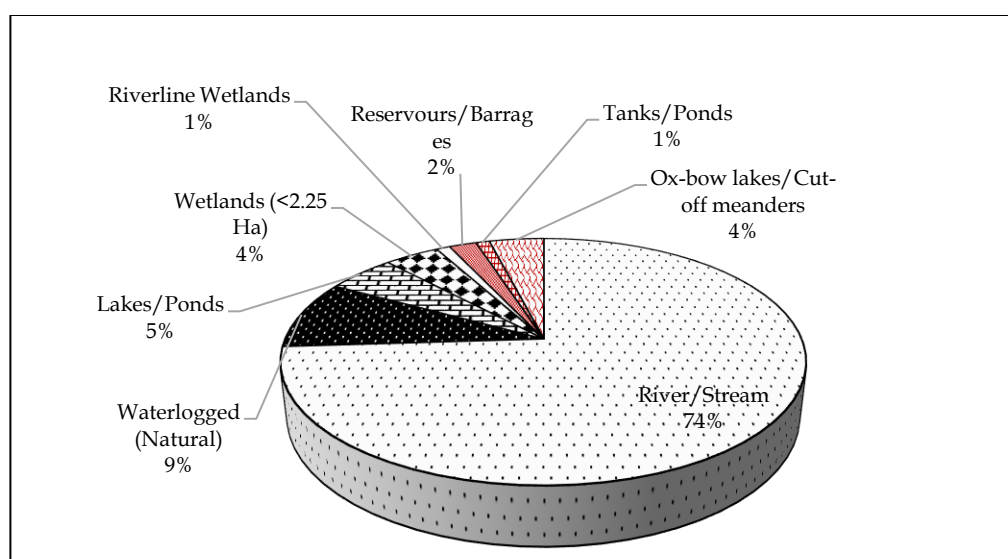


Figure 2.1: Types-wise wetland distribution in Bihar
Source: National Wetland Atlas: Bihar, 2010

2.7 Socio-economic solution of Kosi floods

Engineering solutions have not helped in ruling out devastation but and although structural measures are still not out of consideration flood alerts, real time management of embankments and channels now hold greater promise. The accountability of floods for agricultural backwardness is held with skepticism (see Chapter1) and the menace of droughts too receive increasing recognition. There is some case to believe that planning development to take care of the basic food, water and energy needs of the people with a nexus approach in the region can help because of the

abundance natural resources (Rasul, 2014). However, even with a conservative approach, droughts and floods are found to be real threats in the region that not only endanger life and assets but also discourage private investment on farming. Efficient ways of natural resource management will have to be a vital.

3. Understanding Issues, Concepts and Debates: Theories in Background

This study relates to development of an agrarian economy with a regional focus. Development of course has several facets but each region carries its own constraints, imperatives and directions. Despite wider technological advancements, nature often remains the prime determinant of options and solutions available but when the objective is welfare, human choices at any point of time, decided by social realities of the macrocosm also have to be considered. Moreover, directions laid down by political, economic and administrative structure of governance are binding in the short run.

In the Kosi region of Bihar, agriculture linked activities form the main occupation of the people if not the only one. Agriculture, unlike industry, is an open air operation and is most closely associated with natural resources and conditions. Nature creates several advantages for agriculture in the Kosi basin such as fertility, water abundance and labour power while constraints arising in the aftermath of Bihar's bifurcation from Jharkhand limits industrialization. Agriculture is a major user and re-user of natural resources as also some manmade inputs but at the same time it is contingent on nature's bounty and also vulnerable to its fury. Moreover, social transformations in the wider world and in India has changed human attitude to agriculture as a livelihood too. Choice of agriculture as the mainstay and the prime livelihood may require deliberation in this context.

Agriculture is shown to be a potential victim of global warming but intuitively, Bihar's Kosi basin, already known to be one of the worst victims of natural disasters, will be vulnerable to the implications of climate change for further up-stream regions too. It is essential that adaptation is ingrained and integrated into the development policy for the region. Flood control, mitigation of drought effects and disaster management are necessary

components of development. Migration is a traditional method of adaptation. but it can be a sign of distress and as a solution, welfare compromising especially for women.

Thus as a prelude to the conduct of the study with empirical tools it is important to recognize specific but varied issues that may arise in the course of the analysis in the particular context of Kosi basin. To understand the issues at the theoretical realm taking support of past reflections and findings that went into existing literature can be useful. This chapter is devoted to revisiting such issues summarizing the history of thoughts and events, emerging reality and empirical findings.

3.2. Agriculture as Employment and Economic sector

Agriculture has been seen as a way of life in India for times immemorial engaging family members who allocate their times to farm work and household work with little discrimination. As an employment this primary occupation has enjoyed primacy in the annals of political economy discourses, from early mercantilism to classical economics (Smith, Ricardo, Malthus, Say) and further in neo-classical economic thoughts. Physiocrats of 18th century France laid particular importance of agriculture as a sole generator of net production (*produit net*) but, while not as much pre-occupied, mercantilism viewed land and those in farming worthy of conservation and Smith saw agricultural employment as more durable than others. Neoclassical optimization exercises found application in agriculture where diminishing marginal productivity marked limited land (now water) inputs.

3.2.1. Agriculture in economic theories

In the developing countries of 20th century, farming was seen as gainful employment which included both own account unpaid work and paid farm labour. Any policy that threatened displacement of labour in agriculture was seen with suspicion. As a result, the implications of the green revolution for

employment received critical censure (Thorner, 1956, Alice Thorner, Agarwal, 1983). Evidences of rising labour demand associated with greater application of fertilizer and weeding were reassuring. Implications of mechanization for employment, female labour, family labour and casual labour aroused apprehension but were not so clearly resolved.

Around the same time the significance attached to agriculture as a source of employment was diminishing in the developed western world where the sector's share in national GDP and employment were both declining to miniscule levels and agricultural economists were turning towards the developing countries with interest as the margins of their discipline were undergoing extension into broader issues of development, poverty and more recently towards trade liberalization, resources, environment, greenhouse gas emission and adaptation to climate change.

3.2.2. Economic incentives

The rationality of producers in agriculture as economic agents and their ways of perceiving market potentials were also studied mostly with a focus on developing countries (Schultz, Nerlove, Muth, Raj). With the rise of industries in the western world, the prospect of drawing on the reservoir of surplus labour latent in agricultures for industrializing the developing economies was seen as an option. Transfer of workforce from one sector to another and from one region to another was seen as an economic response to differences in earning potential (Todaro, Lewis). Employment in agriculture in relation to its productivity was viewed skeptically in the conceptual light of disguised unemployment (Joan Robinson).

3.2.3. India's policy transitions

The productive merit of agricultural employment was under discussion in economic thoughts also in India. The celebrated debate over the higher productivity of small farms and the perceived ability of small farms to use

unpaid family labour intensely provided the rationale for India's land reform policy hinged on ceilings on land holdings. Insecurity of land tenure and particularly the dominance of landlords and money lenders on farming decisions in an inter-locked rural market were seen as discouragement to investment (Bhaduri, Bharadwaj, Harriss_White) and motivated many policy reforms such as such as abolition of zamindari and money lending occupations, tenurial reforms, nationalization of banks and creation of state regulated marketing of agricultural produce. Infrastructural development was emphasized to facilitate farming and marketing of farm produce. Building of roads, strengthening village connectivity, creation of regulated markets and expanding irrigation coverage by large and medium scale projects were manifest. Food security continued to be the driving force.

As land holding got further fragmented with population growth the viability of farming based on unpaid work as an occupation has come under questioning (Sheila Bhalla). The green revolution greatly enhanced production capabilities and delivered food sufficiency but helped repressing prices although this was countered and reinforced by government's complex price administration based on minimum support and subsidized distribution. With development process expectedly agriculture lost share in India's GDP which came down to less than 20% (Figure 3.1) as the tertiary sector moved up.

Despite all this support, the low earning potential of agriculture was creating contrasts when other sectors started responding to growth impetuses. Agriculture's performance also affected banks and marketing incentives with compounding and spill over effects on the larger economy. Unimpressed by the low paying and tedious workload younger men migrated in search of non-farm jobs bringing into focus gender issues when women's empowerment is an important goal of the human community. Infrastructure build up continued in the case of roads but stagnated in irrigation. In the post-2010 years the nature of infrastructure is undergoing shifts with the advancements of electronic communication. Agriculture is becoming a leading user of India's space, mobile, internet and other technology based products under State

policy. The post-independence policies in regards to land tenure, banking, subsidies are being reconsidered.

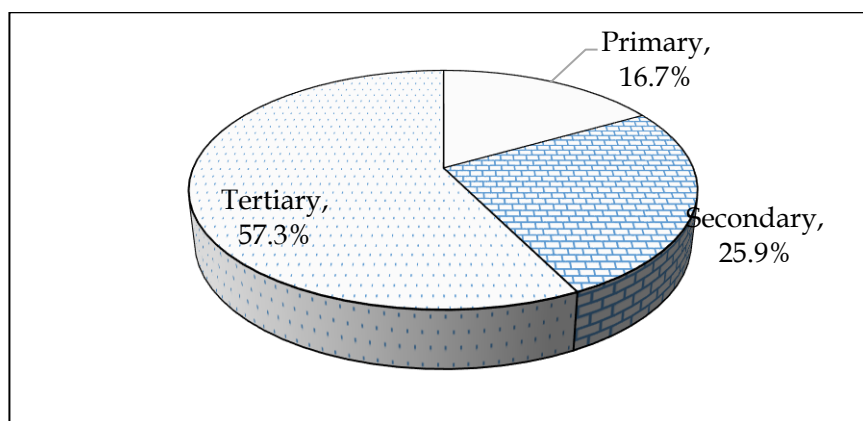


Figure 3.1: GDP at Constant Price (2004-05 Price) in India (2011-12TE)
Source: Economic Survey, 2015

3.3. Food security

According to the World Food Summit of 1996 food security is defined as -“*when all people , at all times, have physical, social and economic access to sufficient, safe and nutritious food to meet dietary needs for a productive and healthy life*” –(FAO and UN on food security). Four main dimensions of food security are – (i) Physical availability of food, (ii) Economic access to food, (iii) physical access to food and (iv) Food utilization, and stability (FAO (2006)). Food security programmes aims to reach adequate food to the public but food is consumed for nutritional intake necessary for human life. It is now argued that nutritional composition of food is also important (Vishwanathan and Minakshi, 2008, Ghosh and Guha-Khasnobis, 2008, Agrawal et. al., 2013).

India has a large population of nearly 1.3 billion and the demand for food is also high. Till recently the staples rice and wheat were recipients of primary policy significance. The food policy was formed when India used to import the staples and get food aid under PL 480 food importer but in recent years India turned surplus with large stocks and exports.

As food sufficiency was attained other dimensions of food security gained urgency. Food production was not the only means of delivering food security. Problems of access remained and government needed to create the appropriate distribution for creating physical availability in all corners of the country. Moreover, the food available has to be affordable given the purchasing power. A government organized public distribution system supported by subsidy is the institution for creating access in India but there are considerations of alternate systems such as private sector driven market distribution (that may lock up large stocks and put upward pressure on market price) and a mixed system based on direct transfer of cash to consumers or a coupon system. Besides the concerns about household access there are further issues of individual entitlements encroaching of intra-household bargaining and distribution, gender discrimination of society, specific needs of people like pregnant or lactating mothers, the sick and undernourished and children. Indian addresses these aspects through various schemes like the Antodaya Anna Yojana (AAY), Mid-day meal scheme in schools, Annapurna for older people, ICDS for children, special benefits for mothers etc. as also propagation of awareness.

In 2013 after much confrontation the National Food Security Bill (NFSA) was passed making food security a legal entitlement of the people and an obligation on the government to distribute large amounts of food to the people. To meet this legal commitment, the government needs to procure and maintain adequate stocks of food regardless of production conditions. Food sufficiency in rice and wheat does not create comfort in this regard as the threat of climate change adds uncertainty on production while implementing the large programme is likely to incur huge costs in logistics and administration. In recent times there are signals of changing this system of food distribution in near future. Also, pulses so long not in policy focus despite their nutrient content are gaining significance but India has shortage in pulses. Uncertainties and transitions therefore loom on the food security front on the distribution side while a number of programmes are in operation to enhance production (Table 3.1)

Table 3.1: Major Agricultural Policies and Programmes related to Indian agriculture

Crop Group	Major Agricultural Programmes
Staples and coarse Cereals	1. Price support through minimum support prices to guarantee high returns for wheat and rice and their assured procurement discouraged diversification to the alternative crops having high production and price risks. 2. Technology Mission on Maize in May 1995, Since then Maize is taken care of under Accelerated Maize Development Programme (AMDP). 3. A sub programme on maize based cropping system for food security wef 1999-00. 4. Integrated Scheme of oilseeds, Pulses, Oil palm and Maize (ISOPOM) from 2004. 5. National Food Security Mission (NFSM), a centrally sponsored scheme launched from 2007-08 to increase the production of rice, wheat and pulses.
Millets	1. All India Coordinated Small Millets Improvement Project (1986) 2. Millet Network of India to promote Millets (2007).
Pulses	1. Technology Mission on Pulses, 1990. 2. National Pulses Development Programme (NPDP) 1999-00. 3. Integrated Scheme of oilseeds, Pulses, Oil palm and Maize (ISOPOM) FROM 2004. 4. NFSM-Pulses-Pulses component of ISOPOM has been merged with the NFSM Since 2007-08.
Oilseeds	1. Technology Mission on Oilseeds 1986. 2. Technology Mission on oil palm 1992-93. 3. Oil Palm Development Programme(OPDP) during 1992-93. 4. Integrated Scheme of oilseeds, Pulses, Oil palm and Maize (ISOPOM).
Fibres and Materials	1. The Technology Mission on Cotton, was launched in February, 2000 to improve cotton production and productivity with internationally competitive fibre quality. 2. Approval of commercial cultivation of Bt Cotton in the year 2002.
Condiments and Spices	1. Trade Policies.
Horticatures	1. Horticulture Mission for the North East, 2001-02. 2. Extended to Himalayan States 2003-04. 3. National Horticultural Mission 2005-06.
Sugarcane	1. Before 2009, Pricing mechanism to recommend statutory minimum support prices based on cost of production 2. Since 2009 fair and remunerative prices for sugarcane taking into account margins for risk and profit of sugarcane farmers.
Livestocks	1. Operation Flood, 1970-1996. 2. Technology Mission on Dairy Development, 1989-1999. 3. Integrated Dairy Development Programme in non-operational flood, hilly and backward areas launched as a centrally sponsored plan scheme since 1992.

Source: Ghosh and Kumari, 2015

Table 3.2: Ways to improve food security in India's transition

Period	Demand	Supply
1960s 1970s 1980s	Public distribution of grains through Fair price shops, ration outlets	Green revolution with intensive use of fertilizer, Procurement at MSP and procurement prices
1990s	Targeted PDS (TPDS), Mid Day Meal and other need based schemes of distribution, focus of mal-nutrition, poverty despite plenty	Attention to resource use efficiency, soil organic quality, excess use of ground water and environmental negative impacts of surface water projects, decentralization of procurement, emphasis on market reforms and pressure to reduce subsidies on food and fertilizer
2000s	TPDS under attack for not reaching the needy, corruption and leakage	Reforms in trade and marketing, concern over stagnation in agriculture, farmer's incomes, Loss of appeal of farming and MNREGA effect on labour availability, problems of large public stocks.
Post-2010	National Food Security Act (NFSA), reports of improvement in TPDS, Technology to curb leakage of food from PDS, Problems with WTO on stocking and permanent solution, Move towards Direct benefit Transfer initiated, added significance to Pulses,	Attention to sustainable agriculture, climate change, food processing, disaster management and early warning, Pulses more important Investment plans in irrigation, promotion of solar power, curbs on misuse of urea, banking inclusion and aadhar for agricultural production, Back to back drought in 2014 and 2015.

Source:

3.4. Migration

Migration for human beings is the movement of people from one place in the world to another for the purpose of taking up permanent or semi-permanent residence, usually across a political boundary where semi-permanent residence would be the seasonal movements of migrant farm laborers. Migration, following the neo-classical stream of ideas, can be viewed as human mobility generated to equilibrate the labour markets among regions

and even countries. When labour flows out from areas of surplus to meet excess demand of other areas it leads to improvement of skill, cultural exchanges and higher aspirations and standards of living. In this perspective, both in-migration and out-migration, linked to types of skill and competitive advantages would result and conventional policies that combat outmigration are deemed to be suffering 'sedentary bias' (Mc Dowell and Haan 1997). Economic history has shown that countries with little nature-based comparative advantage have performed well as producers and exporters (Porter).

However, in reality migration is often found to be uni-directional, from poorer regions to richer one with little feed back effects. Implicit in this migration is the presence of inequality and regional imbalance. In the wider world, political unrest is a major cause of international migration. At the national and also international level, economics is often a drawing force. To combat the problems and instabilities caused by unemployment, nature-caused disruptions, indebtedness and insecure sustenance, male members of households in agriculture dominated regional economies are compelled to migrate periodically to distant urban areas carrying financial appeal.

Migration in Bihar is known to be mostly periodic or seasonal in duration, meant to even out incomes and enable repayment of fixed sums of debts taken under emergency. Members from severely poor households are more inclined to become inter-state migrants than those of intra-state migrants. When the ties with his roots are conserved through the women in an 'unsettled relationship' (Papademetriou and Martin 1991), migration is shown to create social problems including work-load and loneliness, borne disproportionately more by women (Datta et al 2011). Male migrants also become victims of health problems, discrimination and isolation in target destinations due to cultural and linguistic mismatches, deprivations of sufficient housing and health facilities, harassments from law enforcers and municipal administrators and separation from family (Deshingkar et al 2006). Cost incurred in traveling, the period of absence, fear of theft, loneliness of

members, burden of household work and lack of women's safety rise with the distance between target and home sites.

Bihar is known for net-outmigration with a large section of people migrating for informal and low income jobs like rickshaw puller and unskilled factory jobs. The destination was usually a neighbouring city like Calcutta in the east in earlier times but over time it has moved to distant places like Delhi, Punjab and Maharashtra. Those in defense argue that migration is a long-standing practice (Haan 2016) of Bihar associated more with developments elsewhere in the country such as the rise of industries, demand for farm labour, recruitment in public services, need for cheap and unskilled labour, and trade with Nepal and social capital in the form of networks and contacts of migrants rather than specific economic and social features of migrants (Sharma 1997 Oberai et al 1989, Connell et al 1976). However, lack of development and persistent competitive disadvantage in drawing investment will tend to keep wages low sustaining income differential that stimulates migration.

The growing propensity for migration in Bihar (Kumar and Banerji 2010, Deshingkar et al 2009*) has been drawing attention. Bihar remains relatively underdeveloped economically while Maharashtra, Karnataka, the national capital region (Delhi and adjoining areas), and Gujarat are becoming more diversified toward manufacturing and service sectors and have achieved higher per capita incomes. Bihar was once described as an 'extreme case of what has happened to a region subjected to societal stagnation, economic exploitation' (Das, 1992) but the extent of improvement in the situation needs to be assessed.

3.4.2. Labour differentiation

Moreover, migration is no longer a simple outcome of demand and supply mismatches of undifferentiated labour because the quality of labour demanded is also rapidly gaining significance in employment choices. The usual pull and push factors behind migration is supplemented by an added

dimension when the tedium and drudgery of unskilled farm-work loses social esteem and fails to appeal to the youth who increasingly look for non-farm, mechanized and service sector employment. This is a nationwide phenomenon publicized by NSSO 2003 (see Chapter 5) finding that 40% of farm household would like to change occupation if given a chance but Bihar is no exception. Women with lesser mobility are left with little choice but to indulge in manual farm work on family farms or as wage labourers. In turn the employment choice and the migration issue creates issues of gender equity.

3.4.3. Migration and Climate change

Migration has been a constant feature of human community over centuries creating the history of humankind and will possibly remain so. Climate and natural disasters have pushed people (and animals) out of certain regions into more habitable and pleasant ones in terms of weather, food and good life. The realization of climate change also creates anticipations of further dimensions in climate change. Thus while migration owing to failure of governance and administration may be corrected, it may not be eliminated to a large extent given natural constraints and could even form part of policy.

3.5. Gender

In 1993 the UN's Human Development Report found that the quality of life of women was unequal to men across the world but the gender disparities were more sharp in some countries than in others. Feminists complained that social discourses have historically been made with male perspective and missed the gender dimension. While feminists have consistently influenced the course of the history of societal norms especially relating to franchise and employment, many real initiatives actually started from the 1975's World Conference on Women in Mexico. In 2000, the Millennium development Goals set by 191 member states of UN General Assembly and further, the

Sustainable Development Goals of recent times set the agenda for women's development.

Gender roles are deeply tied with tradition. The division of labour was culturally constructed for normal social and economic life and was a subject of onslaughts that brought cultural transitions over years. In agriculture women's role is often associated with technology or the so called plow culture (Boserup 1970). India's Five Year Plans had seen women as objects of welfare and scarcely as human resources, and the events of 1970s changed the course of the Plan history in the country as development began to be monitored with a gender dimension. Meanwhile the recognition of missing women (Sen 2003), biased property norms (Agarwal, 1994) and recently the concept of feminization of agriculture associated with male migration in response to urban bias in development and loss of appeal of agriculture combined with the drudgery of farm labour undertaken by women and the wage bias they suffer brought the focus of development on women in agriculture. The agricultural extension known for its male bias became more gender sensitive. Farming techniques (like transplantation) received attention for improvement and agro-enterprises were encouraged. Besides rural development would be seen as a means of reducing migration which in turn could reduce the burden of work and alienation that women suffered.

3.6. Monsoon and issues of Water management

India as other parts of south Asia has a monsoon dominated climate influenced strongly by seasonal winds that bring rainfall in specific months in the year. South Asia is also one of the leading regions afflicted by hunger and mal-nutrition comparable with Sub-Saharan Africa. Food production and agriculture in general depend on the normalcy of monsoon. Monsoon brings copious rainfall in the sub-continent but with occasional years of scarcity and drought and considerable seasonal variation accompanied by regional scale droughts and floods. Understanding monsoon and coping with its vagaries are a challenge for the human community.

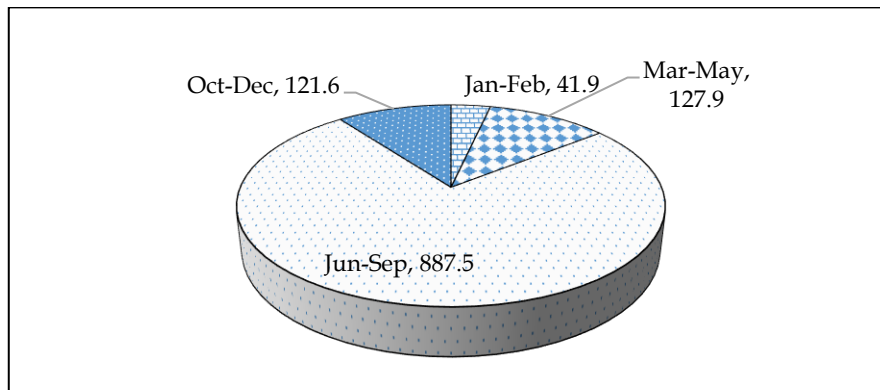


Figure 3.2: Normal annual rainfall distribution in India
(total=1179 Millimeter)

Note: Average of 1901-2013, Source: Open Government data Platform India (Website)

India's annual normal rainfall is 1179 mm of which a large 887 mm (about 75%) is received in the months June-September in the (south-west) monsoon season (Figure 3.2). The monsoon season varies in month coverage from state to state, it starts earliest in Kerala in south but within the month of June much of India is covered by monsoon clouds. Though monsoon is known to retreat from India in September, incidences of heavy monsoon rain is often reported even in October in recent times. Thus arrival, in-season continuity and departure of monsoon are all indeterminate both temporally and spatially. Climate change can make these dates even more uncertain in times to come (Ghosh, 2011). Rest of the annual rainfall in India is received in the immediate post-monsoon, the pre-monsoon months and the winter seasons through western disturbances, local disturbances, cyclonic depressions and reversing monsoon. India's climate is characterized by droughts, floods, tropical storms.

Part of the rain water seeps into the soil to be stored in the aquifers as ground water before it is used up by human extraction or is percolated to other water-bodies. The geology of the terrain and human water management determines the rate of seepage. Agricultural practices including the choice of crops and irrigation methods are important influences on water conservation. Water also

finds its way to rivers and other surface water bodies that further recharge ground water by seepage through soil.

3.6.2. Water use strategies

Ground water is drawn for cultivation by simple dug wells, powered tube wells and other means and the digging gets deeper as the water table sinks. A favourable water table in principle helps to create a rich, intense and diverse cropping system. In the absence of due caution the water table goes down and the situation turns into a catastrophe when rainfall falls severely short of the normal. Caution would warrant a regulation on the type of use made of scarce water resources, type of crops grown in agriculture (such as growing water intensive sugarcane or rice in dry areas), profligate use of ground water owing to low cost energy supply and maintenance of water bodies including tanks and wetlands (see below). The Indian government has a programme National Agricultural Drought Assessment and Monitoring System (NADAMS) to manage drought. India experienced severe departure from normal monsoon creating droughts in 1979, 1987, 2002, 2009, 2012 and 2015. Other indices like aridity index and precipitation index are also used for measuring drought intensity. There are protocols for declaring a district as drought affected making it eligible for States support. Technologies are developed for water conservation. Irrigation by non-conventional means like use of sprinkler and drip is one way while innovative methods of cultivation such as zero tillage and direct seeding of rice and the SRI method are also addressing the problem of drought management (Uphoff).

3.6.3. Use of Surface Water

Extraction of ground water signals overexploitation of scarce water resources but excess surface water can create water-logging when extraction can be useful for controlling land degradation even when utilizing water for agriculture. This is conjunctive irrigation with multiple benefits. The ground water reserve is measured by the water table which is the distance below the

land surface at which water is struck. Although endowed with a network of canals India at one time depended mostly on surface water irrigation for its agriculture, but over time ground water gained prominence and surpassed surface water.

After independence India launched multi-purpose projects not solely for irrigation but combining of flood control, electricity generation, recreation and other objectives. The surface water projects, mostly implemented by government, involved large investments and had vast spatial impacts not all of which were nice or beyond controversy. Many civil society groups protested on environmental effects of surface water projects, possibilities of earthquake and displacement of residents and farmers. As surface water programmes slowed down coupled with poor efficiency of utilization of the existing ones, private efforts picked up. The proliferation of diesel pumps enabled extraction of ground water with private investment. Governments supported these efforts with cheap or even free electricity supply, subsidies on diesel and distribution of solar power harnessing equipment especially to help farmers to cope with droughts.

Agriculture is and will remain to be the most dominant user of water in India (Figure 3.2) but official projections foresee a decline in its claim by 2050. At present agriculture draws over 83% of the water used in the Indian economy and industry and domestic uses each accounts for only 5%. Irrigation is the way to use water to raise crops. Well is the major source of irrigation in India (61%) drawing on extracted ground water while surface water though abundant and once dominant, is becoming difficult to exploit due to environmental complexities.

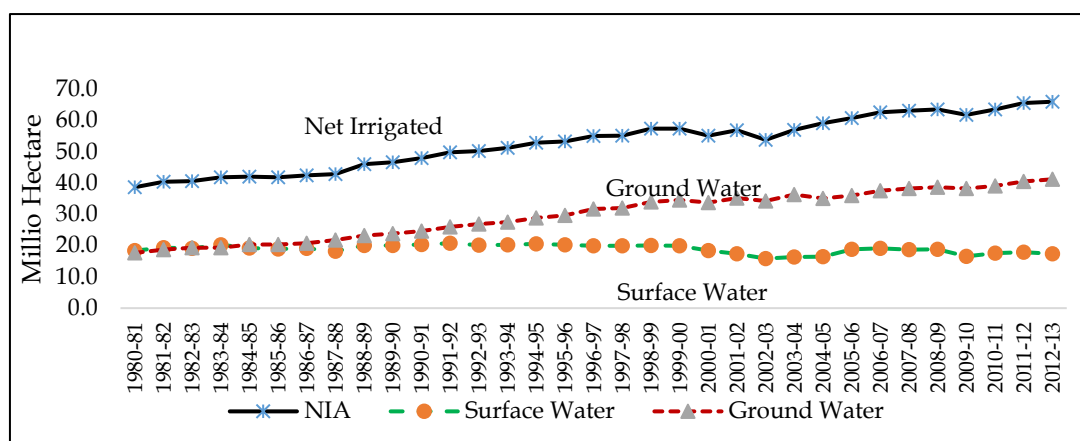


Figure 3.3: Tendencies in ground and Surface Water Irrigation in India
Source: MoA (Website)

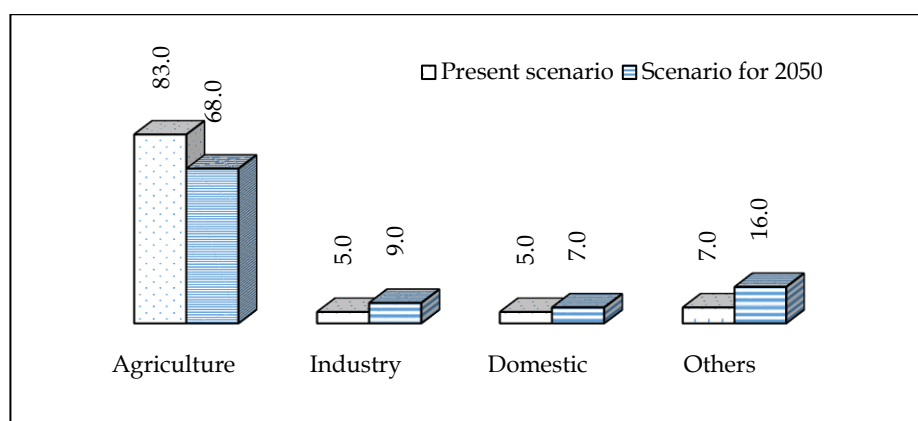


Figure 3.4: Comparing present and future water use pattern in India
Source: Planning Commission, Government of India, New Delhi.

3.6.4. Controlling Floods

Floods are probably the most recurrent, wide spread and frequent natural hazards of the world. They accounted for 28% of the number and 40% of the economic damages due to world natural disaster in the 100 years ending 2004 (Lin, 2007). After Bangladesh, India is the worst flood affected country in the world accounting for 25% of death due to floods, North Bihar being leading in this respect (Bapalu and Sinha)

Apart from extreme weather events natural elements like erosion, composition and consolidation of varied eroded materials, landslides, the consequent silt load and the structure of channel as also human economic activities like deforestation, cultivation on the slopes, construction works related to tourism, electricity generation and flood control itself determine and add to the river flow and load. Farming and habitation at locations close to river beds enhance losses while efficient community or administrative systems for flood alert and rescue restrict the scale of calamity. Engineering of landscapes for artificial flood control is most dominant.

Traditionally, people coped with floods in ways that only minimized the damages. Drawn from experienced communal systems of weather watch, flood alerts and calamity preparedness were essential. They called for strong community bondage. Urban-rural cooperation often through dedicated relief efforts of charity, religious and other social organization, was also important. Since all this could hardly preclude human suffering, this risk management was essentially post-event which was distressing (Ghosh, 2010). Deaths, property losses, crop damages, homelessness, debts, diseases, malnutrition and migration to cities were unavoidable. Support came from relief and charity.

3.6.5. Engineering for flood control

With the development of science and technology, engineering efforts were directed to control floods. Four common methods began to be adopted to engineer river hydrology and mitigate the ill effect of overload in a river. They were (i) building embankments to delay overflows to river plains, (ii) upslope afforestation to minimize erosion, (iii) constructing dams and reservoirs to to manipulate river flows and contain excess water temporarily and (iv) channel improvement by dredging and other engineering means to increase the carrying and discharging capacity. Mostly meant to confine river to a fixed channel, they also included upstream reservoirs that served irrigation needs and generated hydroelectric power. Experience was to show that all the means

have their limitations. Man can tamper with the forces of nature only so far as they do not clash with nature's laws. Preventive efforts, even if effective, were short term flood procrastinations but they could make the impacts of floods when they occur more devastating.

Modern engineering intervention on flood in India dates back to 1850s with the construction of Damodar Embankment. Building embankments is an old method but their evils and unsustainability were well known¹ (Ray, 1954) and in India too embankments have been condemned as wastage of resources but they continue to be the lifeline against Kosi floods even today though with controversies even among local residents.

That structural interventions did not make much difference led to intense technical debate in colonial times but flood control received serious consideration only in 1938 with Nehru's interest so that construction of DVC and Hirakud dams began immediately after independence even after they were termed 'satanic chain' (Willcock, 1930). There is no confirmation that between 1953 and 2005 the structures reduced flood (Sengupta) but developments in USA had set the example.

Following spectacular advances with cements, motor and concrete in early 20th century when the celebrated Tennessee Valley Authority (TVA) inspired flood control projects world-wide, it was hoped that flood would be relegated to history. Dams and reservoirs reduced the flow in river which however were found to adjust its channels according to requirements of the flow. The hope faded away when floods continued in the valley and TVA started to 'keep people away from flood'. Realization dawned that dams and reservoirs could

¹ Embankments caused gradual rise in the river bed and flood heights of China's Yellow river necessitating their further raising so that the river flowed on a high ridge creating catastrophic fear of submergence. Embankments on Mississippi in US had to be raised 17 times between 1777 and 1927 and finally embankments had to be constructed in new sites. That flow of water by gravity is a fundamental truth. A river tends to concentrate in narrow deep channels with discharge capacity.

only moderate floods which were inevitable and construction of dams for flood control lost appeal.

3.6.6. Non-structural protection and Zoning

Studies on water resource management presented different strategies categorized as structural and non-structural, the latter package including early warning, insurance, shelters and also zoning. US regulates land use and development to control losses from floods. This approach presumes that preventing economic development in disaster prone areas is one way of limiting damages.

In India the central water commission (CWC) releases data on damages to crop, houses and public utilities due to flood which helps demarcating vulnerable zones. Central guidelines on 'Flood Zoning' are produced in India to mitigate losses. In 1975 the Centre prepared a model Bill for state governments to follow and the Rastriya Barh Aayog (RBA) recommended zoning of flood prone areas for implementation by states. On an average the RBA marked 7.6 million hectares as affected by flood every year and a total flood prone area of 40 Million hectare now reassessed to 45.4 million hectare. Residents and agencies are advised not to invest in these areas. States could hardly do it. With little exception, states hardly follow the guidelines. Rather than lack of seriousness, impracticality of the solution can be a likely reason for the reluctance of states. Notifying vast areas to be out of bounds of development may not be a sensible option for the states when the residents are poor farmers with little alternatives in land scarce circumstances. Zoning would also serve no purpose if states followed the guidelines but used highly demanding standards to reduce the notified vulnerable zone and if floods continue over region beyond these patches (National Wetland Atlas: Bihar, 2010).

Insurance is a financial means to protect economic wellbeing and diversify risk. Even as the TVA had proved less effective in controlling flood than

expected, the highly developed insurance sector in US too refused coverage, making reconstruction of affected areas difficult and forcing government to step in with outright grants or undertaking its own insurance services. The France government too enacted mandatory 'catastrophy' insurance scheme which also proved to have limitations. In India, after decades of deliberation, a crop insurance programme was launched in the early 2000s. To make it practicable the programme is associated with institutional credit and laced with compulsions but till date success has been limited. It has been modified over time with the latest strengthening effected in recent times.

3.6.7. Disaster management

UN declared 1990 to 2000 as the International Decade for Reduction of Disaster persuading governments to take up disaster management with greater significance. Early success was limited. A Disaster Management Act, 2005 was legislated in India instituting a National Disaster Response Force (NDRF) as a specialized force for response to disaster. Though the responsibility for disaster management lies with the state in India's federal system the Ministry of Home Affairs (MHA) became the 'nodal Ministry' in the central government for providing aid and assistance to the affected state, deploying, at the State's request, service of the Armed Forces, Central Paramilitary Forces and the NDRF and making available communication, air and other assets needed when 'calamities of severe nature' occur.

Disaster management in India is making progress especially with the NDRF in place and with improved communication technology including intense media coverage. Insurance, a financial instrument that is generally useful for managing losses due to unforeseen hazards is clearly unviable as a solution for catastrophes of large dimension in risky zones owing to adverse selection and covariate risk. Private insurers are rather unwilling but a vast improvement of flood prediction and premium rates sharing by the government may interest the sector to develop index-based schemes. Some progress is made in early warning but much more has to be accomplished

which requires Centre-state and international cooperation and information exchange. Public grants and aid are mostly unavoidable but minimizing corruption remains a challenge. Finally, to reduce both hazard and vulnerability, a right mix of two types of strategies by spatial requirement will have to be used. Structural measures are recognized to have moderated the intensity of flood hazards though not they did not eliminate floods. Developments in engineering solutions continued in the post-1950 years but their effectiveness too needs to be assessed with care.

3.7. Climate Change

The awareness of global warming and a possible climate change in planet earth emerged with scientific recordings of temperatures and the resultant Keeling curve that showed the gradual rise over more than century of industrialization of economies. Climate change (CC) is a change in the statistical distribution of weather over reasonably long periods of time. Since climate change has been a natural occurrence and re-occurrence in geological history of the earth, UNFCCC defines it as the same change that can be attributed only human activities that lead to emission of Greenhouse gases (GHG). As CC was admitted even by giant corporate private emitters and gained recognition in policy dialogues at the transnational level, Global Circulation Models (GCM) were constructed to project status at future time periods subject to various scenarios of human activities (Bhatt, 2016).

CC it is agreed is expected to have profound effect on the future of human kind and its welfare. Although not all implications are negative, there are regional and inter-personal differences. The poorer countries and the poorer sections can be more vulnerable to the effects of CC which to an extent is irreversible but their welfare prospects are also likely to be compromised by the act of curbing CC.

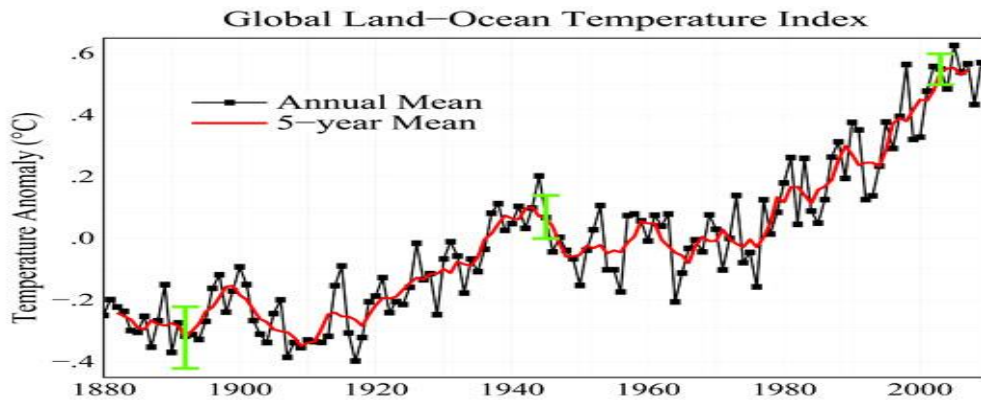


Figure 3.5: Global Warming

Agreements, treaties, conventions were struck to combat both CC and its effects relying on differentiated approach to mitigation and adaptation. The route to climate agreements is hardly smooth as it call for investment and cost which are varying across countries, regions and sections. Initiatives at mitigation met resistances as amply demonstrated by the course of the renowned Kyoto Protocol. Flexible mechanisms were designed to create incentive to cut emission and encourage developed countries or resourceful entities to help in adaptation in poorer countries but progress is slow The closing briefing remark of Doha conference in 2012 was -“The current pledges under the second commitment period of the Kyoto protocol are clearly not enough to guarantee that the temperature will stay below 2 °C and there is an ever increasing gap between the action of countries and what science tells us". Intended Nationally Determined Contributions (INDCs) of the countries provided some hope for 2030 target. The most recent conference of United Nations on Climate Change held in Paris in 2015 negotiated a global agreement on the reduction of emission as part of the method for reducing greenhouse gases.

Agriculture can be deeply affected by CC. Carbon, the dominant component in GHG can help productivity by carbon fertilization but the effects are not accepted in consensus. The temperature rise itself is a serious factor though with both positive and negative effects as crops have their own specific

biological optima of heat demand at various times of crop maturation. The rainfall pattern in the world can be seriously destabilized by CC both spatially and temporally leading to extreme events like floods and droughts in various parts of the world or country or region at the same time as also frequent incidences of the extreme events over time. The impact on monsoon deserves to be understood more clearly given that monsoons are complex phenomena affected by various local factors and pollution. Besides creating water stress, CC is feared to cause melting of ice in polar ice-caps and glaciers in higher altitudes. Natural region will be affected in distinct ways but all this suggests that mountain ecology will be hurt by CC by changing the river hydrology and rainfall distribution. Bihar has been affected by both floods and droughts often at the same time and CC through its direct effect in the region and by its impact at upstream levels can ring a further alarm.

3.8. Ecological conservation: Wetland

Serious concerns being voiced to conserve and preserve the natural resources of the world. Geospatial database of natural resources become important in this context. Wetlands as areas of land that are either temporarily or permanently covered by water, are one of the crucial natural resources. They are neither truly aquatic nor terrestrial and can possibly be both at the same time depending on seasonal variability. Because of their transitional nature, the boundaries of wetlands are often difficult to define. It is the presence of water for a significant period of time which is principally responsible for the development of a wetland. Wetlands exhibit enormous diversity according to their genesis, geographical location, water regime and chemistry, dominant plants and soil or sediment characteristics.

One of the first widely used classifications systems, devised by Cowardin et al (1979), was associated to its hydrological, ecological and geological aspects, such as: marine (coastal wetlands including rock shores and coral reefs), estuarine (including deltas, tidal marshes, and mangrove swamps), lacustrine (lakes), riverine (along rivers and streams), palustrine ('marshy'-

marshes, swamps and bogs). India has 15.26 million hectares of Wetlands constituting 4.6% of total land area (Murthy et. al 2013). Of this the largest component is river and stream (35%).

3.8.2. Wetland conservation Laws

By Indian law, wetland encompass a broad range of ecosystems characterized by bodies of water like lakes, ponds, rivers or marshes, and their surrounding bio-networks. They are breeding grounds for fish and fowl, they store and recharge groundwater, and act as buffers against storms and floods. Wetlands are nature's measures against both droughts and floods, of which India has repeatedly been a victim.

Despite their vital importance to humans, across India, wetlands are seriously threatened by reclamation and degradation through processes of drainage, landfilling, discharge of domestic and industrial effluents, disposal of solid waste, and overexploitation of the natural resources that they offer. In its effort to save and protect wetlands, the Ministry of Environment, Forest and Climate Change has invoked Article 51A of the Constitution, which makes it a Fundamental Duty of every citizen "to protect and improve the natural environment including forests, lakes, rivers and wildlife". India is one of the 169 signatories to the Ramsar Convention on Wetlands, signed in Ramsar, Iran, in 1971, an intergovernmental treaty that provides the framework for national action and international cooperation for the conservation and wise use of wetlands and their resources. There are 2,241 Ramsar sites across the world, including 26 spread across India from Wular Lake in Jammu and Kashmir to Ashtamudi Wetland in Kerala, and from Deepor Beel in Assam to Nal Sarovar in Gujarat.

In 2010 the Centre created the Central Wetland Regulatory Authority (CWRA), headed by the Secretary, Ministry of Environment, and consisting of bureaucrats and experts. Wetlands had to be notified with timelines, activities like reclamation, constructing permanent structures within 50 m, setting up

or expanding industries, throwing waste, etc. were prohibited in wetlands and 12 activities including fishing, boating, dredging, etc. were restricted without prior permission from the state government. The implementation was poor owing to the complications created by the interpretation and opposing interest groups. In March this year, a new set of rules intended to protect wetlands was created. The Draft Wetland (Conservation and Management) Rules, 2016, seek to replace the older Wetland (Conservation and Management) Rules, 2010, at a time when several petitions on the implementation of the 2010 Rules are pending at the National Green Tribunal (NGT). The draft rules propose the removal of CWRA, do away with environmental impact assessment at the centre and transfer much of the power of notification and implementation from the Centre to the state governments.

Chapter 4 : Data Sources, Methods and Specifications

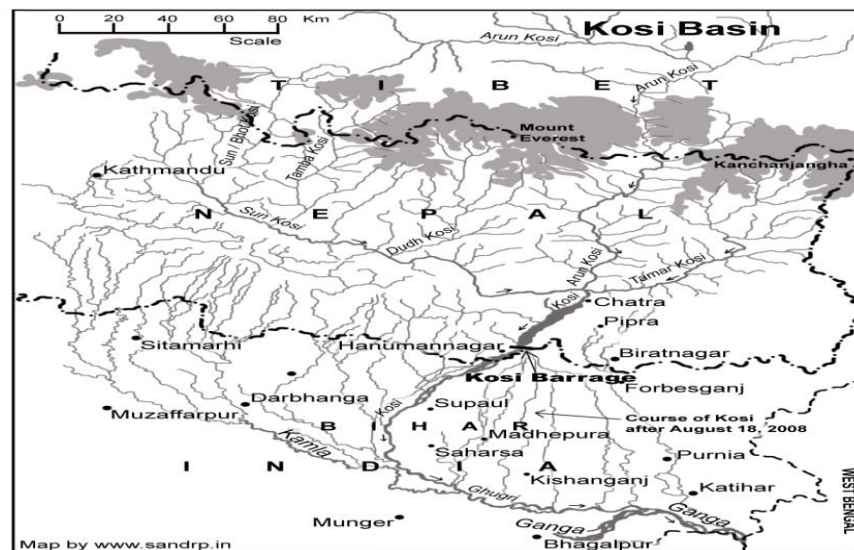
This study is based on both primary and secondary data supported by interactions and insights gathered from field visits. Secondary data is collected only from official Government of India and state government websites. Sample survey of households conducted in two Kosi basin Region (KBR) districts provides the primary data supported by interactions at markets, with traders and processors. Analysis of information is done using tabulation, mapping, simple statistical tests (t-tests) on differences and econometric regressions. Field visits were made to gain ground level insights to enable interpretations and resolution of quantitative results. Data sources, sampling technique, specifications, equations and econometric modelling make up the methodology which is explained in this Chapter.

4.2. Secondary Data sources

Secondary data are collected from different government official sources which are mostly available in public domain. To identify the sub-state basin region, an administrative district has to be the unit of analysis requiring creation of a computerized data base of information collected at district level. State and country level information are also used to compare this region with other regions of the state of Bihar, other states of the country and with the country average.

Different socio-economic and environmental dimensions such as population, income, land use, agriculture, water, poverty and food security are covered. District level data are collected also for weather variables but information on climatic variables particularly Rainfall data is also required for the Nepal part of the Kosi basin to account for upstream and downstream linkages within the

region. Heavy rainfall in Nepal frequently causes floods in the study region (see Chapter2). For this reason, rainfall data from neighbouring district Biratnagar (Map 4.1) in Nepal is required but disabling limitations were encountered. For example while average monthly data of Nepal rainfall is available (Chapter 5) rainfall at the district level is incomplete with crucial gaps. In lieu therefore, based on data availability, the level of annual extreme of Biratnagar is considered as trans-boundary upstream rainfall although admittedly this is an unavoidable limitation of the study. Further upstream rainfall in China is not used but may be explored in future research



Map 4.1: The Kosi basin in trans-boundary perspective

4.3. Classification of region by altitude in River basin

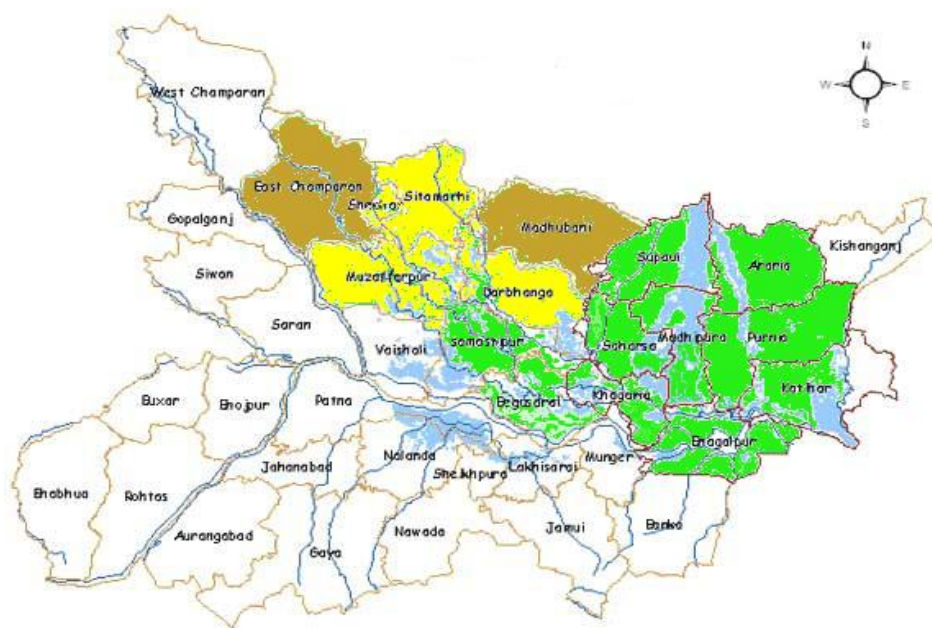
Sixteen districts of Bihar are first identified to be affected in varying degrees by Kosi hydrology. Bihar has a total of 38 districts (Map 4.2) including capital Patna, of which 16 constitute Bihar's Kosi Basin Region (KBR) and the remaining 22 Non-Kosi Basin region (NKBR). In reality there are several river basins in the state and a shock in any basin will spill over to other so that drawing a sharp boundary of any one basin is not easy. The identification of

the 16 districts including Supaul, Araria, Saharsa, Darbhanga and Madhubani is done in consistency and consultation with the larger research work done by ICIMOD.



Map 4.2: Districts of Bihar

The sixteen KBR districts are further categorized into three sub-regions called Upper-stream, Mid-stream, and Lower-stream based on altitude (Table 4.1). The altitudes vary between 70 to 370 feet in which districts in the altitude range 160 to 200 feet are classified as mid-stream (MID) and those above 200 feet as upper-stream (UPPER) while the remaining districts constituting as many as 10 out of the 16 districts located mostly south and east of MID and UPPER are classified as the Lower-stream (LOW) region. The categorization of sub-region is presented in the map of KBR (Map 4.3) and associated major statistics in Table 4.1.



Map 4.3: Classification of Kosi Districts into three sub-region
Note: Prepared by Team using GIMP 2 Software. Brown: UPPER, Yellow: MID and Green: LOW

Table 4.1: Classification of Bihar Kosi Basin region (KBR)

Sub-Basin (altitude in feet)	Districts	Latitude (degree)	Altitude (feet)	Population	Geographical Area
Lowstream (LOW) (70-170)	Bhagalpur	25.24	141	2.90	2.72
	Begusarai	25.42	135	2.80	2.01
	Khagaria	25.50	118	1.60	1.60
	Katihar	25.53	70	3.00	3.11
	Purnea	25.78	118	3.20	3.35
	Samastipur	25.86	118	4.10	2.80
	Saharsa	25.88	135	1.80	1.76
	Madhepura	25.91	148	1.90	1.92
	Supaul	26.11	112	2.10	2.55
	Araria	26.13	154	2.70	2.90
	LOW	25.74	125	26.10	24.72
Midstream (MID) (171-270)	Sheohar	26.51	174	0.60	0.46
	Darbhanga	26.17	171	3.80	2.71
	Sitamarhi	26.59	184	3.30	3.37
	Muzaffarpur	26.12	200	4.60	3.37
	MID	26.35	182	12.30	9.91
Upperstream (UPP) (271-370)	Madhubani	26.34	270	4.30	3.78
	East Champaran	26.60	370	4.90	4.61
	UPPER	26.47	320	9.20	8.39
KBR	Total	26.18	209	47.60	43.02

Source: MoA (Website), Census of India (2011) and Wikipedia.

Although geographically speaking deeply associated with mountains, KBR is largely a plain area. As noted in Chapter 2 the topography of the region is biased towards the flatter segments. The KBR makes up 43% of Bihar's area and is home to 48% of population (Table 4.1). The LOW region has the largest shares of land (25%) and people (26%) though the MID region, where very small farms make up 94% of the holdings, is most densely populated (highest population per unit area) and is intensely cultivated (highest crop area per unit physical land area). The largest region by geographical area is in the downstream location LOW where agriculture is more varied. This region is therefore taken up for primary survey in the next stage.

4.4. Methods for Secondary data analysis

Data permits exploration of both temporal and spatial dimensions of the identity of KBR combining cross section (district level) data and (annual) time series data. For temporal analysis, recent information has been used to maintain contemporary relevance in times of socio-economic dynamics. Past data are also used as necessary. Poor availability of data for remote past (before 1999) is a constraint but lack of updated data to current levels, specifically for post-2011 years at the district level is a greater limitation.

Treating a decade as a temporal distance worth comparison, the year 2010 was compared with the past information represented by the year 2000 using three previous consecutive years' averages (trienniums) of each variable with the reference year as desired mid-point. The reference years are respectively 2009, 2010, and 2011 for the terminal year 2010 and the years 1999, 2000, and 2001 for the starting year 2000. These years were chosen in this study because the most recent information is available mostly till 2011-12 (at the time of analysis) and past information is not available for years before 1999. In some cases information is not available for year 1999 and the year 2002 was considered as its replacement, shifting the triennium for 2000. The choice of the time-frame

enables us to see the impact of a massive agricultural development programme initiated by the state government in 2008 under the Roadmap for Agriculture Development (initially 2008 to 2012 now 2008-2017). Under this roadmap, the state government has been paying its attention on all aspects of agriculture, from inputs to marketing of final products since 2008.

For spatial analysis, intra-KBR comparisons among sub-regions as well as comparisons with Non-Kosi basin region of Bihar (NKBR) and Bihar as reference points. KBR is a large area in Bihar. The NKBR accounts for even a larger share (58 %) of total geographical area of the state but the population density is significantly higher in KBR at 1289 compared to 1036 per square Km in NKBR (see Chapter 5 for details). KBR's population density was high in 2000 compared to NKBR, but over the decade it grew by 29% against 24% in NKBR increasing the exposure and vulnerability of the region. Similar comparison was made among the three KBR sub-regions UPPER, MID and LOW both at levels and temporal changes.

Among the features considered for comparison and distinction we examine natural and social aspects with a major focus on agriculture and access to food. Shifts in cropping pattern and crop yield rates change in the KBR and in its sub-regions are examined. Water is a primary requirement for agriculture. It is also a critical influence on human life in all respect. Rainfall and irrigation are most important and often inter-related influences of agriculture but in an integrated basin level approach where spatial linkages are created by slopes and river flows, weather of upstream areas would also be a determinant of the yield rates of local crops and lives of local people.

Slope-based linkages are explored using spatial crop yield models and available data. Temperature is also known to be an important biological influence on crop maturation. Crop yield in any region or sub-region will also depend on economic incentives and infrastructure that guide resources allocation. The

panel data approach takes account of human interventions affecting people uniformly in the region (food policy and extension) and those impacting them differentially (infrastructure).

Specifications

1. Crops:

Food crops (FC): Wheat (WH) and Rice (RC).

Coarse Cereals (CC): Maize (MZ) and Millets (MLT) consisting of Jowar, Bajra, Ragi, Barley, and others millets.

Major crops (MC): Rice, Wheat and Maize

Pulses (PLS) : Gram, Tur and Other Pulses.

Fruits (FR): Mango, Banana and other Fruits. Vegetables (VG) - Potato, Sweet potato, Onion and others. Oilseeds (OLS) - Groundnut, Castor, sesamum, Rapeseed&Mustard, Linseed, Sunflower, Safflower seed. Others crops – Fibers (FB), Spices (SP), Beverages&Narcotics (BN), Fodder crops (FOD) and other non-food crops.

Source: Directorate of Economic and Statistics, Department of Agriculture and Cooperation, Ministry of Agriculture, Government of India.

2. Animals:

Cattle: Cross-Bred (C-B) Cattle, Indigenous (INDG) Cattle, Buffalo (BFL)

Source: Directorate of Economic and Statistics, Department of Agriculture and Cooperation, Ministry of Agriculture, Government of India.

3. Land use:

NAGR: Area under non-agriculture uses, *FOR:* Area under forests, *BUCL:* Barren & uncultivable land, *PPGL:* Permanent Pasture and other grazing land, *MISC:* Land under miscellaneous tree, crops, & groves not included in net sown area. *FL:* Fallow land other than current fallow land and includes

Current Fallow and Culturable Waste land, *NSA*: Net sown area, *CI*: Cropping intensity, *AVFSZ*: Average farm size, *GA*: Geographical area

Source: Directorate of Economic and Statistics, Department of Agriculture and Cooperation, Ministry of Agriculture, Government of India.

4. Demographic

POP: Population, *POPDEN*: Population density (persons per square Kilometer), *RPOP*: Rural population, *LAB*: Labourer population, *WRK*: Workers in population, *WRKAG*: Workers engaged in agriculture, *LFEXP*: Live expectancy in years, *LITR*: Literacy Rate, *FLITR*: Female Literacy Rate, *SXR*:- No of female per '000 Male, *CSXR*: Sex ratio for population under age of 5 years, *CBR*: Crude Birth Rate, *CDR*: Crude Death Rate, *IMR*: Infant Mortality Rate, *UMR*: Mortality ratio of children under 5 years, *GD*: Gender Disparity

5. Economic

GDP: Gross domestic product of nation, *PCGDP*: Per capita GDP (GDP divided by national population), *GDDP*: Gross District Domestic Product, *GDDPA*: GDDP from Agriculture only, *PCGDDP*: Per capita GDDP (GDDP divided by district population), *GDDPFL*: Gross District Domestic Product of Forestry and Logging, *GDDPF*: Gross District Domestic Product of Fishing, *GDDPMQ*: Gross District Domestic product of Quarrying and Mining, *GDPAGAL*: Share of agriculture and allied sector in GDDP or GDP, *GDPIND*: Share of industry in GDDP or GDP, *GDPSEV*: Share of service sector in GDDP or GDP, *URBN*: Share of population (%) resident in Urban areas.

Population is from census or projected by growth rate as relevant. Other sources.

6. Farm size classes

Marginal: farms below and up to 1 hectare, *Small:* Farms from 1 hectare to 2 hectare, *Semi-Medium:* Farms 2 hectare to 4 hectares, *Medium:* Farms between 4 hectares and 10 hectares and *Large:* Farmers over 10 hectares.

7. Agriculture

Kharif season: The time of southwest summer monsoon or simply monsoon, crops are sown in months of May and June and harvested during October and November, *Rabi season:* starts in month of November and December and lasts till March and April, *Zayad:* March to June, *Yield:* Production in tonnes per hectare, *Fertilizers:* Nitrogen (N), Phosphate (P), Potash (K),

4.5. Field Visits

A field visit was conducted between from February 20th, 2014 and February 25th, 2014 to explore the problems and challenges of the Koshi river basin. The field visit team consisted of representatives of IEG (Institute of Economic Growth, New Delhi), ICIMOD, ANSISS (A. N. Sinha Institute of Social Studies, Patna), BSDMA (Bihar State Disaster Management Authority, Patna), CISRO (Commonwealth Scientific and Industrial Research Organization, Australia), and IWMI (International Water Management Institute Nepal Office, Kathmandu).

In North Bihar, the villages that were visited are from four districts namely Supaul, Saharsa, Madhepura, and Araria. In each village, one focus group discussion was organized which was followed by individual interviews of several villages to understand local issues and their problems. Some open ended questions on flood, life, livelihood, etc. were asked in both the focus group

discussion and individual interviews. A second field visit was conducted by IEG team to district Nalanda of Bihar, a district outside the Kosi basin region. The visit was motivated by two interests, one was to compare the study districts of Kosi basin with a neighbouring district that suffers less disadvantage due to flood proneness and the second was to look for linkages in development between Kosi and non-Kosi districts of Bihar for a possible integrated perspective of planning.

A visit was made to Khagaria in 2015 to interact with residents. Team members had interactions with farmers in Saharsa as well as with traders and processors operating in the entire region. Besides, workshops arranged by ICIMOD and partner ANSSI at Patna also helped in gaining field level and scientific understanding. Visit was also made to Nalanda to get a sense of the contrast.

4.6. Primary data and Study area

Based on the map of Bihar and the listing of districts, blocks and villages obtained from local and state authorities along with information on cropping pattern, living conditions and river channels, the study region for primary data collected was determined. The sample designing was made with the methodology as outlined below.

Primary study is meant for understanding the nuances of the lives of people in the KBR. Focusing on a smaller area within the KBR allows greater details to be accessed than secondary data would permit. The study design therefore has to bring out the typical elements of the geographical region interest in current context with subtle variations that differentiate areas even under similar riverine conditions. At every stage of sampling the proximity to the rivers is kept in view while selections are made also with a view to capture maximum variation subject to this commonality. In a multi-stage stratified random

sampling, 240 farm households were surveyed during 2015 covering both Kharif and Rabi seasons of the reference year 2014-15. The survey coincided with the rabi harvest season of 2015.

At the outset it was decided that two districts which are expected to be affected by river Kosi's dynamics but perhaps in different ways, would be selected. Selection of district followed the criterion of river proximity as also dissimilarity of cropping pattern. Kosi hydrology affects districts Madhubani, East Champaran in UPPER and Sitamarhi, Darbhanga and Muzzaffarpur in MID sub-regions seen in Map 4.3. Due to numerous sudden avulsions, Kosi has broken into many channels and the UPPER districts are rather away from the present main channels of Kosi although this does not any way rule out or lighten the threat and devastation of floods.

A location in the LOW region was preferred. The proximity to the main Kosi channel was an advantage. In this sub-region natural calamities keep larger sections of people and greater share of agriculture under threat as is well known a priori. In the detailed map in Map 4.2 river Kosi can be seen to enter from Nepal into India where its main channel passes through districts Supaul, Madhepura and Saharsa (earlier constituting a same district but later trifurcated). Further downstream and south are districts Begulsarai, Katihar and Bhagalpur (see Map 4.4).

The districts selected for study are Saharsa and Khagaria (see Appendix A4.1 and A4.2 for historical and administrative accounts on the districts) both in LOW sub-region and in same agro-climatic region Z2 (see Chapter 2) but with important differences. Both districts are highly flood prone. Between the years 2000 to 2010 people of Saharsa experienced floods in all ten years and those of Khagaria experienced floods in all years but 2006 (Table 4.2). On the other hand, in 2001 droughts too affected both districts. Khagaria is entirely riverine and low-lying encircled by not only Kosi and its channels but also by the

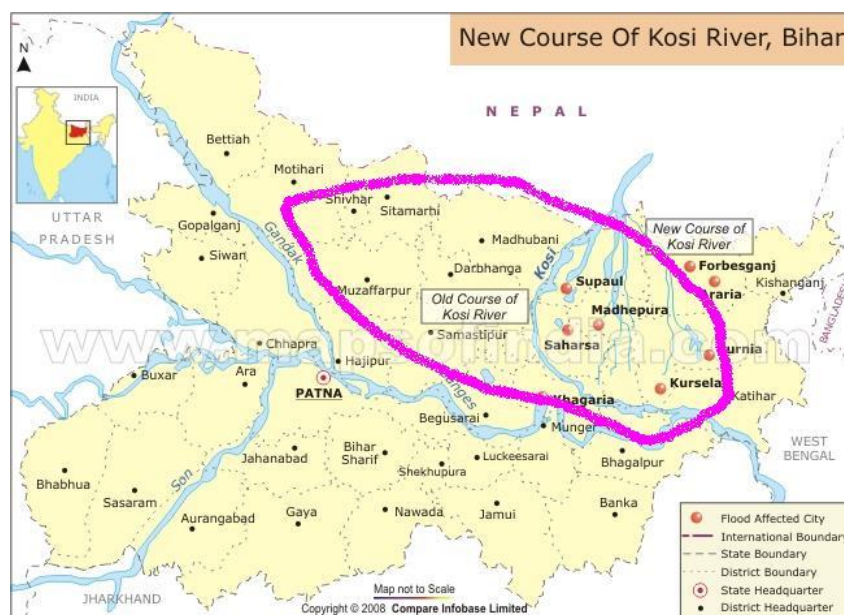
Ganga, whereas Kosi and its channels including tributary Bagmati drain Saharsa. To manage floods, embankments are constructed (A5.2) in both districts but in Khagaria the extent of embankments is relatively less and they are built along river Ganga rather than Kosi.

Table 4.2: Number of years (% frequency) affected by natural calamity in the years 2000-2010 and the years

	Drought	Flood
Khagaria	2001	9 Years except 2006
Saharsa	2001	All Years =10

Source: Farmer Portal (Website)

Field interactions suggested that in KBR maize farming has emerged as a coping strategy against floods although initially it was part of subsistence farming in a community that used to consume maize as a staple. Khagaria and Saharsa are both maize growing districts but the cropping patterns and the significance of Maize seem to differ. Saharsa has more varied cropping pattern with Rice and Wheat being more important than Maize while Khagaria has moved towards greater specialization in Maize but Maize has gained acreage in both districts.



Map 4.4 Sources: www.mapsofindia.com

District Khagaria has seven blocks and Saharsa is divided into 10 blocks. Four blocks from each sample district were selected such that they that are proximate to River Kosi and have the presence of Maize in their cropping patterns. The same approach was also utilized in the selection of villages. Villages in each block were divided into two groups based on their distance from the river with the maximum distance being 13 Km. One village was selected randomly from each group of villages and therefore two villages from each sample block and 16 villages from the two districts. The name of the sample villages, blocks and districts are presented in Table A4.1. From each village 15 households were sampled. Responses were solicited to structured questionnaire developed and discussed in expert forums, refined after focus group discussions and tested with a pilot survey.

The sample farmer households were selected using stratified sampling procedure giving representation to different classes of farmers. Farmers reported land holding in local unit bigha which was expressed in hectares (**conversion of 1 hectare equivalent to 2.27 bigha and 1 bigha= 22 cotta**).

The farm size was small on the average and varied from zero to 10 hectares. One outlier of a very large farm was found and excluded from the sample for distorting the averages. Many in the population operated no land. The farm classes were specified keeping this reality of local land distribution in view and to understand the behavior of each class. For this purpose, five farm classes were specified (i) Landless with no land reported (or actually insignificant land) under operated farm, (ii) Marginal with farm size up to 1 hectare, (iii) Small with farm size above 1 hectare but up to 2 hectares, (iv) Medium with farm size ranging from 2 hectares to 4 hectares and (v) Large with farm size above 4 hectares. This farm size class categorization differs from the official categorization (see 4. 4 specification 5 for secondary data) which is uniform for all India. Size class (iv) 4 -10 hectares is termed as semi-medium in official documents and the designated Large class holds over 10 hectares. Since holdings above 10 hectares is scarce in the study region, 4 hectares is taken as the lower limit of Large holding class in our classification. Thus the Large holder class of this study is a relative standard and is only the Medium holding class by India's standard.

While random sampling was used as the techniques, the intention was to give representation to all farm classes including those reporting as Landless. Local reality, which impacted the availability of households of different sample sizes that were willing to respond, also constrained the number of households sampled in each class. In the study area of Khagaria the sample was fairly well distributed among the size classes. In Saharsa however Large holding households were more difficult to access in the specific study area and received less representation while the marginal class got a relatively large weight. Thus, in the study areas of each of the districts Khagaria and Saharsa, the Marginal class had a lower share at 27% and 51% compared to what is shown in secondary data at 87% and 85% but among the five size classes Marginal had the largest weight in the sample (Table 4.3). Besides households, traders, processors and market agents operating in the study area were also surveyed.

Table 4.3: Sample sizes for Primary survey

	Sample No.	% Share
Khagaria	120	100
Landless	24	20
Marginal	32	27
Small	21	18
Medium	30	25
Large	13	11
Sharasa	126	100
Landless	30	24
Marginal	64	51
Small	15	12
Medium	12	10
Large	5	4

4.7. Household data

Structured schedules were drawn out for collecting household data. The information meant to be conveyed by the data can be classified as socio-economic, environmental and food-agricultural. Social attributes are captured by usual indicators such as caste, community, household size, age and sex of the head, gender and age composition of household, occupations and migration. Environmental information covers distance from river, distance from other amenities, soil quality and water-logging, housing quality. Agriculture would cover cropping pattern, size of sown area and cropped area inclusive of multiple cropping, irrigation, fertilizer use, labour use and farm income.

Prices at which crops are sold are collected as reported by the households along with crop yields, acreages, input use and paid-out costs of cultivation. It was apparent that farming was largely pursued with semi-commercial motivations. Not all food needs of the family are met from farm produced commodities nor are all produced quantities sold. Production, sale and purchase of the food crops were simultaneously practiced though probably at different times of the year.

Imputing the sale prices to home consumed products is taken as a way to assess incomes because the same food could have been sold in the market for a price. On the other hand, the home-consumed food could have been purchased from the open market or the public distribution system (PDS). A more scrupulous definition of commercial agriculture would demand that farmers move away from subsistence production and sell products for cash that provides wider access to food and other goods and services of their choice.

4.7.2. Specifications

1. Household: A unit of society as a collective of family members reported as part of households by the respondent who is usually the head of household.
2. Head of household: Reported as such by respondent
3. Farm size classes of Household: Categories of households engaged in agriculture at the time of survey. They include both land operating (cultivating) households and those with no reported land and fall in five groups based on the range of land holding size.
 - I. Marginal: Land possessed for farming is up to 1 hectare.
 - II. Small: Land possessed for farming is up to 2 hectare but more than 1 hectare.
 - III. Medium: Land possessed for farming is less than 4 hectare but more than 2 hectares
 - IV. Large: Land possessed for farming is more than 4 hectare
 - V. Landless: Reporting no land available for cultivation in the current season. In practice, it was observed from field interactions that some of the households in this category do cultivate some insignificantly small amount by leasing in or reclaiming and crops raised are usually fodder and vegetables for own farm and household consumption. This class may also have in the past held

cultivable land that was lost owing to water and sediment problems, at times passing into government possession as wet land.

4. Household social classes by caste and religion

SC: Scheduled caste, ST: Scheduled Tribe (includes Mahadalits, Dusadh, Musahar), OBC: Other Backward classes (e.g., Yadav, Kushwaha, Kurmi), Forward classes: Other classes not under government reservation (e.g., Bhumihar, Brahmin, Rajput, Kayastha), Minority: All communities excluding Hindus but almost entirely mostly Muslims

5. Migrant: A member of sample household who has, with or without accompaniment of immediate kins like the spouse, moved to places other than the district under survey for at least 1 months for various economic reasons in the last 2 year.

6. Household size (HSZ): Number of members reported by respondent as stakeholders of household returns. They include the migrants.

7. Working age members (WAGE): Members of household in the age bracket (15-60 year) are treated as capable and permissible for economic work. The proportion of such members in the household measured as WAGE/HHSZ determines the earning power whereas children and elder members constitute dependency.

8. Female working Age members (FWAGE): Female members of working age in the household are considered separately as they have a distinction. Certain farm jobs (rice transplantation, dairy) are associated with female labour while many arduous jobs, machine use, farm labour jobs are discouraged due to technical or social reasons. Females also combine farm work more with household chores than do men. The proportion of female workers measured as FWAGE/WAGE or FWAGE/HHSZ may have an impact of level of activity but the direction of impact is indeterminate.

9. Farm size (FSZ): Such land possessed by a farmer through ownership, patta or leasing. It may be sown, re-sown after crop damage, left fallow. Fallowing may occur out of choice or crop damage during the current season. The farm size does not include land lost
10. Total cropped area (TCA): Sum total of the area under all the crops harvested in different seasons. TCA includes multi-cropped area as separate land and is generally expected to be larger than farm size. In cases of severe crop damage or mid-season land loss due to weather events TCA can fall short of NSA.
11. Net sown area (NSA): Land area that was sown and harvested by the farmer at least once in the current crop cycle regardless of the number of times it was cropped.
12. Land lost (LL): Land that became uncultivable or inaccessible prior to sowing owing to natural factors such as low lying status, water-logging, sedimentation or erosion by river. Lost land may become part of the river or may come under wetland protection in which case the government regulates the economic use of the land as a public property¹.
13. Irrigation intensity: Gross irrigated area/Cropped area
14. Sources of Irrigation (IRR): Tanks (TNK), Tubewells (TW), other Wells (OW), canals government and private owned (CNL), Other sources (OS), including sprinkler, drip etc.
15. Farm machines (FM): Tractors, Pump-set Major food crops: Specified as Rice, Wheat and Maize
16. Food crops: Specified as only Rice and Wheat. It may be noted that Rice and Wheat are also commercially sold and other crops including Maize

¹ A wetland is a land area that is saturated with water, either permanently or seasonally, such that it takes on the characteristics of a distinct ecosystem. India has laws to conserve and protect wetlands but the implementation remains incomplete owing to contentions around the Law of 2010 and new law of 2016. The law regulates identified and notified wetlands of the time in terms of uses including agriculture.

and specifically Pulses constitute food but Rice and Wheat are staples treated as essentials for survival.

17. Diversity of cropping pattern (DIVERSITY): Property of land use practice in cultivation which means the choice of growing a number of crops in the same area and in the same crop cycle. Diversity is a move away from mono-cropping indicated by a large share of area allocated to the single most dominant crop in the entire farming year. Higher is this share less is the diversity. A more comprehensive measure is the propensity to varied crops measured by the
18. Rotation (RTN): The choice of crops and their positions in the cropping cycle of the year. For example Rice in Kharif, Wheat in rabi and pulses in the interim may be a possible rotation. Rotation involves the same land but different time periods thus not encroaching on each other's allocation and indicates greater land use by multiple cropping and often synergy though certain rotations may not be favourable for conservation of land and water.
19. Inter-cropping (INTCROP): The practice of growing two or more different crops on the land at the same time. The crops share inputs and land but may not be competitive as there may be favourable synergies in the combination.
20. Family labour (FML): Labour measured as person days supplied by household members and such work may be in the nature of supervision that is not distinguished from manual labour in its pure form. Although family labour is free of cost, in practice cost is imposed on the household by the effort involved especially in inclement conditions.
21. Hired Labour (HRDL): Labour service measured in person-days obtained from hired labourers at given wages rates per labour days.
22. Farm wage rate (FWG): Wage rate per day specified as 8 hours of work paid to laborers hired for farm work. Minimum wage rates are stipulated by the government but the actual FWG may differ from this and depend on the season determining the demand for farm labour, supply of labour

influenced by factors like migration, non-farm jobs in the village or in neighbourhood, public works programme in the study area like MNREGA, other earning opportunities like enterprise and female predisposition and preoccupation.

23. Risk: The state of uncertainty about possible outcome of the effort exerted on farming often signified by the probability of loss owing to weather events. Risk leads to expected value of outcome less than that obtained in ideal conditions. Behavioural results are diversification of crops and crop varieties, low investment on farming, avoidance of certain crops considered more risky or of farming in the risky season and securing safety by minimizing losses.
24. Food insecurity: The state of poor access to food relative to requirement of the households as ordained by nutritional, demographic and cultural demands. Food insecurity can result from farming risk leading to crop loss, low physical availability in markets and poor capability to purchase food from market owing to high retail prices or low income from farming resulting from low producer prices and low productivity. Technically, food insecurity has a broader connotation (taste, quality, intra-household distribution etc.) but insufficiency due to poor purchasing power is more relevant in this context. In technical exercises food security is measured by ratio of available food to food requirement.

4.8. Estimation of Sample averages

1. *Area under a crop (AC)*: Average of the area under any crop of the sample of all land operating farmers including those who do not grow the crop (excluding Landless).
2. *Major crops*: Specified as Rice, Wheat and Maize
3. *Foodgrain crops*: Specified as Rice and Wheat. Maize is treated as a commercial crop.

4. *Area under a crop of growing farmers (AGC)*: Average of the area under any crop of the sub-sample of land operating farmers who actually grow the crop. AGC is greater or equal to AC as the average is over a sub-sample.
5. *Weight attached to each crop (λ_i)* : Share of area under the crop in the entire sample measured as $\lambda_i = A_i / (\sum A_i)$ where i is Rice, Wheat and Maize.
6. *Yield of crop (YLDC_i)*: Average of yield per hectare (Tonnes/hectare) of crop i of the sub-sample of farmers who grow crop i.
7. *Intensity of farming (IF)*: The total cropped area per unit net land possessed helps assess how fully the available land is used. $IF = TCA / FSZ$. IF is different from cropping intensity (CI) defined in official data records as TCA divided by NSA whereas the denominator in IF is FSZ which includes fallow land or sown land with no harvest. Like CI, IF is generally expected to be greater than unity but unlike CI, IF may fall short owing to crop loss or land loss during growing season.
8. *Sale of crop (SALEC)*: Quantity of crop (Kg/farm) sold by crop growing farmers
9. *Fertilizer use for a crop per hectare (FRTCH_i)* : Total fertilizer (N, P, K) used for major crop i by the crop (Kg/hectare) growing farmers in the crop year divided by total area under the crop in the year counting multiple cropping.
10. *Producer price of crop (PPC)*: Price (Rs/Kg) of major crop at which produce is actually sold as reported regardless of the channel of marketing.
11. *Retail price of crop (RPC)*: Retail price (Rs/Kg) in the open market as reported officially for Bihar in secondary sources. RPC is uniform across households but differs among crops.
12. *PDS Price of crop (PDSPC)*: Price (Rs/Kg) at which food crop is available for purchase in public distribution system (PDS) as officially reported in secondary data sources. RPC is uniform across households but differs

among crops. The availabilities in PDS is however limited in quantities per households/members.

13. *Total Fertilizer use per hectare (TFRTH)*: Total fertilizer per hectare (Kg/hectare) used for all three major crops together in the year counting multiple cropping. It is measured as the weighted average of the FRTCH where $TFRTH = \sum \lambda_i FRTCH_i$ where i =Rice, Wheat and Maize and λ_i are weights as defined. FRTCH is a measure of total fertilizer used divided by total cropped area.
14. *Paid cost of crop per hectare (PCCH)*: Paid cost for cultivating major crops (Rs/ hectare) of crops by crop growing farmers. These costs are incurred for inputs like hired labour, machines and animal services, fertilizer, pesticide and all other paid cost reported by cultivating households.
15. *Paid cost of crop (PCC)*: Paid cost of cultivating major crops (Rs/household) where $PCC = PCCH \times A$ where PCCH and area (A) for the major crop is as reported by farmers growing the crop.
16. *Total paid Cost per hectare (TPCH)*: Total paid cost of all three major crops per hectare (Rs/hectare) measured as weighted average of per hectare costs of individual crops $TPCCH = \sum \lambda_i PCCH_i$ where i =Rice, Wheat and Maize across all land operating farmers.
17. *Fertilizer intensity (FINT)*: Total intensity of fertilizer (Kg/hectare) used measured as total fertilizer used for the three major crops in a year per net hectare of land measured as $FINT = TFRTH \times \text{Farming intensity}$
18. *Revenue from crop per hectare (REVCH)*: Indicative revenue (Rs/hectare) per unit area from all production of the crop in the market assuming self consumption as a market transaction (selling to oneself). $REVCH = PPC \times YLDC$ of farmers growing the major crop.
19. *Proportion of produce sold (PRSOLDC)*: Indicative commercialization as the proportion of produce of major crop that is sold. $PRSOLDC = SALEC / (YLDC \times A)$

20. *Revenue from crop per hectare in cash (CSHREVCH)*: Revenue received in cash by selling product (Rs/hectare) without considering unsold produce consumed at home or wasted. $CSHREVCH = PPC \times YLDC \times PRSOLDC$
21. *Income per hectare under a crop (INCH)*: Indicative farm income from a major crop measured as difference between indicative revenue and paid cost. $INCH = REVCH - PCCH$. The farm income per hectare is computed for only farm households growing that crop.
22. *Farm Income from a crop (INC)*: Indicative farm income from major crops (Rs/Farm) measured as product of farm income per hectare from the crop and area under the crop. $INC = INCH \times AC$ where produced quantity and paid cost are considered only for major crop growing farmers.
23. *Total farm Income per hectare from three major crops (TINH)*: Indicative income per hectare (Rs/hectare) from growing three major crops as weighted average of individual crop incomes per hectare. $TINH = \sum \lambda_i INCH_i$ where i =Rice, Wheat and Maize.
24. *Total farm income (TIN)*: Total indicative income of farm from three major crops as the product of per hectare income and total area under the crops. $TIN = TINH \times TCA$ where $TCA = FSZ \times IF$.
25. *Cash income of crop per hectare (CSHINCH)*: Cash income from crops (Rs/hectare) sold without factoring in the value of home consumption while counting all cash expenses of paid cost. $CSHINCH = CSHREVCH - PCCH$. Calculation is for crop growing farmers only.
26. *Cash income from crop (CSHINC)*: Total cash income (Rs/farm) from the major crop considering the area under the crop. $CSHINC = CSHINCH \times AC$.
27. *Total cash income of farm (TCSHIN)*: Total cash income from cultivating three major crops (Rs/farm) without factoring the unsold product but considering all paid cost. $TCSHIN = \sum \lambda_i CSHINC_i$ where i =Rice, Wheat and Maize.

28. *Access to food by growing commercial Maize per hectare (ACCMH)*: Purchasing power of income from sale of maize cultivated per hectare (Kg/hectare) measured by income from Maize divided by retail price of food crops alternatively Rice and Wheat. $ACCMH = INCH \text{ (Maize)} / RPC$ where INCH is for maize growing farmers only and RPC is retail price of food crops Rice and Wheat. Both sold and unsold products are considered (in effect assuming no retention or wastage) for calculating Maize revenue but Maize is mostly a commercial crop grown for sale.
29. *Access to food by growing commercial Maize (ACCM)*: Purchasing power of the income from sale of Maize cultivated by a farm (Kg/Farm). $ACCM = ACCMH \times AC$ where AC is area under crop Maize.
30. *Relative access to food by growing Maize in a hectare (RELACCMH)*: Access to food secured by Maize farming per hectare relative food crop yield per hectare. $RELACCMH = ACCMH / YLDC$ where YLDC and ACCMH relate to the same food crop.
31. *Access to food per capita by growing Maize (ACCMPC)*: Indicative access to food (Rice or Wheat) per head that income from potential sale of Maize brings. $ACCMPC = ACCM / HHSZ$. This is comparable to the nutritional norm of cereals recommended by ICMR and the per capita production of Cereals in India.
32. *Milk Yield (MYLD)*: Milk yield (Litre/animal/day) per animal average over all households possessing milch animals
33. *Number of milch animals per household (NOANI)*: Number of animals (No.) per household.
34. *Milk productivity per household (MPH)*: Daily milk production (Litres/household/ day) per household possessing milch animal measured as product of milk yield per animal and number of animals. $MPH = MYLD \times NOANI$.
35. *Wealth per household as animal (ANIW)*: Value of animals per household (Rs/household) possessing milch animals. Valuation is as reported as

and calves are either bred on farm or purchased and sold in local village fair.

36. *Cost of milk production per household (MILK)*: Paid cost of milk production covering medicinal costs including artificial insemination and fodder/feed purchased by household possessing milch animals.
37. *Price of animal PANI*: Average price of the animal (Rs/animal). $PANI = ANIW / NOANI$.
38. *Producer Price of milk (PPM)*: Average price (Rs/Litre) at which milk is sold to different buyers. This is average of prices of actually sold milk.
39. *Annual Milk production (MPROD)*: Milk production per household in the year assuming an animal produces milk in 27 days and in 8 months in the year. $MPROD = MYLD \times NOANI \times 27 \times 8$.
40. *Cost of Milk production (COSTM)*: Average monthly cost of milk production per household using only paid cost incurred on fodder and medicines. The price of animal is excluded as also minor costs on sheds and utensils.
41. *Income from milk per household (INM)*: Annual income from milk production of household (Rs/household) possessing animals measured as $INM = (PPM \times MPROD) - (COSTM \times 12)$. As in the case of crops all production is considered in revenue imputing home consumed milk using market price.
42. *Milk sold (MSOLD)* : Quantity of Milk sold (litre)
43. *Cash income from Milk (CSHINM)*: Cash income from sale of Milk without factoring in unsold milk and considering all paid cost. $CSHINM = (MSOLD \times PPM) - (COSTM \times 12)$

4.9. Models

Herfindahl-Hirschman index (HHI)

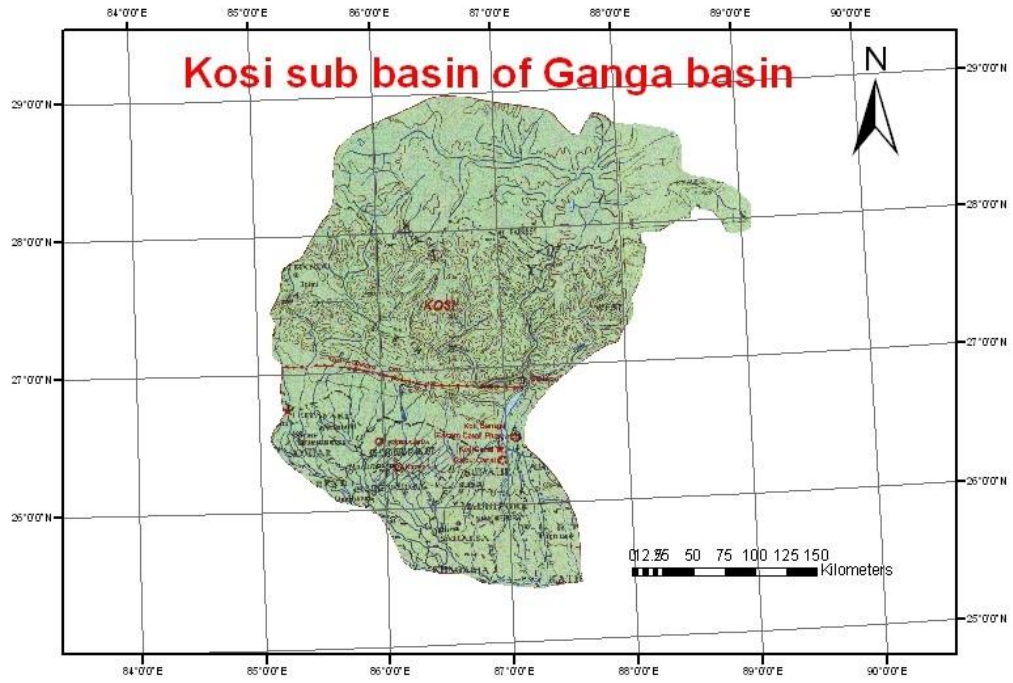
Diversity= $1 - \text{HHI}$ where

$$\text{HHI} = \sum S_i$$

N and s_i is the area share of crop i in cropping pattern with N number of crops or crop groups. HHI measures the probability of choosing the same crop in two random hits with replacement and diversity is the probability of the two outcomes to be different.

4.9.2. Sample selection bias

Omitted variables create specification error in econometrics. Specification error also arises from sample selection bias creating non-randomness (Heckman 1979). Sample selection bias may be created by self-selection by the individual or data units being investigated. For example market wages for working woman may exceed their home wage at zero hours of work or union member may find non unions alternative less desirable Heckman suggest the technique that enable analysis to use simple regression technique to estimate behavioral function free of selection buyers in censored sample with nice asymptotic properties of estimation by using and inverse MILLS ratio (λ) as a regressor where (λ) is a monotone decreasing function of probability that an observation is selecting sampling.



Map 4.5: Kosi sub basin of Ganga basin

5. Overview of Bihar Kosi Region

Creation of Jharkhand undermined Bihar's economy reducing the industrial sector¹ as much of the mining belt went to the new state. Similarly, the forest cover left with Bihar is only a meagre 5.17% of its land cover compared to a desirable target of 22% and Jharkhand's cover of 30%. The bifurcation also affected the demographic character. Jharkhand has 26.2% Tribal (ST) population whereas the same is only 1.3% in Bihar.

Drained by snow-fed Himalayan rivers North Bihar is fertile, water rich, intensely cultivated and densely populated but is prone to floods while the southern part of Bihar is drier and drought prone. The four agro-climatic zones shown in (Map A5.1, Appendix Table A5.1) differ in soil quality, climate and therefore cropping pattern. Rice and wheat occupy about 50% and 30% respectively of the cropped area of Bihar. Pulses, vegetables, though minor, are two of the other major crops. Horticulture, Fisheries and dairy are also in practice in Bihar's agriculture.

5.2. Socio-economic conditions

Development indicators reported in Table 5.1 collected from official documents reflect the weaknesses of Bihar in India's economy. The state is more densely populated and much less urbanized than the entire country. Per capita income reported by the state Government in 2008-09 is only Rs 15000 per annum (at 2004-05 prices) half of Rs 32000 reported at the all India level. It is even less at Rs 11000 as reported by the central agency MOSPI (Website).

¹ Except for Barauni complex entire Industrial base – Bokaro and Jamshedpur Steel Plant and mineral wealth were lost by the state.

The low income of the state may arguably be attributed to the structure of the economy skewed towards agriculture. Agriculture in which over 92% of farm holding are small (less than 1 hectare), has a higher share of GDP than the whole country (27%), generates 81% of employment and claims 57% of state area. Thus the predominance of agriculture in the economy and that of small farms with low income potential are two outstanding features of Bihar. Development theories suggest that the primary sector (in which agriculture is a major component) loses share as a country develops (Chapter 3.2) and the secondary sector gains and the tertiary sector becomes most important. Indian economy has shown such a trend and agriculture's share shrank to 16% in 2011-12. Bihar's service sector however has share of 57% comparable with India (57%) (Figure 3.1).

Table 5.1 : Basic statistics of Economy in Bihar and India

Unit	PCGDP* (Rs)	POP DEN (Per sq km)	RPOP (%)	Small Farms (%)	GDPAGAL (%)	Poverty Rate (%)
Year	2013-14	2011	2011	2010	2011	2011
Bihar	15506	1102	88.7	92.50	27.46	33.7
India	39904	382	68.8	82.29	18.12	21.9

Source: GOI (2008), GOI-MOSPI (website), GOB (2015). **Rs 10994 according to GOI-MOSPI (website) *2004-05price**

5.2.2. Farming as an employment in Bihar: A comparative view

Agriculture has been seen as a way of life in India from immemorial times, engaging family members who allocate their time budget to farm work and household work with little discrimination. Considerable significance was attached to farming in economic theories and policies (Chapter 3) but as its importance diminished in the western world, India's policy too did not remain untouched. Agriculture became embedded in development theory and policy and now in welfare².

² Ministry of Agriculture is now renamed Ministry of Agriculture & Farmers Welfare (MOAFW).

Land under crops declined (net sown area divided by population) declined from 0.17 hectares in 2009 to 0.12 in 2011 and the average farm size diminished from 1.4 hectares to 1.15 in India. Despite failure of land reforms and even with 57% of geographical area being sown, the scene in Bihar is no different and per capita net sown area and average farm size are considerably less in Bihar than in India (Table 5.2). More intriguingly, agriculture employs over 73% of the work force compared to 55% in India as of 2011 although the share has come down from 84% in 1991. Also, owing to skewed land distribution in the state a relatively large section (72%) of agricultural workers are labourers. Between 1991 and 2001 the land- man ratio and the average farm size have reduced significantly while the proportion of labourers in the agricultural workforce rose.

Table 5.2: Land constraint and Labour abundance in Bihar

State/ Country	NSA/POP (Hectare/Person)		NSA/GA (%)		LAB/WRKAG (%)		AVGFSZ (Hectare)		WRKAG/WRK (%)	
Year	1991	2011	1991	2011	1991	2011	1995	2010	1991	2011
Bihar	0.12	0.05	0.44*	0.57	51.27	71.83	0.75	0.39	84.22	73.55
India	0.17	0.12	0.43	0.43	37.99	54.87	1.41	1.15	54.32	54.60

*Undivided Bihar with Jharkhand. Source: Census of India, 1991 and 2011. MoA data

5.3. Agriculture: A Situation Assessment (2013-14)

Although policies that displaced labour from agriculture were severely resisted, increasingly, the futility of upholding agriculture as a key employment generator is seen with skepticism as the land-man ratio turned adverse with population growth, farm size became smaller and unviable with fragmentation and agriculture contributed less to national GDP even when exit from agriculture was low between the years 1991 and 2011. In a path-breaking expose in 2003-04, NSSO's Situation Assessment Survey (SAS), an in-depth study of Indian agriculture showed that 40% of agricultural households would take up a different occupation if given a choice (NSSO, 2005). Tedium of work, poor skill

formation and low income potential of farming along with aspirations instilled by knowledge flow from other societies are changing the choice of occupation.

NSSO conducted another round of SAS in 2013-2014 using slightly changed measurements. The specification of the agricultural household was amended to include those who earn at least Rs 3000 in the reference years with at least one member engaged in principal or subsidiary status in agricultural activities like cultivation of field crops, horticulture, fodder, plantation, animal husbandry, bee-keeping, vermiculture etc. Labour households, artisans and providers of agricultural services are excluded. Also only paid costs are taken into account, with no place for imputation of own resources.

A comparative analysis of specific indicators for agriculture is made for Bihar with the all India averages and with states where dominance of agriculture as livelihood is either more or less. Rajasthan and Tamil Nadu are taken for the purpose because their location and certain relevant attributes are diverse. India has 902 lakhs (over 90 million) agricultural households (AHH) making up 57.8% of rural households. Among the states the share of AHH varies from 40% to 80%. Among the three states Bihar has the largest number of AHH but the share (50% of rural population) is less than the national average. The total number of AHH at 7million is comparable to Maharashtra, a large state, and less than Uttar Pradesh where the AHH constitute 18 million households. Bihar has a large rural sector as seen in Table 5.1 but among rural households Rajasthan has a larger share (78%) of AHH than Bihar and Tamil Nadu has less (39%).

Table 5.3(a): Social profile of Agricultural households (AHH)

	Bihar	Rajasthan	Tamil Nadu	India
Total number ('000)	7094	6484	3244	90201
Share in Rural households (%)	50.5	78.4	34.7	57.8
SC (%)	13.5	19.74	23.5	16.3
ST (%)	1.2	17.9	1.1	13.4
OBC (%)	68.4	48.4	72.3	24.9
MGNREGA Card (%)	21.6	71.3	70.7	44.4
Ration Card (%)	86.1	94.0	99.2	87.7
BPL Card (%)	45.0	32.1	24.3	36.4
Cement (pucca) house (%)	53.0	76.1	71.4	61.7
With debt (%)	42.5	61.8	82.5	51.9

Source: NSSO, 2014

Unlike Tamil Nadu where an overwhelming majority of over 95% of AHH belong to marginalized castes the share is just above 80% in Bihar and unlike Rajasthan the share of SC and ST is relatively low but OBC caste groups constitute nearly 70% of the AHH (Table 5.3(a)). Though a little less than in Tamil Nadu, the share of OBC among AHH is much more than the India average of 25%. Of the AHH in Bihar, only 45% hold BPL cards (receive food at subsidized price) but 86% hold ration cards (receive food at not more than cost price) and a meagre 22% have job cards in MGNREGA. More people live in semi-pucca or katcha houses and the proportion living in pucca (masonry) house is lowest in Bihar, associated partly with frequent floods. Indebted households have lower share.

Table 5.3(b) shows that the gross cropped area per AHH is comparatively small in Bihar in both kharif and rabi seasons. Irrigation intensity is much higher a nearly 70% and the expense on irrigation at Rs 163 per month is far higher than it is in Rajasthan, Tamil Nadu and the all India average. This greater irrigation coverage and cost may be on account of the richer water resources in the state coupled with greater water management problem. Other expenses of cultivation (seed, labour, fertilizer, plant protection) are moderately low and the total cost of cultivation at Rs 1454 is much less than all India average of Rs 2192 suggesting that technology use is not intense in line with the irrigation ratio. About 65% of AHH are engaged in animal husbandry less than all India average of 72%.

Even for this activity the cost, especially expenses on fodder purchases and on hired labour, is less than other cases, the largest share of cost being on 'seed' or genetic material and veterinary services. The farm activities in Bihar are mostly based more on own resources than commercial purchases and subsistence may also be a strong motive of operation.

Table 5.3 (b): Economics faced by agricultural Households (AHH)

	Bihar	Rajasthan	Tamil Nadu	India
Cropped area kharif* (Hectare)	0.57	1.31	0.71	0.94
Cropped area rabi* (Hectare)	0.58	1.25	0.75	0.78
Irrigated area*(%)	68.49	39.47	58.20	47.62
Cost of fertilizer/manure (Rs/month)	376	324	642	526
Cost of plant protection chemicals (Rs/month)	46	47	166	165
Cost of seed (Rs/month)	147	305	337	250
Cost of human labour (Rs/month)	298	255	702	461
Expense on Irrigation (Rs/month)	163	51	35	70
Cost of cultivation (Rs/month)	1454	1730	2538	2192
Livestock farming (%)	64.7	86.1	61.2	71.9
Cost of fodder (Rs/month)	785	1616	1114	1069
Labour cost of animal rearing (Rs/month)	15	34	17	29
Cost of animal husbandry (Rs/month)	1324	1868	1370	1388

Note: *Kharif is July 2012-Dec2012, Rabi is Jan2013-Jun2013, Irrigated are is total of Kharif and Rabi. Source: NSSO, 2014, MoA (Website)

Table 5.3 (c): Income (Monthly) and debts of Agricultural households (AHH)

	Bihar	Rajasthan	Tamil Nadu	India
Income (Rs.)	3558	7350	6980	6426
Animals (Rs.)	279	967	1100	763
Cultivation (Rs.)	1715	3138	1917	3081
Non-farm business (Rs.)	240	710	1061	512
Wages/Salaries (Rs.)	1323	2534	2902	2071
Consumption (Rs)	5485	7521	5803	6223
Net investment (Rs.)	171	453	887	513
Loan outstanding (Rs.)	163	705	1159	470

Source: NSSO, 2014, MoA (Website)

The AHH in India earns Rs. 6,426 per month compared to Rs 3558 in Bihar where the earning falls short of Rajasthan and Tamil Nadu by larger margins (Table 5.3 (c)). The earning from non-farm business is considerably less in Bihar than the other states and so is income from wages and salaries. On the other hand, income from cultivation is less than Rajasthan and the all India average

but nearly comparable to Tamil Nadu and animal husbandry too generates low incomes. The household income is less than consumption expenses by a margin of nearly Rs 2,000 per month although there is net investment though meagre by all India standard. Since only paid expenses are considered the discrepancy reaffirms the place of subsistence farming. The lower investment relative to AHH in the other two states also signals the low income potential of the farming section in the state.

5.4. The Kosi region in Bihar

The focus of this study is on the Kosi region of Bihar. As explained in Chapter 4 we have delineated 16 districts of North Bihar as Kosi basin region (KBR) contingent on the presence of the river in proximity although the specification is made loosely because of overlapping and interacting presence of other rivers like Ganga and the other tributaries of Ganga and Kosi in the flood plains. These districts (Map 3.1) where nearly half of the Bihar's population lives constitute a little less than 40% of state's land area. Of the three sub-regions delineated by altitude, the LOW region is the largest both in area coverage and population. The population of UPPER is least and MID is most densely populated.

The per capita GDP (income) as of 2007-08 is only Rs 10,850 compared to Bihar's Rs. 14,574 and India's Rs. 43,361 (Table 5.4). The 37 districts of Bihar excluding capital Patna (to avoid extremes) are divided into four categories by their per capita income (PCGDDP) of a base year 1999-00. A high level of inequality and a skewed distribution is reflected (Table 5.1 (a)) by the PCGDDP and its growth rate over the period up to 2007-08. The range of PCGDDP is over Rs 12,000 more than the average PCGDDP of Rs 100,000. Munger ranks highest while Seohar in KBR records the least PCGDDP. Only 6 districts fall in the high income category with annual PCGDDP above Rs11, 000 while 12 districts are in the low income category drawing less than Rs 9000.

The KBR has its presence in all the income categories, districts, Begusarai and Bhagalpur ranking second and third and Muzzafarpur³ fifth (Figure 5.2 (a)). However, 7 of the 12 lowest income earning districts of Bihar belong to the KBR. The positive slope implicit in the scatter between the base PCGDDP and its growth rate (Figure 5.1(b)) raises a concern that incomes do not converge and the districts with a head start in PCGDDP has grown faster than the weaker districts. The following sections concentrate on the KBR and its sub-regions with broad comparison against the other part of Bihar (NKBR).

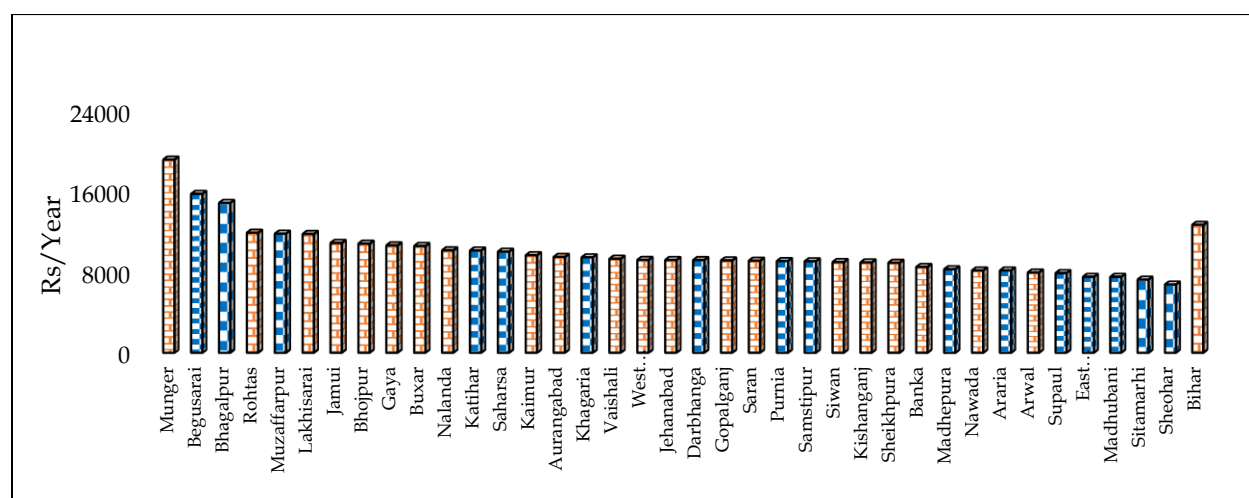


Figure 5.1 (a): Per Capita Gross District Domestic Product at current Prices (2007-08)
Source: GoB, 2011

³ Begusarai has a large a city and is described as the industrial capital of Bihar as it retains the petro-chemical and fertilizer and of Barauni. Bhagalpur, with a long commercial history and known for tassar silk and educational institutions, is a major urban agglomerate. Muzaffarpur is famous for its fruits like mango and litchi and also for wagon, sugar and pharmaceutical industries. All three high income districts have thermal power plants.

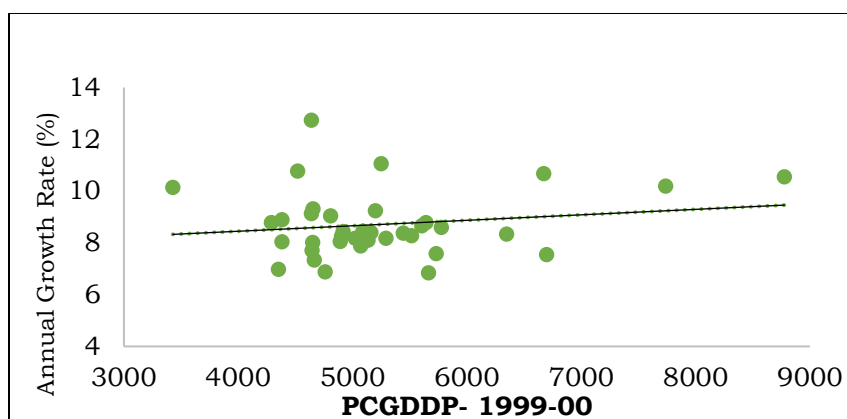


Figure 5.1 (b): Convergence of Income among Bihar districts (1999-00 to 2007-08). Source: GoB, 2011

5.4.2. Economy structure and socio-economic development

The KBR generates only GDP of Rs 36,000 crores out of Bihar's Rs 99,000 crores but its LOW sub-region produces Rs 23,000 crores output compared to only Rs 4,000 crores by UPPER (Table 5.4). A comparison among the sub-regions in KBR also finds agriculture and allied sector to have the largest share (46%) in the UPPER region. Only in the MID region is the structure of the economy skewed substantially towards the service sector (52%). The Industry sector has a share between 17% and 20% with an average share of 19% in the KBR, which is higher than the NKBR. KBR falls far short of remaining Bihar in the place of the service sector in the economy. The economy is highly dependent on agriculture, which has an average share of 34% in the GDP of KBR compared to 22% in NKBR.

Bihar is a largely rural state with only 11 population being urban (Table 5.5 (a)). The KBR is home to 47% of Bihar's rural population and 47.6% of total population. It also claims 47% of the cropped area but the population density is higher (1289) and there are more people (20) per hectare of net sown area than NKBR (19). KBR as a whole generates 37% of the state's GDP and nearly 50% of the state's GDP from agriculture. Per capita GDP is lowest at Rs 7,000 per annum in the UPPER sub-region (Table 5.4) which accounts for only 4% of the state GDP but 11% of GDP from agriculture. The sub-region LOW accounts for higher share

of total and rural population, area, cropped area and GDP than the other two sub-regions.

Table 5.4 : Economy structure in Bihar Kosi basin (2007-08)

Districts	PCGDDP@ <i>Rs. Per Capita</i>	GDDP <i>(Rs. Crore)</i>	GDPAGAL	GDPIND	GDPSEV
			% Share		
LOW	11934	22659	34.9	19.3	45.8
MID	11493	9331	28.1	20.2	51.7
UPPER	6912	3991	46.3	17.2	36.4
KBR	10850	35981	34.4	19.3	46.3
NKBR	16546	63071	22.5	14.7	62.8
BIHAR	14574	99053	26.8	16.4	56.8
INDIA	43361	3896636	19.28	26.28	54.4

Note: @ PCGDDP is 2011 figure. Source: Source: GoB, 2011, GoB (Website)

Table 5.5 (a): Socio-Economic features of the Kosi basin

	POP	SOWN	RPOP	GDDP*	GDDPA@	POP/NSA	POPDEN	URBN
Region	Share to state total (%)			Ratio			Per Sq Km	%
LOW	25.3	27.9	26.2	23.9	26.5	19	1147	9
MID	13.1	11.5	13.7	9.0	12.2	24	1650	7
UPPER	9.2	9.4	7.2	4.0	10.8	20	1280	6
KBR	47.6	48.8	47	36.9	49.5	20	1289	8
NKBR	52.4	51.2	53	63.1	50.5	19	1036	12
BIHAR	100	100	100	100.0	100.0	20	1143	11

Note: * TE 2009, @ TE 2007. Sown land is** NSA is net sown area, Source: Economic Survey of Bihar, 2014-15.

Literacy rate of Bihar is only 53% in 2011 against India's 74% but in KBR literacy is even lower. The MID sub-region has highest literacy. The birth, death and infant mortality rates measured by CBR, CDR and IMR are higher than in Bihar and the under 5 mortality rate is appalling at 87%. The LOW sub-region performs particularly poorly. Gender bias is evident in the GD but the adverse sex ratio is possibly explained by migration given the higher child sex ratio (932). Bihar has higher female literacy and child sex ratio compared to all India average and life expectancy is lower by a year.

Table 5.5 (b): Indicators of Social development (2011)

	LFEXP Year	LITR %	FLITR %	SXR '000 Male	CSXR '000 Male	CBR	CDR	IMR %	UMR	GD
LOW		48	66	907	935	29	7	61	90	141
MID		51	70	902	928	28	8	54	85	137
UPPER		48	70	913	927	28	8	56	74	147
KBR		49	68	906	932	29	8	59	87	141
NKBR		57	78	924	935	26	7	52	70	137
BIHAR	65	53	73	916	934	27	7	55	77	139
INDIA	66	74	65	943	919	22	7	44	55	-

Note: * denotes 2009TE figure and @ denotes 2007TE figure, Life expectancy of India is 66.1 year. Source: Economic Survey of Bihar 2012-13). See Chapter 4 for abbreviations.

Land use pattern reflects development tracked by economic and environmental standards which are sometimes in conflict (Table 5.6). Forests constitute a miniscule 0.1% of land use in KBR. Forest land makes up only 5% of Bihar's land. Non-agricultural land use is higher in KBR at 21% compared to Bihar's 18% and sown area too has a larger share of 63% also used more intensively. Fallow and barren unusable land accounted for nearly 10% of land use though less than in NKBR. There is hardly any change in land use between the years 2000 and 2010 except for a minor increase in fallow land which seems to have happened also in NKBR. Within KBR forest land is reported only in LOW sub-region. Fallow and barren lands are also more extensive in LOW where cropping intensity is lowest, Non-agricultural use ranging around 20% and sown area at 61-65% have similar share in the sub-regions. Land scarcity is evident.

Table 5.6: Changing land use pattern between 2000 and 2010

Table 6.9: Changing land use pattern between 2000 and 2010											
Category	Area (Million Hectare)	Share	NAGR	FOR	BUCL	PPGL	MISC	FL	NSA	CI	
		State	Share in reported Area (%)								Ratio
		total (%)									
2000											
LOW	2.31	22.4	19.9	0.1	6.0	0.1	3.0	5.3	65.9	1.4	
MID	0.83	8.1	24.9	0.0	1.0	0.2	5.1	7.8	61.4	1.5	
UPPER	0.78	7.6	20.8	0.0	1.3	0.3	6.3	5.1	66.3	1.5	
KBR	3.93	38.1	21.3	0.1	4.2	0.2	3.9	5.9	64.8	1.4	
NKBR	5.43	52.5	15.7	8.8	5.2	0.2	1.3	9.6	62.2	1.4	
BIHAR	10.34	100.0	18.0	5.1	4.8	0.2	2.4	8.1	63.3	1.4	
2010											
LOW	2.3	22.4	20.4	0.1	6.0	0.1	3.11	8.1	62.2	1.5	
MID	0.8	8.1	25.4	0.0	1.0	0.2	6.23	5.2	62.1	1.6	
UPPER	0.8	7.6	21.1	0.0	1.2	0.2	6.40	4.8	66.1	1.3	
KBR	3.9	38.1	21.8	0.1	4.1	0.2	4.30	6.9	62.7	1.5	
NKBR	5.4	52.5	15.8	8.9	5.1	0.2	1.35	13.0	55.8	1.3	
BIHAR	10.3	100.0	18.3	5.2	4.7	0.2	2.59	10.4	58.7	1.4	

Note: See Chapter 4 for abbreviations **Source:** MoA (Website)

5.4.3. Water resources: Benefits and troubles

Bihar shares rainfall pattern that characterizes India (Figure 3.1). KBR gets more rainfall in pre-monsoon, monsoon and post-monsoon seasons than NKBR (Table 5.7). The average rainfall is found highest in the densely populated downstream sub-region LOW. Despite lower monsoon rainfall more evenly distributed where rainfall helps multiple cropping of land. Over 10% of rainfall is received in the period February to May before the monsoon strikes and about a quarter in September to November period after the monsoon season. Rainfall distribution is most concentrated within the monsoon season (June-August) in the upstream region (UPPER). The rainfall distribution reflects the spatial dimension of the monsoon with implications for the basin's hydrology.

Table 5.7: Seasonal variation in normal rainfall in Bihar for the period 1901-2012 (mm)

Category	Dec-Jan	Feb-May	Jun-Aug	Sep-Nov	Annual
LOW	16.24	100.87	773.82	307.40	1198.33
MID	15.08	81.39	786.86	294.67	1178.00
UPPER	15.17	86.83	804.30	287.20	1193.50
KBR	15.81	94.25	780.89	301.69	1192.64
NKBR	17.17	70.28	761.47	273.47	1122.38
BIHAR	16.58	80.64	769.87	285.67	1152.77

Source: IWP (Website), IMD (Website). Note: All above values are average values for the period 1901-2012.

Table 5.8 (a): Trend in Average water table in Bihar (meters)

Category	2007	2008	2009	2010	2011
LOW	3.72	3.45	3.83	4.24	4.09
MID	2.50	1.97	2.51	2.88	2.65
UPPER	2.95	2.87	2.59	2.82	3.08
KBR	3.32	3.01	3.35	3.72	3.60
NKBR	3.94	3.75	4.51	5.13	4.99
BIHAR	3.67	3.43	4.01	4.52	4.39

Source: CGWB (Website)

Table 5.8 (b): Seasonal variation in water table in Bihar for the period 2007-2011 (in Meter)

Category	January	May	August	November	AVG	S. D
LOW	4.70	5.44	2.89	3.41	4.11	1.17
MID	3.09	3.66	2.03	1.92	2.65	0.84
UPPER	3.71	3.81	1.83	2.22	2.89	1.02
KBR	4.17	4.79	2.57	2.89	3.59	1.05
NKBR	5.22	7.04	3.22	3.33	4.96	1.81
BIHAR	4.77	6.07	2.95	3.14	4.37	1.47

Note: AVG: Average; S. D: Standard Deviation, The water table is the distance from the ground level down to the water in the well. Source: CGWB (Website).

A favourable water table in principle helps to create a rich, intense and diverse cropping system, also explain the water tables reported by the government. In the study region ground water is used widely for irrigation but surprisingly, the water table is not observed to have declined between 2007 and 2011 (Tables 5.8 (a)). The water table has tended to remain fairly stable at a little over 3 meter in KBR and lower at 4 meter in NKBR because ground water in KBR is recharged by the heavy rainfall during the monsoon and post-monsoon periods when the water table is significantly higher. The water table reaches a minimum in August

before the showers strike (Table 5.8 (b)). Within KBR the average water table is lowest at 4.11 meter in the LOW region and highest (best) at 2.65 meter in the MID region. The LOW region also shows a sign of receding water table (increasing from 3.7 meters to 4.1 meters) raising a need for caution although a higher water table

5.4.4. Upstream rainfall

In a river basin local rainfall is not the only source of water availability. This in Hydrology is not bounded by territorial limits created by geo-politics even with engineering barriers. The Kosi flood of 2008 also demonstrated the role of spatial relations. Figure 5.2 indicates that the year of the great flood was not one of extraordinary monsoon in Bihar.

Tables 5.8 the sub-regions UPPER, MID and LOW are constructions to allow the interaction. Rainfall and its distribution in UPPER is relevant for MID and LOW and that on MID is relevant not only for its agriculture but for that in LOW. Rainfall in further upstream zones outside Bihar and India can also be important for river flows. Rainfall distribution in Nepal (Figure Table A2.2(a)) finds an annual average over a long period to be 228mm in monsoon seasons June to October. The highest average rainfall is being 353 mm in July followed by 313 mm in August. There are year to year deviations the positive deviations (excess rainfall) being 31% and negative being 24% both deviations being largest in October a month of low average rainfall. In the more recent period 2000 to 2012 the positive deviations are found to be lower whereas deficits are more pronounced. The distribution is highly concentrated in July and August. In these 13 years maximum yearly rainfall was received in July in 46% of the years and in August in 31% through June and September too have ranked first in annual rainfall in some years.

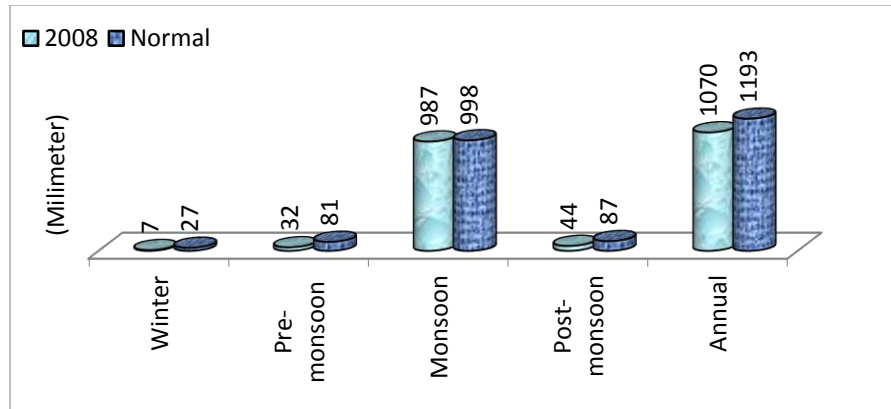


Figure 5.2: Seasonal rainfall distribution in Bihar: Normal and 2008

Source: IWP (Website), IMD (Website).

Note: All normal rainfall values are average values for the period 1901-2012.

5.4.5. Irrigation

Bihar is endowed in having an irrigation intensity of over 62% compared to India's 44 % and Bihar has been no different despite the nature of its endowment. Energy is required for pumping up groundwater (tube well). It may be noted that sources other than mainstream and conventional ones (canals, well, tanks) are also gaining significance at current times. Their share in India's net irrigated area increased from 22% in 1980-81 to 34% in 2012-13. Technology driven precision irrigation aided by implements like the sprinklers and drip are gaining acceptance and are promoted by public policy. Figure 5.1 shows Bihar has not caught up with these need-based and resource saving techniques as compared to India.

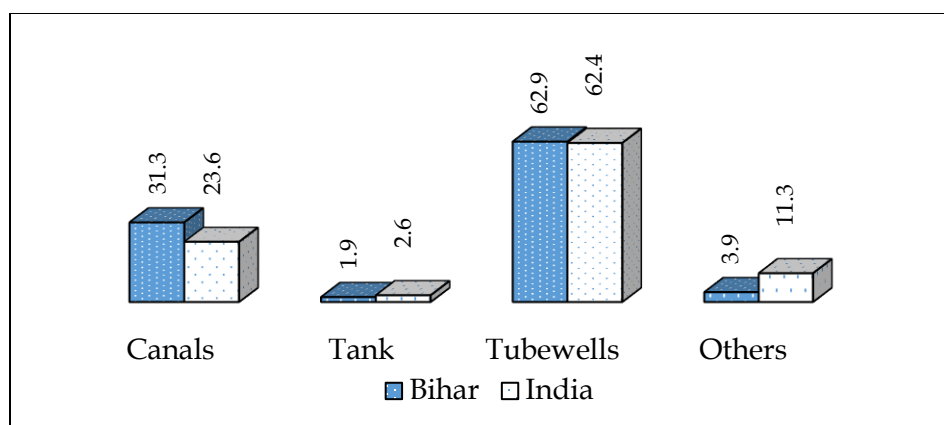


Fig 5.3: Net irrigated area by sources in Bihar and India (% NSA)
Source: MoA, Website

Table5.9 (a): Trend in gross irrigated area in last ten years (%)

Category	Irrigated area in 2000	Irrigated area in 2010
LOW	49.24	56.79
MID	36.97	44.93
UPPER	45.60	48.98
KBR	45.72	52.85
NKBR	67.92	69.57
BIHAR	58.57	62.53

Source: MoA (Website)

Rainfall is the original source of water for any irrigation. Sources like deep tube wells and canals managed systematically draw on past rain water stored scientifically. Tanks and simple dug wells depend on rainfall in the current season or immediate prior seasons. Irrigation projects on rivers in mountainous slopes are also associated with generation of hydel energy which in turn help to power irrigation.

In KBR, about 53 % of cropped area is irrigated which is less than almost 70% in the other region (Table 5.9(a)) despite the high water table. The irrigation intensity has improved remarkably in the reference decade 2000- 2010 in the regions and sub-regions. Within KBR the midstream (MID) region has the lowest percentage of irrigated cultivated area and LOW part has the highest proportion of irrigated area.

The KBR and NKBR are also distinct in their sources of irrigation. Despite the riverine geography, only about 7 % of irrigated area is accounted for by canal irrigation in KBR compared to 31 % in NKBR. Yet the basin is covered by a dense network of canals (Table A5.2 (a) and A5.2(b)). There is no presence of private canals. Tube-wells are the most important source of irrigation in both regions, more in KBR and within KBR it is least important in the UPPER region and has a 92% share in LOW in 2010. Over time, tube-wells have gained more significance but only at the cost of tanks and other sources mostly in upstream districts while canal irrigation remained stable. Tanks play a significant role in midstream region (Table 5.9 (b)).

Table 5.9 (b): Comparing sources of gross irrigation in the period 2000 and 2010 (%)						
	Canals			Wells		Other
Category	Govt.	Private	Tank	Tube-wells	Other wells	sources
2000						
LOW	10.41	0.00	0.91	84.52	0.31	3.84
MID	0.73	0.00	10.23	86.13	0.00	2.90
UPPER	3.88	0.00	21.97	71.96	0.00	2.18
KBR	7.17	0.00	5.88	83.35	0.20	3.40
NKBR	30.38	0.00	1.78	60.16	0.66	7.02
BIHAR	20.61	0.00	3.51	69.93	0.46	5.50
2010						
LOW	5.36	0.00	0.23	92.87	0.00	1.54
MID	4.10	0.00	14.77	76.85	0.00	4.27
UPPER	21.01	0.00	0.08	78.47	0.00	0.44
KBR	7.00	0.00	3.85	87.06	0.00	2.09
NKBR	30.53	0.00	1.62	62.96	0.40	4.49
BIHAR	20.62	0.00	2.56	73.11	0.23	3.48

Source: MoA, Website

5.4.6. Distribution of Operational Holdings

The average farm size is 0.38 hectare in KBR and is marginally lower than in the NKBR. Within the basin the MID region as expected has the smallest size. Over 90 % of land holdings are below 1 hectare in size in both KBR and NKBR but these marginal land holdings do not account for 60 % of total land area (see

Table 5.10) illustrating the inequality of land distribution. The Average size of this dominant marginal land holding group is even less at 0.24 ha in KBR than 0.27 ha in NKBR. In contrast, the average size of large holdings is about 54 times greater.

In the MID region 94% of the holdings and 65% of the land fall in the marginal category with an average farm size of .22 hectares while only 0.02% of holdings are in the large category. The average farm size is smaller in KBR where the proportion of marginal holdings is also larger relative to NKBR.

Table 5.10: Pattern of operational holdings

Category	Item	Marginal	Small	Semi medium	medium	large	All classes
LOW	Number	353773 (90.13)	25046 (6.38)	11190 (2.85)	2395 (0.61)	100 (0.03)	392504 (100)
	Area	92121 (55.76)	31428 (19.02)	29137 (17.64)	12412 (7.51)	106 (0.06)	165204 (100)
	<i>Average</i>	<i>0.27</i>	<i>1.24</i>	<i>2.62</i>	<i>5.17</i>	<i>12.73</i>	<i>0.42</i>
MID	Number	442449 (93.86)	19750 (4.19)	8114 (1.72)	970 (0.21)	86 (0.02)	471369 (100)
	Area	93894 (65.46)	24352 (16.98)	20358 (14.19)	4825 (3.36)	13 (0.01)	143442 (100)
	<i>Average</i>	<i>0.22</i>	<i>1.24</i>	<i>2.51</i>	<i>5.00</i>	<i>16.07</i>	<i>0.30</i>
UPPER	Number	718466 (91.50)	44881 (5.72)	17282 (2.20)	4541 (0.58)	71 (0.01)	785241 (100)
	Area	167393 (58.28)	52602 (18.31)	44219 (15.40)	22877 (7.96)	132 (0.05)	287223 (100)
	<i>Average</i>	<i>0.23</i>	<i>1.17</i>	<i>2.54</i>	<i>5.00</i>	<i>11.85</i>	<i>0.37</i>
KBR	Number	421529 (91.38)	26201 (5.68)	11182 (2.42)	2307 (0.50)	86 (0.02)	461305 (100)
	Area	101973 (57.89)	32306 (18.34)	28827 (16.37)	11823 (6.71)	1208 (0.69)	176137 (100)
	<i>Average</i>	<i>0.24</i>	<i>1.22</i>	<i>2.56</i>	<i>5.06</i>	<i>13.55</i>	<i>0.38</i>
NKBR	Number	363619 (90.80)	24035 (6.00)	10715 (2.68)	2025 (0.51)	79 (0.02)	400473 (100)
	Area	92597 (57.46)	30399 (18.86)	27805 (17.26)	10262 (6.37)	79 (0.05)	161142 (100)
	<i>Average</i>	<i>0.27</i>	<i>1.26</i>	<i>2.59</i>	<i>5.04</i>	<i>16.32</i>	<i>0.40</i>
BIHAR	Number	394892 (91.12)	25214 (5.82)	10998 (2.54)	2190 (0.51)	82 (0.02)	433376 (100)
	Area	97907 (57.92)	31458 (18.61)	28442 (16.83)	11153 (6.60)	83 (0.05)	169043 (100)
	<i>Average</i>	<i>0.26</i>	<i>1.25</i>	<i>2.59</i>	<i>5.07</i>	<i>15.11</i>	<i>0.39</i>

Source: Agriculture Census (Website)

5.5. Remarks

Factors behind the poor state of economic development of Bihar and Kosi basin in particular are complex. Being rich in water resources has not helped develop an agriculture which is high in productivity and income generation. Despite high monsoon rainfall, water table and intense irrigation, the region remains steeped in poverty, migration, poor GDP and excessive dependence on unproductive agriculture.

6. Agriculture in Bihar Kosi Basin Region

Agriculture's share in the Indian economy declined from 38% to 17% between the years 1980-81 and 2011-12. Within agriculture in the period 1980-81 to 2013-14, the share of horticultural crops in India's value of output (current price) rose from 13% to 19% and livestock sector's share increased from 18% to 32%. The 1990s started a post-GR era when food sufficiency attained at the national level diverted concerns towards food distribution and conservation of soil and water resources. A new paradigm in agricultural development is again becoming evident.

Public policy and, increasingly, the actions of the private sector driven corporate sector have supplemented the actions of millions of farmers in the country to set the direction of the evolution. Policy is becoming more oriented towards use of modern technology, the management of disasters, forecasts of weather and global crop outlook to guide farm decisions and food security. Specific interests of the weakest sections are entering the mainstream of development (Chapter 2 and 3). This Chapter presents some glimpses of the agriculture of the vulnerable Kosi basin region of Bihar based on official statistical data.

6.2. Place of Agriculture in Bihar's Economy

Claiming large shares of geographical area, GDP and employment (Figure 6.1(a)) of the state, the importance of agriculture in Bihar's economy is far more than it is in the nation's economy (Chapter 5). Literature informs that farming in India and in Bihar is done in small and marginal holdings, mostly leased land held in scattered parcels undermining efficiency of farming.

As in other parts of India Bihar had a feudal land holding system prior to independence creating exploitation and setting the rationale for government legislated land reforms. As of now leasing laws in Bihar are restrictive, leasing being practically banned with the exceptions for disabled persons. A tenant in Bihar is entitled to purchase ownership right after continuous occupation making it insecure for potential land owners and discouraging leasing. Nevertheless, 21.04% of land is leased in Bihar (see A9.1 for institutions) and share cropping is more popular than fixed rent. Recently as the paradigm is changing Bihar is not untouched by attempts at making land laws towards more flexible. The Bihar Agriculture Land (Conversion for Non Agriculture Purposes) Rules, 2011 is under consideration. Bihar is not one of the hotspots of the green revolution. Yield rates of rice, wheat and sugarcane are lower than all India level (Figure 6.2 (b)) but that of pulses and oilseeds higher. Food security implication does not show Bihar as one of surplus (Figure 6.1 (c)). Per capita production levels of food including milk are less than all India level. Advantage is seen only in maize which can offer power over food through the market.

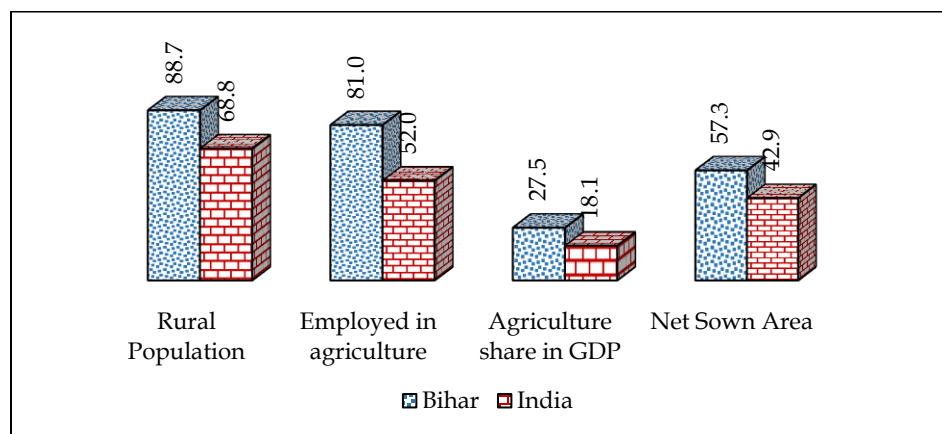


Figure 6.1 (a): Share (%) of Rural/Agricultural sector in Bihar's resources (2011)
Source: MoA (website), Census India (2011) and GOI-MOSPI (Website)

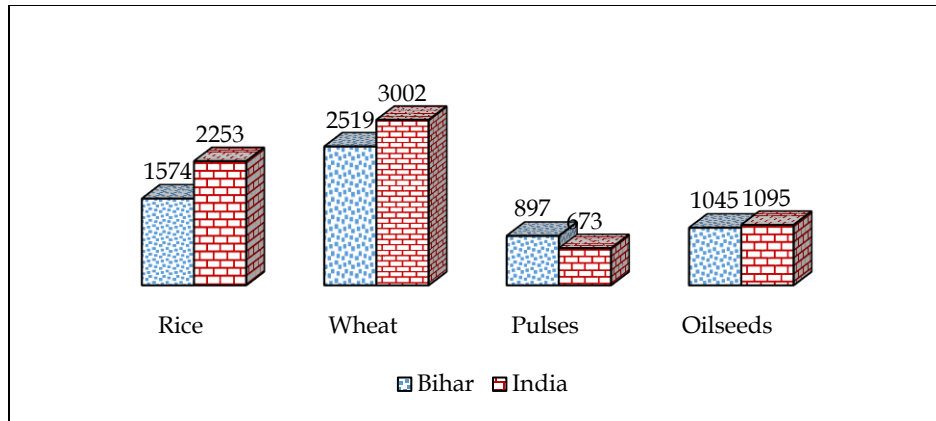


Figure 6.1(b): Bihar's Yield rates of major Crops (Kg/hectare, 2011 TE).
Source: MoA (Website)

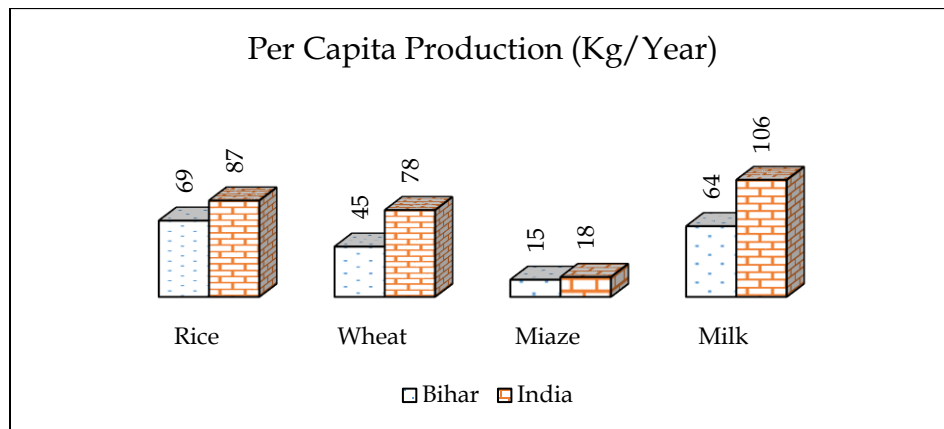


Figure 6.1 (c): Per Capita Production in Bihar and India (2011)
Source: MOA (website), Census India (2011)

6.2.2. Water management

Bihar has many rivers (see A2.3 (a)). Several of them are snow-fed and perennial and enrich the groundwater and surface water reserves and deposit alluvium used by Bihar's agriculture. Because of heavy rainfall and melting of snow in the river catchments, the northern part of Bihar is prone to recurrent floods even though droughts are not rare. In the face of global warming, climate resilience is recognized as a requirement for agriculture and water management is expected to play a crucial role in Bihar's development. In Chapter 2 it was seen that KBR

receives copious rainfall, it is drained by rivers both of local origin and those carrying water discharges from upstream areas which also receive heavy rainfall. Not surprisingly, water table is relatively high in KBR and irrigation intensity is also high (Chapter 5) though despite the surface water resources, the dependence is more on ground water.

6.2.3. Cropping Pattern

Bihar grows crops like Rice, Wheat, Coarse cereals, Pulses and Vegetables. Food crops Rice and Wheat together account for the largest area share over 70% against 44% in India. Preference for Coarse cereals (9%), Pulses (7%) and Oilseeds (2%) is also evident. A small share (5%) allotted to fruits and vegetables is comparable to India. No major change in the cropping pattern is observed from 2000-01 to 2012-13.

Bihar's cropping pattern differs from India in emphasizing Rice and Wheat more, in allotting lower share of area to Pulses and Oilseeds (Figures 6.2 (a)) and in the greater significance. Over time Rice and Wheat together remained stable in acreage share while Pulses lost but that of Maize rose (Figure 6.2(b)).

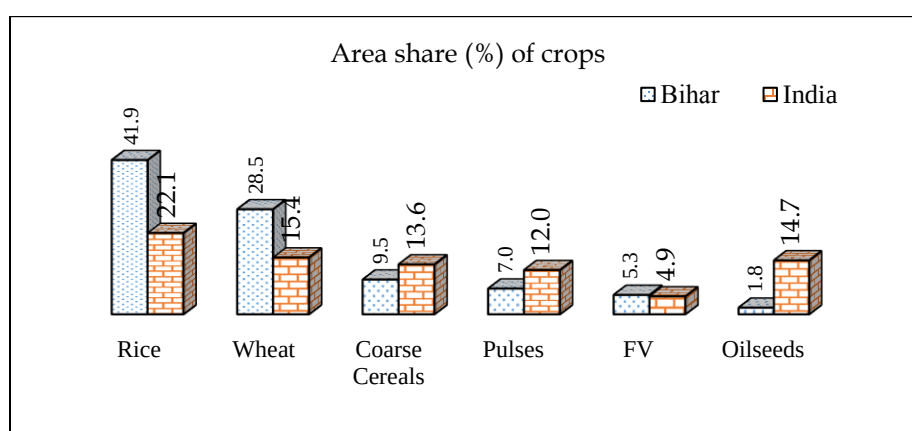


Figure 6.2(a): Comparison of Bihar and India (TE-2012) Source: MoA (Website)

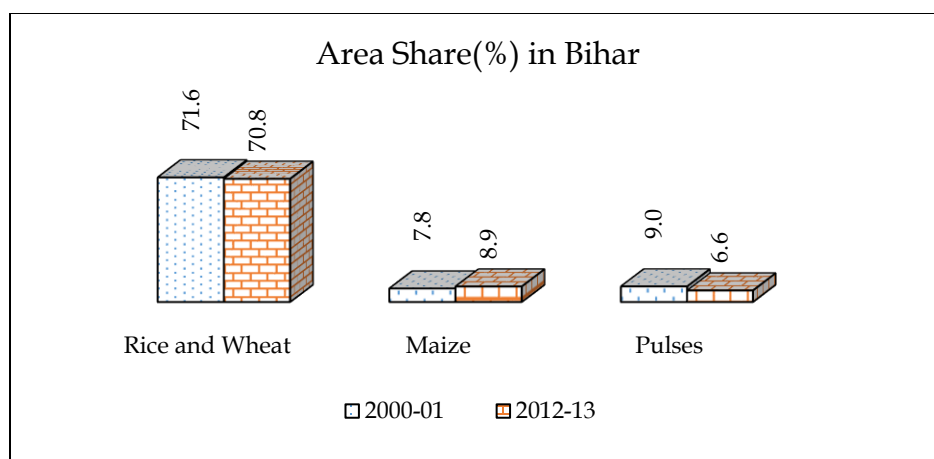


Figure 6.2(b) Temporal changes of crop shares over time in Bihar
Source: MoA (Website)

6.2.4. Steering Committee recommendation for Agriculture in Bihar

For reversing an economic decline, an Action Plan for development was necessary. A Steering committee chaired by Prof V.S. Vyas, formed towards this end noted that agriculture in Bihar was showing signs of diversification, the share of food crops in GDPA having fallen from 66% in TE 2001 to 48% in TE 2006 while the shares of livestock farming, fishery and horticulture improved. Maize was already gaining share. Bihar enjoys a comparative advantage in dairy farming but lacked of facilities for artificial insemination, and poor veterinary services. Fishery also holds promise but a lack of improved fish seeds is a constraint. High cost of cultivation, poor access to inputs and land were also maladies. Excessive or deficient rainfall and floods were other constraints

The Steering Committee suggested strengthening extension and research for improving productivity and welfare in agriculture. To overcome the constraints of small farms, the Committee advocated legalization of tenancy and other institutional engineering like systematic transfer of land ownership to small farmers, encouraging large farmers to shift to non-farm enterprises and distribution of surplus land. To improve crop yields land and water development

programmes, effective extension, proper use of irrigation, timely input supply and strengthening the credit supply institution were needed.

6.3. Crop production in Kosi basin and Sub-regional dimensions

Agriculture in the KBR is described in the following sections in a comparative perspective, comparing the KBR and NKBR and Bihar as a whole and the three sub-regions within KBR. Latest data on acreages available by districts is 2012-13 though for the state as a whole data up to 2014 is accessible in the open domain. Crop yield data is available up to 2010. Abbreviations used for crops are explained in Chapter 4.

6.3.2. Cropping Pattern and Diversification

In Table 6.1 among all the major and minor crop groups accounting for all of the cropped area Rice has the largest share followed by Wheat in Bihar, KBR and non KBR as well as the sub-regions of KBR. Maize is the third most important crop in all cases except in NKBR where Sugarcane and Vegetables are as important. Fruits and vegetables together make up 5.7% of cropped area in Bihar but 6.4% in KBR and within KBR the share of horticulture is least in MID and above 7% in UPPER. Fibre crops like jute, mesta and sunhemp is grown only in LOW sub-region in KBR.

Agriculture in Bihar is marked by crop specialization both in KBR and NKBR. Rice and Wheat are the leading crops accounting for 66 % of total cropped area in KBR and about 77 % in NKBR as of 2012-13 suggesting that KBR is more diversified than NKBR. Coarse cereals like Maize and Barley account for about 15% of cropped area in KBR and only 5 % in NKBR (Table 6.1). Fruits, vegetables, Oilseeds and all other miscellaneous crops get higher share of cropped area in KBR than in NBBR. Pulses, grown in both regions, are more important in NKBR.

Among the sub-regions the UPPER and the MID are more specialized in Rice and Wheat and LOW is more diversified towards other crops.

Table 6.1: Cropping pattern (% cropped area) in KBR in 2012-13

Crop	LOW	MID	UPPER	KBR	NKBR	Bihar
Rice	36.0	41.4	49.8	39.9	44.6	42.4
Wheat	21.1	34.0	28.5	25.2	31.2	28.4
Maize	19.8	7.1	6.5	14.5	4.0	8.9
Millets	0.3	0.2	0.3	0.2	0.5	0.4
Pulses	6.4	6.7	3.8	5.9	7.2	6.6
Sugarcane	0.9	3.5	2.9	1.8	4.5	3.2
Spices	0.2	0.1	0.1	0.2	0.1	0.1
Fruits	1.6	1.9	3.4	2.0	0.7	1.3
Vegetables	5.0	3.1	4.0	4.4	4.5	4.4
Oilseed	2.5	1.6	0.6	1.9	1.5	1.7
Fibre	5.5	0.0	0.0	3.3	0.6	1.8
Minor Crops	0.8	0.5	0.1	0.6	0.7	0.6

Source: MOA (website)

The Herfindahl index (see Chapter 4) calculated over all the twelve crop groups comprehensively finds that KBR is more diversified in its cropping pattern compared to NKBR and within KBR the LOW sub-region is most diversified followed by MID (Figure 6.3.a). UPPER is least diversified even compared to NKBR. The concentration of crops is reflected in the share of cropped area occupied by major food grains which is 78% in UPPER, 75% in MID and is less than 60% in LOW. The relation of crop diversification with natural vulnerability is mixed. While diversification is a conventional coping technique, specialization in food grains is also a way to conserve food security in the face of calamities. Over time the share of maize rose in KBR and in Bihar primarily due to the preference shown by UPPER sub-region (Figure 6.4 (b)) and the share of major cereals rice and wheat diminished.

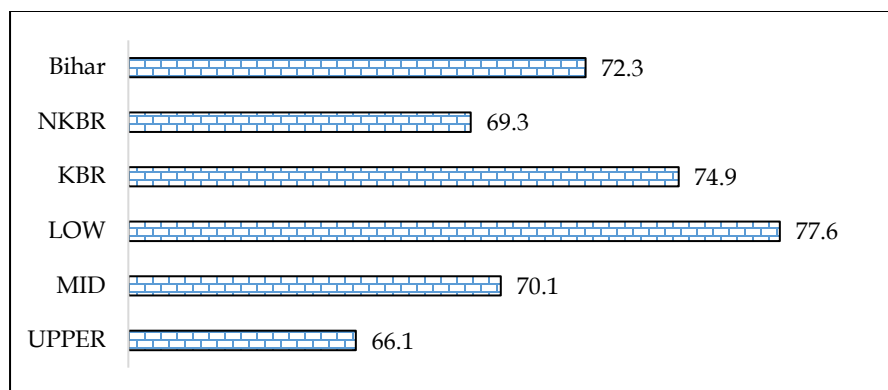


Figure 6.3 (a): Herfindahl Index of Diversification of crops
Source: MOA (website)

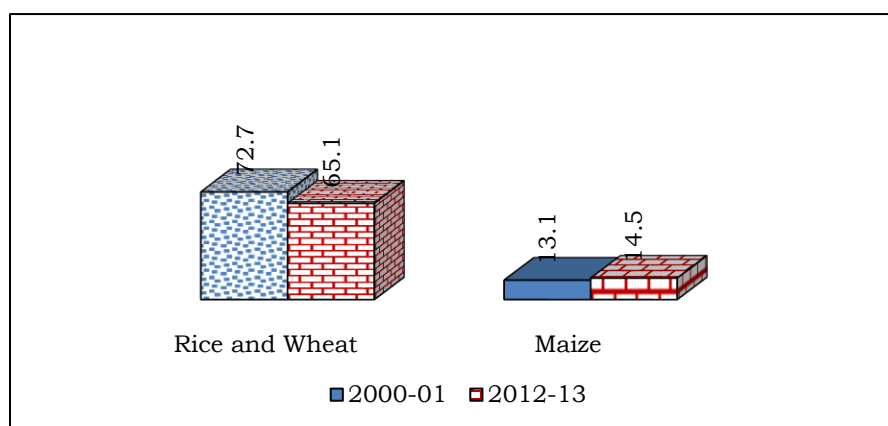


Figure 6.3 (b): Share (%) of major crops in KBR
Source: MOA (website)

6.3.3. Seasonality

Kharif and *rabi* are the two crop seasons in India while a third crop season is also observed in the crop calendar of certain states. In Bihar the third season is called *jayad* mostly for growing a few vegetables but more recently also Maize. On the whole both KBR and NKBR devote their cultivated land mostly to Rice and some Maize in the kharif season and to Wheat, Maize and some Rice in the rabi season. Agriculture in the rabi season is by far more diversified in KBR where Rice, Maize and Moong are kharif crops and Wheat, Rabi Rice, Maize, Rapeseed-mustard and Gram are rabi crops.

Table 6.3 (a) : Cropping pattern in *Kharif* season (2010)

Category	LOW	MID	UPPER	KBR	NKBR	BIHAR
Rice	75.4	97.4	89.3	82.7	91.8	87.9
Jowar	0.0	0.0	0.0	0.0	0.1	0.1
Bajra	0.0	0.0	0.0	0.0	0.2	0.1
Maize	18.0	1.5	5.6	12.3	5.9	8.6
Ragi	0.4	0.5	0.2	0.4	0.1	0.2
other cereals	0.3	0.0	0.0	0.2	0.1	0.1
Groundnut	0.0	0.0	0.0	0.0	0.1	0.0
Castor	0.0	0.0	0.0	0.0	0.0	0.0
Sesamum	0.1	0.1	0.1	0.1	0.1	0.1
Sunflower	2.0	0.0	3.5	1.7	0.0	0.7
Tur	0.4	0.4	0.4	0.4	1.0	0.8
Other pulses	3.5	0.2	1.1	2.4	0.6	1.4

Source: MoA, Website

Table 6.3 (b): Cropping pattern in *Rabi* season (2010)

Category	LOW	MID	UPPER	KBR	NKBR	BIHAR
Rice	7.7	0.6	1.0	5.1	1.5	3.0
Jowar	0.0	0.0	0.0	0.0	0.0	0.0
Wheat	55.2	76.0	83.3	63.9	73.6	69.5
Barley	0.3	0.1	0.2	0.2	0.6	0.4
Maize	17.8	5.8	2.4	12.8	2.2	6.7
Other cereals and millets	0.1	0.0	0.0	0.0	0.0	0.0
Rapeseed &	3.7	2.2	1.2	3.0	2.5	2.7
Linseed	0.6	0.4	0.5	0.6	1.1	0.9
Safflower seed	0.0	0.0	0.0	0.0	0.0	0.0
Gram	0.5	0.0	0.1	0.3	3.8	2.3

Source: MoA, Website

Accounting for more than half of the total cultivated area in both the KBR and NKBR, the *Kharif* season is watered by southwest monsoon. The pattern observed to be similar across each sub-region of KBR is partly accounted for by our considering *jayad* (summer) Maize as a *kharif* crop for convenience as *Jayad* and *kharif* seasons overlap in timing. Large variations are observed within KBR. Paddy (Rice) claims 97 % of cultivated area in MID sub-region compared to only about 75 % in LOW. On the contrary Maize claims almost 18 % in LOW but the share is less than 2% in MID. Sunflower and Pulses too claim fair amount of crop land. Wheat is the leading *rabi* crop but it is less important in KBR, its share (65 %) being less in KBR than in (74 %) in NKBR. Gram is more important

than Maize in NKBR. The downstream part of KBR in particular, diversify the rabi acreage among Maize, Paddy, Rapeseed-mustard and Pulses while the other two sub-regions specialize more in wheat.

6.3.4. Maize cultivation

Bihar and in particular the KBR region stands out in India for Maize cultivation. Maize is a crop of emerging importance gradually gaining place commercially in human food habits, food processing activities, as input in other industries and potentially as a feedstock of ethanol, a biofuel that can mitigate energy crisis and curb greenhouse gas emission.

In Bihar Maize is cultivated in all three seasons, but Jayad Maize is found important only in KBR (Table 6.3(c)) where Maize is grown substantially in all three seasons while *kharif* is the main Maize season in NKBR. Maize also gained acreage at the national level in the post-2000 period from 3.7% to 4.4% of the cropped area. It had a much larger share in Bihar even in 2000-01, which was largely stable up to 2003-04 at 7.8% but witnessed a structural jump to 8.9% in 2004-05 and then reached a peak of 9.1 in the two years 2010-11 and 2011-12 (Figure 6.4 (a)). Within KBR the sub-region LOW is the main Maize growing territory allocating a share of 18 % of area but over time cropping patterns changed. In 2012-13 The LOW sub-region still remains leading Maize growing area and it has increase 18.0% to 18.8% while substantial gains are made by Maize in the UPPER sub-region where the share became 8% from 3%. The MID allots 7% share which is showing a falling trend.

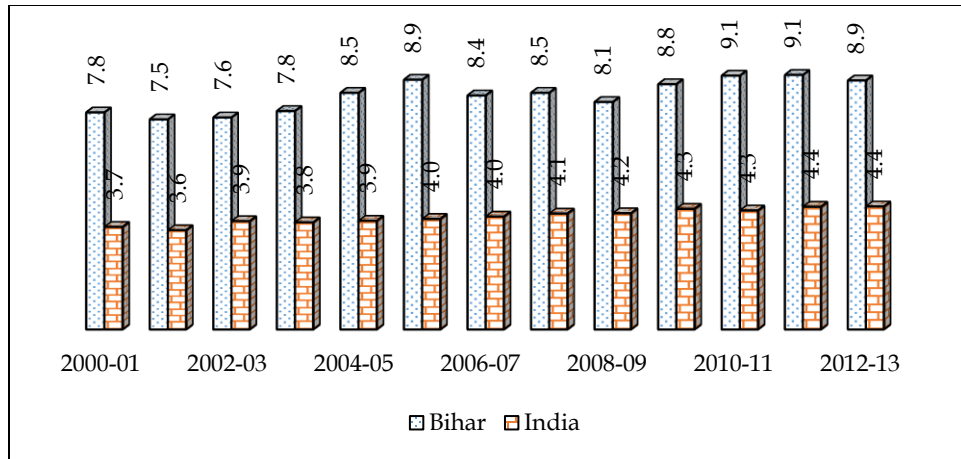


Figure 6.4(a): Comparison of Bihar and India over time in Maize area share (%)
Source: MOA (website)

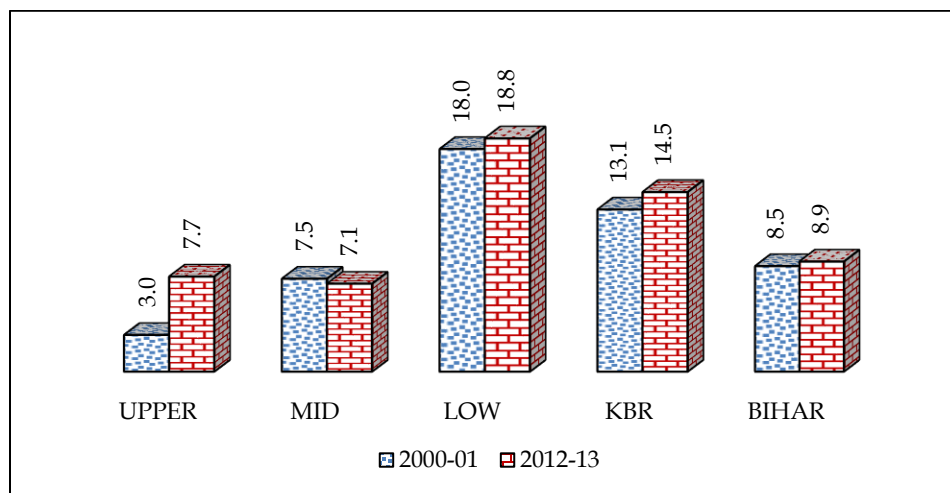


Figure 6.4 (b): Share (%) of Maize Area share over time in KBR
Source: MOA (website)

Table 6.3 (c): Maize cultivation in Bihar across agriculture seasons

Category	Kharif (Monsoon)	Rabi (Winter) (In %)	Jayad (Summer)
2000			
LOW	33.26	33.90	32.84
MID	16.00	50.28	33.72
UPPER	40.05	26.93	33.02
KBR	29.79	37.12	33.08
NKBR	69.43	13.44	17.12
BIHAR	52.74	23.41	23.84
2010			
LOW	29.20	35.02	35.78
MID	12.66	48.83	38.51
UPPER	28.47	36.34	35.19
KBR	24.97	38.64	36.39
NKBR	68.10	14.92	16.98
BIHAR	49.94	24.91	25.15

Source: MoA, Website

6.3.5. Rotations and inter-cropping

The cropping patterns analyzed so far also supported by the information collected from field visit suggest the prevalence of double-cropping (sequential cropping) as well as inter-cropping systems in both KBR and NKBR. Cropping systems of the region which refers to crops, crop sequences, and management techniques used on farms over a period of years are based on the following rotations: (1) paddy-maize rotation, (2) paddy-wheat rotation, (3) maize-maize rotation, (4) paddy-gram rotation, (5) paddy-rapeseed and mustard rotation, (6) maize-wheat rotation and (7) paddy-inter-cropping of wheat and rapeseed and mustard. Rotation 5 and 6 less common in KBR. Maize-maize rotation is found mainly in water-logged and severely flood affected areas. In some regions the land is left fallow or uncultivable in one season (kharif) in the rotation. Rice is sometimes inter-cropped with fodder crop (dhaincha) and wheat and maize with some pulses and vegetables.

6.3.6. Use of Chemical Fertilizers and their balance in use

Table 6.4 shows that agriculture in Bihar and in Kosi basin in particular is highly input intensive. As of 2010, total fertilizer consumption (NPK) in KBR is 213 Kg per hectare in comparison to 192 Kg per hectare NKBR, 201 Kg per hectare in Bihar and 142 Kg per hectare in India (now 125 Kg in India) (Table 6.4.a). The high intensity is observed in both kharif and rabi season. In the kharif season of 2010, consumption of nitrogen and potash was found higher in NKBR compared to KBR. Consumption of nitrogen in kharif season in KBR has fallen from 105 kg per ha in 2000 to 90 kg per ha in 2010. The LOW and MID parts, the major users of this nutrient, are the main contributors to the decline. The steep decline in the MID region can be associated with the change in the share of assured tube-well irrigation.

NPK ratio presented in Table 6.4 (b) shows imbalance in use of fertilizer in both regions but also an improvement towards reaching the ideal ratio '4:2:1' in each region and sub-region. The application of N and K is very high in the rabi season. The LOW sub-region which is most intense in use of fertilizers has shown the most phenomenal improvement in adjusting the nutrient balance. However, the appropriateness of nutrient use would actually depend on the specific soil's quality at the time of application that can be determined only by regular soil testing at micro level.

Table 6.4 (a): Consumption of Fertilizer across agriculture seasons (Kg/ Hectare)

Category	Kharif				Rabi				Total			
	N	P	K	NPK	N	P	K	NPK	N	P	K	NPK
2000												
LOW	101	24	9	134	119	31	15	165	86	24	10	120
MID	141	35	13	189	292	80	43	415	196	52	24	272
UPPER	58	11	3	72	139	47	19	205	84	23	8	115
KBR	105	25	9	139	165	45	23	233	113	31	13	157
NKBR	94	16	3	113	123	22	9	154	103	18	6	127
BIHAR	99	20	6	125	141	32	15	188	108	23	9	140
2010												
LOW	92	31	20	143	258	71	80	409	152	46	42	240
MID	92	29	17	138	159	49	30	238	120	37	22	179
UPPER	77	20	8	105	143	42	29	214	101	28	16	145
KBR	90	29	18	137	219	62	61	342	138	41	34	213
NKBR	120	35	13	168	186	50	27	263	137	38	17	192
BIHAR	107	32	15	154	200	55	42	297	137	40	24	201

Source: Fertilizer association of India, Delhi.

Table 6.4 (b): Comparing N P K Ratio across agriculture season

Category	Kharif	Rabi	Total
2000			
LOW	12:3:1	8:2:1	8:2:1
MID	11:3:1	7:2:1	8:2:1
UPPER	22:4:1	7:3:1	11:3:1
KBR	12:3:1	7:2:1	9:2:1
NKBR	27:5:1	14:3:1	8:2:1
BIHAR	17:3:1	10:2:1	17:3:1
2010			
LOW	5:2:1	3: 1:1	4:1:1
MID	6:2:1	5:2:1	5:2:1
UPPER	9:2:1	5:1:1	6:2:1
KBR	5:2:1	4:1:1	4:1:1
NKBR	9:3:1	7:2:1	8:2:1
BIHAR	7:2:1	5:1:1	6:2:1

Note: According to FAI, the **N:P:K ratio** was **8.2:3.2:1** in 2011-2012 against an **ideal 4:2:1** .

Source: Fertilizer association of India, Delhi.

6.3.7. Crop productivity

Rice yield in Bihar is relatively low at 1.0 tonne per hectare but in KBR the yield is even lower at 0.7 tonnes compared to 1.1 tonnes in NKBR. Within KBR the sub-region LOW has the highest rice yield at 0.9 tonnes followed by UPPER

whereas MID records poorest rice yield of 0.5 tonnes per hectare. In wheat the KBR has a lead over NKBR with a yield of 2.4 tonnes per hectare compared to 2.2 tonnes but the yield is less than all India average of 3 tonnes. Maize yield in Bihar is 2.8 tonnes per hectare, higher than all India. KBR has a considerable lead over NKBR accounting for the state's high yield rate. Within KBR, sub-region LOW has a Maize high yield of 3.9 tonnes followed by MID at 3.1 tonnes and 2.6 tonnes in UPPER. KBR is also more productive in oilseeds. The KBR lags behind NKBR in the productivity of pulses but within KBR, UPPER records higher than state average.

Table 6.5: Yield rates of crops (Kg/Hectare) in 2010-11

	Rice	Wheat	Maize	MLT*	Pulses	Oilseeds
LOW	932	2328	3959	875	739	1162
MID	522	2337	3088	1005	720	1161
UPPER	712	2478	2580	1025	913	1441
KBR	722	2381	3209	968	791	1255
NKBR	1157	2258	2198	997	979	968
Bihar	1024	2283	2797	963	879	1066

Note: * excludes maize crops. Yield rates (Kg/Hectare) at all India level in the same year are 2239 for Rice, 2989 for Wheat and 2540 for Maize.

Source: MoA (website)

Thus KBR displays heterogeneity in productivity. UPPER sub-region is weak in maize productivity but has advantage over the other sub-regions in all other cases (Rice, Wheat, Millets, Pulses and Oilseeds), LOW is fairly productive in Maize, Pulses and Oilseeds only and MID has advantage in Maize, Millets and Oilseeds. A comparison between years 2000-01 and 2010-11 finds impressive gain in Maize productivity from land, moderate rise of Wheat yield, stagnation in Pulses yield and a fall in Rice yield in the KBR (Figure 6.5).

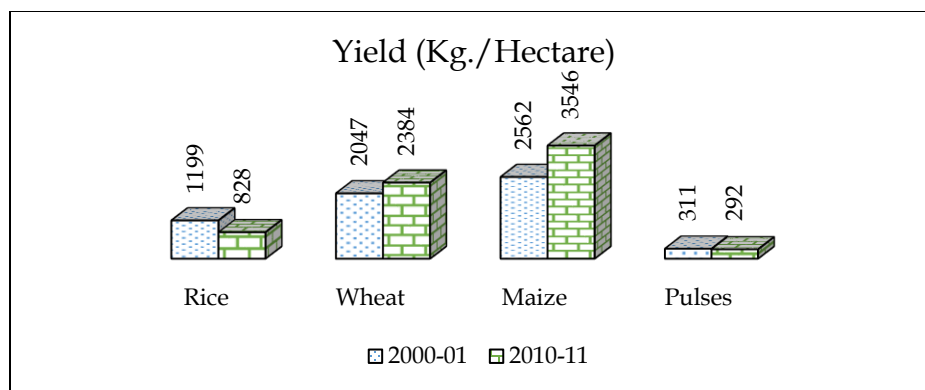


Figure 6.5: Improvement of crop productivity in KBR.
Source: MoA (Website)

6.3.5. Shifts within crop sector and the rise of Maize

Figures 6.6(a) traces decline in crop area, slow under Wheat, faster under Rice and rapid decline under Pulses in KBR. Increase in area is seen only under Maize though at a slow pace. Compared to Bihar, the acreages of KBR moved in tune but Rice area in KBR fell at slower pace, that of Wheat kept pace especially from 2005, Pulses area fell faster. Maize area rose in line with Bihar but has grown faster since 2008.

Yield of Wheat and Pulses showed a pick up in 2005 after a period of stagnation and Rice yield only became more unstable from 2004 while Maize yield improved continuously (Figure 6.6(b)). The yield rates of all four crops moved in sync between the region and the state but yield rates of all the crops excepting Maize are seen to be more variable in KBR than Bihar. Thus Maize has the risk reducing advantage of having higher average yield level with less variability than the state.



Figure 6.6 (a): Indices of crop area in KBR Kosi region and Bihar
Source: MoA (Website)

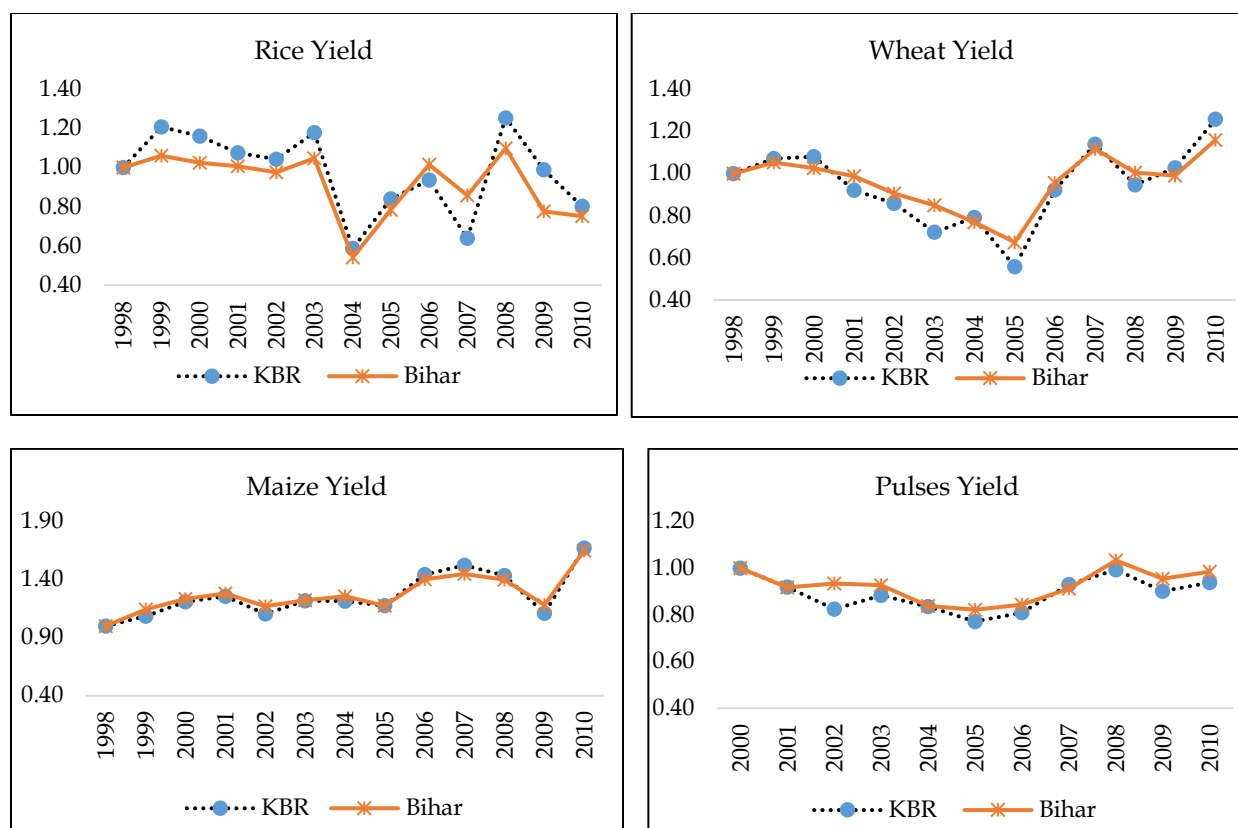


Figure 6.6(b): Indices of crop Yield in Kosi region and Bihar
Source: MoA (Website)

Considering the relative contribution of area and yield in causing production changes between 1998 and 2010 indices with common base show differences across the crops (Figure 6.6 (c)) but yield is the dominant positive force in all cases. In KBR area under food crops was mildly declining or was stagnant at best. Yield rate of Wheat declined in the sub-period 1998 to 2005 followed by sharp rise surpassing past records. Similar decline is noted in Rice in the first sub-period but with no clear improvement in the following period. In Maize both area and yield increased over the period and Pulses witnessed remarkable diminution of area accompanied by rising yield rate. In all cases the yield is more variable.

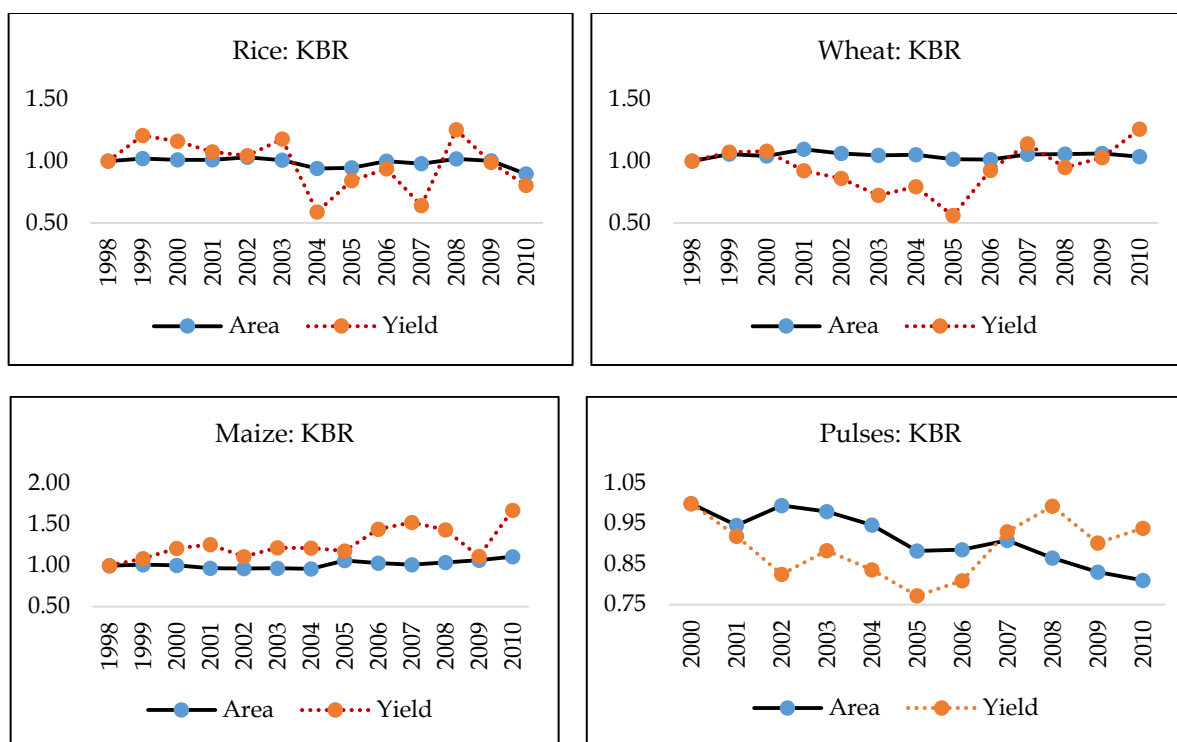
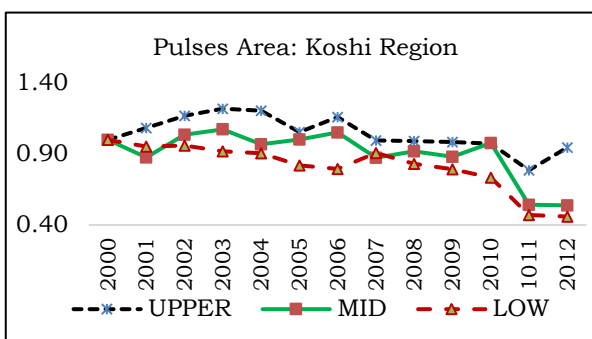
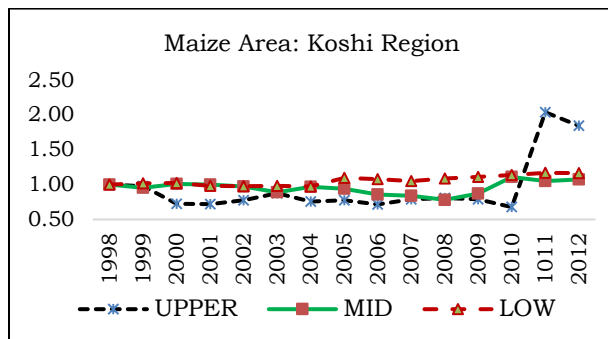
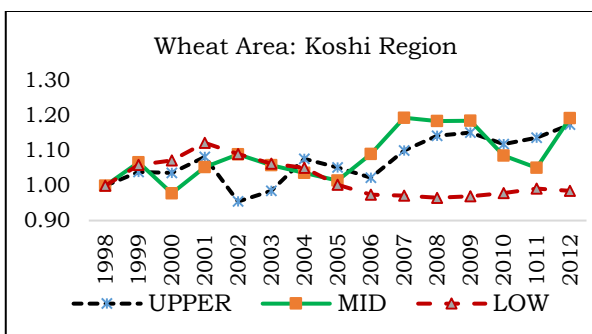
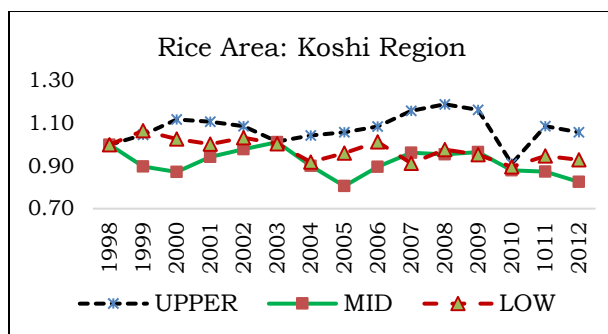
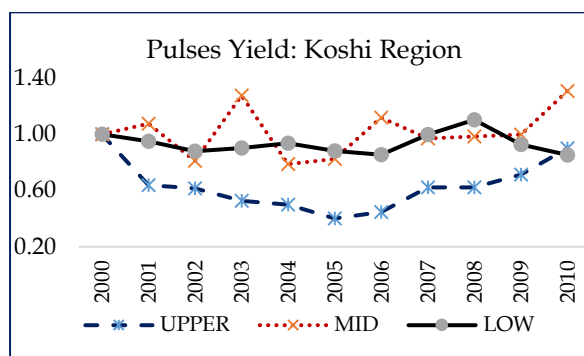
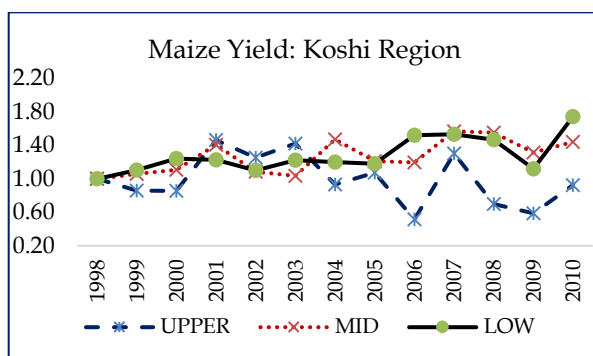
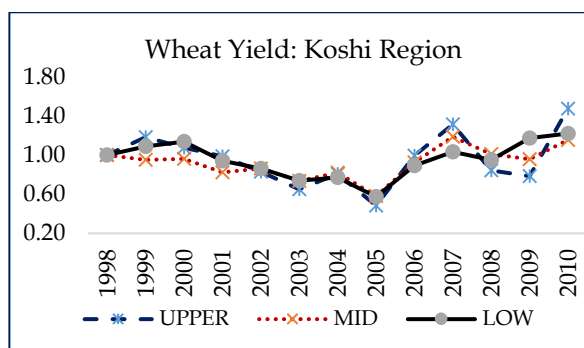
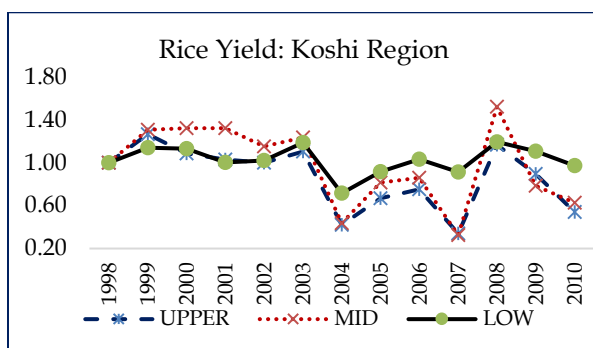


Figure 6.6 (c): Indices of Crop Area in Kosi Region
Source: MoA (Website)

Within KBR, area under Rice increased only in the sub-region UPPER up to 2008. Over all the acreage declined in all cases. Area under Wheat declined only in LOW but increased in the others but only since 2005. Maize area remained stable initially in the sub-regions. In LOW Maize area started rising from 2005, in MID from 2009 and in UPPER there was a spectacular increase in 2010. Pulses area contracted in all three sub-regions but while it showed a small improvement in 2011 in UPPER, a distinct plunge is evident in the other two in 2010 (Figure 6.6(D)). Total area under all four crops in KBR displayed a gradual decline in steps since 1999.



6.6 (d): Indices of Crop Area in different region in Kosi
Source: MoA (Website)



6.6 (e): Indices of Crop Yield in different region in Kosi
Source: MoA (Website)

6.4. Animal Husbandry

In rural Bihar 29% of the households possessed milch animals compared to 44 in India (NSSO, 2014b)). Total number of milch animal (cattle and buffalo) in Bihar is 4.9 Million in 2012-13 and the number (excluding goat) increased 2.5 Million from 2000-01 to 2012 but the number of goat slightly declined from 6.0 Million to 3.0 million. About 23% of the cattle in Bihar are cross-bred (28% in India) and 77% are non-descript. Goat constitute 37.9% of the total animals. Bihar produces 6.8 Million Kg of milk which at the per capita level is 188 grams/day in 2012 but which in 2000-01 was 88 grams/day signifying a good performance.

Milk production grew by 15% per annum in the same period. In 2012-13, Milk yield in Bihar is 3.26 kg/animal per day (excluding Goat: 4.29 kg/day) which has grown by 30% from 2000-01. Fodder land occupies 0.44% of area compared to 4.73% in India. Egg production is 837 Million number which grew annually by 1.4% from 2000-01. The growth of milk production and productivity is shown by indices at base 2000-01 but the relative performance of productivity per animal perhaps indicates a shift in animals towards exotic varieties though fodder required for feeding the animals has shown only modest growth.

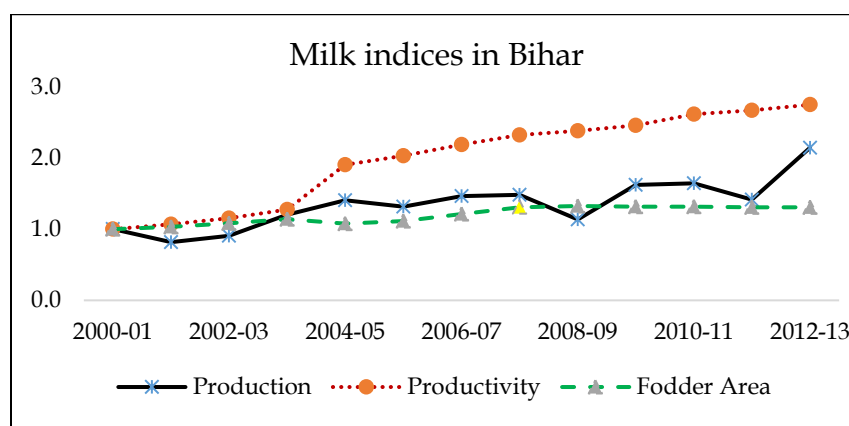


Figure 6.7: Indices of Milk production and fodder area in Bihar.
Source: MoA (Website), DoAH (2014)

In the Kosi basin (KBR) the number of animals add up to nearly 9 million much greater than NKBR's 4 million accounting for 70% of Bihar's animals (Table 6.6(a)). Cattle comprises 66% of the animals in KBR and Bihar. The majority of the animals (5 million) are found in the LOW sub-region which also possess the largest number of both cattle (4 million) and buffalo but the proportion of cattle in the stock is far higher at 78% in LOW than Mid(57%) and UPPER(61%).

Table 6.6 (a): Animal Scenario in Kosi Basin and Bihar (2011-12)

Region	Cattle	Buffalo	Total	Cattle	Buffalo	Total
	'000 No.			Female (%) Share		
LOW	3841	1052	4893	72.02	80.01	73.74
MID	1186	890	2076	80.14	67.51	74.73
UPPER	1060	678	1738	60.05	60.17	60.10
KBR	6087	2620	8707	71.52	70.63	71.25
NKBR	2188	1575	3764	71.31	76.43	73.45
Bihar	8275	4195	12471	71.46	72.81	71.92

Source: Agriculture Census (Website)

Table 6.6(b): Milk Production scenario in Koshi Basin and Bihar

Region	Production			C-B cattle	Ind. Cattle	Buffalo	Others
	2005-06	2013-14	ACGR	2013-14			
	'000 Tonnes			Share (%)			
LOW	1126	2043	7.73	29.9	35.3	31.7	3.0
MID	482	623	3.26	16.2	27.3	53.4	3.1
UPPER	271	423	5.72	4.2	39.3	53.0	3.5
KBR	1879	3090	6.42	23.6	34.2	39.0	3.1
NKBR	1089	1018	-0.84	6.7	33.6	59.4	0.3
Bihar	2968	4107	4.14	19.4	34.1	44.1	2.4

Note: ACGR: Annual Compound Growth Rate. Source: GoB-CD (Website)

Milk production in 2013-14 was a little over 4 million tonnes in Bihar and KBR contributed 75% of this and NKBR had a share of 25%. The growth rate of production was negative in NKBR but high (6%) in KBR. Interestingly, of this production 39% and 34% respectively came from buffalo and indigenous cattle while the C-B cattle produced only 24% of the total milk. In NKBR that share was even smaller and indigenous cattle were the largest contributors. The major

attributes of the NKBR and KBR of being drought prone and flood prone probably affected dairy performance as moisture availability is an advantage for the sector.

6.4.2. *Natural advantages and disadvantages*

Given the higher feed demand of exotic varieties, the potential of improving the indigenous varieties of animal is recognized in current policy. An expert committee set up by NDDDB 2007 for suggesting a breeding policy categorized the state by 'harsh' and 'better' conditions. Climatic conditions like average temperature, humidity and rainfall were taken into account in reference to milk production. Consideration was also given to the sentiments of the farmers some of whom were found to be risk averse, preferring small stocks of sustainable indigenous stock while others are willing to take the risk of rearing cross-bred and even pure-bred exotic varieties.

KBR districts in LOW sub-region Supaul, Saharsa, Madhepura, Katihar fall in the harsh climatic zone but with better resources. Seohar, Sitamarhi, Darbhanga and Muzzaffarpur constituting the MID are all resource poor but the first two districts also fall in the harsh climatic zone while the remaining two along with Khagaria of LOW region are privileged climatically. Bhagalpur, Begusarai and Samastipur are fortunate with better climate and better resources. Madhubani and East Champaran both in the UPPER region and Purnea and Araria in LOW suffer harsh climate as well as poor resource base. Thus the correspondence with sub-region classification is partial. Seohar, Sitamarhi, Araria and Purnia are least privileged for livelihood.

6.5. Marketing

India initiated marketing reforms in agriculture in the early 2000s and circulated a model Act for the states to legislate. This required the earlier regulated marketing rules Agricultural Product Marketing Act (APMC) to be changed. The

earlier APMC Act used to mandate a state recognized, representative or administered APMC committee to supervise marketing practices and create facilities and infrastructure for the market yards, regulate fair pricing and license the traders. The system has been functioning poorly leading to exploitation of producers by licensed agents and bureaucratic market committees.

Bihar repealed its APMC Act but did not legislate a new Bill due to disagreements and therefore has an open marketing system not regulated by the APMC Law. This has created greater freedom for farmers to sell and traders to join business and even corporate and other larger business entities from within and outside the state to enter the agricultural marketing scene though unfair practices can also arise in the process. Farmers are free to sell to traders, in the mandi (including the APMC premises) and to larger powerful buyers. They also sell in cooperative or collective unions that are forming. Primary agricultural cooperative Society (PCAS) is strengthened in Bihar for procuring grains at state fixed priced and the cooperative Sudha formed by NDDDB as part of operation flood is a successful agency for buying milk. Both have started facing competition from private sector (see Institutions A9.1).

Table 6.6: District Wise Procurement Summary (2015) (PACS and Vyapar Mandal)

District	Societies	Farmers	Paddy from Farmers	Paddy to Rice Mill	CMR from Rice Mill	To-SFC, CMR
	No.	'000	000 Tonnes			
	1840	47	291	226	161	150
LOW	(24)	(17)	(16)	(15)	(16)	(15)
	980	17	91	56	36	40
MID	(13)	(6)	(5)	(4)	(4)	(4)
	797	18	100	92	49	49
UPPER	(10)	(6)	(5)	(6)	(5)	(5)
	3617	81	483	374	245	239
KBR	(47)	(30)	(26)	(25)	(25)	(25)
	4101	194	1347	1142	747	733
NKBR	(53)	(70)	(74)	(75)	(75)	(75)
	7718	275	1829	1516	992	972
Bihar	(100)	(100)	(100)	(100)	(100)	(100)

Note: CMR: Custom Milled Rice, SFC: State Food Corporation, PACS: Primary agricultural cooperative Society. Figures in Parentheses are (%) share in Bihar. Source: Government of Bihar (website). Co-operative Department. <http://online.bih.nic.in/cooperative/ProgressRptToViewQTY.aspx>

7. Spatial inter-linkages of Crop yield functions

Kosi basin, with a dominantly agrarian economic structure falls short in land area relative to its population even while considerable land is lost to agriculture because of the ferocity of the river descending on the plains. The productivity of crops on the available sown land is therefore an indicative measure of resource use. Both economic and natural variables are drivers of crop yields but for policy making, it is important to understand the underlying ways the impacts occur. In the case of a river basin, we hypothesize that while economic incentives, infrastructure and weather are decisive factors, ignoring the spatial dimensions of yield determination can be misleading.

Crop yield functions are conventionally estimated econometrically by using either cross section (CS) data or time series (TS) data or pooling both types of data. While Panel data-based methods are replacing these methods for their superior efficiency (Hsiao, 1986, Baltagi, 2001, Quah, 1996 a,b), as geo-referenced data become increasingly available, even the conventional Panel data methods are criticized for treating observations as purely independent. The Tobler's Law of Geography 'everything is related to everything else and near things are more related than distant things' is a summary of the non-randomness in nature, but in reality 'neighbourhood' is an even more complex concept than proximity. The presence of slope in a river basin, creates spatial linkages that transcend ties within close proximity. Panel data estimates that do not account for spatial inter-linkages are likely to be inefficient or biased leading to unreliable results and erroneous conclusions.

Natural sciences accept nature's orderliness and spatial clustering of ecology, measuring patterns and gradients using Simpson index, the Herfindahl index and the Bray-Curtis index of (Hugh, 1973). First propounded by Moran (Moran, 1948, Dray, 2011, Dray et al., 2008), the data-based classical method of computing Moran index (MI) is becoming a

powerful tool for measuring spatial correlations. Earlier anticipated theoretically by Barron, 1940 and Mahalanobis, 1944 measurement was immensely advanced by the contributions of Geary 1954, Anselin 1995, Wartenberg, 1985, Getis and Aldstadt, 2004, Dray et al. 2008 and Dray, 2011 and computationally facilitated by the exposition of Anselin, 2001. The method remains yet in evolution (Chen et al, 2013) as interpreting and use still eludes simplicity.

Interests in spill-overs from technology practiced elsewhere, convergence of income across countries (Barro and Sala-i-Martin, 1995, Arbia and Piras, 2005) and industrial clusters and urban agglomeration with continuous night light (Fujita and Thisse, 2002, Wang et. al., 2012, World Bank, 2015) manifest the recognition that spatial dimensions are as much a feature of economic performances. While a spatial statistic is data driven, spatial econometrics, also under active development, is theory backed (Anselin 1995, Paelinck and Klassen 1979). Use of advanced spatial methods is relatively scarce in agricultural economics¹ except for a handful of studies conducted in context of a core-periphery relation of agriculture with urban centers (Chan 1994, Baldwin, 2001, Wang et. al., 2012). Agglomerations within agriculture (Wang et al., 2012), climate change scenarios (Baylis et al., 2011, Kumar, 2013), and comparative advantage (Arfa et al., 2010) of the sector are also interests that bordered on spatiality.

Many of the natural determinants of yield such as sub-soil attributes are geographically clustered and interactive. Not likely to be captured explicitly by the regressions, they get subsumed in the residual errors that in turn become mutually correlated. Spatial autocorrelation of yield is also not surprising given that there is a strong role of social exchanges and demonstration effects among close neighbors who are not bound by administrative or political demarcations. In context of a river basin more

¹ The idea of the economic man created by Adam Smith and the theory of the isolated state of Johann Heinrich Von Thunen in the 19th century, possibly the earliest treatments of spatiality or location were associated with the economics of agriculture.

important is the role of the sloping terrain along which the river carries down water and solid discharges from upstream regions. That rainfall, not just local but also in upstream regions, can have a serious role in deciding crop productivity is not hard to conceptualize. Clustering based on proximity need not be the leading basis of spatiality (Arfa et al., 2010)² and 'Neighbourhood', by this argument may not be symmetric.

For effective policies towards a productive agriculture, yield functions need to be modeled without unreasonable restrictions on spatial dependences. When geographical linkages are ignored, omission of important exogenous explanatory variables makes the estimator of the remaining coefficients inconsistent and biased (LeSage and Pace, 2009, Elhorst, 2011). While autocorrelation of error makes conventional yields function estimates less efficient, the presence of spatial dependence of yield and of influential spatially lagged exogenous variables is more damaging resulting in erroneous policy implications (Elhorst's (2003)) and flawed advocacies (Florax et al., 2002).

Econometric and pure statistical methods can be combined for addressing spatial relations by testing the significance of the MI over yield rates and their residuals after correcting for known explanatory variables in models with and without spatial corrections. While natural variability of yield is partly explained by exogenous driving variables it is expected that spatial correction will reduce the unexplained variability further through improved specification. In studying the convergence of per capita GDP across Europe, Arbia and Piras, 2005 found similar tendencies with both non-spatial and spatial models but the residual autocorrelation measured by MI got reduced but not eliminated.

However, the character of spatiality in crop yield determination cannot necessarily be associated with spatial concentration.

² An OECD study (Arfa et al., 2010) on France found no relation between regional concentration and spatial autocorrelation that may be created by policy such as the Common Agricultural Policy of EU.

7.2. Objective, Model and Data

Use of panel data is most common in all sub-areas of agricultural economics (Baylis et al, 2011, Mythili, 2008, Gulati and Kelly, 1999, Kumari and Ghosh, 2014) because it offers larger degrees of freedom and considers differences over both time and cross section. Since predictable cross sectional heterogeneity cannot always be ruled out, the simple OLS regression estimated with pooled data generally is not the best option. Use of dummy variables for CS units or the conventional Panel regression using Fixed Effect (FE) or Random Effect (RE) are commonly advocated.

In this chapter estimates of yield functions of four major food crops Rice and Maize in the kharif season and Wheat and Maize in the rabi season are made using secondary panel data of 16 Kosi basin districts of Bihar in India for the year 1998-99 to 2010-11 and applying econometric models that are corrected for spatial inter-relations.

The variables considered are Economic (E) including price of the study crop and the possible substitute crops, Local Weather (LW) including monthly rainfall and temperature, Infrastructure (INF) like Road and Energy, Irrigation (IRR) and upstream Rainfall (URF). The Basic Model (BM) for any crop in a general form taking account of spatial dimensions is expressed by the equation (1).

$$Y_{it} = \beta_0 + \mu_i + \rho \sum w_{ij} Y_{jt} + \beta_1 E_t + \beta_2 INF_t + \beta_3 IRR_{it} + \beta_4 LW_{it} + \gamma_k \sum m_{ik} URF_{kt} + \lambda \sum w_{ij} v_{jt} + \epsilon_{it} \dots (7.1a)$$

where subscripts i and t are for the district and the year, parameter μ_i is the fixed effect of the i^{th} district, ρ is spatial auto-regression of Y , λ is spatial autocorrelation of the error, γ is the effect of the exogenous upstream rainfall, w_{ij} is the spatial weight for neighbourhood defined as contiguity and m_{ik} is the weight for k^{th} upstream region and ϵ is iid. Thus neighbourhood is

broadly specified both by proximity (contiguity) and by geography (slope) specified only as an upstream region. The model is flexible in allowing ρ , λ and m to take values restricted to zero as also to take non zero values. In the most restricted case with no spatial correlation when all the three above parameters are zero, the model degenerates to the simple Panel Fixed effect (PFE) model in which the unexplained variances of yield are likely to be high.

Further, if $\rho=0$ and $\gamma=0$ and $\lambda=0$, $\epsilon_{it} = u_{it} \dots (7.1b)$

which stands for the residual in the non-spatial PFE model

$$\text{and } \sigma_i^2 = \sqrt{\sum_t \left(\frac{u_{it}}{y_{it}} \right)^2} \dots (7.1c)$$

is the variance of yield if the spatial inter-linkages are ignored and signifies yield volatility corrected for exogenous economic and local weather effects only but

$$\sigma_i^2 = \sqrt{\left(\frac{\epsilon_{it}}{y_{it}} \right)^2} \dots (7.1d) \text{ is the variances if the spatial inter-linkages are considered.}$$

Economic variable expressed as E_c corresponding to crop c (c = Kharif Maize (KMZ), Kharif Rice (KRC), Rabi Maize (RMZ) and Rabi Wheat (RWH)) refers the expected returns measured by the price of the crop or its revenue (priceX yield) of crop per unit area to allow for possible dynamics of driven yield to be part of the economic incentive. All the economic variables are deflated by average price of fertilizers (Nitrogen, Phosphate and Potassium) taken as a proxy of cost of production. The returns need to be anticipated in advance much before the harvest. Based on Nerlovian approach (Nerlove, 1958) the returns during the previous harvest are taken as proxy of farmers' expectations of future price. Prices impact on crop yield by rationally guiding inter-crop resource and input allocation. The impact of own price is expected to be positive. The effect may be negative if the alternative crop competes for inputs and resources but there could be other binding objectives for the

choices of crops such as food security and cash generation for purchase of farm inputs.

Local rainfall and temperature at different times within the crop season have varying implications on crop growth and grain formation depending on biological requirements of the crop at different stages of growth. Infrastructure includes road and energy use per unit of total crop sown area that help in mechanization, post-harvest processing and marketing of crops that adds to the incentive for growing the crop. Irrigation, also expressed as a ratio to cropped area, is supplementary to natural rainfall as a source of water but it has deep interaction with rainfall which enriches the soil, recharges the reservoirs and decides the volume of flow in rivers. Rainfall in upstream regions not only govern river flows and discharges leading to floods, waterlogging, water adequacy and scarcity but it also modulates the efficacy of irrigation that draws on reservoir storage and effective drainage.

The model allows economic variables to interact with infrastructure ($E \times INF$) and irrigation to interact with local rainfall ($IRR \times RF$) and with upstream rainfall ($IRR \times URF$) to determine crop yield. The sub-regions in KBR are supplemented with an additional extra-sample and trans-boundary region in Himalayas (TBH). Figure 7.1 describes possible geographical inter-linkages of yield determination through rainfall. Thus crop yield in all districts in region LOW may be affected by average rainfall in MID, UPPER and TBH, those in MID by rainfall in UPPER and TBH and in UPP only by rainfall in TBH. Keeping this in mind, the weight m_{ik} for i -th district is defined to be one for each district j in the k -th upstream region and zero otherwise where $k=1, 2, 3$ and 4 (LOW, MID, UPP, TBH) when upstream rainfall effect is accounted for. For yield auto-regression and error auto-correlation weight w_{jj} is one only for contiguous in-sample districts and zero otherwise where contiguity is constructed using Queen criterion (Anselin, 2013)³.

³ Most analysis of spatial auto-correlation adhere to a common definition of contiguity either Queen or Rook type determining what polygonal units in the grid are selected as neighbours. Following Chess analogy, neighbourhood by Queen criterion covers polygons with common

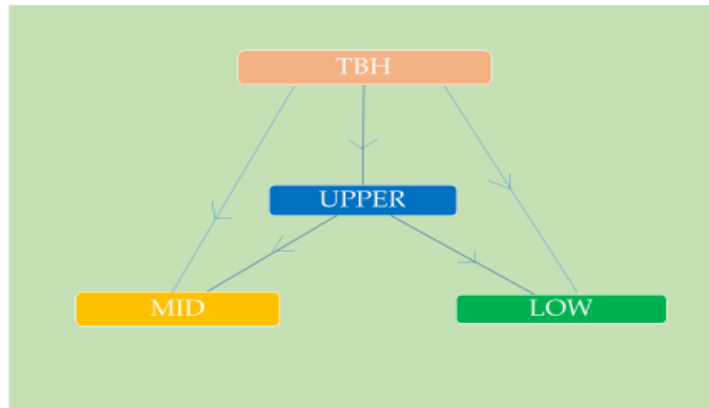


Figure 7.1: Connecting the river basin through rain effect on crop

The method followed in this analysis involves first estimating non-spatial models of simple pooled regressions (OLS), regressions with district dummy variables (RDD) and Panel regressions with Fixed effects (PFE) and statistical tests conducted for equality of parameters and heterogeneity among districts. The PFE is preferred as appropriate to the alternative Panel data RE model using the rationale that all Kosi districts are chosen purposively rather than randomly. The BM (equation 7.1(a)) as explained above and given in equation 7.1(a) along with five other different models nested in the BM are estimated. The six models testing for effects of exogenous upstream rainfall, error auto-correlation and yield auto-regression are (i) BM: with all three types of spatial dimensions, (ii) SPEXAC: upstream rainfall effect and auto-correlated residuals, (iii) SPEXAR: upstream rainfall effect and auto-regressive yield, (iv) SPEX: only upstream rainfall effects, (v) SPACAR: auto-correlated residuals and auto-regressive yield but no upstream rainfall effect and (vi) PFE: simple panel fixed effect with no spatial dimensions. The best equation is selected on the basis of statistical tests on spatial tests on the spatial parameters. In all selections only variables that appear with t-statistics exceeding one are retained.

boundaries and vertices while common vertices are excluded in Rook construction. Higher order contiguity can also be specified.

The yield rates at level, as residuals of non-spatial PFE model and as residuals of selected spatial model are examined for remnant spatial autocorrelation using global Moran coefficient (MI) Where

$$MCG(t) = (n / \sum_{i=1}^n \sum_{j=1}^n w_{ij}) (\sum_{i=1}^n \sum_{j=1}^n w_{ij} (y_i - \bar{y})(y_j - \bar{y}) / \sum_{i=1}^n (y_i - \bar{y})^2) \dots (2)$$

n= no of spatial units (or observations) and y_i and y_j are values of the variable of interest at locations i and j, $\bar{y}$ is average over all locations of the variable and t is a time point, i and j are proximate by Queen Criterion. The variances of PFE and selected spatial model are computed.

A strong Panel is constituted by pooling 13 years with 16 Kosi districts making up 208 observations. Ministry of Agriculture (MoA) of Government of Agriculture, Central Statistical Organization (CSO) and India Meteorological Department (IMD) are Indian official sources of data for all aspects except rainfall data TBH which is trans-boundary data obtained from website (GoN, website). Consistent monthly data required for constructing this variable was not available given the remote mountainous character of the region, the weak data collecting protocol and poor weather monitoring infrastructure of the time. Only option was to use the available data on the maximum yearly rainfall recorded in weather station in Biratpur in Nepal (GoN, website) though this is a serious limitation.

Mapping of the districts by three different ranges of actual crop yield suggests some degree of clustering especially for kharif rice (Figure 7.2). The MI of uncorrected yield rates reported in Table A7.2 shows and a rise in the spatial autocorrelation over time in all cases with the spatial auto-correlation coming out to be insignificant in 2000-01. On the whole indications on the spatial dimensions are rather inconclusive. The outcome is found similar (Figures A7.2) by plotting the autocorrelation of actual yield against yield of close neighbours.

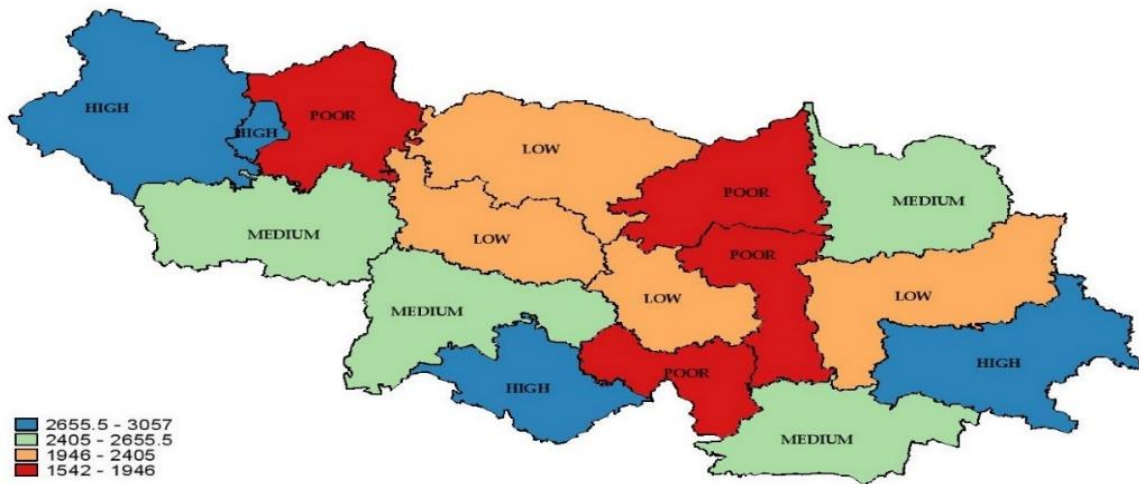
Rice Kharif Yield: Kosi Districts (2010-11)



Maize Kharif Yield: Kosi Districts (2010-11)



Wheat Rabi Yield: Kosi Districts (2010-11)



Maize Rabi Yield: Kosi Districts (2010-11)



Figure 7.2: Spatial distribution of Crop yields in Koshi districts in 2010-11

7.3. Results

The t-statistics of non-spatial models tabulated in Appendix Table A7.3 shows that between the PFE and the REGDD local rainfall in July and September and temperatures in June and July are determinants of kharif Rice yield. The indications are similar from OLS. The outcomes of comparisons are not much different for the other three study crops. Tests for restrictions on parameters leads to selection of BM for Rice and Wheat and

SPEXAC for Maize in both seasons (Table 7.1) suggesting spatial dimensions in yield determination.

Table 7.1: Probability based on F-statistics for testing Hypotheses					
Hypothesis	Model	Kharif rice	Wheat	Kharif maize	Rabi maize
Equal intercept	RDD	.0008	.000	.000	.35*
Independent residuals	PFE	.000	.000	.000	.000
Equal heteroscedasticity.	PFE	.011	.499*	.0002	.000
No error auto-correlation or yield auto-regression	SPEX	.000	.000	.000	.000
No auto-regression	SPEXAC	.000	.006	.321*	.797*
No error auto-correlation	SPEXAR	.000	.000	.000	.000
No upstream exogenous weather effects	SPARAC	.000	.000	.000	.000
Selected		BM	BM	SPAC	SPAC
Note: BM is the benchmark for testing the Spatial Models. * insignificant at 10%					

Upstream rainfall is a common impact on all cases supporting the specific significance of spatial inter-linkages of a river valley. Auto-correlation of errors among contiguous spatial units is another commonality suggestive of clustering of omitted determinants or common shocks on yield. The absence of auto-regression of contiguous yield of maize in both seasons may arguably have something to do with demonstration effect related to possible concentration of extension efforts in staple crops rice and wheat which enjoy disproportionately more consideration in public food policy.

Table 7.2: Effect of Upstream rainfall on crop				
	Kharif Rice	Kharif Maize	Rabi Wheat	Rabi Maize
Positive marginal effects				
UPPER Rainfall				June (pre)
UPPER Rainfall Interactions with IRR	April(pre), July (in)	June (in)	June (pre)	August (pre), September(pre)
MID Rainfall	July (in)			
MID Rainfall Interaction		August (in)		
TBH	Extreme (in)			Extreme(pre)
Interaction TBH				
Negative Marginal Effects				
UPPER Rainfall	June (in), July (in)			
UPPER Rainfall Interaction				
MID Rainfall	August (in)		July (pre)	
MID Rainfall Interaction	March(pre), October (late)	July (in)	August (pre)	July(pre), September(pre), October (in)
TBH		Extreme(in)	Extreme(pre)	
Interaction TBH	Extreme (in)			
Note: Kharif season is specified as period June to September-October and Rabi season as October-November to March-April with a slight overlap. The period of Rainfall is indicated by the month name and interactions are between basin level rainfall and local irrigation intensity. In brackets ‘pre’, ‘in’ and ‘late’ indicate pre-season rainfall, in-season rainfall and late season rainfall in relation to growing periods.				

7.3.2. Upstream rainfall effects on crop yield

Table 7.2 lists the different spatially lagged rainfall variables that are found significant in the different selected yield function equations. Rainfall in UPPER region directly has no effect except for a positive effect of June rainfall for rabi Maize yield and a negative effect of June-July (average) rainfall for kharif rice yield. Upstream rainfall mostly influences yield by its interaction with local irrigation.

7.3.3. Weather effects

The estimates suggest that rainfall in any month including a pre-season month can also be decisive. Rabi yield seems to be sensitive more to pre-

season (previous kharif season) rainfall. The effect of rainfall can however be favourable or adverse depending on the period of incidence and the needs of the crop. Local temperatures in different months have different implications depending on biological demands for grain formation. Interaction between rainfall and irrigation is common. Adverse interaction between irrigation and local rainfall at various periods is observed in a number of cases in the MID region suggesting drainage problems.

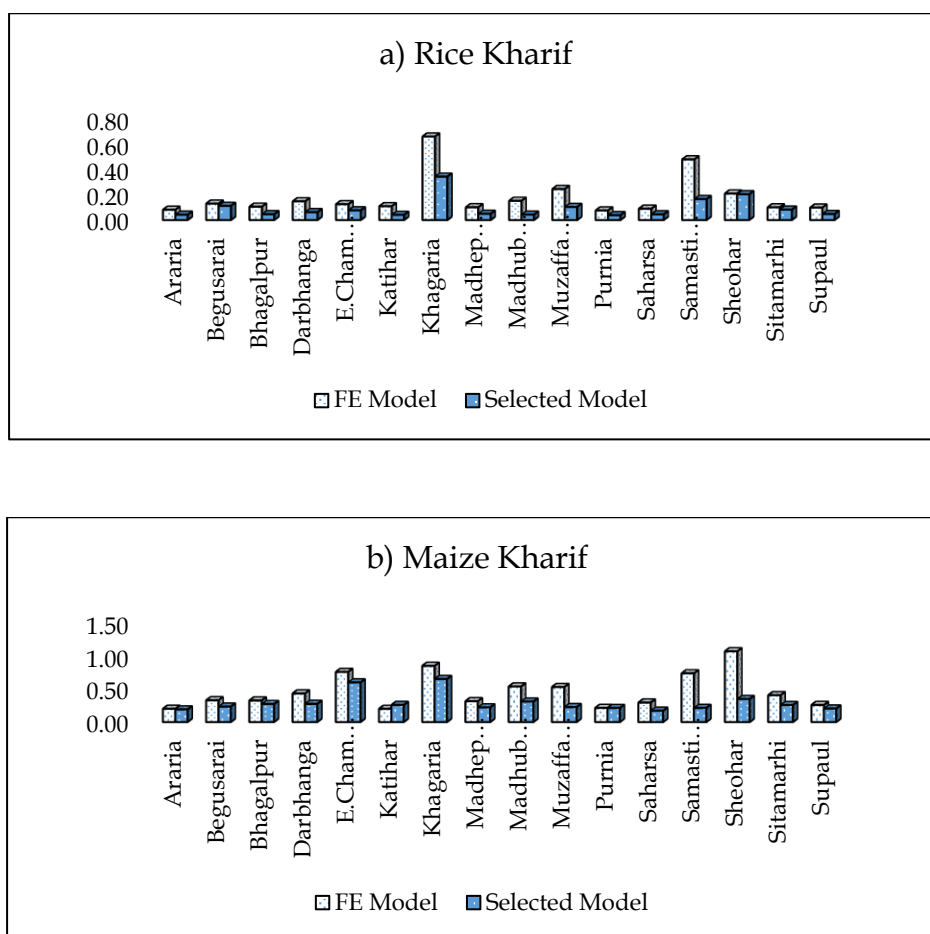
7.3.4. *Economic variables of consequence*

Significance of the economic variables (Table A7.2) show that markets do matter even in the vulnerable and underdeveloped region under study. In all cases farmers evince rationality in choosing to use inputs and resources for relatively more profitable crops. Expected negative signs of parameters of competing crops as Maize and Urad for kharif Rice, rabi Rice for Wheat and Wheat for rabi Maize are observed. Interestingly the responses of Maize yield are particularly sensitive to infrastructure especially energy. Also returns from maize appears to have a positive effect on wheat yield indicative of a non-competitive relation. Possibly wheat is grown as food crop but Maize is a commercial crop that helps to finance farm inputs.

7.3.5. *Comparison with Non-Spatial Models*

When estimating the non-spatial models we found improvement of the fit and moderate changes in the estimates when moving on from simple pooled regression treating all observation as homogeneous to the regressions that allow for district effects and further to Panel PFE regression. Tests on RDD suggested that the districts have differential implications for yield. The PFE model gave lower standard deviations than the RDD models though the coefficients are same as expected. Hypotheses of independent errors and equal heteroscedasticity among districts are also rejected (Table 7.1). The t-statistics of local rainfall of the OLS and the PFE model are given in Table A7.3 in appendix.

A number of divergences with respect to local rainfall effect are noted in relation to spatial selected models. For wheat local rainfall in September, December and January show negative effect in the spatial model as against October and December in non-spatial PFE model. Only March rainfall has a positive effect in both models as commonality. For rabi maize only February rainfall is favourable for yield in the spatial model against June and March in the PFE model.



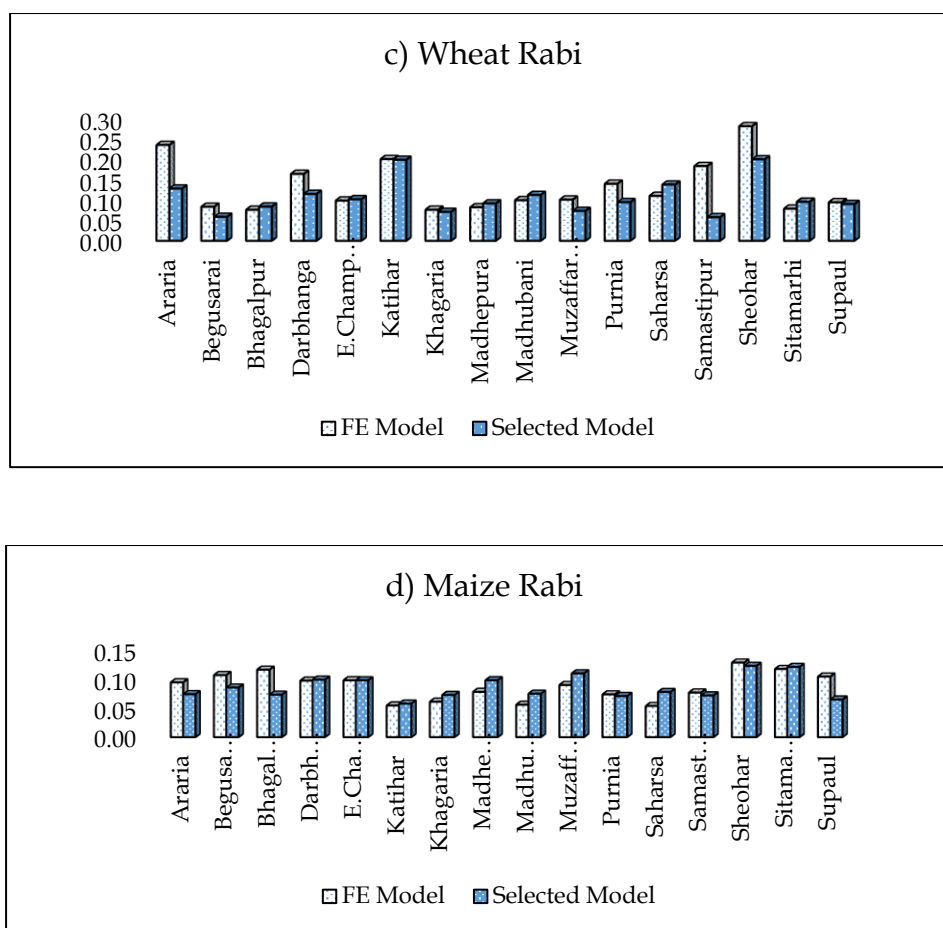


Figure 7. 3: Variance of Residual yield from Panel Fixed effect Model and Selected Spatial Model

The variance of yield after accounting for both local and spatial factors appears low compared to that produced by non-spatial model though there are exceptions. While this is not a statement about the variability of yield, the reliability of estimation can be deemed to be better. Yield of kharif crops such as rice in Khagaria and Samastipur and maize in Samastipur and Seohar, without factoring in the effect of upstream rainfall, is highly variable. The variance remains relatively high even after spatial correction in Khagaria for Maize and rice grown in the kharif season and for Seohar for rabi wheat and maize (Figure 7.3). The Moran coefficients provided in Table A7.3 showed some improvement in terms of reduced autocorrelation of the residuals compared to PFE. Elimination of autocorrelation however did not emerge from the correction.

7.4. Conclusions and Suggestion

Econometric crop yield function estimation is useful for making inferences about the drivers and constraints on farm productivity and for forecasting production, both of which can potentially facilitate policy making. This exercise was aimed to bring the spatial dimension into modeling taking account of three types of spatial dimensions arising from social exchanges, clustering of soil attributes and the geographical implications of slope in river basins of rainfall effects. There is a firm indication that spatial dimensions may not ideally be disregarded.

In the present study auto-correlation among residuals is a common form of spatiality as is the dominant effect of upstream rainfall leading to misrepresentation of the role of local rainfall, economic factors and temperatures in the exercises with non-spatial modeling. Moran coefficient or mapping, suggestive of mild spatiality, are inconclusive representations even when corrected for local effects but the residual autocorrelation is mitigated but not eliminated even when the model is corrected for spatial linkages (as in the study by Arba). The lack of data on mountain rainfall is a serious limitation of the study that would be overcome in medium future going by the efforts taken on data generation, management and sharing.

Empirical evidences that rainfall in upstream regions have a strong impact on crop can help in making more meaningful outlook. Overlooking rainfall in upstream areas could be a source of serious distortion and injustice in crop insurance programmes.

Chapter 8: Results of Primary survey data analysis

This chapter and the following one are devoted to analyzing the findings from primary household data collected from two Kosi basin districts. In this chapter assessment is made of general socio-economic aspects of rural agrarian life of different classes of farmer households in the study region with special attention to migration, income, agriculture, food security and gender roles.

Primary survey was conducted in 16 villages of the two districts Khagaria and Saharsa covering 8% of KBR and 3% of Bihar. The study region is designed not to be representative of the state or even the districts due to its riverine specificity. The design also made the sample cover all sections of rural households including both the landed cultivators and the landless farmers involved in farm work. The cultivators raised crops on leased and owned land and are categorized as Marginal, Small, Medium and Large while the Landless sometimes had owned and lost land due to natural calamities. Some of them were also found to lease in some minimal land of low quality to grow fodder, millets or vegetable for their animals and self-consumption. The details of the sample and specification can be found in Chapter 4. The results of the analysis are reported below. The two samples are designated as Sample 1, drawn from district Khagaria and Sample 2, drawn from district Saharsa in the discussions.

8.2 Social attributes

Between the two selected districts, the sociology is more varied in the Saharsa sample. Female-headed households as reported constitute 25% households in Sample 2 compared to 9% in Sample 1 of Khagaria. Secondary data places this proportion at 7.4% (Census), less than 10% at the all India level. The implication of the prevalence of female heads is complex. It may not mean women's empowerment and on the contrary imply greater burden for the

women. Minority households make up 10% in Sample 2 compared to 3% in Sample 1 (see Chapter 4 for specifications). On the other hand, the caste composition is less varied in the Saharsa sample. Sample 1 has higher share of SC households at 14% and also has 3% ST households that are completely absent in the Sample 2 (Table 8.1). Nearly 80% of the households in the Saharsa sample are OBC by caste. In general these casts have major share in Bihar's population. The number of educated members (high school) in an average household is 2.7 in Sample 2 and only 2 in Sample 1. The Ministry of Panchayat places Khagaria among the country's 250 most backwards districts.

Table 8.1: Societal characteristics of sample households

	Female headed	Size	Educated	SC	ST	OBC	Religious minority
	%	No./household				%	
Khagaria							
Landless	12.5	7.7	1.7	37.5	8.3	41.7	8.3
Marginal	12.5	7.0	2.2	15.6	0.0	46.9	0.0
Small	4.8	7.5	2.1	4.8	0.0	61.9	4.8
Medium	10.0	7.2	2.2	6.7	3.3	66.7	0.0
Large	0.0	9.5	1.8	0.0	7.7	69.2	0.0
Sample 1	9.2	7.6	2.0	14.2	3.3	55.8	2.5
Saharsa							
Landless	23.3	6.9	2.6	16.7	0.0	43.3	13.3
Marginal	35.9	6.4	3.0	3.1	0.0	89.1	9.4
Small	13.3	7.3	2.4	0.0	0.0	93.3	13.3
Medium	0.0	8.0	2.2	0.0	0.0	83.3	8.3
Large	0.0	10.0	1.8	0.0	0.0	100.0	0.0
Sample 2	25.4	6.9	2.7	5.6	0.0	78.6	10.3

Note: See Chapter 4 for specification details. Secondary data from Bihar: SC-13%, ST-1.2%, OBC-68% Literacy rate is 63%.Female headed-7.4%, Minority religion-17%. Source: Computed from Sample data, Census of India (2011)

The sample averages may be compared to data from secondary data on State Bihar but it may be kept in view that the study areas are chosen purposively from within two particular districts and the average of classes may be biased because of the representation given to each size class including a Landless class reflected in higher (or lower) weight in sample compared to their share in real life economic composition. The sample drawn in Khagaria has a higher representation of large farmer but marginal farmer finds largest representation in both samples (Chapter 4 Table 4.3).

The social profile can be distinguished among the 5 farmer household classes. The size of the average household is large at 7.6 members in Sample 1 and 6.9 in Sample 2 with little variation among the household classes, except that the size appears to increase with the size of land holding. The household size is very large at about 10 members in the Large class in both districts. Field interaction (see chapter 10) showed that the tradition of joint family prevails in the area to prevent land fragmentation.

ST households are found only in Sample 1 but mostly among the Landless. The share of SC households is also highest in the Landless class in both cases but in Sample 1 this community is represented more widely among the land classes of the sample. The share of OBC households is higher in the larger farm classes. In Sample 2 the Large class is entirely represented by OBC. The Minority religious group has larger share among the Landless but in Sample 2 these households are more evenly distributed than in Sample 1. The number of educated members (at least class X) varies from 1.7 to 2.2 in Sample 1 and from 1.8 to 3.0 in Sample 2 but the Large class has least number of educated and Marginal class has the highest in both cases.

The Landless and the Marginal classes have relatively high proportions of female headed households in both districts. In the Saharsa sample 36% of Marginal class households are headed by women and 13% of the Small holding class households has a woman head. Among the Medium class households, the head is a woman in 10% of households in Sample 1 whereas all heads are males in Sample 2. None of the sample households in the Large class has a female head in both districts.

8.2.2. Migration

Migration is historically a major characteristic of Bihar's sociology (see Chapter 3). As discussed migration can mean a sign of distress when triggered by 'push' when an exodus of unskilled labourers characterizes it. The distress

can be incident disproportionately more on the women if they are left behind alone to take care of households and manage farms even as 'heads' bearing the burden of work and the pain of social loneliness.

In Sample 1, 52% of the households experienced migration of at least one member in the last 2 years, the share being larger among the Landless (71%) and the Large (70%) classes. However not all migrants moved for earning. We found that 15% of the migrants are students who went to colleges outside the district. Field interactions and data collected revealed that with few exceptions the students went for higher studies in Patna or other districts endowed with colleges. Over 49% of the migrants are students in the Large holding class in Khagaria sample but in the Saharsa where 58% of the sample migrants in the Marginal class households and 28% in the Small class were students, it appears that smaller farm classes had also sent their children away from home for higher education (Table 8.2(a)).

Migrants who went for work to earn and are classified as migrant workers, constitute 80%-85% of the migrants. The average number of migrant workers is 0.8 per household (inclusive of households in which no migration is observed), but the number is above one among the Landless (Table 8.2(a)). Many migrant workers went to work outside the state. Inter-state migrant workers who went to other states make up 86% of the total number of migrant workers in Sample 1 and 80% in Sample 2 (Table 8.2 (a)).

Of the migrant workers 53% in Sample 1 and 41% in Sample 2 worked as labourer and all the migrant labourers in Sample 2 and 96% of them in Sample 1 have migrated to other states. This same ratio is 100% among the Marginal farm households and 91% among the Landless classes in Sample 1 and is 100% among the Small and 85% among the Landless classes in Sample 2. The Large class in Sample 1 and the Large and Medium classes in Sample 2 have relatively lower share of inter-state migrant workers and in all other classes in the samples the proportion is over 75%. Almost all the migrant

labourers in sample households work outside the state except among the Landless group in Sample 1 in which only about 10% are labourer in Bihar.

Table 8.2 (a): Statistics on Sample households with one or more migrants

	Migration	Students	Per household	Interstate	Labourer	Interstate
Classes	Samples % Households	% Migrants	No	Migrant Workers % Worker		% of Labourers
Khagaria						
Landless	70.8	2.94	1.46	91.4	54.3	89.5
Marginal	31.3	10.0	0.53	100.0	76.5	100.0
Small	66.7	9.52	1.00	75.0	52.4	100.0
Medium	40.0	18.51	0.53	75.0	37.5	100.0
Large	69.2	49.26	0.54	42.9	28.6	100.0
Sample 1	51.7	15.3	0.80	86.5	53.1	96.1
Saharsa						
Landless	70.0	14.29	1.33	85.0	65.0	100.0
Marginal	59.4	14.47	.91	78.3	30.0	100.0
Small	53.3	58.33	33	100.0	14.3	100.0
Medium	66.7	28.13	.92	63.6	27.3	100.0
Large	60.0	33.33	.80	50.0	50.0	100.0
Sample 2	61.9	21.05	94	80.0	41.0	100.0

Source: Computed from sample data

Both KBR districts experience migration intensely and many of the migrant workers are actually labourers in other states suggesting that migration among the Landless and smaller holding classes are signs of economic distress. Table 8.2(b) shows that the largest section of sample migrant workers went to Delhi and National Capital Region (NCR) reported as 29% of migrant workers by 29% households in Sample 1 and as 48 % of migrant and by 51% of households in Sample 2. Other north Indian states follow. Seventeen percent of migrant workers in Sample 1 and 13% in Sample 2 worked within Bihar representing intra-state migration but destinations for migrants in Sample 1 were spread across the country. The destination reported by households in Sample 2 was confined to northern India and to Bihar where migrants were mostly accompanied by their spouses. In Sample 1 too, migration to north India and intra-state locations was associated with spouse accompaniment. Although large annual remittances up to Rs 1.6 lakh from north-east were received, migrants in NCR and other northern regions

tend to send smaller amounts owing to higher cost of living and larger financial commitment to accompanying family.

Table 8.2 (b): Destination of Migrant workers by Households and Migrant attributes and their Remittances

Destination	Landless	Small *	All	Migrant Worker (%)	Spouse accomp anying (%)	Remittance Rs. '000/Year
<i>% Household</i>						
Khagaria						
NCR	47.6	21.7	29.2	28.6	21.9	5.4
North West	4.8	8.6	6.2	7.5	6.3	10.0
PJB/HRY/UP/JKH/C HH	23.8	30.4	23.1	23.5	25	15.9
WB/ORS	4.8	4.3	4.6	5	3.1	13.0
South	4.8	21.7	15.4	16	9.4	55.0
NE	0.0	4.3	3.1	2.5	3.1	165.0
BHR	14.3	8.7	18.5	16.8	31.3	20.1
Sample 1	100	100	100	100	100	12.1
Saharsa						
Delhi/NCR	54.5	46.7	50.7	48.5	55.6	29.9
North West	9.1	8.8	8.2	7.3	11.1	22.4
PJB/HRY/UP/JKH/C HH	9.1	24.4	23.3	24.3	22.2	26.9
WB/ORS	13.6	2.2	1.4	1.5	0	65.0
South	4.5	6.7	6.8	5.1	0	23.0
NE	0.0	0.0	0.0	0	0	23.0
BHR	9.1	11.1	9.6	13.2	11.1	25.6
Sample 2	100	100	100	100	100	14.8

Note: *Small includes also Marginal land holding households. Sources: Computed from survey data

8.3. Leasing practices, Land Use intensities, Land Loss and Cropping pattern

Farmers were asked on how much land they leased or shared in (or out), how much land was owned and what the procedures of leasing were. Not all such land could be actually sown. Farm size thus includes fallow and sown lands possessed by farmers while total cropped area and total production take into account all crops harvested in single and multi-cropped area. The farmers were asked about the extent and reasons of land loss they suffered during the last ten years to speculate on how much land could have been sown in addition to the current NSA.

The average farm size of our Khagaria sample is fairly large at 2.2 hectares compared to the district's average of 0.45 hectare reported in secondary data of year 2010 (collected at 5 year interval). The large difference may be attributable to sample design, definitional differences, the dynamics of land lost or even the lack of information on legitimate use of leased or common (such as river bed, river bank, water-logged) land. The size is smaller at 1.2 hectares in the Saharsa sample in which moreover, the average sizes are also smaller than in Sample 1 in all the farm classes. In the Large class the average farm size is barely 6 hectares in both samples.

8.3.2. *Leasing*

Leasing is nearly prohibited by law in Bihar (Chapter 6) though the institution of land tenure is undergoing a change (see A9.1) Leasing is therefore unrecorded and oral agreement between the owner and the user of land which is not officially recognized. Based on household responses it appears that all classes of farmers are in the habit of leasing in land informally on oral agreement. The proportion in the region is far higher than the 7% cited for Bihar (Chapter 6). Although systems are informal and unstructured general impression was that the lease is usually for one year at a time, though renewable and both cash rents (approximately about Rs. 8000-12000) cost (except labour cost) and produce sharing on 50:50 basis are in practice in the region. Subsidies and support are received by land owner. Dispute and even violence are the only ways of redressing violation of terms. Leased land constitutes 31% and 27% of the farmlands in Sample 1 and Sample 2 respectively (Table 8.3 (a)). The Large class leases the least portion in Sample 2 in which the Small class shows higher proclivity for leasing land. All classes in Sample 1 lease in uniformly (about 30%-33% of farm size).

Table 8.3 (a): Land endowments: available Farm size, cultivated land, use intensity, tenure, Irrigation and loss

	Farm size	Cropped area	Intensity of farming	Leased	Irrigated	Land lost
	Hectare		%		% Farm size	
Khagaria						
Marginal	0.61	0.93	152	32.14	87	0.00
Small	1.52	2.21	145	33.22	87	9.84
Medium	2.77	3.36	121	15.83	86	0.66
Large	5.86	6.07	104	32.14	98	16.25
Sample 1	2.19	2.66	121	30.74	88	14.61
Khagaria						
Marginal	0.55	1.19	216	25.57	89	34.48
Small	1.45	3.50	241	37.78	88	0.00
Medium	2.42	4.17	172	27.96	97	92.73
Large	5.04	4.99	099	10.00	91	51.72
Sample 2	1.16	2.12	183	26.97	90	19.80

Note: See Chapter 4 for specifications and definitions. Land lost is not part of Farm size.

Source: Computed from survey data

Only 6% and 9% of households from Sample 1 and Sample 2 were found to lease out land. Their corresponding average farm sizes were 1.1 Hectare and 1.8 Hectare. None of the households in Landless class in sample 1 were among them but 8 of the 30 Landless farmer in Sample 2 leased out some land. Majority of them had members working as non-farm labour but others were self-employed in trading, dairy and business. Few such households also earn from service. It is not evident that leasing was a localized transaction but interactions suggested that households who have insufficient supervisory capacity or had greater interest in off-farm jobs within or outside the area leased out land.

5.3.3. Farming Intensity

Information from Sample 2 suggests that farming is more intensive in Saharsa study region. As a result, the total cropped area of an average household is larger in this sample despite smaller farm size in all classes except the Large class for which farming intensity falls short of one. In field interactions we learnt that poor drainage leading to water-logging owing to monsoon rains prevented rabi sowing in some of the available farm land.

The relation between farm size and intensity of farming is ambiguous. It seems clearly inverse from data in Sample 1 but in the Saharsa sample, the Small class farm more intensely than the Marginal but then the inverse relation follows. Irrigation intensity is high in both samples but higher in Sample 2. The Large and Medium classes are relatively privileged but sensitivity of irrigation endowment to the scale of farming is weak.

5.3.4. Land loss

In the Khagaria sample, another 0.47 hectare per household over and above the 2.2 hectare of farmland reported could also have been cultivated if land loss could be averted. In the Saharsa sample where the average farm is smaller, land loss is larger at 0.76 hectares. All classes of households are victims of land loss and interestingly, even the Landless class of households actually have degraded lands which either lie idle or come under wetland status¹. They sometimes report leasing some insignificant quantum of poor quality land for growing fodder crops. In both samples the river has devoured land. Sedimentation and waterlogging are also significantly accountable in Saharsa. Among man-made interventions, canals have not claimed any land, highways made a small encroachment into 0.02 hectares in Sample 2 and embankments caused land loss of only 0.04 hectares and 0.01 hectares in Sample 1 and Sample 2 respectively.

Among all causes, the river has been the major usurper of land in both districts (Table 8.3(b)). The damage has been up to 1.16 hectares reported by the Large holding class in Sample 1 and 1.15 hectares by Medium class farmers in Sample 2. The implication of the loss of course is related to the total holding size and the lost land amounts to as much as 92% of the remaining farm size of the Marginal farm class in Sample 2. Flood is also reported as a cause of land loss by all classes except the Marginal class in both cases and the Large class in Sample 2.

¹ A wetland is a land area that is saturated with water, either permanently or seasonally, such that it takes on the characteristics of a distinct ecosystem.

Sedimentation is reported by all classes in Sample 2 and the Small and Medium classes in Sample 1. Man-made embankment to control flood also reduced land holding sizes of Marginal, Medium and Large classes in Sample 1. Water-logging affected smaller classes in Sample 1, sparing the larger classes. In Sample 2, all classes are victims of water-logging. Building public Highways resulted in small land losses mostly reported in Sample 2. Going by our sample, Saharsa lost more land (35%) compared to Khagaria (15%) and river movements proved to be a major factor. The loss seems to be biased against smaller classes though no class is spared. Operating farmers are reduced to landlessness by natural hazard

Table 8.3 (b): Land Loss (Hectare) due to natural and man-made conditions

	Flood	River	Water-logging	Sedimentation	Embankment	Canal	Highway	Others	Total Loss
Khagaria									
Landless	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Marginal	0.00	0.06	0.01	0.00	0.02	0.00	0.00	0.00	0.09
Small	0.06	0.01	0.01	0.02	0.00	0.00	0.01	0.00	0.11
Medium	0.11	0.45	0.03	0.01	0.07	0.00	0.00	0.00	0.67
Large	0.27	1.16	0.00	0.00	0.11	0.00	0.00	0.00	1.54
Sample 1	0.08	0.32	0.01	0.01	0.04	0.00	0.00	0.00	0.47
Saharsa									
Landless	0.00	0.26	0.08	0.10	0.01	0.00	0.03	0.00	0.47
Marginal	0.01	0.51	0.10	0.12	0.00	0.00	0.00	0.00	0.74
Small	0.05	0.75	0.31	0.48	0.04	0.00	0.02	0.00	1.64
Medium	0.02	1.15	0.24	1.01	0.00	0.00	0.02	0.00	2.44
Large	0.00	0.26	0.08	0.10	0.01	0.00	0.03	0.00	0.47
Sample 2	0.01	0.40	0.12	0.20	0.01	0.00	0.02	0.00	0.76

Source: Computed from survey data as reported.

5.3.5. Crop choice

Information about area and production was asked for a range of crops but in practice some crops were allotted miniscule area. Rice, Wheat and Maize are the major crops in Sample 1 accounting for over 96% of cropped area and 80% in Sample 2 (Table 8.3 (c)). The share of area under Rice, Wheat and Maize at 40%, 25% and 15% in KBR and 42%, 28% and 9% for Bihar based on secondary data (see Table 6.1) make Sample 2 more representative of the region while the corresponding shares highlight the specificity of in Sample 1.

The HHI at 75% by secondary data presents agriculture of KBR as more diversified than the primary data though closer by Sample 2 at 70%.

A cropping pattern heavily biased towards commercial Maize (47% of the cropped area) was observed in Sample 1 explaining the lower diversity. In contrast farmers grow more number of crops in the sample drawn from Saharsa where Pulses and Vegetables account for 6% of the area in addition to the same three crops as in Khagaria study region. Unlike in Sample 1, Rice and Wheat are more important crops than Maize in share of area in Sample 2, but even here the Large holding households allocate far higher share of area to Maize compared to the average.

Table 8.3 (c): Important crops of the study area as Shares (%) in total cropped area of all Land operating households and Herfindahl Hirschman index (HHI) of diversity

	Rice	Wheat	Maize	Pulses	Veg	Important	Diversity
Khagaria							
Marginal	21.5	23.7	49.5	-	-	94.7	64.98
Small	19.9	26.7	44.8	-	-	91.4	68.10
Medium	23.5	28.9	44.3	-	-	96.7	66.39
Large	17.8	29.7	50.2	-	-	97.7	62.76
Sample 1	21.1	28.2	47.0	-	-	96.3	65.37
Saharsa							
Marginal	38.7	27.7	14.3	4.2	2.5	87.4	73.72
Small	72.6	13.7	10.3	1.7	1.7	100	44.35
Medium	43.2	18.0	21.3	4.6	1.9	89	72.35
Large	24.0	20.0	44.7	8.0	1.2	97.9	70.22
Sample 2	46.7	20.8	18.9	4.2	1.9	92.5	69.73

Note: See Chapter 4 for specifications and equations of HHI. HHI is computed taking account of area under minor crops.

To understand crop diversification practices we looked for the tendency of farmers to favour of a single crop over others (mono-cropping). The share and identity of the largest crop in the cropping patterns and an index to measure diversification are used for the purposes. In both cases the largest crop occupied a little less than half the cropped area though the crop was different. Maize was the largest crop with 47% share in Sample 1 (not less than 44% in any class) while in Sample 2 almost the same share was taken up by Rice but there are variations across classes. All classes allocated 44-50% of area to the largest crop in Sample 1 but in Sample 2 the Small category devoted

73% of the area while the Marginal and the Large classes avoided monocropping.

To summarize, in Sample 2, farmers except in the Large class, allocated 60% to 80% to food grains Rice and Wheat but all the classes also grew Pulses and Vegetables. Also to be noted, Rice the dominant crop in Sample 2 had the smallest share in Sample 2 and Maize its leading crop got the smallest share in Sample 2 showing sharp contrast within the KBR. Diversity measured in terms of Herfindahl-Hirschman index (HHI) too suggested greater incidence of diversification in Sample 2 where the probability of finding different crops in two random trials of selection was higher at (69%) than in Sample 1 where the probability works out to 65%.

8.4 Crop choice, Inputs use, Productivity and Income of Major Crop growers

The following sections report the indicators of farming computed using the primary data both agronomic and market-based at the level of crops. The indicators are then compared across the classes and between the two samples.

8.4.2. Acreage, Yield and Price of three Major crops

Rice, Wheat and Maize are specified in this report as major crops (see Chapter 4) but not all cultivating farmers grow these crops. In the sample drawn from Khagaria, out of the 96 cultivating farmers, only 44 grow Rice, 83 grow Wheat and as many as 89 farmers grow Maize. Out of the 96 cultivating farmers in the Saharsa sample, 89 grow Rice, 86 grow Wheat and 69 grow Maize. Thus, the area yield, price and economic indicators of the major crops can be analysed by considering the average of the sub-sample of farmers who grow the specific crops.

Among only Rice growing farmers (Table 8.4) the average area allocated to Rice is 1.21 hectares in Sample 1 which is larger than the paltry 0.78 hectares sown by sample households in the more Rice dominated district of Saharsa.

Table 8.4: Samples Mean and Coefficient of Variation (CV) of Acreage, Yield and Producer price of Crop growing farmers only

Producer price of Crop growing farmers only							
Variable	Unit	Rice (Paddy*)		Wheat		Maize	
		Mean	CV	Mean	CV	Mean	CV
Kharagia District							
Sample 1	Number *	44		83		89	
Area	Hectare	1.21	.83	0.86	.82.	1.34	.75
Yield	Tonnes/Hectare	2.5	.35	2.5	.27	6.8	.40
Price	Rs./Kg	10.90	.07	12.80	.12	10.10	.09
Cost	Rs.'000/Hectare	24.0	.65	34.0	.42	39.5	.44
Saharsa District							
Sample 2	Number *	89		86		69	
Area	Hectare	0.78	.85	0.49	.76	0.55	.78
Yield	Tonnes/Hectare	2.0	.67	1.9	.30	5.6	.27
Price	Rs./Kg	10.00	.16	12.00	.08	9.50	.09
Cost	Rs.'000/Hectare	35.6	.68	26.8	.83	39.7	.51

Note: *Number of farm households in the sub-sample of households growing each crop. Price is producer price reported by the sellers of the crops. Area and yield are annual averages over seasons of the sub-samples. Conversion: 1.5Kg Paddy equivalent to 1Kg Rice. Source: Computed from sample.

The relatively large average Rice area among rice growing farmers implicitly suggests a preference for Rice even in Sample 1 though not all farmers are privileged with land suitable for the staple crop. Field interactions uniformly showed that farmers are reluctant to grow rice or other crops in the kharif season for fear of flood. If all sample cultivating farmers are considered Rice area is only 0.2 hectares. The yield rates are found lower in Sample 2 than in Sample 1 for all major crops.

The yield rate is also higher at 2.5 tonnes per hectare for Rice (paddy) and Wheat and 6.8 tonnes for Maize in the Sample 1 than in Sample 2. Yield rates of rice (converted) at 1.7 tonnes, wheat at 2.5 tonnes and Maize at 6.8 tonnes in Sample 1 and 1.3 tonnes, 1.9 tonnes and 5.6 tonnes in Sample 2 are higher than what is quoted for KBR in 2010-11 (Table 6.5). In 2014 the corresponding yield rates per hectare of Rice, Wheat and Maize were 1.9, 1.9 and 3.3 in Bihar and 2.4, 2.8 and 2.6 in India.

For Rice and Wheat, the yield is not only lower but also more variable in Sample 2 but yield of Maize is more variable in Sample 1 (40%) than in Saharsa (27%). The crop growing farmers fetch prices higher by Rs 900, Rs 800 and Rs 600 per tonne for Rice (paddy), Wheat and Maize respectively in Sample 1. Variability of prices is moderate, more in Sample 2 and less for Maize. In Sample 2, the cost of farming is remarkably higher Rice, Wheat and Maize compared to Sample 1. The cost is highly variable across farms.

At the disaggregate level, Table A8.1 fails to find any notable privilege of larger scale measured by farm size. As would be normally expected, the area under all crops increase with the size of holding but yield rates of crops are not sensitive to farm size. In fact, the smaller classes appear marginally more productive in Wheat. There is also no pattern of producer prices across the classes to indicate superior bargaining power of larger classes. In most cases the Marginal class obtains relatively higher prices. Most sensitive to scale of farming is the cropping pattern which moves in favour of Maize in larger farms.

8.4.3. Fertilizer use

Sample 1 registers higher average fertilizer use intensity in all three major crops (Table 8.2(a)). It is highest for Maize in both cases and are respectively 211 Kg, 202 Kg and 284 Kg for Rice, Wheat and Maize in Sample 1 and 107 Kg, 148 Kg and 150 KG in Sample 2 as against secondary data available for Bihar state averages of 99, 145 and 138 (MoA, Website).

8.4.4. Cost of cultivation, Labour cost and Labour hiring

Except for Wheat, labour cost per hectare is higher in Sample 2 than in Sample 1 (Table A8.2 (a)). Hired labour has the highest share in Rice in both samples at 44% and 69% in Sample 1 and Sample 2 respectively, but for Wheat, the share of labour is 26% in both samples while for Maize labour has a higher share of over 46% in Sample 2 than in Sample 1. Indeed, Table 8.2

(b) indicates that Labour hiring is more common in Sample 2 from Saharsa where the hired laborer's share in total person days employed is 74% for Rice and 69% for Maize compared to 65% and 59% for the same crops in sample 1. The reported wage rates averaged across households are however found lower in Sample 2 in all three crops, being Rs 150-156 per person-days compared to Rs 180-Rs 191 in Sample 1 (Table A8.2 (b)). The average farm labour wages reported in the surveys are Rs 180, to Rs 193 in Sample 1 higher than Rs 150-Rs 156 in Sample 2. Secondary official data reports farm wages to be Rs. 181 in Bihar (MoA, website) closer to Sample 1 findings but higher than in Sample 2. Stipulated wage paid for MNREGA in the same year is Rs 162 (PTI, 2016) which matches with Sample 2 findings. Incidentally MNREGA is found to be in operation drawing people from sample villages in Saharsa and also reportedly engaging a few members of Sample 2. Our field visit interactions in Khagaria study area gave the impression that MNREGA work is not operation at the time of visit mostly because work is not identified on account of paucity of usable land for roads, water-works etc. A few households however reported earnings from NREGA.

Among the farm-classes, labour hiring increases with farm size in Sample 1 but displays no specific pattern in Sample2 (Table 8.2(b)). Wage rates paid by households vary with the time of demand, type of tasks given and availability of jobs in the area at the time of hiring. The average wages therefore differ but with no visible sensitivity to the farm size class except that the Marginal class in Sample1 seems to pay relatively higher wages for Wheat and Maize.

Labour's share in the cost of Rice cultivation is uniformly high between 53%-72% in Sample2 but peaks at the Medium class level in Sample 1 in which Large farmers spend less (Table A8.2 (c)). For Maize, the labour cost share rises with farm size. Family labour constituting 35% to 42% of total person days in Sample 1 and 26% to 55% in Sample 2 is used more profusely for Wheat among the crops. Hiring is more extensive for growing Rice (TableA8.2 (b)). Hired labour cost per hectare of the average farmer is Rs 11,000 and Rs

24,000 for Rice in Sample1 and Sample2 respectively as against Rs. 8169 obtained from secondary data. Labour cost is relatively low for Wheat in both samples.

Among the other costs are costs of seed, machine services, pesticides and fertilizer. Sample 2 reports lower average costs on non-labour inputs. Total cost of cultivating Rice per hectare is remarkably higher in Sample 2 but is more comparable between the two samples in cases of the Wheat and Maize. Per hectare, the cost is Rs 24,000 and Rs 36,000 for Rice, Rs 34,000 and Rs 27,000 for Wheat and Rs 40,000 for Maize in the samples of the two districts of Khagaria and Saharsa respectively. Corresponding figures taken from secondary data (MOA-Website) are found to lie in the range (Table 8.2 (c)) but only the cost of cultivating Maize appears higher in the samples possibly attributable to the disproportionate weightage allotted to the Large class (Table 8.2 (d)) in the sample.

Total cost of cultivation per hectare of crop area rises with farm size in Sample 1 but no clear pattern emerges in Sample2. Total paid cost on inputs works out to Rs 16,400 for Rice, Rs 34,500 for Wheat and Rs 30,400 for Maize among the numerically most dominant Marginal farm households of Sample1. The same costs are higher in Saharsa sample. The intensity of labour hiring in the migration prone Saharsa may be responsible for the higher cost in Sample 2.

8.4.5. *Incomes from major Crops*

Given the presence or absence of patterns in producer prices, crop yield and costs of crops in favour of larger farms, the same in incomes per hectare may be examined by imputing home consumed output values and deducting only the paid costs (see Chapter 4). Computing the difference between revenue at reported producer prices and the paid cost, positive income per hectare is found in cases of Rice and Maize for the average household of Sample 1 in Khagaria and only of Maize in Sample 2 in which Rice incurs a large loss of

Rs 15,000 per hectare. Income from maize is far higher than Rice even in Sample 1 (Table 8.5).

Table 8.5: Annual Crop Incomes and Paid cost of Crop growing farmers and Weights from cropped Area of Land operating Sample households

Districts	Revenue			Paid Cost			Income		
	Rice	Wheat	Maize	Rice	Wheat	Maize	Rice	Wheat	Maize
	Rs. '000/ Hectare								
Sample 1 (Khagaria)	27.2	31.6	68.9	24.0	34.0	39.5	3.2	-2.4	29.4
Sample 2 (Saharsa)	20.3	23.5	53.1	35.6	26.8	39.7	-15.3	-3.3	13.4

Note: Source: Computed from Sample data

In Sample 1 the income per hectare from Rice is highest for Marginal holding households at Rs 10,260 and negative only for the Large class at Rs12,000 (Table A8.2 (d)). Incomes per hectare from Wheat and Maize too are largest for Marginal class and falls with farm size. Only Wheat shows this declining tendency of income with farm size in Sample2. Thus except Maize which is a far more paying crop than food crops Rice and Wheat, agriculture is hardly profitable even if self-consumption is viewed as part of returns and is loss making in terms of cash generation.

If only the sold product is considered for computing revenue the cash income from Rice and Wheat are negative in both cases where the excess payment in cost over revenue can be accounted for as a premium paid for assured food for consumption (Table 8.6).

Table 8.6: Cash Incomes earned from production of Major crops

Farmers	Rice	Wheat	Maize
	Rs. '000 /Household		
Marginal	-4.93	-8.69	23.34
Small	-11.64	-17.54	-1.62
Medium	-22.12	-18.11	28.33
Large	-58.17	-38.23	37.97
Khagaria	-16.20	-17.96	27.22
Marginal	-16.22	-7.29	-2.87
Small	-15.34	-17.68	-0.36
Medium	-45.48	-17.54	-0.29
Large	-30.27	-14.96	8.45
Saharsa	-22.11	-10.24	-0.97

Note: Cash income is revenue from only sold produce less paid cost. Source : computed from Survey data

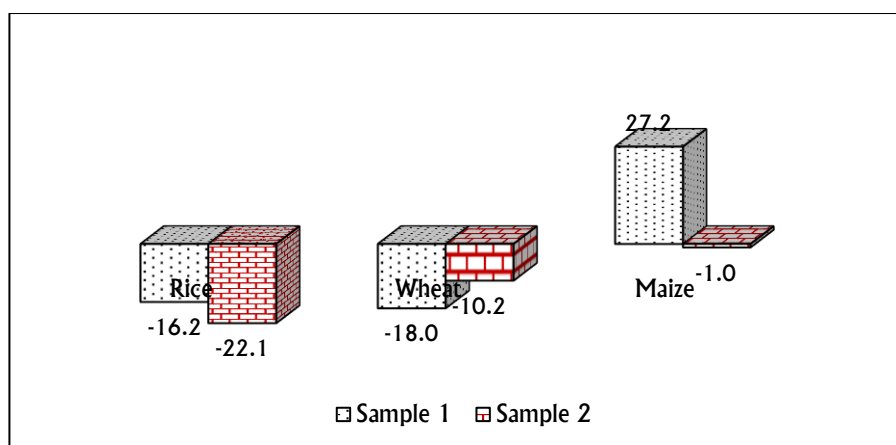


Figure 8.2: Cash Incomes (Rs. '000)

8.5. Commercial farming of Maize

In Bihar 34% of people are estimated to live below the poverty line (BPL) and 45% of the agricultural households hold BPL card (See Table 5.3(a)). Poverty is evident in Khagaria and Saharsa both from secondary data and from actual field observation. MGNREGA the public works programme for providing minimum employment for subsistence is weak in the area (see Chapter 9). Food security is a serious concern not only because of the low productivity and low incomes from crops but also for the fear that heavy rains and floods can disrupt physical food supply from market. The government steps in to counter the low purchasing power over open market supplies through subsidized public distribution (See Chapter 3) but some mount of subsistence farming is not avoidable. In reality we found that farmers resort to all the three sources namely farm production, open market and PDS to meet food needs (more in Chapter 9).

It is therefore of interest to enquire on who the commercial Maize farmers are and their motivation for growing for commercial Maize as observed in some areas of KBR. Rice and Wheat are the main staple food crops of the people and Maize is mostly a way to acquire purchasing power over food and other commodities through the market using cash as the medium. Maize therefore provides access to food though it also brings cash and the benefit it provides

through these two means may vary across farmers of different regions and classes.

8.5.2. Identifying and distinguishing Maize farmers

While Maize is a popular crop in the study region, the question arises: Who is a Maize farmer and in what way is he or she distinct? Firstly, intuition suggests that more are will be sown under Maize or any other crop if the farmer possess more land although there can be exceptions. This makes identifying those with outstanding preference for a crop. The plot of Maize area planted by a household against its farm size shows a clear positive relation though the exact form of the relation is ambiguous (Figures 8.3). A quadratic effect is suggested by the plots in both cases, peaking in one and rising in the other. Further a regression of maize area against the farm size as an explanatory variable in a polynomial relation also confirms that larger holdings allocate greater absolute area to maize. This may be nearly tautological relation but the line of best fit can be used to separate out the farmers who show preference for maize (MZF) from those who do not (NMZF). Observations that lie above the line or show positive sign of error in the regression (u) are deemed to be MZF who sow more area under Maize taking account of their limitation of land

Where $AREA_i = a + b_1 FSZ_i + b_2 FSZ_i^2 + \dots + b_n FSZ_i^n + u_i$

Farmer i is specified as a Maize farmer (MZF) if $u_i > 0$ and otherwise the farmer is specified as Non_Maize farmer (NMZF).

Comparison of the means of different social and economic attributes of the two types of households are made with statistical tests. In Sample 1 the NMZF and MZF households differed little. Although the NMZF group had a higher mean of cropped area, irrigation intensity and Maize productivity the differences were not statistically significant. The only difference observed was the significantly larger household size of the NMZF than MZF suggesting the

role of greater food needs. In Sample 2 household size did not differ significantly between the two groups though it was larger for NMZF as in Sample 1 but the NMZF had a larger cropped area and was more commercialized in respect of marketing food grains. Less intensive and curiously less market oriented food crop farming were associated with Maize preference. MZF have greater access of food (rice) through maize sale but the difference is not significant in Sample 1 but in Sample 2 maize farming is Maize farmer is identified with greater access over food.

Table 9.3: Maize cultivators compared with other cultivators

	Units	Khagaria				Saharsa			
		NMZF	MZF	All	T-stat	NMZF	MZF	All	T-stat
Cropped area	Hectare	2.9	2.4	2.7	1.26	3.1	1.6	2.1	2.36*
Irrigation	%	87.8	86.5	87.3	0.31	86.7	88.5	87.8	-0.34
Foodgrain Production	Tonnes	3.1	3.2	3.1	-0.28	2.3	2.0	2.1	0.97
Foodgrain Marketed	%	25.4	75.5	47.6	-1.26	30.9	17.0	22.3	2.60*
Maize Productivity	Tonnes/Hectare	6.6	5.9	6.3	1.15	2.4	2.5	2.5	-0.79
Farm Size	Hectare	2.1	2.3	2.2	-0.37	1.1	1.2	1.2	-0.08
Household Size	No	7.8	7.2	7.5	-1.94*	7.1	6.8	6.9	0.39
Land	Hectare/Person	0.3	0.4	0.3	-0.69	0.2	0.2	0.2	-1.16
Worker	%	31.0	28.6	30.0	0.61	10.6	10.9	10.8	-0.21
Access Foodgrain (Retail)	Tonnes/Hectare	0.9	1.3	1.1	-1.5	0.4	0.9	0.6	-3.56*

Note: NMZF: Non-maize farmer, MZF: Maize farmer. Differences in mean (NMZF-MZF) are subjected to t-test. Source: Compute from Survey data

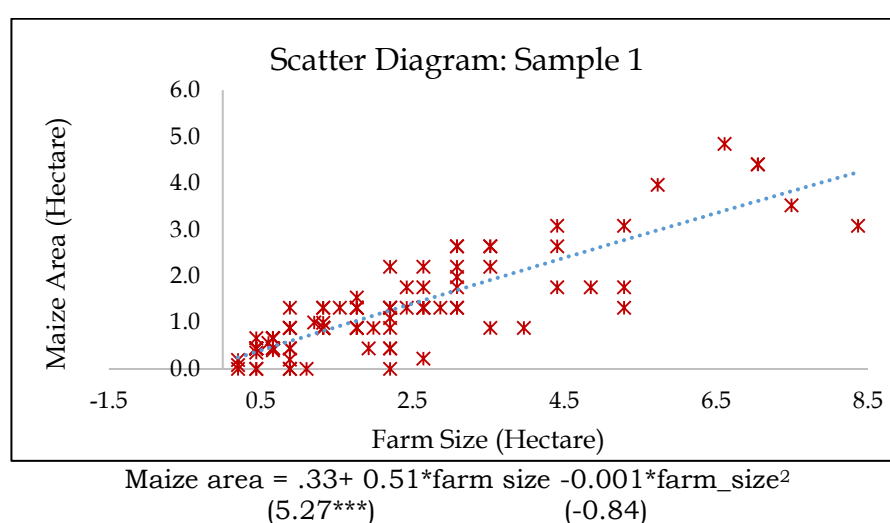
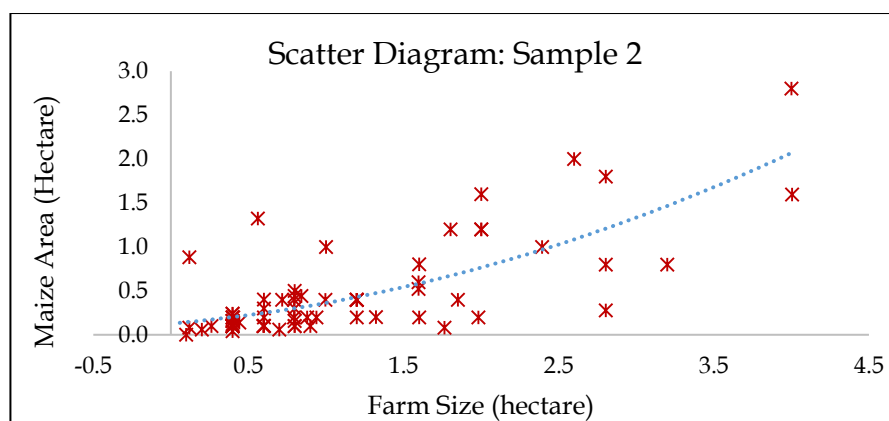


Figure 8.3(a): Scatter plots of Maize area against Farm Size



$$\text{Maize area} = -.11 + 0.36 \cdot \text{farm size} + .004 \cdot \text{farm_size}^2$$

(5.9***) (1.79*)

Figure 8.3 (b): Scatter plots of Maize area against Farm Size

8.5.3. Prices and other incentives

Prices fetched by selling the major crops Rice, Wheat and Maize as reported by the sample households are not uniform (Table A8.1). They vary depending on their bargaining power, quality of grains and nearest market facilities. Decisions to produce and to acquire food from different sources would rationally depend on various relative returns and prices.

Table 8.8 (a): Ratio of producer price of maize to other prices

	Producer Price		Subsidized PDS Price		Retail Price		Retail Price	Subsidized Price
	Rice	Wheat	Rice	Wheat	Rice	Wheat	Food grain	
Marginal	0.96	0.77	3.46	5.18	0.35	0.40	0.31	1.01
Small	0.91	0.77	3.39	5.09	0.34	0.39	0.29	0.83
Medium	0.91	0.83	3.32	4.98	0.33	0.38	0.29	0.90
Large	0.93	0.78	3.31	4.96	0.33	0.38	0.28	0.84
Sample 1	0.93	0.79	3.37	5.06	0.34	0.39	0.29	0.91
Marginal	0.94	0.79	3.20	4.80	0.32	0.37	0.27	0.82
Small	0.99	0.79	3.12	4.69	0.31	0.36	0.26	0.69
Medium	1.03	0.80	3.17	4.76	0.32	0.37	0.26	0.67
Large	0.93	0.75	2.96	4.45	0.30	0.34	0.24	0.60
Sample 2	0.95	0.79	3.17	4.76	0.32	0.37	0.27	0.76

Source: Computed from sample data and MoA data.

Wheat price that producers fetch is highest in both samples though the retail price is lower. The ratio of rice to wheat prices is thus 0.85 and 0.83 in samples 1 and 2 respectively. The price at which producers sell maize is lower

than not only wheat but also rice in both samples (Table 8.8 (a)). With respect to retail prices of food crops it falls even below 40%. Compared to prices of subsidized grains in PDS, it is nearly 3 to 5 times higher.

The producer prices of Rice and Maize are not much different (Table 8.8 (b)). Taking account of relative yields and costs of production, the income (loss) ratio between Rice and Wheat is negative in Sample 1 but positive in Sample 2. For commercial maize cultivation the food security motivation can also be important. The assessment can be done by measuring the access it gives to food and comparing this access either to direct access via subsistence cultivation or to what is required for meeting nutritional needs. Purchasing power over food from PDS can be a relevant motivation but since PDS supplies are bounded, access in the open market is a greater consideration.

Access to Rice from growing Maize is poor, less than 1 tonne per hectare in Sample 1 and less than half a tonne in Sample 2. Access to Wheat is 1.1 tonne and 0.5 tonnes in Sample 2. The access from commercial Maize is found 60% of rice yield in Sample 1 and 34% in sample 2. In Wheat the access is only 45% in Sample 1 and 27% in Sample 2. The access to food is less than half of the yield of food from the same land. It cannot be a motivation for choosing the commercial crop if agrarian conditions such as land quality are uniform. Rice equivalent of the income earned from maize by the household at the per capita is less than both the per capita national production and also the nutritional norm set by ICMR in both samples and in all classes except the Large class in Sample 1. Thus, land use decision is made not simply in regard to food security objective but also other restraints such as weather and soil.

Table 8.8 (b): Indicators of remuneration and incentive from Maize

	Sample 1			Sample 2		
	Rice	Wheat	Maize	Rice	Wheat	Maize
Producer Price (Rs. /Kg)	16.4	12.8	10.1	15.0	12.0	9.5
Retail Price (Min-Max) (Rs. /Kg)	26-30	21-26	-	26-30	21-26	-
PDS Price (Rs. /Kg)	3.0	2.0	-	3.0	2.0	-
Income (Rs. 000 /Hectare)	3.2	-2.4	29.4	-15.3	-3.3	13.4
Yield (Tonnes/Hectare)	1.1	2.5	6.8	0.88	1.9	5.6
Access to Food (Tonnes/Hectare)	0.98	1.13	-	0.45	0.51	-

Source: Survey data and MoA data.

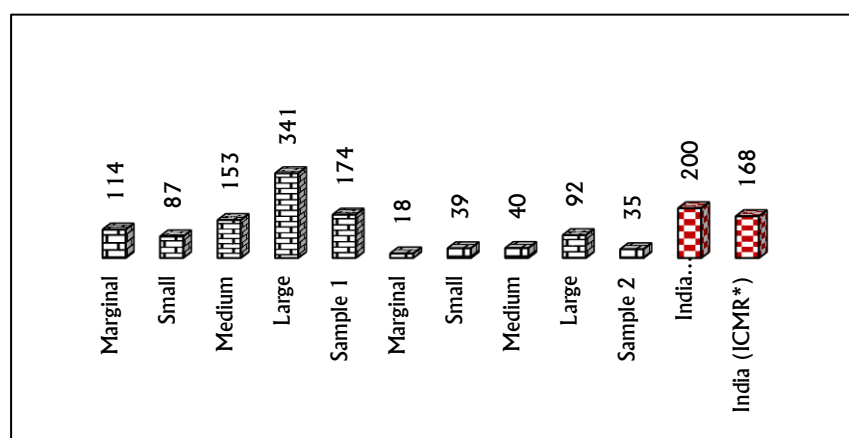


Figure 8.4: Per Capita access Rice (Retail) (Kg/year)

Note: * denotes Cereal and Millets

8.5.4. Econometric analysis of acreage decisions

Econometric modeling is explored with the cross-section household primary data to identify what incentives drive decisions on what crops to grow. In a model that takes care of the complexity of food insecurity, commercialization and public intervention the incentive variable needs to be specified with flexibility and the equation needs to take account of other influential variables to bring out the effect of the incentive with due correction. The identification of the incentive variable can help in gaining at least some understanding of the farmers' motivation of farming specific crops.

Samples of 120 and 126 households comprise the cross-section data. Acreage functions of Maize as also food crops Rice and Wheat are estimated using linear regressions. Households vary by workforce capability and food needs

depending on family size, age and gender composition and by the attributes like education, gender and age of household head acting as the supervisor, may be influential decision making. Social factors like caste may also influence crop choice as a part of culture and tradition. Irrigation intensity of land is important as crops differ in their demand for water but of possession of pump-sets can influence the quality of water use. The location and topography of the farm land can also influence crop choice. Rainfall and other weather specificities are likely to affect all farmers equally and are ignored for possible lack of variation.

Incentive is the economic variable explaining crop area of interest. Not all of the farmers grow each crop but the reason for not growing a crop may be economic or non-economic. In the former case the relevant incentive may be insufficient enough to justify growing the crop. While the dependent variable i.e., area of this crop is zero in this case it does not warrant exclusion from the sample because this censoring at zero occurred because of the incentive value. The sample selection bias created by this complexity is corrected by a Heckman model.

The incentive variable INCTV is specified in four alternate ways before a choice best on diagnostics is made.

$$INCTV_{ij} = RELP_{ij}$$

where $RELP = \text{producer prices of crops } i / \text{producer prices of crops } j$ and the coefficient is expected to be positive and significant if i and j are substitutable crops for the land and if farming of both crops is fully commercial and yield differences across households make no difference.

$$INCTV_i = INCHECT_i$$

Where $INCHECT = \text{absolute income per hectare expected from the crop}$ and the coefficient is expected to be positive and significant so long as the income

gives access to food and other commodities and the alternative to cultivating crop i is to resort to other livelihoods.

$$INCTV_{ij} = INCHECT_i / INCHECT_j$$

Where $INCHECT$ = income per hectare expected from the crops i / income per hectare expected from the crops j and the coefficient is expected to be positive and significant so long as the crops are substitutable and grown commercially for earning.

$$INCTV_j = ACCHECT_j$$

Where $ACCHECT$ = income generated by growing maize in a hectare of land. It denoted purchasing power over food crop j which is not necessarily a substitute crop. The coefficient is expected to be positive in maize area function in the presence of severe food insecurity but may be negative in the proximity of food security when there is no utility from cash from maize cultivation. The alternative to maize cultivation would be outside agriculture. In yield function of a food crop the sign is expected to be negative as the crop gives way to maize for commercial purchasing power.

Crop area (AREA)

$$AREA = (INCNTV, HWRK, FHWRK, FSZ, HHSZ, IRRG, DPUMP, DTRACT, DMIGR, DHHGEND, DHHEDU, DWLOG, DISTRD, DISTRIV)$$

In equation $INCTV$ is the economic incentive for growing the crop which can be specified in different ways in the complex situation. Maize competes with rice in the kharif season and with rice and wheat in the rabi season. Explanatory variables are both continuous and categorical and the prefix D is used for denoting dummy variables. Specifications and theoretical expectations are detailed below.

HWRK: Working age members of sample household divided by farm size.

FHWRK : Working aged female family members divided by number of working aged household members.

FSZ: Farm size (inclusive of uncultivated land in possession).

HHSZ :Household size and its sign possibly signals the pressure of food and income security.

IRRG : Share of possessed land under irrigation. Positive sign shows the irrigation facility as a facilitator of this crop but a negative sign indicates diversion of irrigated acreage towards more preferred crops.

IRRGRT: Ratio of total irrigation area to total cropped area

IRRG RTP: Interaction of Irrigation ration and DPUMP

DISTRD : Distance of the residence from main road. A positive coefficient would suggest the importance of marketing and commercialization.

DISTRIV : Distance of residence from the river. A negative sign would be sign of the threat and fear of flood.

Dummy variables that take the value 0 by default but is unity if certain conditions are met.

DMIGR: The household has at least one migrant work. The sign if positive can be indicative of the significance of investible funds for the crop while a negative sign would divert funds and acreage to another crop. It can also be a proxy of labour availability.

DTRACT: The household possesses a tractor and the coefficient shows the significance of technology

DPUMP: The household possess a pump-set for irrigation and the coefficient shows the significance of technology

DHHEDU: The household head is educated at least up to Class X

DHHGEND: The household head is a female

DWLOG: Some part of the farm land is low lying and water-logged

Price ratios did not serve as incentive variable for explaining Maize area. In the case of Sample 1 none of the incentives produced a positive coefficient. The chosen equation has relative access to Wheat as the incentive with a negative significant coefficient. This suggests that maize is grown only to meet

fixed food security needs but the motivation for cash income is weak possibly owing to other alternatives like migration. In Sample 2 it was difficult to identify an incentive variable but the best one is the income relative to wheat which shows maize as a commercial crop competing with wheat as a source of income. Rice and wheat seem to be grown for commercial motive in Sample 1 as crops competing for resources. In Sample 2 rice is presented as a commercial crop competing with maize but area under wheat is compromised by the purchasing power of maize as an economic alternative. Thus in Sample 2 commercially rational motives appear important and wheat is a major competitor for both rice and maize.

Among other variables farm size is an important factor in all cases for deciding to opt for the crop and the area to sow. Larger farm sizes encourage growing each of the crops with a quadratic (peaking) effect as also the area sown though there is a scale inverse relation with wheat area. Irrigation, mechanization, distance from the road are decisive for choosing wheat while irrigation, distance from road, female workers, farm size and water logging are important factors for rice. Longer distances from the road which means poorer access to markets encourage food cultivation. Water-logged soil discourages rice cultivation but favour maize cultivation in Saharsa. Migration has a negative effect of taking up crops but has no effect on acreage of grower. Female dominance in the worker strength is a positive factor in rice cultivation. Smaller farmers are drawn more towards rice cultivation than larger farmers.

Table 8.9 (a). Regression results of Area equations of Sample 1 (with Heckman correction)-Sample Period: 2014-15 (Cross Section)						
	Maize		Rice		Wheat	
Independent variable	Coef.	Z -Stat	Coeff.	Z -Stat	Coeff.	Z -Stat
Incentive variable	ACCHECT (Wheat)		RELP (Wheat)		INCHECT (Rice)	
Constant	-1.158	-2.58	-3.624	-2.81	2.54	2.56
INCTV	-0.0003	-2.44	3.52	3.15	0.05	2.50
IRRG	0.27	7.31	0.58	7.85		
DTRACT					0.921	2.96
DPUMP					-1.75	-3.00
IRRGRT					-2.11	-2.93
FSZ	0.20	4.09	-0.38	-3.35	0.19	4.55
DISTRIV	0.19	2.71				
DISTRD			0.44	3.28	0.28	2.37
FHWRK			1.71	1.35		
DWLOG			-0.55	-1.28		
Dummies:						
Village 1	-1.25	-1.28	1.06	1.88	0.21	0.36
Village 2	-0.01	-0.02	1.06	1.84	0.486	0.88
Village 3	1.09	3.12	1.74	2.69	-0.06	-0.10
Village 4	2.16	5.75	0.58	0.81	-0.40	-0.59
Village 5	0.66	1.56	0.33	0.60	1.12	1.95
Village 6	2.42	6.68	0.00		0.00	
Village 7	1.54	3.85	0.00		0.00	
Select						
FSZ	0.98	5.47	0.29	3.41	0.94	4.04
FSZ ²	-0.05	-3.93	-0.02	-2.62	-0.04	-1.85
DMIGR	-0.25	-1.97			-0.25	-1.93
HWRK					-4.23	-2.27
FHWRK			-1.86	-2.46		
Constant	-0.66	-2.30	-0.20	-0.51	-0.40	-0.78
Lambda	-0.24		-0.78		-0.68	
Number of observations	120		120		74	
Censored observations	31		76		37	
uncensored observations	89		44		37	

Table8.9 (b).Regression results of Area equations in Sample 2 (with Heckman correction) -Sample Period: 2014-15 (Cross Section)						
	Maize		Rice		Wheat	
Independent variable	Coeff.	Z -Stat	Coeff.	Z -Stat	Coeff.	Z -Stat
Incentive variable	INCHECT (Wheat)		INCHECT (Maize)		ACCHECT (Wheat)	
Constant	-2.47	-1.71	0.447	2.35	0.406	1.82
INCTV	0.002	1.54	0.123	9.15	-0.0002	-2.38
IRRG			0.226	8.42	0.170	8.54
DTRACT	0.778	3.89	0.626	4.61		
DPUMP						
IRRGRT	-0.361	-1.02				
IRRGRT ²	0.07	2.18				
FSZ	0.41	12.98	0.15	2.66	0.047	2.41
HWRK	-0.72	-1.81				
DISTRIV	0.11	1.28	-0.02	-3.47		
DISTRD						
FHWRK						
DWLOG			0.13	1.10		
Dummies:						
Village1	2.05	2.21	-0.17	-0.41	-0.12	-0.63
Village 2	1.44	1.22	0.91	2.52	-0.67	-1.62
Village 3	1.68	1.37	0.32	1.40	0.27	0.52
Village 4	1.58	1.32	-0.12	-0.62	0.09	-3.50
Village 5	2.42	2.34	0.11	0.60	-0.68	-1.98
Village 6	2.36	2.14	-0.09	-0.49	-0.35	0.94
Village 7	0.42	1.65	0.06	0.36	0.16	1.82
Select						
FSZ	0.394	3.69	1.43	5.10	0.836	5.33
FSZ ²	-0.019	-2.31	-0.126	-3.55	-0.025	-4.06
DMIGR			-0.377	-1.81	-0.206	-1.47
DWLOG	0.719	2.72				
HWRK			1.099	2.78	0.172	1.32
Constant	-0.899	-4.14	-0.941	-2.97	-0.553	-2.11
Lambda	0.612		-0.238		0.302	
Number of observations	121		100		104	
Censored observations	57		37		40	
uncensored observations	64		63		64	

Chapter 9: Prospects of Agriculture: Village profiles, Farm Incomes, Food security, and Allied activities

Agriculture as an occupation is still important in Indian economy but it has diminished its share in the economy and within agriculture, crop cultivation has yielded share to animal husbandry. Kosi region has very specific features (Chapter 2) where agriculture faces severe vulnerabilities and yet the scope for industries is even more limited. Secondary data showed that agriculture remains the leading sector in terms of both employment and income generation.

A composite impression of the agriculture sector in the Kosi region is portrayed in this chapter based on averages of the three major crops discussed in Chapter 8 to assess farm income prospects, marketing practices, food security implications of farming, allied activities like animal husbandry and trading. In the process the chapter provides glimpses of the typical Kosi region villages that were covered in the study as well as the marketing facilities available as privilege or constraint to the farmers and the traders who serve agriculture.

9.2. Village profile

The primary survey covered 8 villages in Khagaria and 8 villages in Saharsa. The villages are located 4 km to 50 km away from the respective district headquarters. Most of the villages have Kosi and its tributaries flowing past some also had Ganga. The furthest distance of the river from a village is 0.5 Km. A few villages were also located between two rivers, accessible to food shops and other facilities only by boat. 1% to 100% of village area were observed to be water logged.

All the villages had the facilities of one or more primary schools, anganwadis, PDS shops and a private clinic with an untrained doctor but 7 and 8 villages

in Khagaria and Sharsa also had middle level school education available. Villages with Muslim residents generally had Madarsa.

Not all villages had a police station and public health facilities. Electricity and Telephone (Mobile) was available not to all household nor for 24 hours. They were available to 30% to 100% of the households in a village and two village are not electrified at all. Hand-pumps were the most common source of drinking water but quality was not always good. In some villages residents purchased water in jars. Cow dung and crop residue are used mostly for as cooking fuel and gas supply (LPG) was rare. Solar plants which were not always effective had been distributed by the state government. Also not all village were accessible by paved all-weather road. For transportation to the nearest town or district headquarters modes like private hired vehicles were used and by cycle or rickshaw if distance were short, as bus transportation was reported by only 10 villages.

Cultivation is the major occupation in all villages but not the only one. People reported drawing livelihood from wage-labour, small business (shop, vegetable vendor, hawker), semi-skilled labour (mason, driver, cycle repair, trading) inside and in some cases also outside the village. Migration is also common in all villages.

In the 5-year period 2011-2015, 6 of the 16 villages had flood and 11 villages had at least in 4 years and only 2 villages reported having no flood in the last 5 years. Flood affected at least 52-80% of the area of a village. In most of the cases movement of Kosi river or heavy rainfall in Nepal was associated with the flood which caused huge damages of crops and property as also loss of communication and destruction of roads. Drought was reported mostly in the years 2011, 2012, 2013, or 2014. Two villages faced drought in 4 of the 5 years and 6 villages did not report a drought. Early warning of floods was duly given by the district administration and media and victims also received public relief for the calamities.

9.3 Weights of crops to derive Farm level indicators

As a proportion of the total acreage under all these three crops together, the weight (λ) of the major crop also presented by household classes (Table A8.2 (d)) and the derivation is explained in Chapter 4. In Sample 1, Rice has a weight of 0.20 - 0.25 but in the Maize inclined Large category the weight is only 0.18. Corresponding weights are higher at above 0.5 except among the Large class in Sample 2. Considering all cultivating households (Table 8.5), the average physical area under Rice is higher in Sample 2 although this is not so if Rice growing farms are only considered.

9.4. Input use and Farm incomes

In Chapter 8 various agricultural measures like yield, income, cost and input use are computed based on sub-sample data of the specific crop growers only and averaging across them. This avoids bias imposed if averaging is done across all sample households. For example, yield rate of any crop will be grossly underestimated if the sample includes farmers who do not grow the crop. In other words, the averages of the indicators are area weighted where the weight is zero if there is no area under the crop concerned.

The above approach helps comparison across the major three crops but the average farm indicators including income from crop cultivation regardless of what crops are grown, can be computed only using all the land operating farmers. The farm household indicators are obtained in this Chapter as weighted averages where the weights are the shares of the crops in the total area under the three crops in all the sample (λ) as explained in Chapter 4.

9.4.2. Fertilizer use

Despite significantly lower intensity of farming, the fertilizer (NPK) use per unit area under the three major crops is much higher (244 Kg per hectare) in Sample 1 than (130 Kg per hectare) in Sample 2 (Table 9.1). The use intensity

per net land area is also higher (295 kg per hectare) in Sample 1 compared to (238 Kg per hectare) in Sample 2 (Table A8.2 (a)). A clear positive scale response is marked in both samples in respect of fertilizer use per cropped area but when fertilizer use intensity per net land unit is observed, the rise in intensity with farm size is disturbed at the Large farm level which also marks a plateauing of the farming intensity. The same patterns hold at the level of the individual three crops except Maize in Sample 1, of which the fertilizer intensity reaches a peak at the level of Large farmer.

9.4.3. Income from farming

Weighted by the area shares of the three major crops total income, cash income, paid cost and fertilizer intensity per hectare as also the intensity of labour hiring are reported in Table 9.1. Income Per hectare considering all produced quantity at imputed value is found positive in Sample 1 of Khagaria at Rs 14,360 per hectare and is negative at a loss of Rs 4,560 per hectare in Sample 2. Cash income if only commercial production for sale is recognized, is lower by Rs 10,000. It is still positive in Sample 1 and suggests a large loss in Sample 2. Despite the higher intensity of farming in Sample 2, the difference in income potential of agriculture observed over Sample 1 may be on account of the place of the highly commercial and profitable Maize in the cropping pattern and partly the greater use of family labour and fertilizer.

There appears to be inverse returns per hectare to scale as Marginal households fetch higher incomes (or lower losses) per hectare and even a positive cash income in Sample 1. If total farm incomes are taken into account given that farm size is variable, Marginal households earn the least income in Sample 1, in which there is a clear positive relation of household farm (crop) income with farm size class (Figure 9.1). In Sample 2, the Large class (which grows Maize) earns the highest income although Medium class is the biggest loser. The average household income of the major crops is Rs 38 thousand and –Rs 9000 annually in the two samples working out to a positive but paltry Rs 2600 and a negative Rs 416 per month in the two samples. Obviously the

outcome would be worse if only cash income was taken account of excluding self-consumption. The loss possibly is borne because of the uncertainty regarding food access despite public distribution and to make use of available land instead of letting them be idle.

Table: 9.1: Annual total Income, Paid cost, Fertilizer and Irrigation intensity and Labour hiring of Major crops Rice, Wheat and Maize

	Income		Paid Cost	Fertilizer	Irrigation	Hired Labour
	Total Rs.000/ Hectare	Cash Rs.000/ Household	Rs. 000/ Hectare	Kg./Hectare	% cropped area	% of person days
Khagaria						
Marginal	25.96	8.91	28.26	176	87	48.26
Small	9.29	-8.45	36.16	231	87	57.35
Medium	9.84	2.21	34.94	281	86	64.18
Large	11.86	-2.67	43.09	319	98	73.82
Sample 1	14.36	4.49	34.51	244	88	59.84
Saharsa						
Marginal	-5.38	-10.79	34.69	111	89	66.07
Small	-5.56	-12.79	28.74	141	88	50.43
Medium	-7.95	-27.70	33.60	193	97	58.42
Large	4.28	-7.32	31.72	178	91	55.29
Sample 2	-4.56	-13.34	34.18	130	90	64.50

Note: Weights are in Table 8.2 (d). See Chapter 4 for details of derivation and Chapter 5-6 for secondary data on Bihar. Source: Computed from sample data

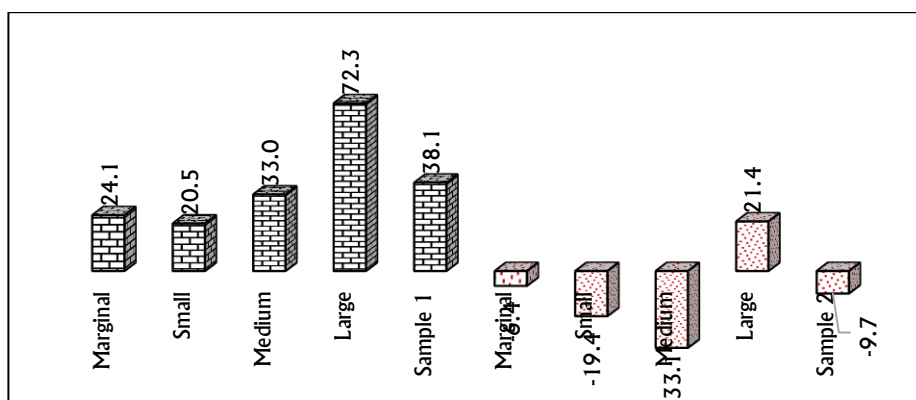


Figure 9.1: Farm Household Income (Rs. '000)

Source: Computed from sample data

9.5. Marketing, Sources of Food and Access to Food

The producers both consume the products at home and sell in the market although the former has a smaller role in Maize production. Maize was traditionally a staple in Bihar (rolled out as Makka ka Roti) but over time food habits have changed to more convenient to prepare food crops while Maize has gained commercial appeal. Not all marketed Rice and Wheat truly represent marketable surplus as seen in Table 9.3. Selling takes place after harvest to raise cash, repay loans and save storage space but sometimes the retained portions do not suffice and the seller buys back from the market at a later date. So far as the sold amount exceeds the purchased amount the producer may be perceived as a net seller. Farmers purchase from the PDS at subsidized prices, an important welfare instrument in the regions but when even this availability does not suffice excess needs are met by purchases from the open markets.

9.5.2. Marketing of farm produce

Traders are the most important primary purchasers of raw produce serving over half of the sellers of all the major crops and as much as 82% of Maize sellers in Sample 2. (Table A9.1). Sale in the *mandi* and sale through landlords or moneylenders is also common among households in Sample 1 but the dependence on traders is more in Sample 2. Sale to PACS is not reported by any of the sample households though diverse other channels including producer companies find mention in the reports. The respondents did not show interest in PACS. At around the time of survey there were reports of confrontation between farmers and PACS over payment default, delayed procurement and fraud. Large farmers depend less on traders. Not all producers sell. Sellers constitute 41%, 51% and 97% of the producers of Rice, Wheat and Maize in Sample 1 and 47%, 35% and 72% in Sample 2. Thus, nearly half the producing farmers in the sample sell Rice and Wheat in market while nearly all of the Maize producers at least in Sample 1 are commercial.

Table 9.2: Marketing and Choice of channel (%) of crop growing households

	All Seller	Landlord	Village Trader % Producer	Mandi	Others
Sample 1(Khagaria)					
Rice	40.91	9.09	25.00	4.55	2.27
Wheat	50.60	12.05	30.12	2.41	6.02
Maize	96.63	17.98	48.31	25.84	4.49
Sample 2 (Saharsa)					
Rice	47.19	1.12	35.96	6.74	3.37
Wheat	34.88	1.16	26.74	5.81	1.16
Maize	72.46	1.45	59.42	10.14	1.45

Note: Landlord includes Sahukar, Mahajan, Mandi is village wholesale market and others Includes states cooperatives producer company etc. Source: Computed from Sample data.

9.5.3. Subsistence and Commercialization

Households in all classes are both buyers and sellers of food crops in the market (Table 9.3) generating a small net surplus or a deficit. Considering both Rice and Wheat Sample 1 households emerge as net food surplus producer while those in Sample 2 face a net deficit both due to wheat. The results varied across classes (Tables A9.2 (a) and (a)) but the marginal classes were in deficit. In Sample 1 only 12% of the Rice and as much as 65% of Wheat produced is marketed by the farmer while in Sample 2, marketing is more important in Rice than Wheat. On the consumption side too the open market is important but dependence on PDS is also clear. Most farmers reported having some make-shift storage capacity improvised in their residences.

Table 9.3: Dependence on Market and sources of food of Sample households in Khagaria and Saharsa districts

Districts	Khagaria		Saharsa	
Average Annual quantity (Tonnes)/household	Rice	Wheat	Rice	Wheat
Production	0.73	1.41	0.66	0.62
Marketed	0.29	0.61	0.27	0.10
Consumption	0.74	0.84	0.63	0.75
Farm Produced	0.25	0.52	0.30	0.36
Wage for Food	0.02	0.02	0.01	0.03
Public Distribution	0.13	0.09	0.15	0.10
Market	0.34	0.20	0.18	0.26
Surplus	-0.01	0.57	0.03	-0.13
Consumption/Production (%)	100.78	59.32	95.49	121.46

Source: Calculated from survey data

Both producer prices and the retail market prices are likely to be depressed by the prices of food that consumers can access in the PDS but they are also influenced by government support though the significance of the state run cooperative organization PACS was not revealed in the survey. The average of reported producer prices is found moderately less in Sample 2 and the prices are moderately variant among households. The producer prices are not averse to smaller farmers but are far less than retail prices quoted officially.

9.5.5. Sources of Food and food security

Households require food not just to meet needs of members but also for feeding animals and sometimes for providing food to farm labour as wage in kind. It is difficult to separate out the components of food need and no such data could be availed. In practice cooking for farm labourers is part of regular household kitchen chore. Table A9.2 (c) shows over 80% of the households in both samples hold a ration card. Only 3% in Sample 1 and 1% of sample households in Sample 2 qualify for the cheapest food in AAY category. Whereas 39% in Sample 1 hold BPL cards, the privilege is available to only 11% in Sample 2. The Large class does not find its place in the BPL and AAY categories. The share of the Landless in BPL group is 72% in Sample 1 and only 4% in Sample 2.

In household consumption the dependence on open market is lower in case of Wheat (24%) than Rice (46%) in Sample 1 while the reverse is true in the Saharsa sample (Tables 9.2 (a) and (b)). The PDS has a greater role in Sample 2 in meeting consumption needs of both rice and wheat. In both districts higher share of rice consumption (17% and 23%) is met from PDS than wheat (10% and 13%). Farm production amounting to subsistence agriculture is a most dominant source of consumption. In addition, the Landless class also avails some food as wages in kind. Thus the tie with open market is more as a seller in Wheat and buyer of Rice among the sample 1 households. In Sample 2 the tie is as a buyer of Wheat and seller of Rice. Pulses are minor

crops in the area especially in Khagaria and all of the produced pulses are retained for home consumption. Farmers depend on open market substantially for pulses. Household consumption levels of rice and wheat are relatively high in the landless class but consumption of pulses is lower. Total consumption from all sources at the per capita level exceeds food security norms as recommended by ICMR (Figure 9.2) but it may be noted that the consumption by its derivation is a representative one and the actual consumption of a member may be less than the reported figure if adjustments for wastage, feed and food of guests and labourers are made. Moreover, the farmers and the regions especially in Khagaria specialize in the major crops so that the cereal security may not necessarily translate to nutrition security.

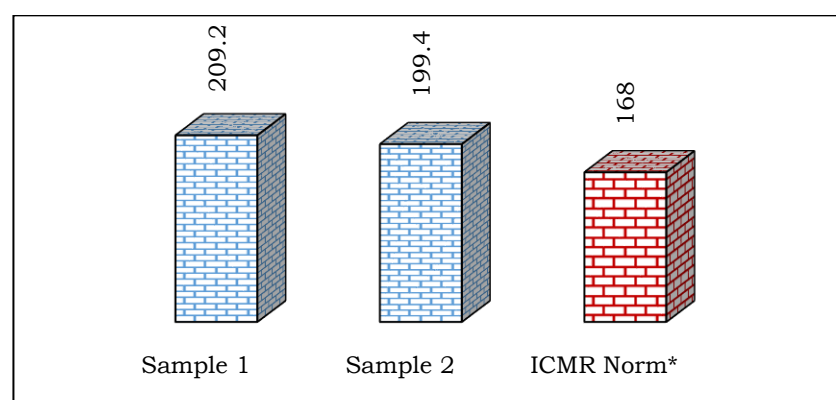


Figure 9.2: Per capita consumption of Food (Rice and Wheat) (Kg/Year)
 Note: * denotes Cereals and millets. Source: Computed from sample data, MoA data and ICMR data

9.6. Markets, marketing facilities and Trading

As part of the study the research team visited markets and trading agencies that serve the farmers in marketing grain with a special focus on Maize. Information as well as perception based views expressed by proprietor and manager of trading enterprises and authoritative stake-holders of structured market premises are collected. Although subjective and liable to bias the views make the assessments of agriculture in the region more balanced for holistic planning. Interviews at three markets, nine village traders and two large traders all dealing with Maize made up the investigation.

9.6.2. Markets

Three markets were visited in the Khagaria district located at varying distances from the headquarters. All three markets were regulated before the APMC law was repealed (see Chapter 6) but the committee was reportedly ineffective. Now the markets remain unregulated with no official supervision. However, two of them were found to be managed by local agencies but the third market which is located furthest from the district headquarters has no management. All the three markets function on daily basis. The marketing route of the product was in all cases from farmers to village traders to the mandi (market) and then to the large trader. No processor involvement was observed at the village level. Value addition inclusive of transport cost in the local supply chain measured as the difference between the farmer's prices and the large trader's price varied from Rs 1100, Rs 1300 and Rs 3000 per tonne in the three markets.

The nearest market is situated in the Khagaria headquarters at 0 Km distance, has a market area of 800 square meter with a with a 200 square meter paved yard. The catchment of this market is 8 Km. It does not have a road or parking place but situated on a main road the lack of these facilities does not create a problem. There is one constructed office building. Only rice (10 tonnes a day) and maize (1 rack a day) are transacted daily. Transactions in wheat are not reported. A local food business manages it.

The second market is 25 Km away from the headquarters and has a catchment of 2 Km. It is managed by a local charity organization. The market is 2 square Km in size which is fully utilized and all of it is paved. It has no road or parking space and no administrative offices. Rice (40 tonnes a day) and Maize (1000 tonnes **OK?a day) are transacted daily and wheat is also sold seasonally (50 tonnes a day). Transactions in pulses (10 tonnes) also take place.

The most remote market 45 Km away from headquarters is unsupervised by any agency with an area of 500 square meter of which only 15% is utilized. It has a small paved yard but no administrative offices or trader stalls and sheds. The catchment is 2 Km. Rice, wheat and maize are all sold and purchased here.

Facilities: All the three markets have paved yards, roads, weighing machines and moisture meter. Drinking water is available by hand-pump in two of the markets and latrines are available in only one of them. Only one market in the headquarters reported having an office building. Refrigerators or freezers, grading equipment, canteens and cow sheds are not found in any of the markets. The facilities are generally underutilized. Loading platforms and open storage can be trader's own resort.

Fees: Weighing fees, imposed on the basis of the number of bags was the one major item of cost of trading. There were no market fees, grading fees, taxes or commissions.

Deficiencies: All markets have shortfalls in physical amenities and infrastructure that traders want to be upgraded. Functionaries of two of the markets reported shortage of support services or inadequate staffing and want these lacunae to be addressed. Lack of rules and regulation to guide the marketing is not felt to be any weakness by anyone and no one wanted any change in the rules. While the lack of regulations has brought a sense of freedom, in one of the markets marketing practices and pricing were identified as problems. These functions were part of the market regulation although there may be other ways of addressing these important concerns without resorting to old systems depending state power or vested powers.

9.6.3. Traders

Investigators spoke to Nine village traders operating in Maize in Khagaria. All the respondent traders are male, seven of them of age 50-65 and two younger in the range 25-35 years. Uniformly, the education of the trader is school level, at the most Class 10 and some of them up to only Class 4. The traders are all found to be sole proprietors of the business except two cases where they reported as working for Farmers' groups.

The operations were mostly in rabi season with average volume of 200-300 tonnes **is this per day?**but a majority reported some transactions in off-season at higher prices. The ratio of peak season to lean season prices paid was found to range from 0.74 to 0.89 among the none traders but the modal value was 0.83. They operated in competition with other agencies operating within 30 Km of their stores but did not complain that the competition was unfair.

All the respondents purchased Maize from different sources but farm-gate purchase was common. The catchment of this sourcing was generally 3-5 Km and the total volume was 1500 tonnes but only one trader who reported buying a large volume of 4000 tonnes reported collecting produce from villages that were 20 Km away from their stores. This trader also procured from the stores of other smaller traders. The purchases from farmers were made by cash payment. Only the larger trader reported payment in both cash and cheque to farmers and transactions only in cheque with traders.

All the traders reported that farmers also delivered at their stores. These sellers constituted about 150-1000 and came from distances up to 5 km. In this case too there were aberrations as one trader reported farmers' groups coming from 20km away with larger volumes. Only four of the nine traders reported that other traders too delivered at their stores. These traders about 150 in number in each case, hailed from a radius of up to 5 Km., some of these traders were landlords themselves.

In farm-gate purchases the transport cost, depending on the distance and road quality was borne by the traders. Loss in transportation was perceived low (2-3%) of volume and mostly unavoidable by the traders. The loss was attributed to moisture content in the grain, faulty packaging and fracture in the journey. Careless practices of farmers were also cause of grain damage and this could be improved by training. They had access to storage space generally open or as semi-permanent structures and owned by them as also weighing machine (Dharmakata).

Considerable though low quality local employment was generated in the operations. Permanent help (3-5) was provided by family members but different handling operations required casual employment for about 10 persons at daily wage but two traders reported employing over 30 persons in the season. Loading and off-loading required help in all cases and involved cost. Stationery was another item of expense. Some spent on storage. License cost was reported by three traders while others were not required to pay and one trader even paid income tax. None reported fumigation as a cost. Among the problems faced by the village traders, lack of cash and credit and its high cost (implicit corruption) was the most extensive report but lack of transport facilities, poor roads, cumbersome checks on the way were other problems. Five of them reported facing unfair competition.

9.6.4. Large traders

Two large traders were interviewed. Both had the business model of a company. The operator was male and relatively young (35 years and 40 years). Both were graduates and one had a degree in Commerce. They reported business experience 8-30 years. They mostly buy from traders' stores but also from farm-gate away at 40-80 Km distance paying traders in cheque and farmers mostly in cash. Farmers (or farmer groups) also deliver at their stores. Larger numbers (1000) come from distances 50-70 Km but about 300 of them come from greater distances of over 100 KM. Traders too deliver at

their stores 150 of them from 100 Km, 70-100 from nearby villages 15-20 Km away. These resourceful traders also have open or semi-permanent storage facility and operate with competition. They attributed grain losses to moisture, faulty packing, and also to theft and pests. Permanent employment with higher salary of Rs 8000 monthly is given to 10-20 persons apart from casual seasonal employment. The cost structure of these large traders is different from the village trader. Cost is incurred on loading and off-loading, weighing (Dharmakata), storage mostly at railway facilities, fumigation, stationery, accounts and auditing, license, security and taxes though accounts, auditing and taxes are handled at the corporate level. They made no complaints on unfair state taxes. Buyers reported by both respondents are urban traders outside the state or district.

Table 9.4: Reporting (Number) of attributes from Village Traders

Problems	Casual Labour	Buyer	Cost
Character	Number	Type	Items
Lack of credit (7)	1-5 (3)	Rural trader (6)	Loading, offloading (9)
High cost of credit (6)	6-10 (3)	Urban trader (1)	Storage (5)
Poor rural roads (7)	11-15 (1)	Processor (0)	Taxes (1)
Lack of transport (8)	16-20 (0)	Institution (0)	Stationery (8)
Check posts (3)	21-30 (0)		License (3)
Unfair competition (5)	30-60 (2)	Consumer (1)	Clearing, security Fumigation (0)

Source: Computed from sample data. Figures in brackets are number of traders reporting affirmative.

9.6.5. Processor

In villages rice and wheat mills operated using electricity from public supply and private generators. Some households were involved in threshing maize for farmers at a cost using diesel threshing machine and local labour. Micro-enterprises were found in town of Khagaria and Saharsa, where rice products (Chura, Muhri), wheat products (Biscuits, Bread), Oilseed (Mustard Oil), Poultry and Dairy feed (Maize and Mustard) were manufactured in sheds. They were family enterprises sometimes led by women entrepreneurs and using family labour commonly child labour. Loans were taken from private sources and from bank via self-help groups.

9.7. Allied farming among farmers

Cultivation of crops is not the only agricultural occupation in the study area. Despite abundance of water, the landed and landless farm households of the sample region are not found to be engaged in fisheries. Only 2 respondent households from each district reported fishing as a source of earning. Rearing animals for milk is however a much more common practice in which both landed and landless farmers participate.

The State cooperative formed under Operation Flood is making strides though traditional traders are still the most popular buyers of milk produced. The sample milk producers sell to COMFED the state cooperative which markets Sudha brand of milk and milk products from its outlets. More remarkably, private dairies are also emerging. In our field visit, to the specific area we found the preponderance of Ganga dairy which specializes in storage of milk and manufacture of milk products as also construction of infrastructure for the supply chain. Many respondents reported selling to the collection centre of Ganga or COMFED. Direct selling at door-step and home consumption remain important components of the vast milk marketing system.

Animals for milk production are held by 65% of sample households in Sample 1 and 69% in Sample 2 (Table 9.5). Among the landless class a considerable section (30%-40%) possesses animals but proportion is higher among the landed. The value of animal wealth is Rs 54,000 in Sample 1 and Rs 30,000 in Sample 2 but obviously increases with farm size. Similarly, the income from milk production and the cash income is found scale sensitive though they peak at the level of small or medium farmer (Figures 9.3). The number of animals held per household is about 2.1-2.2. The value depends on the quality and the number of animals.

Khagaria has climatic conditions more favourable than Saharsa as noted by an NDDB expert Committee (see Chapter 6) Milk productivity per day is 5

litres in Sample 1 and less than 3 in Sample 2. The sample from Khagaria has a lead over that from Saharsa in most aspects of dairy such as the number and the value of animals, productivity, milk price and income. Dairy in the Saharsa sample where 58% of milk is retained, is more subsistence oriented and that in Khagaria is more market oriented. Among the classes the Landless is least market oriented. Considering the milk retention per capita household milk consumption is higher than the per capita national production and the ICMR norm (Table 9.3 (c)).

Table 9.5: Animal husbandry and Milk production in sample households

	Animal		Milk			
	Possessed	Value	Yield	Price	Home Consumpti on	Income
	%	(Rs.000)	Litre/Day	Rs. / Kg	%	Rs.'000/Year
Khagaria	65	54.0	4.8	30	42.0	35.91
Saharsa	69	29.5	2.6	28	57.5	0.93

Note: Assumed milk production for 27 days in a month and 8 months in a year. Feed cost covers fodder and medicine including artificial insemination. Source: Computed from sample data

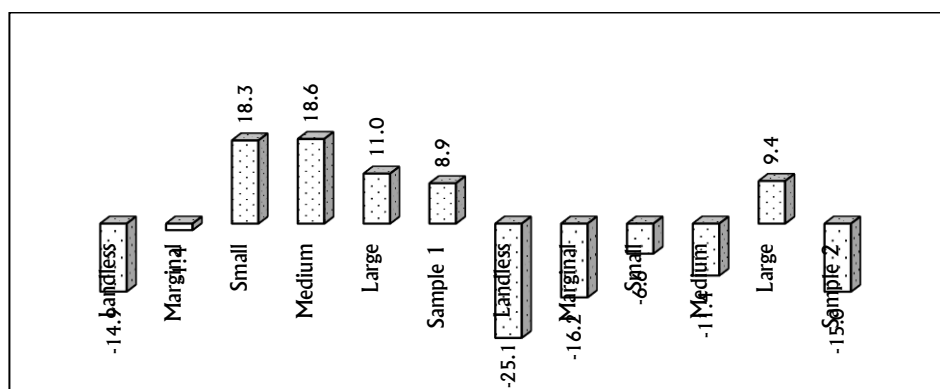


Figure 9.3 (a): Dairy Cash Income (Rs. '000)

Source: Computed from sample data

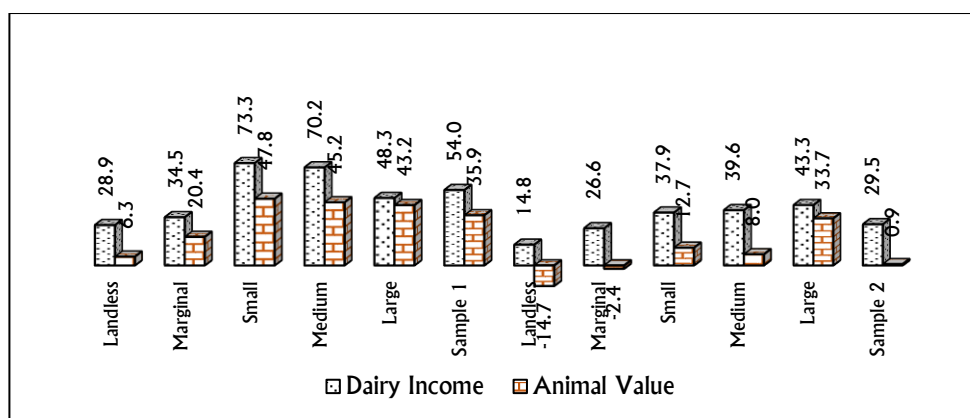


Figure 9.3 (b): Income and Animal value by farm class ('000)

Source: Computed from sample data

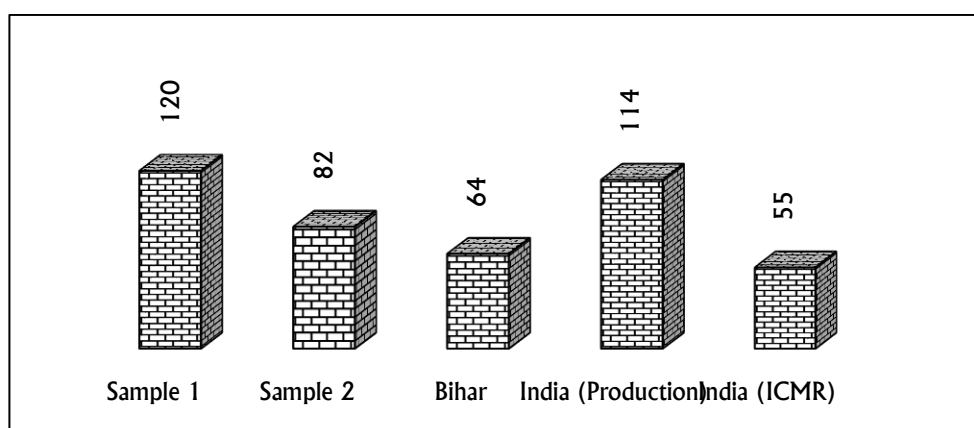


Figure 9.3 (c): Per Capita Milk Consumption (Kg/year)

Source: Computed from sample data

9.7.2. Case study in Khagaria district

BOX 1

A landless farmer found to be tending three cows, a couple of goats and a sick calf under a make-shift thatched shed on the river bank. The location is remote, between two rivers. All facilities are accessible by crossing the rivers by boat. He found it necessary to indulge in farm-labour at times (MNREGA is rare in the area) but expressed satisfaction that milk was picked up by a trader at Rs30 per lota (vessel holding 1.25 litre) at his farm to be sold to local sweet shops. The farmer leases in 16-17 cotta (almost equivalent acres) to produce wheat, maize and jay mostly for feed purpose and also has to purchase the remaining feed from the market. Green fodder is a dominant cost measured as Rupees per cotta for specific period. At the time of delivery pulses are feed to animal. The total paid cost, inclusive of medical costs and AI, works out to about Rs 1,12,327 but taking account only of sale (Rs 1,03,950) and the paid cost there is no profit loss in Rs. 8,377. The business however provides milk for the family, cowdung and calf for possible sale.

9.8. Gender dimensions

Women in the study region are social victims of male migration from the family. As a result, the proportion of households headed by a female is fairly large (see Table 8.1) specially in the landless and marginal category. However, the role of a head comes with greater responsibility and work-load. The wife of male head usually reported their occupation as housewife (34%) but 32% also reported it to be family unpaid farm work, another 11% also reported farm work as independent cultivator. Agriculture wage labour was reported by 8% followed by non-farm wage labour by 3% and 2% as services. Most women also had a secondary occupation though mostly reporting household work but 30% reported family farm labour, 10% reported farm wage labour, 8% reported MNREGA (all in Saharsa) and 4% reported in non-farm work. Wage disparity was significant at over Rs. 50 person/Day.

Considering all sample households in the set, the contribution and participation of a typical woman in livelihood activities is measured by the time allocation to and participation in decisions on such activities, obtained as an average over all the adult women in the sample.

Women devote their available time among household and economic tasks but given that farming is informal and non-professional (unpaid family farm work) in most cases, a part of their routine life conducted in conjunction with household roles, the tasks are not clearly distinct nor is the division of time sharp. Interactions with women reveal that they manage multiple tasks such as cooking, child care and animal care intermittently and in continuity. So the time allocations have serious overlaps. Moreover, the average woman in Table 8.7(a) and (b) includes women who are part or full wage-earners, family farm workers, supervisors, students and housewives.

An average woman spends 6 hours in cooking and other core household work, 3 hours in tasks that lie between household and economic work, such as fetching water and making dung cakes, 7 hours in farm jobs though they are combined with household work in cultivator households. An average of 5 hours is spent as worker in other farms. The tasks are performed concomitantly and the women keep aside about 3-4 hours clearly for work of non-agricultural nature.

Women are found to participate in decision making in farming but their say is more confined to feeding animals and hiring farm labourers, the former being entirely in female domain. This domain is marked in all classes and it also claims full rather than partial participation of women (Table 8.7 (b)). Decision on household expenditure are also made by them fully or partially. The least participation of women is observed in decisions relating to technology adoption, purchase of machines and leasing of land in all of which their participation in decision making is marginal.

10. Insights from Literature review and Field visit

Understanding complex realities that keep Bihar Kosi basin region (BKRB) underdeveloped and its inhabitants poor, at the mercy of nature and prone to migration would hold the key to the solution of the problem. While data, both secondary data collected official sources and primary data from the field would be instructive in scientifically revealing problem, causes and some of the solution, to initiate the process a groundwork was necessary. First, although the region has not been a topic of extensive economic literature, limited insights emerge from studies undertaken in other disciplines relating to the broader region and a few studies, some of them by local scholars do exist. Moreover, the present research project too, in addition to hearing what the numbers in the data speak facilitated by science based methods, supplemented the understanding by personal visit to the sites and interactions with stakeholders.

10.2. Field Visit in North Bihar

ICIMOD (International Centre for Integrated Mountain Development) organized a field visit under its regional research programme called Koshi Basin Programme which is funded by Australian Government, Department of Foreign Affairs and Trade (DFAT). For detailed information, the web link (<http://www.icimod.org/kbp>) can be referred. The Programme has a major focus on the downstream part of the Koshi River which includes Tarai area in Nepal and North Bihar in India. Within the downstream region, the main focus of the study is on again on North Bihar in India primarily because of two main reasons – (1) the dense population of the region, the estimated number being more than 1000 persons per square kilometer in the 16 vulnerable districts of the basin and (2) its risk of natural disaster which relates to water induced

hazards in the basin. Water risk information used in the paper was taken from Koshi Basin Information System developed by ICIMOD. (For detail information, I have written an email to concerned person in ICIMOD. As I get such information, I will make available to you.) Figures 1 shows that the population is the highest in the lower and middle part of the Koshi river basin. The highest water risk is also observed in this region (ICIMOD).

The field visit was conducted between from February 20th, 2014 and February 25th, 2014 to explore the problems and challenges of the Koshi river basin. The field visit team consisted of representatives of IEG (Institute of Economic Growth, New Delhi), ICIMOD, ANSISS (A. N. Sinha Institute of Social Studies, Patna), BSDMA (Bihar State Disaster Management Authority, Patna), CISRO (Commonwealth Scientific and Industrial Research Organization, Australia), and IWMI (International Water Management Institute Nepal Office, Kathmandu).

In North Bihar, the villages that were visited are from four districts namely Supaul, Saharsa, Madhepura, and Araria (see Figure 1). They are the worst Koshi flood affected districts of the river basin although they are affected in different ways. The profile of the districts in terms of vulnerability to floods as conceptualized from experience is outlined in Table 1. In each village, one focus group discussion was organized which was followed by individual interviews of several villages to understand local issues and their problems. Some open ended questions on flood, life, livelihood, etc. were asked in both the focus group discussion and individual interviews. Their responses are summarized in the subsequent sections.

Table 10.1: Name of Visited Villages along with their features			
S. No.	Name of Visited Villages	District	Characteristic of sample villages
1.	Bichauli	Darbhangha	Exposed to all three main problems of the river - flood, water logging, and siltation
2.	Kabira Dhap	Saharsa	Situated within the river embankments
3.	Fulaut	Madhepura	This village falls outside of river embankment and faces recurrent flooding
4.	Chhatapur	Supaul	This village was badly affected by 2008 flood. This village throws light on damage and rehabilitation activities
5.	Pachira	Araria	This village falls under relatively flood free zone. It is surrounded with many paleo-channels of the river.
6.	Bhaptiyahi	Supaul	This village is very closely situated to <i>Maha Setu</i> (Big Barrage).
7.	Kusaha	Sansari district in Nepal	The east embankment breached in 2008 at Kusaha village.

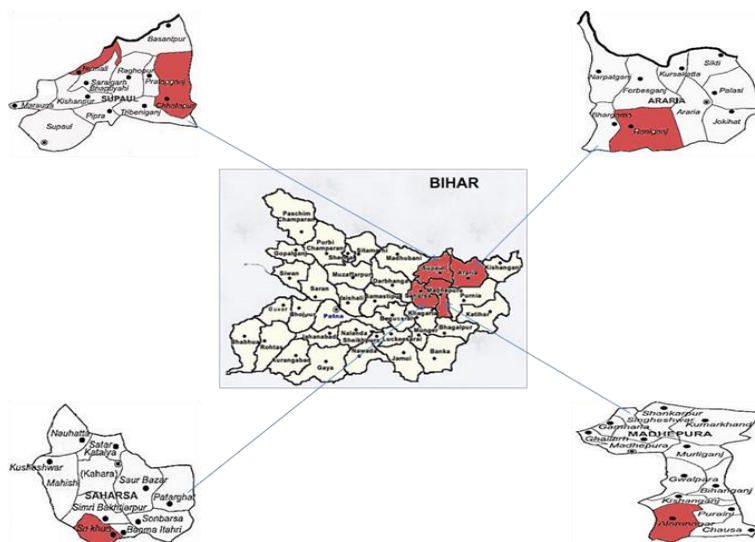


Figure 1: Sample Blocks and Districts visited in **Koshi** River basin

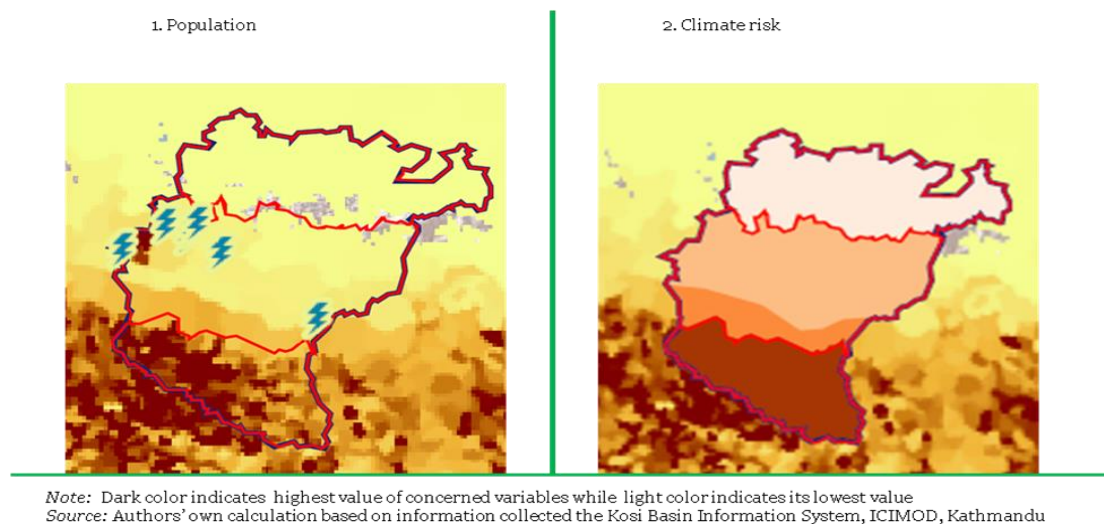


Figure 2: Spatial Pattern of Population and Climate Risk in The Kosi River Basin

10.2.2. Flood plains, siltation and Embankments

Flood has long been considered as a major problem in the Koshi river basin due to the dynamic nature of the river. The Koshi River frequently changes its course and transports high sediment loads eroded from mountains upstream. All transported sediments are finally deposited in the plains which partially lie in *Terai* Nepal and partially in North Bihar in India. The downstream part therefore faces a significant issue of siltation with intense implications for agriculture. This silt is a mixture of both sand and clay. The villagers also pointed out that silt deposition in farm lands actually reduces crop productivity significantly. The extent of decline in crop productivity depends on the thickness of deposition. In order to calibrate the perception based implication, we further, interviewed a soil scientist working in *Krishi Vigyan Kendra*, Madhepura district. According to him, the deposition of silt initially has an adverse effect on crop productivity by undermining the balance in the soil's organic content and by reducing the water retaining (holding) capacity of soil. But, in the longer run, this adverse impact turns to positive. The long-run waiting period may be five to ten years depending on the location of the farm and the thickness of silt deposition. Villagers reported that because the silt is not just sand, it cannot be used in construction work.

Although the siltation due to Koshi floods appears to be beneficial to agriculture in the long-run, its immediate effect is so very severe that apprehensions inevitably surround farming and livelihood decisions taken by the residents. The following photographs from both Nepal and Indian parts of the basin taken during this field visit (February 20th to 25th, 2014) capture the state of a farm land almost six years after the devastating flood of 2008. The photographs portray how sand and other sediments deposited by the 2008 flood have made a barren landscape of what was once fertile land used for agriculture production. This silted land has now become infertile. In the photograph from Nepal, we can now observe emergence of some vegetation across the destroyed landscape even in red zone area but, vegetation, even if meager is barely perceptible in the similarly destroyed land in Bihar, India.

Embankments, though much critiqued as a solution to floods are still important structures observed in region visited. They are seen along both sides (East and West) of the river extending from Nepal territory (afflux) to Saharsa in Bihar. The total lengths of both east and west embankments are 125.5 and 117 km, respectively (For details, please refer to Table 2). The embankments are closer to the river basin in Bihar to allow space to the river for flooding in times of excessive precipitation. This space however is far from vacant. It is occupied by farmlands and *kutcha* dwellings accommodating inhabitants with poor means. Undoubtedly they are highly vulnerable to disaster and evacuation readiness is an essential part of the adaptive strategy to reduce losses. The specific regions we visited however escaped major flooding in most recent event (2013)¹. In this year, severely flood affected districts were Katihar, Bhagalpur, Bhojpur, Saran, Patna, Buxur, Vaishali, Begusarai, Khagaria, Munger, Samastipur, and Lakhisarai.

¹ Flooding 2013 in Bihar was primarily caused by continuous heavy rainfall in Ganga basins resulting in excess water level in Ganga and its tributaries in Bihar overflowing embankments in July, August, and September 2013.

Table 10.2: The Detail of Koshi Flood Embankments in Both Upstream (U/S) and Downstream (D/S) of Koshi Barrage					
Sl No	Name of Embankment	U/S or D/S of Kosi Barrage	Length in Nepal portion	Length in India portion	Total Length
1	Eastern afflux bundh	U/S	32.0 Km	-	32.0 Km
2	Western afflux bundh	U/S	12.0 Km	-	12.0 Km
3	Eastern Kosi Embankment	D/S	0.5 Km	125.0 Km	125.5 Km
4	Western Kosi Embankment	D/S	16.0 Km	101.0 Km	117.0 Km
Total length:					286.5 Km
<i>Source: www.indianembassy.org.np</i>					

10.2.3. Flood and flood control

Flood is a major problem in the Koshi river basin but we were reported from field interactions that its severity has increased also by certain man-made factors. In fact, all major flooding events that occurred after the construction of the Koshi Barrage and embankments were caused by embankment breaches. There has been a history of breaches² – the most recent one dates back to 2008 which took place upstream of the barrage (*i.e.* Nepal) while the years 1963, 1968, 1971, 1980, 1984, 1987 and 1991 that saw other events all took place in downstream of the barrage (*i.e.* Bihar, India). Though, heavy rain during the monsoon season is a main cause of floods, the poor maintenance of embankments can possibly be held more responsible for the breaches that actually magnified the repercussion of the floods. This causality has been already cited by few researchers for example Mishra (2008), Dixit (2009), Shrestha et al. (2010), etc. In fact Dixit (2009) clearly stated that the breach and related flood in year 2008 were not the result of monsoon-induced floods and cannot be attributed to climatic causes. He has further discussed possible causes which are summarized as: inappropriate technology for the high-sediment carrying river, poor management of embankment infrastructure, lack of warning mechanism, institutional dysfunction and government deficit and poor capacity of respond to the humanitarian crisis.

² Wikipedia (Website)

More interestingly, the same causality between embankment and floods is widely perceived by people living around embankments where we visited. The hypothesis is evidently validated by the river discharge data. According to data collected from Department of Hydrology and Meteorology in Nepal, the river discharge at the time of the embankment breach in 2008 was 4,270 m³/sec at Chatara, which is slightly higher than the long-term average of August (1985-2010) and much less than the maximum discharge that was recorded at the Chatara station between the years 1985 and 2010 which was 9,610 m³/sec on 11th August 1987.

The embankment breach of 2008 and its subsequent devastation could have been averted if some concentrated efforts would have been taken timely. There were prior indications that an embankment breach was imminent (Shrestha et al., 2010). That the embankments were defunct and needed strengthening was even agreed upon at a meeting of the Joint Committee on Water Resources between Nepal and India in 2001. Further, it was pointed out that the above maintenance could not be started because of the hierarchical communication mechanism in place and the trans-boundary character of management of the Koshi Project.

Not just the maintenance of embankments, the morphology of the river is also a significant issue which was unfortunately ignored during the implementation of Koshi Project³ through which construction of the barrage and the embankments started. It is thus evident that hard engineering

³ Following disastrous floods in 1954, Government of India and Government of Nepal jointly made a plan to control floods through a series of dams, embankments and river training works. This plan was further known as The Koshi project. The work under this project was divided into three continuous interlinked stages – the first was a barrage to anchor the river that had migrated about 120 km (75 mi) westward in the last 250 years laying waste to a huge tract in north Bihar and to provide irrigation and power benefits to Nepal and India. The second part was to build embankments both below and above the barrage to hold the river within the defined channel. The third part envisaged a high multipurpose dam within Nepal at Barakshetra to provide a substantial flood cushion along with large irrigation and power benefits to both countries. This was followed by the Kosi Agreement between Nepal and India signed on 25 April 1954 and revised on 19 December 1966 to address Nepal's concerns. Further letters of Exchange to the Agreement between the two countries identified additional schemes for providing benefits of irrigation. While the first two parts of the plan were implemented by the Government of India, the Koshi High dam, the linchpin of the whole plan, for various political reasons has yet precluded any action for several years but has since been revived under a fresh agreement, in a modified form for further investigations and studies.

solutions have failed to control flood and this failure has been discussed in a high level international workshop on 'River Dynamics and Flood Hazard assessment with special reference to the Koshi river' organized on 23rd March, 2011 in Patna, Bihar. In this workshop, it was emphasized that the problem of Koshi flooding is a problem of sediments and therefore a doubt was raised on hard engineering solutions.

In contrast and quite curiously, there is also a strong demand for enlargement of embankments by locals particularly who are living in areas beyond the embankments, such as in the Alamnagar Block in Madhepura District. It reflects a knowledge gap and the lack of awareness among people. The fact that the people living within embankments are opposing them while people living outside the embankments are demanding them also indicates a stark lack of spatial integration. In Khapur, Kisanpur Ratwara, Alamnagar, Alamnagar East *Panchayats* located in Alamnagar Block, people are in favour of embankments because they have been struggling with floods for many years. Even, a *Sarpanch* from one of above *panchayats* offered that gram panchayat could build embankment at their own cost using labour working under Mahatma Gandhi National Rural Employment Guarantee act, if the state government had no objection and provided a 'no objection' certificate.

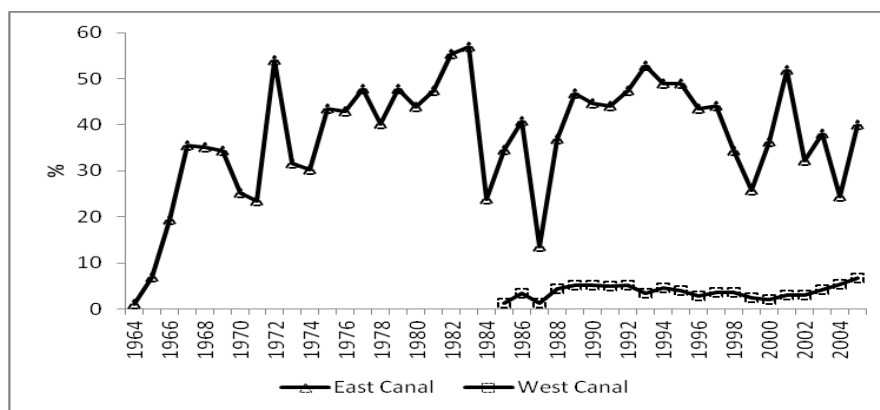
10.2.4. Irrigation

Besides flood control, irrigation is another objective of the Koshi project. Two canals were dug out in the river basin to carry water from the river to farm lands. They are built on the eastern and western sides of the river and are therefore known commonly as East canal and West canals. The construction work on east canal was completed in year 1964 while work on west canal is still going on. The East canal (length 42.30 km) irrigates agriculture lands in Bihar only whereas the West canal which is longer in size (length till date 91.8 km) irrigates agriculture lands in both Terai Nepal (length in nepal 35.1 km) and Bihar (length in Bihar 56.1 km). In Bihar, the East canal and its branches extend to four districts (Supaul, Madhepura, Purnia, and Araria)

and the West canal and its branches reach out to Madhubani and Darbhanga districts.

Despite an impressive project, these canals have failed in their objective, as is reflected from graph 1. It shows poor performance of both East and West canals of the Koshi River in providing irrigation to agriculture. At its best and only in very rare cases, both the above canals of the Koshi River have achieved 50 % and 10 % of their targets, respectively. There are several reasons of failure of its irrigation objective. Among them, two reasons appeared to be relatively important on the basis of the discussion with local people – *one*, lack of field channels, and *two*, the incidence of silt deposition. The Koshi canals were constructed with an assumption that the farmers will construct field channels on their own to take canal water into their agriculture fields, but this has not happened because farmers lack resources. Also while canal water flowed from one farm to another farm, 20 to 30 percent of water was lost by leakage into the ground (Mishra, 2006). The water loss taking place in this way can be a cause of both moisture stress and water-logging as we presently found to be confirmed.

Besides, the canals carry high sediment load originating from the river and deposit them finally into the farms when they are irrigated. The high sediment content in the canal water destroys soil fertility as already discussed and therefore, people refrain from using canal water to irrigate their farm lands. That is why a majority of the farmers use ground water for irrigation which, in turn, depletes water table in the region.



Graph 1: Achievements of Both Canals (%Target irrigated area)

Note: For detail and season wise information, please refer to Tables A1 and A2 in Appendix.

Source: Authors' own calculation using information collected from Department of Water Resource, Government of Bihar, Patna.

Interestingly we found Bamboo boring, a local and low-cost type of shallow tube-well, very popular for irrigation among farmers across the regions. We found almost 10 thousand bamboo boring in a panchayat called Kamalpur in Rani Ganj block of Araria district. Further, we noticed that all these shallow tube-wells are diesel powered tube-wells. Not a single electric tube-well was observed in the sample villages we visited. The shortage of electricity appeared as the main reason for the popularity of diesel tube-wells.

Deep tube-wells are not considered suitable by a majority of the farmers of Bihar because most of them are poor and their holdings are very small. They can hardly afford to install these costly tube-wells and even if they cannot make full use of them, because their holdings are not only small but the small holdings are also divided and fragmented into several pieces. Thus a deep tubewell is not an economic option for the farmer in the region. On the other hand, the Koshi embankments have created water logging problem especially in villages located behind the embankments. A number of villages have been facing these problems since the construction of embankments, whose implications are largely adverse for the life and livelihood of the inhabitants of these villages. The beneficial role of embankments is far from confirmed even by our observation. However, the structural and engineering interventions for flood control are only part of the problem that causes perpetuation of underdevelopment.

10.2.5. Governance

The above discourse certainly reveals not only a serious element of technological deficit but also poor governance and lack of determination of both Government of Bihar (GOB) and Government of India (GOI) to address developmental issues of the region. Other evidences observed in this field visit also strengthen this conjecture. Corruption in the delivery of flood relief and in the rehabilitation programme as reported by the local people is a leading indicator of the failure of governance. In his book written in Hindi language, Mishra (2006) has presented number of evidences of corruption involved in construction of East and West canals and their maintenance. The situation is also observed worse in Tarai Nepal where a large number of people who lost their lands in construction of the Koshi barrage and embankments are still waiting for compensation, even after almost five decades of barrage and embankments. Since the GOI had to pay compensation, according to agreement between above two countries signed in year 1954, the GOI can also be held responsible for above delay in disbursement of compensation. Political instability in Nepal and poor determination of the Government of Nepal (GON) are also equally responsible for the delay because the GON has failed to put strongly their case before India. We have observed some civil societies working in both India and Nepal part of the Koshi basin but they too failed to sensitize the governments of the above issues. Dialogue gap among civil societies within a country and between the two countries could be one of the reasons for this problem. During this field visit we met one activist Mr. Prahlada Ji in Tarai Nepal and another activist named Mr. Ranjeev Ji in Bihar, India. Both have been working in the Koshi river basin for the last four decades. Both were asked whether they knew each other or they were in touch with any civil society from the other country. Their responses strongly suggested that they neither knew each other nor was each in touch with any civil society from the other country.

10.2.6. Livelihood

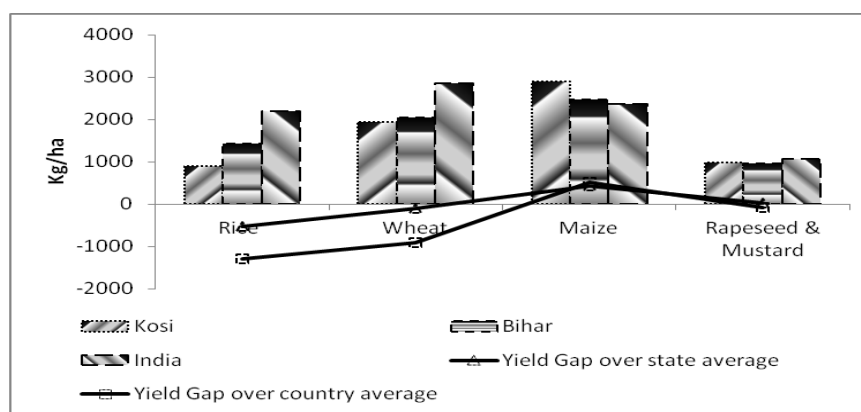
Agriculture: Agriculture is found to be the main source of livelihood in the visited parts of the Koshi river basin. People are engaged in agriculture either as cultivator or as agriculture labour. But the share of agriculture labour is significantly higher than that of the cultivators. Despite the country's lofty land reforms programmes, agriculture is still suffering from skewed landownership. Few farmers have large sized holding varying from 50 *Bigha* to 100 *Bigha*⁴ while a large number of farmers have very small sized, even less than one *Bigha*, of land holding,

A significant number of sharecroppers were observed to be operative in the villages we visited indicating that the majority of landlords owning the farm lands in these villages do not want to do cultivation on their own. Such a situation is known to create poor incentive to increase productivity. Those who own the land and have the capacity to invest are little interested in doing so while those who have considerable interest in investing in agriculture lack in rights, security and capacity. This skewed landownership is one of the most important reasons of low productivity in agriculture in the Koshi river basin.

Thus, numerical dominance of farm labour, the skewed land distribution and the tenorial practices all suggest inequalities and the lack of access to land that discourage agricultural progress. The yield rates of the leading crops (*i.e.* Rice, Wheat, and , Rapeseed & Mustard) is lower in this region in relation to other parts of the country, even while the soil quality is much better in this region compared to many other regions (see graph 2). Land reform has been a great challenge for the government in entire Bihar state, not only in the Kosi river basin. Political influence of landlords is mainly responsible for poor implementation of land reform policy in the state. Even, the present state government headed by Mr. Nitish Kumar constituted a committee on land

⁴ The bigha is a unit of measurement of area of a land, commonly used in the study area.

reform in 2006 which has submitted its report on April 2008 with some recommendations. But, the state government has not taken any action, till today after six years. Even for a government that claims itself as development oriented the lack of political will in this matter is implicit.



Graph 2: Yield and Yield Gap of Selected Crops

Source: Authors' own calculation using information collected from Department of Agriculture and Cooperation. Ministry of Agriculture, GOI, New Delhi.

Like elsewhere in India, agriculture in the Koshi river basin is also dominated by the crop sector, as is reflected from peoples' interactions in study villages. The share of the livestock sector appeared to be very low, possibly its flood prone nature proving to be a discouragement. There is enormous scope for aquaculture particularly fisheries and Makhana cultivation. More than 85 % of total production of Makhana in India is accounted for by the Kosi region. However, the performance of both the economic activities is observed unsatisfactory. Ownership of a pond seems to be big constraint for fisheries and Makhana cultivation. A special community called *Mallah* is attached with these professions, but curiously, it is rarely found that they have any ownership of the ponds. Majority of the fisherman and Makhana growers conduct fish farming and makhana cultivation in leased- in government or private ponds or land and hence they have no assurance of profit. Makhana is a high value crop but its growers belong to very poor community facing exploitative traders. This paradox highlights another constraint of the makhana growers, that of the poor makhana marketing system in which long chains of middlemen are involved.

While paddy was the only major crop grown earlier in the region (PC, 1979), wheat, maize, paddy and rape seed & mustard are found predominant crops today. Villagers said that agricultural production condition is slightly better in *Rabi* cropping season than that in *Kharif* season. When they were asked about its reason, most of them blamed floods and the consequent water logged conditions of the season. Waterlogging is one of the big problems facing farmers. On our way to the field particularly from Darbhanga to Saharsa via Supaul on February 21st, 2014, we saw several water logged fields on both sides of road, even in month of February after six months of rainy season. It is not hard to imagine the intensity of the water logging problem that people would have faced in the rainy season and the initial months (*i.e.* October, November, and December) of winter season. Nevertheless, farmers grow paddy by adapting to the inclement conditions through early and late sowing strategy and sowing such variety of paddy which can survive in water logged conditions. Farmers are adapting to adverse weather condition by changing the varieties of seeds and some traditional varieties of rice have almost disappeared, as is reflected from conversation with Mr. Ranjeev ji who has been working as activist in the Koshi river basin for last four decades. Unfortunately these varieties were able to survive almost flooded situation, for example *ghiya*. It is evident that traditional varieties of paddy can survive better than modern varieties of paddy in flooded fields. Overall, water logging is not only gradually changing the nature of agriculture but also causing a loss of biodiversity. Effect of water logging on acreage of paddy was evidenced in a Planning Commission's study (PC, 1979) on evaluation of the Koshi embankments conducted in 1979. In this study, it was observed that cultivated area under paddy has reduced significantly in villages facing waterlogging problem.

Strangely the effect of water logging is also implicit on wheat and maize crop grown in the rabi season because of the fact that we observed different sizes and stages of these crops among different locations at the same time. In some locations we observed flowering and grain filling stage of these crops while booting and heading stage of above crops was observed in some other location.

However, one may question the above speculation because there are so many factors that affect crop growth. However the impression we got was that such significant differences observed in crop growth in not so significantly different locations could be possible if the crop sowing timing varied among the locations. Farmers' adaptation to water-logged situation to avert its negative effect is a strong suggestion.

As we visited the villages during *Rabi* season, we were able to understand more about *Rabi* crops than *Kharif* crops. Growing three crops (wheat, maize, and rapeseed and mustard) has been reported widely while growing one crop (peas) has been reported marginally. Wheat and Maize seem to be the primary crops while rapeseed and mustard and peas seem secondary crop. We observed from fields in Bihar, which was later also affirmed by the farmers, that rapeseed and mustard is grown in mixed cropping systems with wheat. Such a pattern was rarely observed in Tarai Nepal. It seemed also apparent from the fields in Tarai Nepal that Rapeseed and Mustard is considered as a primary crop there. One other difference was also observed between Nepal Koshi and India Koshi that relates to crop diversification. The cropping system was found more diversified in Nepal Koshi than in India Koshi basin. Demand can be responsible for above difference. This speculation is being justified here with an example of sugarcane. There are no significant geographical differences between Tarai Nepal and Bihar Koshi but farmers cultivate sugarcane significantly more in Tarai Nepal while it is not cultivated in Bihar Koshi, where even a single field of sugarcane was not observed during the visit. Sugarcane cultivation in the former is made appealing by huge demand for sugarcane generated by the sugar industry. There are several sugar mills in Birat Nagar, Nepal, but, there is no sugar mill in Bihar Koshi. Within Bihar, the crop sector was observed more diversified in Non-Koshi Bihar compare to Koshi Bihar. This could be clearly observed by looking at farmers' fields in both regions. In Hazipur from Vaishali District and Samsatipur District, both located non-Koshi Bihar, number of crops (*i.e.* wheat, maize, rapeseed & mustard, peas, tobacco, cabbage, cauliflower, potato, Banana) were sown while only maize, wheat and rapeseed & mustard were sown in Kosi Bihar. A

distinct pattern of crop specialization was observed also within Koshi Bihar. In relatively low-land areas, maize was found dominant whereas wheat was found more important in relatively up-land areas. When the reason for this spatial nature was enquired from farmers, they reported that low-land areas remain water logged till the end of November and December when it becomes too late for sowing wheat. Since maize is a longer duration crop than wheat is and it requires relatively more soil moisture at the time of sowing than wheat the farmers prefer maize in low-land areas. This behavioural pattern can also be seen as farmers' adaptation to weather conditions.

During this visit when farmers were asked about their agricultural problems, most of them actually mentioned the poor marketing system as the main constraint. They do not get fair price of their produce. According to farmers, the biggest drawback is poor transportation facility and the involvement of a large number of middlemen in the marketing chain. Due to lack transportation facility and poor road connectivity, the transportation cost is high. Farmers have to bear this cost if they choose to bring their produce to market yards themselves. Given their poor means, they are forced to sell their produce to middlemen at very low prices whereby the farmers get less income. Assured price and good agriculture marketing system were strong felt needs in almost all visited villages. Measures to meet these needs would probably help to increase farmers' income and accelerate agricultural development, in line with the experiences gained from Haryana, Punjab, and Western Uttar Pradesh. Assured marketing system is one of the major reasons of agricultural development in these states. Besides, an improved marketing system will also generate some employment opportunities for locals. However, the specialization in agriculture that these developments could generate may have negative feed- back effect on natural resources about which the government will have to be cautious.

10.2.7. Off-Farm activities, Social and Gender issues

The region visited was historically well-known for skilled rural artisans such as Mithila painting. Nowadays, these activities have almost disappeared. Neither state nor the central government has taken any worthwhile initiative to revive these arts. There is still a lot of scope for off-farm activities and for the rural artisanship in the area but such activities were not reported during our field visit. Yet, the lack of opportunities forces people to migrate to other places. Out migration was found very high in both Nepal and Bihar part of the Koshi river basin. Both seasonal and permanent migrations were reported in the villages visited. Only difference between Nepal and Bihar part is that more international migration was observed in Nepal compared to Bihar. Push factors like the lack of availability of employment opportunity and natural disasters seem to be mainly responsible for the above reported migration. Tragically, we saw that survival is still the main concern of people rather than their living standard.

In Bihar, we further found that people's mobility is more inter-state than rural to urban within the state, possibly reflecting the meagre employment opportunities available in the urban centers of Bihar state. We did observe people who are working locally in brick kiln and pulling rickshaw but their number is nearly negligible. It was also observed that the majority of migration was among the male working age populations and the migrants rarely were accompanied by their wife and children. Their wives alone have to look after children and domestic welfare.

The intensity of the tendency for out-migration in the Bihar part of the river basin was also evidenced by a field survey conducted in three villages (Parsauni, Lilja, and Gopalpur Khurd) during 2003. The first two villages are located in Madhubani district while third village is located in Supaul district. This field survey was carried out by a NGO (Non-Governmental Organization) called Barh Mukti Abhiyan. Out-migration was observed in around 80 % of

inhabitant households of these three villages. Further, Punjab and Delhi were reported as popular destinations of migrated people in this survey.

As already outlines siltation and floods are a constant threat to the lives of the households in the region. The risk affects not only the livelihood reflecting in lower productivity, poor income and out-migration but also the everyday lives of the people. Scarcity of water is a problem even if physical adequacy or abundances could be associated with this difficulty. The same contamination of water with silt that afflicts the people in the conduct of farming makes the affects the quality of water for drinking or other domestic usages in the perception of the people although ideally the potability factor can subjected to further scientific assessments. There is no supply of electricity in the area visited. Small solar panels are used by a few privileged households for powering a limited number of bulbs or fans in the house. Waterlogging is a problem not only for agriculture but also for communication and possibly is a leading cause of the prevalence of diseases especially malaria reported in the area.

Some poor women were found to be engaged in select and very low paid off-farm activities like fetching *sarpat and kush* (wild grasses) from river bank part of which is used for own consumption and the rest is sold to others. Some women were found to be caring for other people's cattle and bringing them to grazing on open lands. The cattle owners pay the women only two square meals for their services. Women were also found collecting cow dung and making dung cake for fuel purposes. Dung cake and wooden sticks and twigs are largely used for burning the hearth as was reported to us.

Women are considered as one of the most vulnerable sections of a society, especially in less developed regions and this was clearly the impression we also gained from visiting the villages. In each of the villages we visited, the condition of women was perceived to be worse than it was for men. In the most of cases, we observed that husbands of the women migrate to other places for livelihood, and they visit their native place only occasionally. The

wife alone has to take care of the family and manage all household related responsibilities. Despite, the work-load, she is hardly allowed to take any decision. Further, we noted a strong son preference and gender differences in wage. Male gets around 200 Rupees per day while female gets 80 to 100 Rupees per day for the same type of work. This wage difference was observed across the region. When people were asked about its reason, most of them said that labour productivity is the main reason of wage difference between male and female workers.

In this field visit, we came across two types of cases that confirmed the 'son preference' behavior of the local people. The first one is rooted in the economics of the region. In most of cases, parents encourage sons only to pursue his education. It is very rare that a daughter is encouraged to continue her education after primary or junior high-school level. From our interactions with the local people the poor access to education especially after primary level and the low income of the people are responsible for this gender imbalance. The local inhabitants are not able to bear the expenses on education of more than one child and the girl child resultantly suffers this behaviour induced deprivation. Primary schools are available locally but schools for further higher levels of education, even secondary and higher secondary schools are located very far from these villages. The second case is more a reflection of the social mindset in which married women are pressurized to give birth to a baby boy. From our interactions with women in the villages it seemed apparent that people particularly the husbands, want to have more than one son rather than a daughter.

10.2.8. Field Visit to Nalanda

Nalanda is a historic site located close to Patna, the Capital of Bihar. Nalanda is famous all over the world for the ancient international Monastic University established in 5th Century BC, which taught Vedas, Logic, Medicine, Meta Physics, Prose Composition and Rhetoric. Nalanda District is popularly known as Bihar Sharif. The river Phalgu and Mohane flows through the

District of Nalanda. These rivers are almost dried river and these fill with water during monsoon season only. The district is divided into three sub divisions (Biharsharif, Rajgir and Hilsa) and 20 Blocks (Giriyak, Rahui, Nursarai, Harnaut, Chandi, Islampur, Rajgir, Asthawan, Sarmera, Hilsa, Biharsharif, Ekangarsarai, Ben, Nagarnausa, Karaiparsurai, Silao, Parwalpur, Katrisarai, Bind and Tharthari). However, Nalanda is geographically divided into two parts – *One*, Hills Land of Rajgir and Giriyak and *Two*, Cultivated Agriculture Land.

Biharsharif is Sadar Sub- Division and most Urbanized. Hilsa Sub – Division is predominantly rural and agricultural. Rajgir smallest Sub-Division of the three is most important from tourism perspective. The District spread over the area of 2,355 sq.km.as per census 2011 and its total population is 28,72,523. Nalanda District basically based on Agriculture and Tourism. The farmer mainly grows Paddy, apart from it they grow Potato and onion. Few people of the District are also involved in Handloom Weaving. Since the District is a famous tourist destination, tourism plays a vital role in the economy of Nalanda.

We have visited Bihar Sharif of Nalanda District during February 2016 and interacted with farmers. More interesting, it appears to us that these farmers are very innovative. Farmers in this district have widely adopted System Rice Intensification. After success of application of system intensification approach in rice, some farmers have started applying this approach in wheat without any government support. Rice, wheat and vegetables are major crops in this district. Vegetables are grown in both Kharif and Rabi season. Groundwater is the major source of irrigation in Nalanda. Nalanda does not fall in the Kosi basin while it has a very interesting relationship with the Kosi basin. It provides agricultural employment to labours coming from the basin. Such opportunity is likely to increase as farmers in Nalanda district are increasingly adopting SRI and SWI techniques.

10.2.10. Field Visit to Khagariya

Earlier, Khagariya was a part of the district called Munger and it came into existence as a district in year 1981. The district is bounded by Saharsa in the north, Munger in the south, Madhepura and Bhagalpur in the east and Begusarai and Samastipur in the west. There are seven community development blocks in the district namely, Alauli, Khagariya, Mansi, Chautham, Beldaur, Gogari, and Parbatta. Its geographical area is 1486 sq. km and total population is 1666886 with sex-ratio of 886, population density of 1122 per sq. km, and urban area of 12.46 sq. km. Out of total population, about 58 per cent people are only literate in the district. Only 33.64 per cent of total population are workers including both main (18.37 per cent) and marginal workers (15.25 per cent). The major chunk of workers (58.08 per cent) are agricultural labours. This district can be inferred as rural district because only 5.23 per cent of total population reside in urban area of 12.46 sq. km (0.84 per cent of district's total geographical area).

Agriculture is the major occupation in the region and maize and wheat crops are majorly grown. Paddy is grown in some upland parts of the district. Maize is grown in all three agriculture seasons (Kharif, Rabi and Jayad). Cropping patten is increasingly shifting toward maize in the district and it opens many avenues of rural employment, for example, threshing, marketing, food-processing etc. Beside crop sector, livestock is also observed a major livelihood option in the district. However, dairy sector seems unprofitable in the district as is reflected from our interactions with farmers. Milk is mainly procured by Sudha dairy and local sweet makers. Local sweet makers collect milk at door while farmer has to deliver milk at collection of Sudha dairy. Few private dearies are also observed in the districts, that collects milk from farmers.

10.2.11 Field Visit to Saharsa

Saharsa, which was constituted as a district in year 1954, is one of three districts of the Koshi division. This division comprises of three districts namely Saharsa, Madhepura and Supaul. Earlier, Madhepura and Supaul districts were parts of Saharsa district and both districts were separated out in year 1981 and 1991, respectively. Its total geographical area and population is 1687 sq. km. and 1900661, respectively. Though Saharsa district also faces challenge of frequent flooding in the Koshi River, it is slightly better in sex-ratio (906) than Khagariya district. Population density (1127 persons per sq. km.) of the district is also higher than Khagariya. Saharsa is surrounded by the districts of Supaul and Madhubani in north, Khagariya in south, Madhepura in east and Darbhanga in west.

Saharsa is divided into two subdivisions – Saharsa and Simri Bakhtiyarpur that are further divided into blocks. Saharsa subdivision located at northern part of the district has seven blocks (Kahra, Sattar Kataiya, Nauhatta, Mahishi, Saur Bazar, Sonbarsha, and Patarghat) whereas there are three blocks in Simri Bakhtiyarpur (Simri Bakhtiyarpur, Salkhua, and Banma Itahri) located in southern part of the district which is close to Khagariya district. Saharsa district is one of the most affected districts by floods in Kosi River. The main stream of river flows through Nauhatta, Mahishi and Salkhua blocks of the district. However, the river other than the above blocks of the district too through its minor streams. These blocks are Saur Bazar, Patar Bhat, Sonbarsa, Simri Bakhtiyarpur, and Banma Itarhi. Both east and west embankment built up under Kosi Project are situated in the district. Even, some villages of the district fall within both the above embankment. These villages are largely located in Nauhatta and Mahishi blocks. Situation of these villages are pathetic, as is appeared from field visits.

Agriculture is the prime source of occupation in Saharsa district. Rice, wheat, maize and vegetables are grown in this district. Water logging and siltation are major problems that farmers face in the district. Land right is

also a very critical issue here as right on flooded or water logged area goes to the government as per government rule. A large number of working population are found migrated to Delhi, Punjab, Gujarat and many more developed regions for the livelihood. Only non-working population for example women and old-aged men live in some villages of the district. Remittances is key source of income of the majority of population in the district, as is appeared from the field visit.

11. Concluding Remarks and Deliverables

As one of the most disaster prone and poverty stricken regions of the world the Kosi basin is at least partly responsible for keeping the state of Bihar underdeveloped. Separation of industrially richer part as Jharkhand and a history of poor governance are other factors. Sometimes described as an inland delta, the region is subjected to ravaging floods and unexpected lateral shifts of the river. The dynamics of river Kosi are related not just to the local rainfall but to rainfall in the upper courses of the river in the Himalaysa lying even outside India's territory. Moreover, Kosi basin adjoins other major basins with hydrological linkages.

The floods have profound implications for the life of the residents. Fighting the floods and planning for their prevention have been a conventional thrust of an engineering character. The havoc of Kosi floods is alternatively identified with river movements, water-logging and debilitating sedimentation of soil. However, given the historical legacy of backwardness and the the atmosphere of risk that envelops economic decisions, socio-economic status of the region also needs to be studied and addressed for development planning.

Poverty, migration and disasters are regular features of Bihar where urbanization still remains low and agriculture is the dominant economic sector. Rice, Wheat and Maize occupy greater share of the cropped area in Bihar than at the national level but Bihar falls short of all India level in the per capita production of rice, wheat and milk and also in the yield rates of most agricultural products, pulses being the exception. NSSO (2014) found that the average agricultural household earns less than it spends. Our analysis from secondary data finds that inequality exists among the districts of Bihar unfortunately with no signs of convergence and many of the low income districts belong to the Kosi basin. Periodic migration is a traditional practice of the state but received wisdom suggests it is distress induced.

The conclusions drawn in this chapter are based on combined findings from secondary data analysis, primary survey and insights obtained from field visits. Secondary district level data from official sources are used to construct the description of Bihar's Kosi region and its sub-regions delineated by altitudes. Primary survey is conducted in two districts Khagaria and Saharsa both in the low altitude sub-region LOW. The study area is purposively designed to be close to the rivers in consistency with the study objective of studying people affected by Kosi dynamics and the finding may vary from possible district level conclusions. Different classes of farmers are studied ranging from landless to relatively large holders by the standard of the specific region though not by the nation's standard because the largest farmers in the region rarely possess over 10 hectares of land for farming. Important differences also come out between the two study regions despite their location in the same sub-region.

The Saharsa study region (or Saharsa) has a more varied sociology with higher share of female-headed and minority class households, more number of educated members in a household but greater caste homogeneity than in Khagaria. The average household size in the study areas are large, joint family system) being prevalent to prevent land fragmentation. Both districts experience intense migration. Higher education also stimulates migration which is a sign of skill formation but the preponderance as labourers among migrants especially going to in other states suggests distress as causative.

11.2. Natural vulnerability

The state experiences both floods and droughts. Although north Bihar which mostly comprises of the Kosi basin is known more for floods and south Bihar for droughts, in reality the Kosi basin experiences both floods and droughts. However, floods are more recurrent and devastating and only a few years were relatively uneventful in the recent history of floods. To mitigate the adversities of floods, the disaster management forces have to be on high alert in the

season supported by efficient weather forecast and international cooperation. Evacuation is a contingency in many vulnerable areas evoked before the catastrophe strikes. There is no consensus in literature on the negative effect of floods on the socio-economic conditions and on agricultural productivity. Bihar has been criticized for not doing enough for drought-proofing. There is a system of subsidizing irrigation in times of drought but the protocol for declaring drought in a district is not above questioning. In recent times despite poor rainfall there was no support on this account even as poverty intensified and crops wilted.

11.3. Water, Agriculture and spatial Linkages: Secondary data

Kosi basin made up of 16 districts and about 40% of the area and population of Bihar, is densely populated. The LOW sub-region is the largest by area, population and sown area but the MID sub-region is more densely populated. The UPPER constituted of only two districts has the least urbanization and lowest per capita income (GDDP). Demographic indicators are poor in the region which show that state failure is also a weakness in development. Per capita GDP calculated for the Kosi districts is even lower than the Bihar average. Agriculture is more important in resource use and the economy (land, labour, GDP) in the Kosi basin than in non-Kosi basin and so development of agriculture holds a key for the region's development.

The basin is relatively rich in water resources received from copious monsoon rainfall, elevated water table and higher irrigation intensity which is mostly ground water dependent despite the surface water abundance and preponderance of riverine wetlands in the region. It has a more diversified agriculture with higher share of maize than the state or the country. Maize area shows a rising trend in recent times. Rainfall in LOW sub-region is both abundant and well-distributed to enable year-round agriculture but MID and UPPER have rainfall more concentrated in monsoon.

Rivers are major components of the water scenario but flows and heights in rivers are determined not only by local rainfall but also rainfall upstream, even beyond the border in Nepal and China. Not only rainfall, human maintenance of embankments and barrages and timely management of releasing or holding water are powerful influences. This applies also for Ganga which is closely connected with Kosi and other rivers in the area. Tube-wells are the most important sources of irrigation in the basin but they are supplemented by government canals in the LOW sub-region while not being so privileged, the other two sub-regions especially the UPPER also depend on tanks. Field interactions gave the impression that the preference for ground water arises due to the lack of field channels and the sediments present in surface water.

Most villages visited in Khagaria and Saharsa experienced floods in the previous five years associated with rainfall in Nepal or embankment breach or river movements. The floods led to heavy losses of crops, affected sowing in next season and cut off communication lines. Droughts were also experienced though rather less frequently than floods causing crop loss of low quality crops. Farmers irrigated land using diesel run pumps.

Yield rates of food crops rice and wheat are low in Kosi basin though productivity of Maize is higher but the estimated regression model based on secondary panel data with spatial correction showed that yield will respond to prices, infrastructure and effective water management backed by information on rainfall. Rainfall of upstream regions with its managed use by irrigation is important for yield. Disregarding spatial implications of yield determination can produce erroneous conclusions.

11.3.2. Land loss

Both study regions under primary survey are victims of land losses. River movement is the most dominant reason for land loss other than flooding , water-logging and sedimentation while manmade reasons like embankments,

canals and roads had a very small role. Even the landless reported having lost land they had once possessed.

11.4. Cropping pattern, productivity and Risk management

In Kosi basin Rice is dominantly raised in the kharif season and Wheat and Maize in the rabi season. Rice and some pulses are grown in both seasons and Oilseeds in winter. The minor crops in a season compete for resources and time. Maize is grown in both seasons and also in the third zayad season. The LOW sub-region has a more diverse cropping pattern especially in the rabi season. The share of rice is highest in the UPPER sub-region. Farmers allocate maximum (78%) of crop area to food crops rice and wheat in UPPER followed closely by MID whereas their share is below 60% in LOW where maize is also important. In the UPPER and the MID the dominant kharif and rabi crops respectively are rice and wheat. The Herfindahl index showed Kosi basin to be more diversified in crop choice than the rest of Bihar and within the Kosi basin the LOW sub-region to be most diversified and the UPPER least so.

While rice area made slow descent in the last decade, wheat area showed stability. Maize area specifically in the LOW sub-region showed steady but slow increase but more recently area under maize in the other two sub-regions is displaying remarkable increases. The agronomic suitability of Maize in these conditions and the commercial demand from industries is responsible for this rise of Maize. Area moved away from pulses. Horticulture and animal husbandry are also practiced more commonly in Kosi region than in the non-Kosi belt.

Yield rate for rice is volatile and did not show much gain over time, that of wheat showed improvement since 2005 and maize yield consistently rose and with less volatility. Pulses showed remarkable gains in productivity.

Within the basin there are sharp differences. Agriculture is most diversified in the LOW sub-region and maize has a higher share than in other sub-

regions. All three sub-regions have their specific advantages in producing certain crops. The higher altitude UPPER sub-region is weak in maize productivity but has advantage in Rice, Wheat, Millets, Pulses and Oilseeds and the MID has advantage in Maize, Millets and Oilseeds. Cultivation of pulses is not extensive and is declining but the productivity is high. Maize yield is considerably high in Kosi basin and especially in the LOW sub-region than Bihar. Popularity of maize cultivation in this sub-region has made Bihar recognized for maize cultivation.

11.5. Risky environment for decision making

Cropping pattern is related with agro-climatic advantages, market demand and risk management. Risk management is also closely linked to food security because under conditions of natural risk, poverty and poor market access farmers minimize their vulnerability to food insecurity. While the effect of flood on agriculture leaves room for debate, it appeared in the field survey that regardless of the actual occurrence of a flood, the risk of flood pervades the decision making environment in agriculture.

Diversification of risk is a well recognized instrument of risk management practices in conventional and structured systems. The Kosi basin is more diversified in its cropping pattern than rest of Bihar and the LOW sub-region is more diversified than the other two sub-regions possibly as part of risk coping but within the LOW region our primary survey found that flood prone Khagaria was more specialized by the Herfindahl index. The high share of food crops in Kosi basin is another dimension of risk management even if that amounts to lack of diversification. This is found in the UPPER and MID sub-regions though shifts are already apparent. Khagaria however is more commercialized by concentrating on Maize while Saharsa diversifies among crops and emphasizes food crops for food security. Khagaria is a unique model which shows risk coping strategy as not just risk diversification or growing low value crops but concentration of resources to land use that is safest though compromises on food security are addressed in different ways.

Most farmers are reluctant to sow kharif rice for fear of flood and excess rainfall but rather they let the land lie fallow or preferably grow fodder crops that do not demand resources. However, rice is required for subsistence and farmers privileged with upland plots grow rice willingly in Khagaria. Flooding and monsoon rains affect not only kharif decisions but even spill into the rabi season if water-logging persists. Wheat is an alternative staple in Bihar to Rice but it makes rigid demands on climatic conditions and crop calendar. It can be grown only in upland areas. On the other hand, Mize in its hybrid varieties requires more soil moisture at its sowing time and is more flexible in its crop calendar and the farmer can wait till the water subsides. This flexibility of hybrid maize is one reason cited by farmers for choosing this crop in our field visits.

The preference for food crops is however pervasive in the data. In Saharsa study region the cropping pattern is still rice dominated followed by wheat another food crops but an average farmer even in Khagaria who grows rice allocates more acreage (1.21 hectare) to rice than the same farmer in Saharsa (0.78). Acreage allocation to rice is inhibited by lack of land suitable for rice in terms of water-logging and low risk by other farmers who move away from rice. Only the privileged farmers grow rice. Leasing is restricted by law.

In Khagaria sample average farm size is larger and so is the area under the each of the three crops. The yield rates of crops obtained by growing farmers also exceed the the same in Saharsa where the yield is more variable. Rice yield is low by the standard set by the state in both cases but that of maize is high.

11.5.2. Food security constraints

Risk coping has pushed farmers in the water-logged Khagaria to commercial crop maize but rice and wheat remain dominant in Saharsa. The risk taking behavior in Khagaria in terms of food security can be explained partly by the

State's proactive intervention of assuring a minimum quantum of food to be available at low prices in the PDS. Since this supply does not suffice to need family needs, the need for farm production remains.

11.5.3. Farm profitability

If all produce is valued at producer prices reported (or if all produce were sold) farming is notionally profitable in the Maize dominated study area of Khagaria but loss making in Saharsa. In Khagaria rice brings notional profit to all classes except the Large class but Wheat is loss making barring the marginal classes in both regions and Maize is profitable to all class cases. Overall farm income based on the three major crops is positive only in Khagaria sample but records a net loss in Saharsa. If only sold product is considering farming does not generate net cash income in Saharsa despite maize.

11.5.4. Subsistence, Commercialization and Food security

The study regions present a semi-commercial character of production where farmers grow not only maize but also food crops rice and wheat both for home consumption and marketing. Also food is sourced from multiple sources including purchases from market and farm production but availability of subsidized food through PDS outlets enhances the freedom to sell even the same food crops in the open market at a higher price. In other words, farmers sell food at higher prices and buy food in the PDS at low prices though PDS quota being limited, they also buy from open market. Farmers hardly have a net surplus of food but they market considerable portion of produce probably to obtain cash incomes but buy back from open market or PDS.

Despite the losses or the meagre profit, production makes sense because the cost of food security is lower from farm production than open market so far as the PDS is insufficient. For example if rice yield from a hectare is 1.1 tonnes (Khagaria), the retail price is Rs 26000-30000 a tonne and the cost of cultivation is Rs 24000 per hectare (Rs 22000 per tonne), the saving from

producing a tonne of rice rather than buying is Rs 4000-Rs 8000 even though the revenue from selling the same rice may be lower than the cost at producer price of Rs 10000. The farmer on the average is found to buy back food grain and is likely to pay the higher (off-season) retail price of the range.

The wedge between the producer price and the retail price reported officially is uniformly high. In the case of PDS purchase at say Rs 2 a kg there is huge saving over open market purchase as also farm production and so a choice between PDS and other sources does not appear. Home produced consumption preempts use of cash but cash is required for paying standing debts, buying from PDS and procuring other necessities. Sale of food and maize can only be justified by these needs for cash. So far as other avenues for earning cash are open the choice is not among the specified crops only. Thus in the presence of off-farm jobs and MNREGA the incentive for farm production can in an extreme case be limited to food security needs.

11.5.5. Input use and farming intensity

Agriculture is input intensive and fertilizer use is higher in Bihar and especially in the Kosi basin than in India but over time a tendency towards nutrient balance is observed. The higher fertilizer use intensity especially in Khagaria may be because of washing out of soil nutrients and infertile sediment deposit. Labour is a major component of the paid cost of production. Farming of Rice and Maize is more dependent on hired labour in Saharsa where the average reported farm wage is also low despite greater migration and MNREGA but the direction of causality is not clear. The share of labour in the cost of production is as high as 70% in rice production and 45% in maize production in Saharsa. The total costs of production of Rice per hectare is much higher in Saharsa compared to Khagaria. Farming is more intensive in land use in Saharsa but in Khagaria farmers fail to harvest from part of the farm land due to crop damage or infertility.

11.5.6. Scale sensitivity

The analysis being done at the level of farm class provides a glimpse of differential implications of scale. In socio-economic aspects, smaller or landless farm households are found to be smaller in size, more prone to migration, be female headed, belong to minority caste and religion groups and are inclined to education. They are also vulnerable to land loss though other classes are not spared either. The small farms have greater affinity to food crops while larger farmers allot more land to maize even in Saharsa. There are no signs of scale favourable economies. Indeed, mostly, smaller farms are more intensive in land use, more productive of crops despite greater fertilizer use by larger farms. Cost of cultivation rises with farm size leading to smaller income per hectare. The advantage of the small farm is mostly due to use of family labour.

Thus only the large class makes a loss in rice cultivation in Khagaria, only the marginal class makes a gain in wheat in both study regions and even the computed cash income is positive for the marginal class in Khagaria. Only the income from maize increases with farm size suggesting returns from scale. However, although the smaller farms enjoy economic advantages at the per hectare level, due to the small size of their holding there is a distinct positive relation between farm size and farm income. While in Khagaria smaller farms earn less from farming than the larger counterparts, in Saharsa the larger farmers suffer greater losses. As imputed and cash incomes are not sole driving forces for cultivation, the larger farm classes may be gaining in terms of food security which is a major concern in joint families.

11.5.7. Leasing of Land

Bihar has rigid legislation on land leasing nearly banning the practice but for exceptional cases. Yet informal land leasing is very common, more in Khagaria than Saharsa but the agreement is oral and unrecorded. Causal interactions revealed practices of cash rent as well as share cropping but conducted in

non-systematic processes. The income generated would work out to be less than what is computed in our exercises if the rent could be taken as a cost. Very few respondents reported leasing out but those who did earn their living by cultivation as well as off-farm jobs like trading. In Saharsa most of those who reported leasing out were landless.

11.5.8. Marketing system

Bihar disbanded its market legislation and has open markets. Village traders are the single most dominant buyers of produce though farmers in Khagaria sample also sell to landlords and money lenders. Producer companies are also reported but the state cooperative did not buy from any of the sample farmers. Village traders buy both from farm gate and at their own stores. In the absence of State supervised markets they are required to have their own facilities of storage, weighing and grading.

Large traders from corporate organization also operate. They buy from farmers sometimes from farm-gate and from traders' stores over larger distances than the village traders but farmers' groups and traders also deliver at their collection centres. Unlike village traders they transact in cheques (with traders) and also in cash (with farmers), pay taxes, spend on fumigation, security and accounts. Both types of traders generate considerable employment casual worker in case of the village traders and salaried persons by large traders. Both traders do not invest of storage infrastructure but use temporary semi-permanent arrangements. Large traders also make use of railway facilities. Moisture content in grains is reported by all as the main cause of product loss though loose and defective packing, farmers' carelessness, pests are also reported. All traders operate under competition and no one complained on the absence of rules nor desired them.

Markets were once regulated through dysfunctional committees. Their precincts were found to be to be moderately endowed. Some of the markets are supervised by local agencies like a food business enterprise or a charity

organization. Transactions in maize and rice are more common than wheat. Rice and wheat milling and maize threshing are primary processing activities within the village relying on mechanization through bank credit. In the towns there are formal enterprises also for further processing of rice, wheat and maize to biscuits, puffed or flat rice, bread and popcorn and other snacks. The raw material is not always procured locally.

11.5.9. Dairy farming

Similar results are observed in the case of milk production. Most households including the landless keep animals and grow some fodder but no net cash income is generated though part of the milk is sold but the part retained for home consumption possibly justifies the occupation and the cost incurred. Milk is sold to informal milk vendors, cooperative societies and private dairy but there is no buying back of milk. The yield is poor but Khagaria shows a much better performance than Saharsa where the performance is poor. The wet ambience of the region favours dairy but lack of land also affects fodder cultivation though most farmers in our field visit reported growing some fodder in otherwise unutilized land in the kharif season. Breeding exotic varieties will need to consider the constraints on fodder production.

11.5.10. Non-farm employment and amenities

Apart from cultivation and dairy other occupations are semi-skilled work like mason, cycle repair, tailoring, driver besides unskilled labour. MNREGA employment was rare in Khagaria but relatively more common among Saharsa households. Failure to identify jobs is a reason that had held back the initiative. People worked within and sometimes outside the villages. For transportation, bus service was rare and all weather road access was found only in few cases. Although barring three villages all had electricity supply and no village was deprived of tele-connectivity, in practice not all houses had electric connection of mobile phone services. Electricity even if available was not for 24 hours. Solar cells were distributed by the government but there

were complaints on their quality. Drinking water was taken by hand pump and in some villages there was market for sale and purchase of quality water. Use of LPG was reported in some districts but all reported use of dung cake and crop residue. Nearly all the surveyed villages experienced flood in at least 3 of the last five year caused by heavy rainfall in Nepal, embankment faults and mostly movement of river and most villages reported experiencing droughts in the two three of the five years. The administration or the media had provided early warning of flood.

11.5.11. Gender

Women in the farm households divide their time among farm and household chores without sharp boundary and have a significant say in matters of livestock. Women work primarily in household work, farm work (unpaid), wage labour in agriculture and other sectors. Some women only in Saharsa reported employment in MNREGA. Farm wage is found considerably lower for women than men despite labour scarcity. Women entrepreneurs are found in good processing associated with Self help groups. Migrants who work in northern India generally are accompanied by their wives who also work in cities. In the

11.6. Summary highlight

The broad highlight is the problem created by water-logged and sedimented conditions prevailing beyond kharif season into the rabi season and the persistent risk of floods under which the farmers take decisions. Water-logging and river movements also reduce land available for productive cultivation pushing rice and wheat growers to upland areas and giving space to winter and summer maize after residual water from monsoon reduces. This shift is allowed in the face of food insecurity because of the availability of subsidized grains in PDS. Some food is sold in the open market for raising cash but unfortunately part of this does not represent any surplus and needs to be purchased back at higher prices. Like maize, rice and wheat too provide

the same opportunity for commercial production to be sold at prices higher than the PDS price at which farmers buy food but their cultivation is constrained by suitable land and high cost of cultivation. Besides, the market for maize is expanding. However econometric regressions indicate that food security target is important for deciding maize acreage in Khagaria though commercial motives work in other cases.

Scarcity of labour, risk of floods, lack of cultivatable land for sowing, difficult circumstances of water-logging and siltation and high cost of cultivation make farming unprofitable as also unappealing but nevertheless worth cultivating to meet food needs of family. Despite all the weaknesses and the economic losses farmers in Saharsa are still found inclined towards food crops. Soil quality being more amenable for rice and cheaper local labour being available compared to Khagaria are possible reasons but relatively lower access to cheaper grain in PDS may also be pushing farmers to the recourse of subsistence farming. Our primary data shows that lower share of households in the study region holds a ration card in Saharsa and lower share of the card holders have access to the public subsidies given to poorest sections (BPL, AAY) compared to Khagaria. The shares of rice consumption met by PDS and farm production are greater in Saharsa while farmers in Khagaria depend more on open market for procuring rice for consumption.

11.7. Policy

Risk of flood is a real constraint creating aversion for farming in some seasons and some locations. Waterlogging and sediments make it harder to farm and the scarcity of labour and disinterest in farm labour jobs make matters worse. Expanding maize in the disadvantaged areas and contracting rice cultivation are coping mechanisms evident but only with support from government through the PDS food.

There is a limitation to farmer's expansion of acreage. Given that natural disasters are nearly unavoidable under present facilities and the risk faced by

farmers, zoning needs to be implemented for safety of human, property and crops and reduction of public burden. Cultivation of food crops need to be confined to safe zones while commercial maize cultivation promoted in other areas but the limitation of land cannot be avoided. Even in the case of maize expansion of acreage will hit a limit as the motivation seems to be partly to raise cash for food security.

Farmers also seem to prefer other avenues of earning to cultivation among which migration is one. Yet food security remains a concern even as food crop acreage contracts. In this context PDS plays a vital role in this area. Replacement by cash transfer is not likely to take care of food security at the current position especially if the retail price cannot be controlled. While the area may deserve special treatment in continuing public distribution of grains in kind in the long run, it is not sustainable especially for the distortions that can be accepted at best as a short term measure.

Comparisons between the retail price, the cost of cultivation, the producer price and the PDS price display paradoxical relations when prices are beyond any purview of price discovery. The PDS ensures minimum food security for the poor farmers and is important in the vulnerable region but farmers grow the same crop for sale because they can buy it at subsidized price. The large difference between the retail price and the producer price needs to be examined for possible sources of market imperfections. While there is no complaint about the unregulated marketing but the state run cooperative procurement is resented for irregularities. The sources of food in the open and public channels require attention. If the retail price was closer to the producer price (vertical integration) production incentive for food crops would be lost because of losses or low profits. Similarly, if the PDS were expanded production would be eased out. Importance of cash remains but given the tendencies, farmers might look for other avenues of earning rather than cultivation. Food inflation has a serious effect on farmers' decisions and welfare.

All development planning can be done keeping in mind the ongoing and anticipated institutional changes in water, food and subsidy policy. There are also broader precepts of experts at the national level that guides planning. However, both institutional changes and expert guidance will have to be tailored for specific situations. It is strongly suggested that cultivation of rice, deemed to be water demanding needs to be shifted to the water rich areas especially in the east. Similar statements are made about sugarcane. However, in practice there are constraints. Policy needs to be finely sensitive to farmer's preference and reluctances in regard to acceptance or rejection of crops in certain conditions and decisions on irrigation and fertilizer, that are formed from experience in real life. However scientific information may be conveyed to them and steps taken ensure that their decisions are not distorted by public actions. Similarly, any plan of promoting dairy will have to be made keeping in view fodder area constraints and climatic conditions.

With a longer perspective we suggest zoning the area to safer confines, emphasizing productivity over acreage for expanding production, promoting yield enhancement by infrastructure and cooperative water management and bringing down labour cost by possible mechanization. Development of agro-processing industries in the surrounding regions outside the riverine areas and even beyond the Kosi region may be considered for generating employment with gender balance and without causing the family alienation that migration causes. Land acquisition laws for the purpose may have to be worked out in ways acceptable to farmers. Banning of leasing is not of much relevance. It only helps to deprive the poor tenants of their due support. The reports on disaster management are positive but has to be strengthened as a routine matter while also optimizing the risk through efficient land use.

While maize is a source of optimism, it cannot be a total solution to the poverty of the region. Agriculture has its limitation in such vulnerable areas. It is not only largely un-remunerative but the occupation with farming imposes huge costs on human being and public authorities. The wet-land aspect of the region also demands recognition for developing other aquatic activities such

as fishing, makhana cultivation, algae culture for energy and tourism. At the same time the regions can serve as a reservoir of water resources which require scientific intervention to prevent losses to evaporation and seepage and possible use in emergency circumstances of water scarcity in the area or elsewhere. The eco-system service of the region and the compromises of the people of the nature challenged region for restricting agriculture need to be capitalized in market compatible or other ways. For example a special status for food insecurity may be set off against water services and wet land conservation.

11.8. Deliverables

Baseline data with baseline scenarios: Data available at the time of the research are provided in Appendices to Chapter 5 and 6 and the discussions are presented in the corresponding chapter of the report. The data set consists of district level data of Bihar. The Kosi basin region (KBR) is identified by districts affected by the river's dynamics. The unit of analysis is the KBR. District level data are aggregated to the regional and sub-regional level of the Kosi basin based on altitude.

Analysis covered development indicators at the district level, such as gross domestic product, demographic details, land use, water use and agriculture. Input use mainly fertilizer for agriculture and cropping patterns, diversification, source wise irrigation, rainfall and water table and crop yield rates are covered. Comparison is made between KBR and the state and within KBR. This discussion is backed by a chapter on the specifics of Kosi river and the Kosi basin combining engineering and environmental dimensions in Chapter 2. Theoretical discourses of aspects and their recent updates that are likely to come up in the research such as development, food security,

migration, gender issues and flood control are also given as background in Chapter 3.

Historical data: Data presented in Chapters 5 and 6 are cross section as also time series covering recent decade to bring out historical tendencies for possible comparison within the time span and in tendencies in future. Comparisons of the tendencies is made between KBR and the rest of Bihar and among the three sub-regions of KBR.

11.8.2. Agriculture and Water related institutions

Institutions are bound by sets of rules that change over time with policy and paradigm. India is undergoing transition in many of these institutions. Details of the systems and agencies governing food supply, food marketing, flood control, irrigation and water use, cooperative, disaster management and government policies on transferring support are discussed in Appendix to Chapter 10.

Yield Function: Econometric yield functions of three major crops in the Kosi basin of Bihar are estimated and presented in Chapter 7 using panel data and corrections for spatial dependences. Explanatory variables include economic variables like relative prices or relative revenues with respect to possibly competing crops for facilitating policy making based on parameters. The yield functions are corrected for effects of climatic variables rainfall and temperature (maximum and minimum) at the monthly frequency to capture seasonal dimensions. Moreover, linkages with upstream areas are taken into account so that effects of rainfall during specific times of the year in different altitudes on crop yield are allowed for. The spatial model which also allows for spatially autocorrelated errors and spatial dependence of yield owing to demonstration effects of unobservable interlinkages was found to perform better than conventional fixed effect panel regression model. The model helps to assess effects of price changes as well as effects of infrastructural development on price responses. Given the specification climate change

effects can be worked out based on expected scenarios of temperature. Climate change is also expected to render monsoon more uncertain and prone to excesses and deficits with differentiated spatial implications. The yield model therefore can make projections for crop yield under differing rainfall scenarios in the mountains and the plains and facilitate designing efficient irrigation for making good use of upstream rainfall.

11.8.3. Household questionnaire

At the outset field visits were organized by the sponsor which provided a first hand initial view of the Kosi basin and its economic and natural problems to set the research in motion. Pilot surveys were made along with further field visits to gain insight and improve the questionnaire. The questionnaire was addressed to the household represented by a member respondent covered questions on various social, demographic, economic and agricultural aspects. The survey was purposively designed to study two specific regions in the plain area of KBR which are close to rivers.

Moreover, a few market functionaries such as traders were asked about their mode of business, problems and constraints and amenities for marketing. Visits were made to units of agro-processing on primary and secondary nature. The answers were both quantitative and qualitative based on subjective impressions of respondents. The results of the analysis of the data are reported in Chapters 8 and 9 of the report.

11.8.4. Land use decisions in context of commercialization and food insecurity

The primary data collected were analyzed to assess the cropping pattern decisions of farmers, diversification, commercialization and self-consumption. The focus was on productivity, input use, paid cost of production and income potentials and cash generation from three major crops as also options of marketing versus subsistence and buying back of food, sources of food from market, PDS and farm production and implications of

subsidized food, commercial sale and open market purchase. Cross section data from primary survey in the two regions were modeled Econometrically to understand decisions on acreage under each crop and the decision of not sowing the crop. Subject to limitations of cross-section data the estimates are expected to identify the relevant economic incentive in each case and the main motivation for cultivation along with subtle implications of seasons, land quality and work capacity. The analysis is done in the

Bibliography

1. Agrawal, Bina (1983). *Mechanization in Indian Agriculture*. Delhi: Allied Publishers, Reprinted 1986.
2. Agriculture Census (Webiste). <http://agcensus.nic.in/> Government of India.
3. Agriculture Census (Website). Input Survey. Government of India.
4. Arfa, N.B., Rodriguez, C., Daniel, K. and Shonkwiler, S.J. (2010). Spatial Structure of Agricultural Production in France: Role of the Common Agricultural Policy, OECD Workshop on the Disaggregated Impacts of CAP Reform, 10-11 March 2010.
5. Arfa, N.B., Rodriguez, C., Daniel, K. and Shonkwiler, S.J. (2010). Spatial Structure of Agricultural Production in France: Role of the Common Agricultural Policy, OECD Workshop on the Disaggregated Impacts of CAP Reform, 10-11 March 2010.
6. Baldwin, R. E. (2001). The core-preiphery model with forward looking expectations. *Regional Science and Urban Economics* 31, 21–49.
7. Baltagi, B. H. (2001). *Econometric analysis of panel data*. 2d ed. Chichester, UK: Wiley.
8. Barro Robert J. and Xavier Sala-i-Martin (1995). *Technological Diffusion, Convergence, and Growth*. NBER Working Papers 5151, National Bureau of Economic Research, Inc.
9. Barron, C.D. (1940). Papworth research bulletin.
10. Baylis, K., Nichlas, D.P. and Gianfranco, P. (2011). Spatial approaches to panel data in agricultural economics: a climate change application. *Journal of Agricultural and Applied Economics*, 43(3), 325-338.
11. Bhatt, Yogesh (2016). *Econometric Analysis of Biofuel Prospects using Classical and Bayesian Techniques with Implications for Food Security in India*. PhD thesis submitted to Kumaun Univesity, Nainital, Uttarakhand India.
12. Bihar Agriculture Management & Extension Training Institute (BAMETI) (Website).
http://www.bameti.org/pdf/agriculture_profile_of_the_state.pdf
13. Census India (2001, 2011). Office of the Registrar General and Census Commission, Ministry of Home Affairs, Government of India. <http://censusindia.gov.in/>
14. Centre Ground Water Board (CGWB) (website) <http://gis2.nic.in/cgwb/Gemsdata.aspx>
15. Chan, H.W. (1994). *Cities with Invisible Walls*. Oxford University Press: Hong Kong.
16. Chen, S., Pernetta, J.C. and Duda, A.M. (2013). Towards a new paradigm for transboundary water government implementing regional frameworks through local actions. *Ocean and Coastal Management*, 85(B), 244-256.
17. Choithani, C., & Pritchard, B. (2015). Assessing Bihar's Coupon-based PDS. *Economic & Political Weekly*, 50 (3). pp. 61 – 68.
18. Command Area Development and Water Management Programme. Retrieved from <http://www.wrd.bih.nic.in>.
19. Department of Animal Husbandry (DoAH) (2014). *Basic Animal Husbandry Statistics*. Ministry of Agriculture, Government of India.

20. Dray, S., (2011). A new perspective about Moran's coefficient: spatial autocorrelation as a liner regression problem. *Geographical analysis*, 43,127-141.
21. Dray, S., Daid, S. and Debias, F. (2008). Spatial ordination of vegetation data using a generalization of wartenbeg's multivariate spatial correlation. *Journal of vegetation science*, 19, 45-56.
22. Dreze, J. and Khera, R. (2015) 'Understanding Leakages in the Public Distribution System', *Economic & Political Weekly*, 50(2), pp. 39-42.
23. Drèze, J., Khera, R., & Pudussery, J. (2015). Food Security. *Economic & Political Weekly*, 50(34), pp. 44 – 52.
24. Economic Survey of Bihar (2014-15). Finance Department, Government of Bihar (GOB), Patna Directorate of Economic and Statistics, Department of Agriculture and Cooperation, Ministry of Agriculture, Government of India
25. Elhorst, J.P. (2003). Specification and estimation of spatial panel data models. *International regional science review*, 26(3), 244-268.
26. Farmer Portal (Website). <http://farmer.gov.in/> .Department of Agriculture & Cooperation and Farmers Welfare, Ministry of Agriculture and Farmers Welfare, Government of India.
27. Fertilizer association of India (various). Fertilizer Statistics of India. Government of INDia.
28. Flood Management. Retrieved from <http://www.ward.bih.nic.in>.
29. Florax, R., Voortman, R.L. and Brouwer, J., (2002). Spatial dimensions of precision agriculture: a spatial econometric analysis of millet yield on Sahelian coversands. *Agricultural Economics*, 27, 425-443.
30. Fujita, M. and Thisse (2002). *Economics of agglomeration: cities, industrial location, and regional growth*. Cambridge university press, Cambridge.
31. Geary, R. (1954). The contiguity ratio and statistical mapping, *The Incorporated Statistician*, 5, 115-145.
32. Getis, A. and Aldstadt, J. (2004). Constructing the spatial weights matrix using a local statistic. *Geographical Analysis*, 36, 130-140.
33. Ghosh, Nilabja and Anita, Kumari (2015). Sustainability in Indian Agriculture. "*Reviving Growth in India*" Book (Eds) by Pradeep Agarwal, Cambridge University Press.
34. Ghosh, Nilabja (2008). Problems and Prospects of Crop Insurance: Reviewing Agricultural Risk and NAIS in India. Report submitted to Ministry of Agriculture, Government of India
35. Ghosh, Nilabja (2011). Role of Climate change and Agricultural Performance. Report submitted to Ministry of Agriculture, Government of India.
36. Government of Bihar (2005). Report of Expert Committee on Impact of Interlinking of Rivers on Bihar. Water Resources Department, Government of Bihar, Patna.
37. Government of Bihar (2012). Agriculture Road Map 2012 – 17. Department of Agriculture, Government of Bihar, Patna.
38. Government of Bihar (2013). Standard Operating Procedure for Flood Management, 2013. Water Resources Department, Government of Bihar, Patna
39. Government of Bihar (GoB) (2011). State Domestic Product 2004-05 to 2010-11 (Adv.) & District Domestic Product 2004-05 to 2007-08 Bihar, Directorate of Economics and Statistics Department of Planning and Development, Patna.

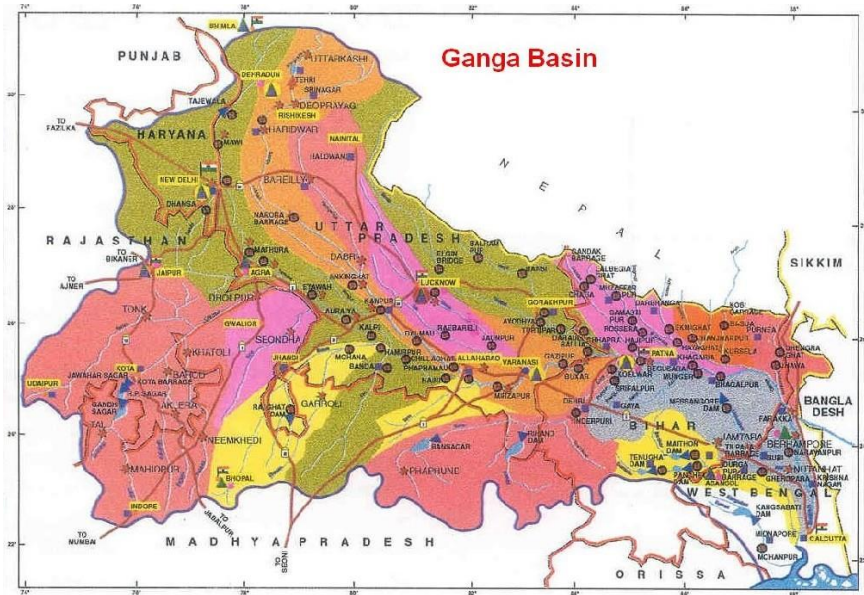
40. Government of Bihar (GoB) (Website). <http://gov.bih.nic.in/>.
41. Government of Bihar (GoB-CD) (Website). <http://cooperative.bih.nic.in/>
42. Government of Bihar, Department of Agriculture (GoB-DoA) (Website). <http://krishi.bih.nic.in/>
43. Government of Nepal (GoN). (Website). Department of Hydrology and Meteorological, Ministry of Science and Technology Department. <http://www.dhm.gov.np/>
44. Ground Water Irrigation Scheme. Retrieved from <http://www.mwrd.bih.nic.in>.
45. Gulati, Ashok and Kelly Tim (1999). Trade Liberalization and Indian Agriculture, Oxford University Press.
46. Heckman, J (1979). Sample Selection Bias as a Specification Error. *Econometrica*, 47, PP-153-161.
47. Hsiao, Cheng. (1986). Analysis of Panel Data. Cambridge: Cambridge University Press.
48. Hugh, G.G.J. (1973). A quantitative evaluation of the Bray-Curtis Ordination. *Ecology*, 54(4), 82-836.
49. Indian Meteorology Department (IMD) (Website) (<http://www.imd.gov.in/>).
50. Indian Water Portal (IWP)(Website) (http://indiawaterportal.org/met_data).
51. Interlinking of Rivers. Retrieved from <http://www.ward.bih.nic.in>.
52. Kale, V S (2008): "Himalayan Catastrophe that Engulfed North Bihar", *Journal of the Geological Society of India*, 7: 713-19
53. Kumar, Ganesh (1992). Falling Agricultural Investment and Its Consequences. *Economic & Political Weekly*, Vol. 27, Issue No. 42, 17 Oct, 1992
54. Kumar, K.S. 2013). Climate sensitivity of Indian Agriculture, (Editors: Nilabja Ghosh and C.S.C. Sekhar), *The future of Indian Agriculture*, 156-169.
55. Kumari, Anita and Ghosh, Nilabja (2014). The trade in temperature and rainfall and the possible vulnerability of Indian agriculture to climate change, *Food studies: An interdisciplinary Journal*, Vol.2 (4), 42-59.
56. LeSage JP and Pace RK (2009) Introduction to spatial econometrics. CRC Press Taylor & Francis Group, Boca Raton.
57. Mahalanobis, P. C. (1944). On large-scale sample surveys. *Phil. Trans. Roy. Soc. London, Series B*, 231, 329-451.
58. Ministry of Statistics and Programme Implementation (GOI-MOSPI) Government of India. (Website). http://mospi.nic.in/Mospi_New/site/home.aspx
59. Mishra, D. Kumar (2009). Drought in Bihar. *Frontier Weekly*, Vol. 42, <http://frontierweekly.com/archive/vol-number/vol/vol-42-2009-10/vol-42-5%20Aug%2016-22%202009/bihar-42-5.pdf>
60. Mishra, DK (2008). TRAPPED! Between the Devil and Deep Waters. Dehra Dun: Peoples' Science Institute and Delhi: South Asia Network on Dams Rivers and People SANDRAP (A YUVA Project)
61. Mooij, J. (2001) 'Food and power in Bihar and Jharkhand: PDS and its functioning', *Economic and Political Weekly*, 36(34), pp. 3289-3295.
62. Moran, P. (1948). The interpretation of statistical maps. *Journal of the Royal Statistical Society Series B-methodological*, 10, 243-251.

63. Mythili, G., (2008). Acreage and yield response for major crops in the pre-and post-reform periods in India: A dynamic panel data approach, Mumbai: Indira Gandhi Institute of Development Research (Report prepared for IGIDR-ERS/USDA Project: Agricultural Markets and Policy).
64. National Sample Survey Office (NSSO) (2005). Situation Assessment Survey of Farmers Some Aspects of Farming, Report No. 496(59/33/3), NSS 59th Round, Ministry of Statistics and Programme Implementation, Government of India.
65. National Sample Survey Office (NSSO) (2014) Key Indicators of Land and Livestock Holdings in India. Government of India. 70th Round
66. National Sample Survey Office (NSSO) (2014). Key Indicators of Land and Livestock Holdings in India. NSS KI (70/18.1), NSS 70th Round, (January – December 2013), Ministry of Statistics and Programme Implementation, Government of India.
67. National Wetland Atlas: Bihar (2010). SAC/RESA/AFEG/NWIA/ATLAS/23/2010, Space Applications Centre (ISRO), Ahmedabad, India
68. Open Government data Platform India (Website). <https://data.gov.in/>
69. Paelinck, J.H.P. and Klaassen, L.H. (1979). Spatial Econometrics. Saxon House Farnborough
70. Participatory Irrigation Management. Retrieved from <http://www.wrd.bih.nic.in>.
71. PTI (2014a). Kosi floods: Bihar orders forcible evacuation. The Hindu, August 3, 2014, <http://www.thehindu.com/news/national/other-states/kosi-water-level-rises-15-lakh-evacuated-in-bihar/article6276935.ece>
72. PTI (2014b). Ganga Reaches Danger Level Mark in Many Places in Bihar India. Business Standard, August 22. http://www.business-standard.com/article/current-affairs/ganga-reaches-danger-level-mark-in-bihar-114082201351_1.html
73. PTI (2016c). Revised MGNREGA wages put States in a quandary. Indian Express, 13 April. <http://indianexpress.com/article/india/india-news-india/mgnrega-wages-states-in-a-quandary-over-new/>
74. Quah, D.T. (1996a). Empirics for Economic Growth and Convergence. European Economic Review, 40, 1353-1375.
75. Quah, D.T. (1996b). Aggregate and regional disaggregate fluctuations. Empirical Economics, 21, 137-162.
76. Rasul, Golam (2014). Food, water and energy security in south asia: a nexus perspective from Hindu Kush Himalayan region. Environmental Science & Policy, 39, 35-48.
77. Rasul, Golam (2014). Food, water and energy security in south asia: a nexus perspective from Hindu Kush Himalayan region. Environmental Science & Policy, 39, 35-48.
78. Rawls, J. (1971) A Theory of Justice Cambridge: Harvard University Press.
79. Sinha, R., Bapalu, G. V., Singh, L. K., & Rath, B. (2008). Flood risk analysis in the Kosi river basin, north Bihar using multi-parametric approach of analytical hierarchy process (AHP). *Journal of the Indian Society of Remote Sensing*, 36(4), 335-349.
80. Somanathan E and Rohini Somanathan (2009). Climate Change: Challenges Facing India's Poor. Economic and Political Weekly, 44(31): 51-58, August 1.

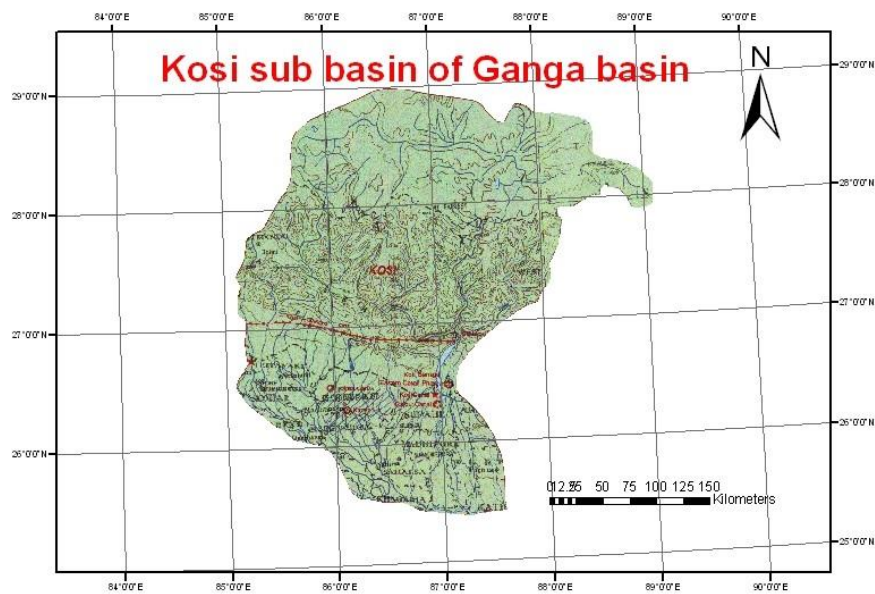
81. Somanathan, E. (2013). Are embankments a good flood control strategy? A case study on Kosi River. *Water Policy*. Vol. 15. pp. 75 – 88.
82. Surface Water Flow Irrigation Schemes. Retrieved from <http://www.mwrd.bih.nic.in>.
83. Surface Water Lift Irrigation Scheme. Retrieved from <http://www.mwrd.bih.nic.in>.
84. Thorat, S. and Lee, J. (2005) 'Caste discrimination and food security programmes', *Economic and Political Weekly*, 40(39), pp. 24–30.
85. Thorner, D., 1956. *The Agrarian Prospect in India*. New Delhi: University Press.
86. Uphoff, N. (1999). Agroecological implications of the System of Rice Intensification (SRI) in Madagascar," *Environment, Development and Sustainability*, 1: 3-4.
87. Uphoff, N. (2005). „The development of the System of Rice Intensification," in J. Gonsalves et al. (eds.), *Participatory Research and Development for Sustainable Agriculture and Rural Development*, Vol. 3, 119-125, International Potato Center-UPWARD and International Development Research Centre, Ottawa.
88. Uphoff, N. and Randriamiharisoa, R. (2002). „Reducing water use in irrigated rice production with the Madagascar System of Rice Intensification," in B. Bouman et al. (eds.), *Water-Wise Rice Production*, IRRI, Los Baños.
89. Viswanathan, Brinda and J.V. Meenakshi (2008). *Changing Pattern of Undernutrition in India: A Comparative Analysis across Regions*. Book (Eds.) by Shabd S. Acharya, Benjamin Davis, and Basudeb Guha-Khasnobis "*Food Security: Indicators, Measurement, and the Impact of Trade and Openness*, UNU-WIDER Studies in Development Economics, Oxford University Press.
90. Ghosh and Guha-Khasnobis (2008). "Measuring the Efficacy of Targeted Public Programmes: Public Works Programmes in India" Book (Eds.) by Guha Khasnobis, Acharya and Davis *Food Security Indicators, Measurement, and the Impact of Trade Openness*, (ICSSR, UNU-WIDER and FAO), Oxford University Press, 2008.
91. Agrawal et. al., 2013
92. Wang, j., Zhang, Z. Su, B. and Zhang L. (2012). A case research on economic spatial distribution and differential of agriculture in China. *Agricultural science*, 3(8), 996-1006.
93. Wang, j., Zhang, Z. Su, B. and Zhang L. (2012). A case research on economic spatial distribution and differential of agriculture in China. *Agricultural science*, 3(8), 996-1006.
94. Wartenberg, D., 1985, *Multivariate spatial correlation: A method for exploratory geographical analysis*, *Geographical Analysis*, 17, 263-283.
95. Wikipedia (Website). <https://en.wikipedia.org/>
96. World Bank (2015). Can Indian cities take better advantage of urbanization to become more prosperous and livable?. November, Vol 14(3), 1-23.

Appendix 2

Appendix A2.1 Maps of River basin in Bihar



Map 2.1(a)



Map 2.1(b)



Map 2.1 (c)

Appendix 2.2. Tables

Table A2.2 (a): Monsoon Rainfall Scenario from Nepal

	Unit	Jun	Jul	Aug	Sep	Oct	Average
Normal Average	Mm	216.0	353.61	313.58	200.4	54.3	227.59
Average Normal Positive deviation	%	36.3	19.92	14.84	22.7	61.6	31.06
Average Normal Negative deviation	%	-28.2	-17.88	-12.86	-19.0	-42.9	-24.17
Sample period: 2000-2012)							
Average Positive deviation	%	33.2	1.6	14.10	29.0	38.6	-32.73
Average Negative deviation	%	-52.0	-24.2	-15.09	-24.7	-47.6	227.59
Maximum Month	%	15.4	46.1	30.77	7.7	0.0	-
Rainfall Actual (2008)	mm	382.8	354.6	317.11	122.0	31.0	241.50
Deviation of 2008	%	77.2	0.3	1.12	-39.1	-42.9	-0.69

Note: Normal is the long term period of 1901-2012. Source:

Table A2.2 (b): State-wise Number of Districts declared by States as Drought affected

State	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014
Andhra Pradesh	13	4	0	0	13	0	13	6	3	6
Assam	0	20	0	0	27	0	0	0	0	0
Bihar	0	0	0	0	26	38	0	0	33	0
Chhattisgarh	0	0	0	0	0	0	0	0	0	0
Gujarat	0	0	0	0	0	0	0	16	0	0
Haryana	0	0	0	0	0	0	0	0	0	21
Himachal Pradesh	12	0	0	0	12	0	0	0	0	0
Jammu And Kashmir	0	0	0	0	18	0	0	0	0	0
Jharkhand	22	0	0	0	24	24	0	0	0	0
Karnataka	0	23	0	21	20	0	24	26	22	9
Kerala	0	0	0	0	14	0	0	14	0	0
Madhya Pradesh	0	10	38	0	37	0	0	0	0	0
Maharashtra	0	0	0	0	28	0	15	17	0	26
Manipur	0	0	0	0	9	0	0	0	0	0
Nagaland	0	0	0	0	11	0	0	0	0	0
Odisha	0	0	0	0	17	16	0	0	0	0
Punjab	0	0	0	0	0	0	0	0	0	0
Rajasthan	22	22	0	12	26	0	0	12	29	0
Tamil Nadu	0	0	0	0	0	0	0	31	0	0
Uttar Pradesh	0	0	9	0	56	0	0	0	0	43
Uttarakhand	11	0	9	11	0	0	0	0	0	0
West Bengal	0	6	0	0	0	11	0	0	0	0
Total	80	85	56	44	338	89	52	122	87	105

Source: Farmer Portal (Website)

TableA2.2 (c): Wetland in Bihar

District Name	Area Hectare	District Share %	Area Share	Open Water	
				Postmonsoon Hectare	Premonsoon
Gaya	11422	2.83	2.31	3979	626
West Champaran	21697	5.38	5.11	11924	10118
East Champaran	12477	3.09	3.00	8915	5119
Rohtas	18641	4.62	4.86	9259	4040
Madhubani	8958	2.22	2.58	2411	2280
Aurangabad	8116	2.01	2.39	3428	1436
Purnia	12401	3.08	3.87	5279	3365
Patna	20678	5.13	6.61	11943	7570
Muzaffarpur	10490	2.60	3.36	6984	4048
Banka	9895	2.45	3.28	5151	3847
Katihar	31011	7.69	10.30	17135	14574
Jamui	7351	1.82	2.45	3593	2734
Supaul	19285	4.78	6.46	9004	9021
Araria	4157	1.03	1.49	2245	1930
Sitamarhi	2601	0.65	0.99	906	588
Saran	21170	5.25	8.07	12118	7950
Samastipur	15022	3.73	5.82	10867	7133
Darbhanga	8709	2.16	3.48	5171	2467
Bhagalpur	24171	5.99	9.66	16237	10273
Nawada	5464	1.36	2.19	2445	1241
Nalanda	1589	0.39	0.67	756	283
Bhojpur	11154	2.77	4.77	5700	2435
Siwan	7105	1.76	3.21	4295	2117
Gopalganj	7122	1.77	3.56	5128	3783
Vaishali	17148	4.25	8.60	11405	5970
Kishanganj	10954	2.72	5.65	5542	4886
Begusarai	20365	5.05	10.78	10628	7703
Kaimur	796	0.20	0.43	296	174
Madhepura	3539	0.88	1.97	1589	967
Buxar	3717	0.92	2.27	2449	1408
Jehanabad	4345	1.08	2.77	1843	564
Khagaria	11645	2.89	7.84	9060	5807
Munger	11979	2.97	8.44	7001	5498
Lakhisarai	4177	1.04	3.40	1759	1447
Saharsa	12086	3.00	10.11	7202	4125
Sheikhpura	296	0.07	0.43	163	73
Sheohar	1476	0.37	3.33	845	782
Total	403209	100.00	4.40	224655	148382

Source: National Wetland Atlas: Bihar, 2010

A 2.3 (A). Rivers of Bihar

Bihar has a maze of rivers and channels of complex variety. The state is a major basin of River Ganga that arises from Northern Indian Himalayas and flows past states Uttarakhand and Uttar Pradesh before entering Bihar and then going into West Bengal. The various rivers either confluence with the

Ganga or one or more its tributaries or their ditributaries. The ributaries join Ganga from both its left bank (Kosi, Ghagra, Gandaki) and from the right (Son). Some of these rivers have short courses originating within Bihar or in Jharkhand (Ajay, Durgavati, Karmanasa, Punpun, Burhi Gandak, Phalgu and Mohana). Other are longer arising in Madhya Pradesh (Son) or West Bengal (Mahananda). Major rivers are trans-boundary arising in Nepal (Baghmati, Kamala, Kankai and Lekendei) and others further higher in Tibet (Gandaki, Ghagra and Kosi). Some rivers are of glacial origin while others are created by rainfall in hills. The rivers tend to join, split and rejoin and create channels.

Ajay: Rises near Munger in Bihar, passes through Jharkhand and West Bengal before joining Bhagirathi (Ganga).

Bagmati: Begins from Nepal, separating Patan from main Kathmandu. It enters Bihar and braids into branches with a main branch joining Burhi Gandak.

Burhi Gandak: Arises in West Champaran Bihar –spills into many channels in zigzag courses to drain into Ganga in Khagaria in Bihar

Durgavati: Originated in Kaimur district in Bihar, creates water falls and is a tributary of Karmanasa. It is known for the dam built on it.

Gandaki: It arises in Tibet and is a major river of Nepal running through gorges and having considerable hydel potential. There are many tributaries and joins two other rivers at the entry point to India at West Champaran, Gopalganj and Muzzaffarpur in Bihar. The Gandaki joins the Ganga near Patna.

Kamala: It arises in Nepal and is known for the numerous confluences, floods and erosions caused in Nepal before entering India in Madhubani where a barrage is created. It joins Kareh of Bagmati in Khagaria and together they

flow into the Kosi while another branch of Kamala leads to Bagmati in the lower course as Kamala-balan

Kankai: Arising in Nepal, it enters Bihar to join Mahananda in Kishanganj in Bihar.

Karmanasa: This short river arises Kaimur in Bihar flows through Uttar Pradesh and Bihar as a tributary of Ganga.

Lakhendei: This tributary of Bagmati arises in Nepal and enters Bihar to form a basin in Sitamarhi and ends in Muzzaffarpur.

Mahanada: This Himalayan River has a complex course. Arising east of Darjeeling in West Bengal it enters Bangladesh but returns to West Bengal in West Dinajpur in India flows into Kishanganj and Katihar in Bihar then again reverses to Malda in West Bengal and to Bangladesh where it joins the Ganga.

Mohana: A river in Jharkhand and Bihar, it joins Phalgu.

Phalgu: It is formed by the confluence of 3 rivers in Bihar's Gaya region and falls into Punpun.

Punpun: A river flowing from Palamau Jharkhand and joins the Ganga in Bihar

Son: One of the largest rivers in India and the world this 784 Km long river rises in Amarkantak in Madhya Pradesh near Narmada, passes up Jharkhand and Bihar to join Ganga at Patna. It is a wide and shallow river with disconnected pools and changes its course. Dams and Barrage are built in Madhya Pradesh to control the river.

Ghagra: It arises from a Tibet glacier then crosses Nepal as river Karnali. It splits into two rivers but rejoin in India in Uttar Pradesh and enters Bihar to join the Ganga near Chhapra.

Ganga: The Ganga originates Uttarakhand with the confluence of headstreams and drawing from melting snows of higher Himalayas. In India the Ganga flows through Uttarakhand, Uttar Pradesh, Bihar and West Bengal. In Bihar the Ganga passes cities Patna, Bhagalpur, Buxar and Saidpur and branches in Pakur as Bhagirathi-Hooghli, confluencing further with Jalangi and Damodar which is a long 541 km river. It enters Bangladesh as the Padma to be joined by tributaries of Brahmaputra called Jamuna and Meghna. The waters of the trans-border basin is managed by the Farakka barrage which diverts some flows into feeder canals to Hooghli and keeps the channel silt-free. Damodar flows from Jharkhand and was also known as the sorrow of Bengal for its floods which are now tamed by dams.

Appendix 2.3 (B): Kosi Multipurpose project

The project was planned with three continuous interlinked stages. (1) A barrage to anchor the river that had migrated about 120 km (75 mi) westward in the last 250 years and to provide irrigation and power benefits to Nepal and India. (2) Embankments, both below and above the barrage, to hold the river within the defined channel. (3) A high multipurpose dam within Nepal at Barakshetra to provide a substantial flood cushion along with large irrigation and power benefits to both countries. While the first two parts of the plan were implemented by the Government of India, the Koshi High dam, has been on hold for various political reasons but has now been revived under a fresh agreement, in a modified form for further investigations and studies, A Kosi Agreement between Nepal and India was signed on 25 April 1954 and revised on 19 December 1966 to address Nepal's concerns. Additional schemes for providing benefits of irrigation were negotiated between the two countries.



Picture 1: Koshi Barrage

The Bhimnagar Barrage, built between 1959 and 1963 at the border was meant for irrigation, flood control and hydropower generation. Built under a bilateral agreement, its entire cost was borne by India though the catchment area is in Nepal at the barrage site. The Eastern Canal and the Western Canal taking off from the barrage were designed to irrigate land and a hydropower plant was built on the Eastern Canal. A bridge over the barrage opened up the East-West highway in Nepal.

In 1976 Nepal took over the project which was renovated with IDA assistance. Kosi embankment system.



Picture 2: Kosi embankment

Embankments on both sides downstream of the barrage with a length of 246 km constructed to check the westward movement of the river and to serve as a silt trap. Earth dams across river, afflux bunds and embankments above and below the river confines the river to flow within embankments.

A2.3(C): Sapta Kosi High Multipurpose Project (Indo-Nepal)

The governments of India and Nepal agreed to conduct joint investigations and other studies for the preparation of a detailed project report of Sapta Koshi High Dam Multipurpose Project and Sun Kosi Storage-cum-Diversion Scheme to meet the objectives of both countries for development of hydropower, irrigation, flood control and management and navigation.

Appendix 4

Table A4.1 : Sample Design with lists of Districts, Blocks and Villages selected				
Sample District	Blocks	Sample Blocks	Sample villages	Types of Sample households and numbers
Saharsa	Banma Itahri	xxxxxxxxxxxxxx	xxxxxxxxxxxxxx	
	Kahra	xxxxxxxxxxxxxx	xxxxxxxxxxxxxx	
	Mahishi	Mahishi (within river)	Dharampur & Rajhanpur	30 FH (15 FH from each village) and 4 ALH (2 ALH from village)
	Nauhatta	Nauhatta (within river)	Hati, & Kadil Patti	30 FH (15 FH from each village) and 4 ALH (2 ALH from village)
	Patarghat	Patarghat (outside river)	Dhabauli, & Kishunpur	30 FH (15 FH from each village) and 4 ALH (2 ALH from village)
	Salkhua	Salkhua (within river)	Koparia, & Bankatti	30 FH (15 FH from each village) and 4 ALH (2 ALH from village)
	Sattar Kataiya	xxxxxxxxxxxxxx	xxxxxxxxxxxxxx	
	Simri Bakhtiarapur	xxxxxxxxxxxxxx	xxxxxxxxxxxxxx	
	Sonbarsa	xxxxxxxxxxxxxx	xxxxxxxxxxxxxx	
	Sour Bazar	xxxxxxxxxxxxxx	xxxxxxxxxxxxxx	
Khagaria	Alauli	xxxxxxxxxxxxxx	xxxxxxxxxxxxxx	
	Khagaria	xxxxxxxxxxxxxx	xxxxxxxxxxxxxx	
	Chautham	Chautham (within river)	Borne, & Balkunda (671)	30 FH (15 FH from each village) and 4 ALH (2 ALH from village)
	Mansi	Mansi (within river)	Thamta, Dharhara, Chak Kamal, & Heyatpur	30 FH (15 FH from each village) and 4 ALH (2 ALH from village)
	Gogari	xxxxxxxxxxxxxx	xxxxxxxxxxxxxx	
	Beldaur	Beldaur (outside river)	Dighuan, & Dudraja	30 FH (15 FH from each village) and 4 ALH (2 ALH from village)
	Parbhatta			30 FH (15 FH from each village) and 4 ALH (2 ALH from village)

Table A 4.2: Sample villages in Khagaria and Saharsa

Khagaria			Saharsa	
Sl No	Villages Name	Sample No	Villages Name	Sample No
1	Beldaur	15	Dharampur	12
2	Pachaut	15	Rajhanpur	23
3	Borney Sthan	16	Hati	16
4	Nawada	13	Kadli patti	15
5	Balha	14	Bankatti	14
6	Saidpur	14	Hansari	14
7	Kanhayachak	18	Balhampur	17
8	Naya Gaon	15	Raipura	15
Total		120	126	

Appendix A.4.1 Khagaria

Bounded by Saharsa in the north, Munger in the south, Madhepura and Bhagalpur in the east and Begusarai and Samastipur in the west, Khagaria occupies an area of 1,486 square kilometres comprising of the sub-Divisions Khagaria and Gogari. There are seven community development blocks in the district namely, Alauli, Khagariya, Mansi, Chautham, Beldaur, Gogari, and Parbatta. According to the 2011 census, Khagaria district has a population of 1,657,599, ranking 300th out of a total of 640 districts in India. Its population density is high at 1,115 inhabitants per square kilometre and population growth rate over the decade 2001-2011 was 29.46%. The sex ratio is 883 females for every 1000 males compared to 918 of India and the literacy rate of 60.87% against India's 74%.

Before it became a district in the year 1981, Khagaria was a part of Munger district. As an administrative district with its headquarters located in Khagaria town, Khagaria still remains a part of Munger Division. Locally, the district is also known as Farakiya based on an anecdote that reflects its flood prone character. The Mughal emperor Akbar's revenue minister Todarmal while mapping the entire empire was said to have failed to map this area due to its difficult terrain, rivers

and dense forests and found it 'separate' (Farak in Hindi) from the rest. Indeed, the district is surrounded by seven rivers namely Ganges, Kamla Balan, Koshi, Budhi Gandak, Kareh, Kali Koshi and Bagmati that cause floods every year. The Koshi and its two tributaries (Bagmati and Kamala) are present and confluence with the Ganga in this district.

Khagaria has seen development through the passage of time. The small town of Khagaria has a major railway junction on the Barauni Guwahati section of the Indian Railways. The National Highway No.31, which connects rest of India to North Eastern region also passes through this town. It is also connected to other regions of north Bihar by another rail line to Saharsa and Samastipur. Building of a major rail bridge across river Ganges at Munger is underway to connect it to south Bihar and Jharkhand directly. On the economic front Khagaria's state remains poor. In 2006 the Ministry of Panchayati Raj named Khagaria one of the country's 250 most backward districts and it is one of the 36 districts of Bihar that are currently receiving funds from the Backward Regions Grant Fund Programme (BRGF). Out of total population, about 58 per cent people are only literate in the district. Only 33.64 per cent of total population are workers including both main (18.37 per cent) and marginal workers (15.25 per cent). The major chunk of workers (58.08 per cent) are agricultural labours. This district can be inferred as a highly rural district because only 5.23 per cent of total population reside in urban area of 12.46 sq. km (0.84 per cent of district's total geographical area).

Khagaria is the most downstream district of the Kosi basin where the river meet Ganga. It is surrounded by four rivers – the Ganga in South, Burhi Gandak in the North and The Kosi and its two tributaries (Kamla Balan and Bagmati) in North side of the district. Flood is almost an annual phenomenon though less associated with heavy rainfall in the district than the river flow and its water height. Different areas of the

district are affected by the surge of water in different rivers in Khagaria (Table A**). Though, none of Kosi embankments are extended to Khagariya districts, but many other embankments (Table A*) were constructed in the district to protect people from regular floods. Because of frequent floods, paddy is only grown in upland areas of the district and is considered third crop. Since enough water is available in the district, farmer has started growing banana. It is grown in about 1500 ha area in the district.

Table A4.3(a): Rivers that pass through Khagariya and Flood affected areas

Rivers	Effected Areas
Ganga	Khagaria (South), Gogari, Parbatta
Burhi Gandak	Khagaria, Gogari, Mansi
Bagmati	Alauli, Mansi, Chautham
Koshi	Chautham, Gogari, Beldaur
Kareh	Alauli, Chautham
Kamla	Alauli Chautham

Table A4.3(b): Embankments in Khagariya district

Sl No.	Embankment	Length (Km)	Sensitive Points and Nearer Village
1	Burhi Gandak Left Embankment	19	Jalkoura, Chainpura, Olapur
2	Burhi Gandak Right Embankment	20	Terasi, Hardiya, Chakkai
3	Khagaria Town Protection Embankment Left	3	Aghori Stahan Khagaria
4	Karachi Badlaghat Embankment Right	47.5	Saidpur, Amni, Chatar, Rasouk, Amba-Alauli
5	Badlaghat-Nagarpara Embankment Right	40	Malpa, Lalpur, Choutham, Poura, Basua, Painkat, Paharpur, Dewaka
6	Nagarpara Embankment Right	16	Bhawanipur, Jayrampur
7	Gogari Narayanpur Link Bandh Left	51	Narayanpur

Appendix A4.2: Saharsa

Saharsa is surrounded by the districts of Supaul and Madhubani in north, Khagaria in south, Madhepura in east and Darbhanga in the west. The Koshi River flows through the district and this district is situated within east and west embankments of the River.

Saharsa, which was constituted as a district in year 1954, is one of the three districts of the Koshi division which comprises of three districts namely Saharsa, Madhepura and Supaul. Earlier, the other two districts Madhepura and Supaul districts were parts of Saharsa district but both districts were separated out in year 1981 and 1991 respectively. Saharsa's total geographical area and population are 1687 sq. km. and 1900661, respectively. Though Saharsa district also faces a severe challenge of dealing with frequent floods just as Khagaria does, it is slightly better in social indicators of development such as the sex-ratio (906). Population density (1127 persons per sq. km.) of the district is also higher than Khagariya.

The district of Saharsa evolved with administrative restructuring. It was part of Munger and Bhagalpur districts successively at one time to be consolidated as an independent district only in 1954. The administrative district of Saharsa was under Bhagalpur division but was later replaced under the Kosi division in 1972. Subsequently, it was broken up to form two additional districts Supaul and Madhepura in 1981 and 1991 respectively. The region is associated closely with India's long history including the Mauryas, Guptas, Buddha. Akbar's administration and the freedom movement from British rule. Saharsa district now consists of 2 subdivisions, viz. Saharsa Sadar and Simri Bakhtiarapur. Saharsa subdivision located at northern part of the district has seven blocks (Kahra, Sattar Kataiya, Nauhatta, Mahishi, Saur Bazar, Sonbarsha, and Patarghat) whereas there are three blocks in Simri Bakhtiarapur (Simri Bakhtiarapur, Salkhua, and Banma

Itahari) located in southern part of the district which is close to Khagariya district. Saharsa town is the administrative headquarters of this district. The district consists of 10 development blocks and anchals each. The area of the district is 1,696 km².

Saharsa district, located upstream of Khagaria, is one of the most affected by floods in the Kosi River and heavy rainfall in upstream Nepal. The main stream of the river flows through Nauhatta, Mahishi and Salkhua blocks of the district. However, the river affects other blocks of the district too like Saur Bazar, Patar Bhat, Sonbarsa, Simri Bakhtiyarpur, and Banma Itarhi through its minor streams. Both east and west bank embankment built up under Kosi Project are situated in the Saharsa district. Even, some villages of the district located largely in Nauhatta and Mahishi blocks fall within the two embankment.

Agriculture is the prime source of occupation in Saharasa district. Rice, wheat, maize and vegetables are grown in this district. Water logging and siltation are major problems that farmers face in the district. Land right is also a very critical issue here as the rights on flooded or water logged area goes to the government as per government rule. A large number of working population are found migrated to Delhi, Punjab, Gujarat and many more developed regions for the livelihood. Only non-working population for example women and old-aged men live in some villages of the district.

Appendix 5

Map A5.1: Agro-climatic Zones in Bihar

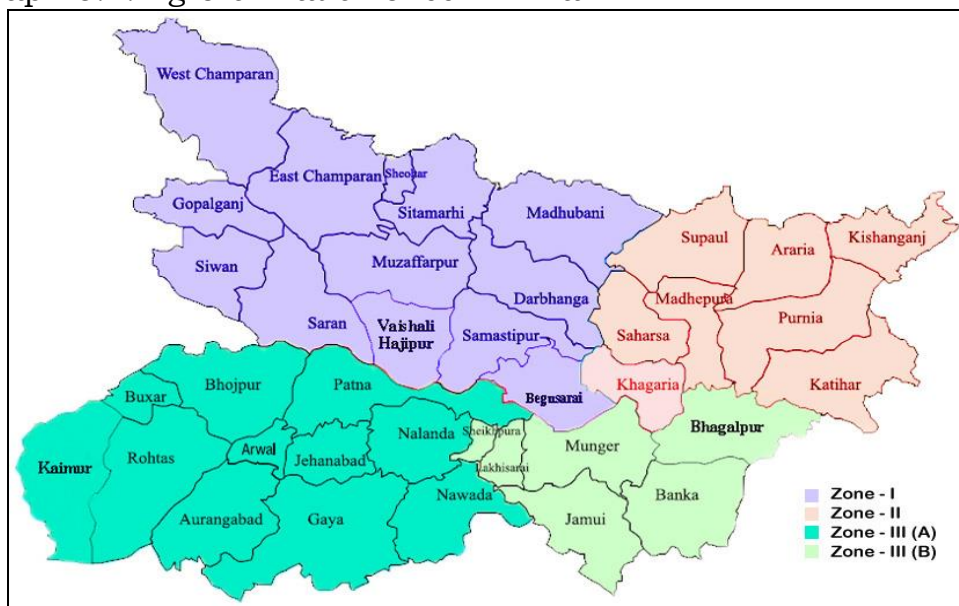


Table A5.1: Physiographic features of the Agro-climatic Zones in Bihar

S. No.	Agro-climatic zone	Soil	pH	Total Rainfall (mm)	Temperature (°C)	
					Max.	Min.
1	Agro- climatic zone I (Northern West)	Sandy loam, loam	6.5-8.4	1040-1450 (1245.00)	36.6	7.7
2	Agro- climatic zone II (Northern East)	Sandy loam, Clay loam	6.5-7.8	1200-1700 (1450.00)	33.8	8.8
3	Agro- climatic zone IIIA & IIIB (Southern East & West)	Sandy loam, Clay loam, Loam, Clay	6.8-8.0	990-1240 (1115.00)	37.1	7.8

Source: BAMETI (Website)

Appendix A5.2 (a): East Koshi Canal and Its Branches

Canal System	Length (Km)
1. East Koshi Canal (Supaul, Madhepura, Purnia, and Araria)	41.3
1.1 Shivanagar Distributary (Supaul)	4.88
1.2 Phulkaha Distributary (Supaul)	14.32
1.3 Rajpur Canal (Supaul)	9.85
1.3.1 Supaul Sab-branch (Supaul and Saharsa)	74.71
1.3.2 Saharsa Sab-branch (Supaul and Saharsa)	90.24
1.3.3 Madhepura Sub-branch (Saharsa and Madhepura)	80.25
1.3.4 Gamhariya Sub-branch (Saharsa and Madhepura)	81.7
1.4 Murliganj branch (Supaul)	59.2
1.5 Pariyahi Distributary (Supaul)	15.85
1.6 Ranipatti Distributary (Supaul)	47
1.7 Lalpur Distributary (Supaul)	17.07
1.8 Jankinagar Branch (Madhepura, Purnia and Araria)	77.43
1.8.1 Murliganj Distributary (Supaul, Madhepura and Araria)	58.35
1.8.2 Parihari Distributary (Araria)	40.24
1.8.3 Banmankhi Distributary (Purnia)	28.2
1.8.4 Dhamdaha Distributary (Purnia)	52.74
1.8.5 Sukhasan Distributary (Madhepura)	13.1
1.8.6 Chausa Distributary (Purnia)	34.39
1.8.7 Kisunganj Distributary (Purnia)	28.65
1.8.8 Aalamnagar Sub-distributary (Purnia)	26.52
1.8.9 Narpatganj Distributary (Araria)	9.15
1.8.10 Kharsahi Distributary (Araria)	46.34
1.8.11 Simarbani Distributary (Araria)	22.99
1.8.12 Bhangraha Distributary (Araria)	21.77
1.9 Purnia branch	60.73
1.9.1 Kasba Distributary (Purnia and Araria)	35.82
1.9.2 Shrinagar Distributary (Purnia)	48.78
1.9.3 Kursela Distributary (Punia)	56.62
1.9.4 Karagola Distributary (Purnia and Karihar)	36.89
1.10 Araria branch	54.32
1.10.1 Mahisasur Distributary (Araria)	19.96
1.10.2 Matiyari Rahikakur Distributary (Araria)	20.18
1.10.3 Phulpur Sub-distributary (Punia and Karihar)	16.77
1.10.4 Katihar Distributary (Purnia and Katihar)	68.37
1.10.4.1 Hasanganj Sub-distributary (Purnia and Katihar)	31.1

Note: Parentheses indicate districts irrigated by canal and its branches

Appendix A5.2 (b): West Koshi Canal and Its Branches

Canal System	Length (in km)
1. West Koshi Canal (Nepal and Madhubani and Darbhanga in Bihar, India)	35.10 in Nepal and 56.70 in Bihar, India
1.1 Rajpur Distributary (Madhubani)	12.8
1.2 Mahadeva Distributary (Madhubani)	8.4
1.3 Ghoghardiha Distributary (Madhubani)	19.09
1.4 Chikana Distributary	17.68
1.5 Jhanjharpur branch (Madhubani)	41.77
1.6 Ugranath branch (Madhubani)	36.57
1.6.1 Videshwarsthan Canal (Madhubani)	27.44
1.7 Sakari branch (Madhubani and Darbhanga)	44.33
1.8 Kaakarghati branch (Madhubani and Darbhanga)	38.11
1.9 Sahaarghat branch (Madhubani and Darbhanga)	20.18

Note: Parentheses indicate districts irrigated by canal and its branches

Table A5.3: Per Capita Gross Domestic Products (PCGDDP) and Annual Growth Rate (AGR)

Districts	PCGDDP (Rs./Year)	AGR (%)
<i>High</i>		
Munger	19193	10
Begusarai	15781	10
Bhagalpur	14885	11
Rohtas	11932	8
Muzaffarpur	11831	7
Lakhisarai	11808	11
<i>Medium</i>		
Jamui	10912	13
Bhojpur	10846	9
Gaya	10673	9
Buxar	10610	9
Nalanda	10183	8
Katihar	10157	9
Saharsa	10043	7
<i>Semi-Medium</i>		
Kaimur	9696	8
Aurangabad	9526	8
Khagaria	9478	8
Vaishali	9346	8
West chamoaran	9228	7
Jehanabad	9221	8
Darbhanga	9211	9
Gopalganj	9174	8
Saran	9148	9
Purnia	9112	8
Samstipur	9087	8
Siwan	9010	9
<i>Low</i>		
Kishanganj	8977	8
Sheikhpura	8943	8
Banka	8524	9
Madhepura	8295	8
Nawada	8178	8
Araria	8175	9
Arwal	7976	8
Supaul	7911	7
East champaran	7540	7
Madhubani	7539	11
Sitamarhi	7269	7
Sheohar	6777	10
<i>Average</i>		
High	14238	19
Medium	10489	24
Semi-Medium	9270	12
Low	8009	16
Total	10501	18

Source: Government of Bihar (GoB) (2011).

Table A5.4: District wise share (%) of Bihar in 2011

Districts	POP	GDDP*	GDDPA\$	TCA	RPOP	URBN	POPDEN
<i>KBR</i>							
Araria	2.7	1.9	2.8	3.9	0.7	6	992
Begusarai	2.8	3.7	2.5	2.3	2.6	19.2	1540
Bhagalpur	2.9	3.6	2.6	2.1	2.6	19.8	1180
Katihar	3	2.4	3.4	3.4	3	8.9	1004
Khagaria	1.6	2	2	1.8	1.7	5.3	1115
Madhupura	1.9	1.9	2.5	2.6	4.7	4.4	1116
Purnea	3.2	2.2	3.4	3.4	3.2	10.4	1014
Saharsa	1.8	2.4	2.1	2.6	1.9	8.2	1125
Sitamarhi	3.3	1.9	2.4	2.8	3.5	5.6	1465
Supaul	2.1	1.9	2.9	2.9	2.3	4.7	919
Darbhanga	3.8	2.3	2.7	2.8	3.8	9.7	1721
Muzafarpur	4.6	3	4.4	4.3	4.7	9.8	1506
Samastipur	4.1	2.3	4.6	3.8	4.5	3.5	1882
Sheohar	0.6	1.5	0.5	0.6	0.7	4.3	1491
E.Champaran	4.9	2.1	5.4	4.9	5.1	7.9	1281
Madhubani	4.3	1.9	5.4	4.5	2.1	3.7	1279
<i>NKBR</i>							
Araval	0.7	1.9	0.5	0.7	2.9	7.4	1099
Aurangabad	2.4	2.1	2.7	3.4	2.5	9.4	760
Banka	2	2	2.5	2.1	2.1	3.5	672
Bhabhua	1.6	2.2	2.2	2.6	1.7	4	488
Bhojpur	2.6	2.6	2.6	3.2	2.5	14.3	1136
Buxar	1.6	2.3	1.9	2.3	1.7	9.6	1003
W.Champaran	3.8	2.5	5.5	5	3.8	10	750
Gaya	4.2	2.4	3.3	2.1	4.1	13.1	880
Gopalganj	2.5	2.2	2.6	3.1	2.6	6.3	1258
Jahanabad	1.1	2.2	0.9	1.1	1.1	12	1206
Jamui	1.7	2.6	1.1	0.7	1.7	8.2	567
Kishanganj	1.6	2.2	2.0	2.1	1.7	9.7	898
Lakhisarai	1	2.7	0.8	1.1	0.9	14.3	815
Monghyr	1.3	4.5	0.9	0.8	1.1	28.3	958
Nalanda	2.8	2.4	2.3	2.7	2.6	15.9	1220
Nawadha	2.1	1.9	1.7	1.8	2.2	9.7	889
Patna	5.6	12.8	3.2	2.7	3.6	43.5	1803
Rohtas	2.8	2.9	4.0	4.2	2.7	14.4	763
Saran	3.8	2.2	2.6	2.9	3.9	8.9	1493
Sheikhpura	0.6	2.1	0.5	0.7	0.6	17.1	922
Siwan	3.2	2.1	3.5	3.3	3.4	5.5	1495
Vaishali	3.4	2.5	3.4	2.5	3.5	6.7	1717

Note: * denotes 2009TE figure and \$ denotes 2007TE figure. Source Economic Survey of Bihar, 2014-15.

Table A5.5: Land Use in 2010TE

	GA	Share	NAGR	FOR	BUCL	PPGL	MISC	FL	NSA	CI
Bihar	Million Hectare	State Total (%)	Share in reported Area (%)							Ratio
KBR										
Araria	0.27	2.6	18.9	0.3	1.8	0.1	7.0	3.5	68.4	1.5
Begusarai	0.19	1.8	22.1	0.0	9.6	0.0	1.9	4.9	61.5	1.5
Bhagalpur	0.25	2.5	27.5	0.0	8.8	0.2	2.6	9.2	51.6	1.2
Katihar	0.29	2.8	19.7	0.6	7.6	0.0	3.8	10.0	58.2	1.5
Khagaria	0.15	1.4	20.7	0.0	9.1	0.1	2.0	5.6	62.3	1.4
Madhupura	0.18	1.7	17.4	0.0	2.2	0.0	4.0	8.6	67.9	1.6
Purnea	0.31	3.0	14.6	0.0	3.9	0.0	2.8	17.0	61.7	1.3
Saharsa	0.16	1.6	17.5	0.0	6.6	0.7	2.6	6.5	66.0	1.8
Samastipur	0.26	2.5	24.2	0.0	1.5	0.0	3.1	4.0	67.3	1.6
Supaul	0.24	2.3	21.6	0.0	8.5	0.1	1.3	11.2	57.3	1.6
Darbhanga	0.25	2.5	23.8	0.0	0.5	0.1	4.8	5.3	65.6	1.2
Muzafarpur	0.32	3.1	20.0	0.0	1.7	0.0	5.5	4.7	68.1	1.5
Sheohar	0.04	0.4	29.4	0.0	0.9	0.0	8.3	8.0	53.4	1.9
Sitamarhi	0.22	2.2	28.4	0.0	0.8	0.6	6.3	2.6	61.4	1.5
E. Champaran	0.43	4.2	17.8	0.0	1.9	0.1	6.3	6.7	67.2	1.2
Madhubani	0.35	3.4	24.5	0.0	0.6	0.4	6.5	2.9	65.0	1.5
NKBR										
Araval	0.06	0.6	16.3	0.0	3.5	0.2	1.4	16.7	61.8	1.3
Aurangabad	0.33	3.2	16.7	5.7	5.0	0.2	0.2	10.8	61.5	1.3
Banka	0.31	3.0	13.8	15.2	14.1	0.4	2.4	15.4	38.7	1.3
Bhabhua	0.34	3.3	10.1	33.0	5.6	0.0	0.2	5.9	45.2	1.3
Bhojpur	0.24	2.3	14.3	0.0	2.8	0.0	0.8	3.5	78.5	1.3
Buxar	0.17	1.6	10.4	0.0	1.3	0.0	0.4	7.8	80.0	1.3
W.Champaran	0.48	4.7	19.5	18.9	0.6	0.2	1.3	3.7	55.7	1.4
Gaya	0.49	4.8	14.8	15.8	5.6	0.4	0.8	36.6	26.0	1.2
Gopalganj	0.20	2.0	16.3	0.0	2.7	0.1	3.6	4.6	72.7	1.6
Jahanabad	0.09	0.9	15.8	0.7	3.5	0.1	0.7	17.0	62.2	1.4
Jamui	0.31	3.0	14.0	30.4	9.4	0.5	0.7	30.9	14.1	1.2
Kishanganj	0.19	1.8	18.4	0.2	5.9	0.2	2.7	11.0	61.5	1.3
Lakhisarai	0.13	1.2	11.2	10.5	5.5	0.0	0.2	28.2	44.3	1.5
Monghyr	0.14	1.4	22.5	20.4	8.2	0.1	0.4	13.9	34.5	1.2
Nalanda	0.23	2.3	19.0	2.0	0.5	0.0	0.5	8.2	69.8	1.2
Nawadha	0.25	2.4	14.4	25.6	4.5	0.4	0.2	13.3	41.7	1.3
Patna	0.32	3.1	24.4	0.0	3.9	0.0	0.3	19.1	52.2	1.2
Rohtas	0.39	3.8	12.2	17.1	4.3	0.0	0.7	2.6	63.0	1.3
Saran	0.26	2.6	12.9	0.0	6.8	0.1	3.2	11.6	65.5	1.3
Sheikhpura	0.06	0.6	16.9	0.0	1.6	0.0	0.5	19.0	62.0	1.3
Siwan	0.22	2.2	14.3	0.0	3.9	0.1	3.8	2.7	75.1	1.4
Vaishali	0.20	2.0	18.6	0.0	12.0	0.2	4.8	2.6	61.9	1.5

Source: MoA (Website)

Table A5.6: Districtwise share (%) of Bihar in 2011

Districts	LITR	FLITR	SXR	CSXR	CBR	CDR	IMR	UMR
<i>KBR</i>								
Araria	45.2	64.2	921	954	30.9	7.8	61	87
Begusarai	57.1	74.4	894	911	26.6	6.5	46	65
Bhagalpur	56.5	72.3	879	934	26.3	5.5	54	69
Katihar	45.4	61.0	916	956	28.8	6.5	59	85
Khagaria	52.2	68.5	883	912	30.7	9.4	66	103
Madhupura	42.8	63.8	914	923	30.1	7.5	71	101
Purnea	43.2	61.1	930	953	27.6	7.2	62	102
Saharsa	42.7	65.2	906	928	32.1	7.8	62	91
Sitamarhi	43.4	62.6	899	932	28.0	9.6	67	106
Supaul	46.6	71.7	925	942	28.5	6.5	64	89
Darbhanga	46.9	68.6	910	928	26.5	8.8	51	85
Muzafarpur	56.8	73.6	898	917	25.5	8.6	60	89
Samastipur	53.5	73.1	909	941	28.7	6.9	54	77
Sheohar	47.3	63.7	890	925	31.2	7.6	50	87
E.Champaran	47.4	68.0	901	923	30.4	8.1	57	75
Madhubani	48.3	72.5	925	931	24.7	7.4	54	73
<i>NKBR</i>								
Araval	56.9	81.3	927	941	NA	NA	NA	NA
Aurangabad	62.1	82.5	916	945	25.6	6.4	48	61
Banka	49.4	69.8	907	939	25.8	6.1	48	63
Bhabhua	59.6	81.5	919	939	25.0	6.0	56	73
Bhojpur	60.2	84.1	900	915	24.6	5.8	48	61
Buxar	59.8	82.8	922	925	25.1	6.8	55	74
W.Champaran	46.8	68.2	906	950	28.6	8.9	57	81
Gaya	55.9	76.0	932	959	24.9	7.4	55	70
Gopalganj	56.0	78.4	1015	945	25.7	6.5	51	66
Jahanabad	56.2	79.3	918	918	24.9	6.2	53	67
Jamui	49.4	73.8	921	956	26.4	6.7	NA	78
Kishanganj	48.0	65.6	946	966	30.5	6.5	61	90
Lakhisarai	54.9	74.0	900	915	24.0	6.5	53	70
Monghyr	65.5	80.1	879	925	25.0	6.7	51	63
Nalanda	54.8	77.1	921	929	26.3	8.0	52	80
Nawadha	51.1	71.4	936	985	25.0	5.9	49	61
Patna	63.7	80.3	892	899	21.8	5.2	39	53
Rohtas	65.0	85.3	914	925	25.9	7.0	52	65
Saran	56.9	79.7	949	922	24.5	7.7	52	70
Sheikhpura	54.9	76.1	926	940	28.9	8.1	58	76
Siwan	60.4	82.8	984	934	26.1	7.5	49	73
Vaishali	59.1	77.0	892	894	27.3	7.6	50	70

Source: Economic Survey of Bihar, 2014-15.

Table A5.7: Percentage of Total Irrigated Area In Vulnerable Districts Of Bihar (% Share)

Districts	2000	2010
<i>KBR</i>		
Araria	38.8	46.3
Begusarai	39.5	58.3
Bhagalpur	42.9	42.5
Katihar	48.0	57.0
Khagaria	62.4	61.7
Madhepura	63.6	69.9
Purnea	54.0	61.0
Saharsa	50.8	53.2
Samastipur	43.9	57.6
Supaul	48.5	60.4
Darbhanga	42.1	57.6
Muzafarpur	41.2	49.9
Sheohar	27.5	35.2
Sitamarhi	37.1	37.1
East Champaran	50.5	49.9
Madhubani	40.7	48.1
<i>NKBR</i>		
Araval	82.6	85.7
Aurangabad	77.2	81.1
Banka	69.7	76.2
Kaimur	82.0	86.5
Bhojpur	81.6	82.6
Buxar	78.9	87.7
West Champaran	51.2	44.1
Gaya	82.4	82.2
Gopalganj	45.5	47.8
Jahanabad	80.9	78.5
Jamui	67.5	33.8
Kishanganj	23.2	28.6
Lakhisarai	56.6	73.3
Monghyr	76.4	72.1
Nalanda	84.8	84.4
Nawadha	84.2	87.5
Patna	68.4	70.7
Rohtas	91.3	94.2
Saran	49.3	54.7
Sheikhpura	67.8	74.2
Siwan	49.6	51.0
Vaishali	43.1	53.6

Source: MoA (Website)

Table A5.8: Sectorwise Gross District Domestic Product (GDDP) at Constant (1999-00) Prices

Districts	GDDPA		GDDPFL		GDDPF		GDDPMQ	
	2000	2006	2000	2006	2000	2006	2000	2006
<i>KBR</i>								
Araria	93.8	92.6	4.7	5.2	1.5	2.2	0.0	0.0
Begusarai	87.0	88.7	6.2	6.1	6.9	5.3	0.0	0.0
Bhagalpur	88.7	88.2	6.1	5.7	5.0	6.0	0.3	0.1
Katihar	91.3	89.8	3.9	4.5	4.8	5.8	0.0	0.0
Khagaria	85.9	86.9	5.0	4.1	9.1	9.0	0.0	0.0
Madhupura	91.8	89.6	3.6	4.0	4.6	6.4	0.0	0.0
Purnea	92.9	89.5	4.1	4.8	3.0	5.7	0.1	0.0
Saharsa	90.2	85.4	3.9	4.4	5.9	10.2	0.0	0.0
Samastipur	91.5	91.9	4.3	5.0	4.2	3.1	0.0	0.0
Supaul	93.9	93.1	3.9	4.0	2.3	2.9	0.0	0.0
Darbhanga	89.3	83.9	5.4	7.4	5.4	8.7	0.0	0.0
Muzafarpur	91.8	89.2	5.0	5.4	3.2	5.3	0.0	0.0
Sheohar	91.5	88.9	5.1	6.2	3.5	4.9	0.0	0.0
Sitamarhi	89.8	86.8	5.8	7.1	4.4	6.2	0.0	0.0
E.Champaran	91.9	90.7	4.4	4.8	3.7	4.5	0.0	0.0
Madhubani	92.7	90.5	3.8	4.4	3.5	5.1	0.0	0.0
<i>NKBR</i>								
Araval	86.5	90.2	6.1	6.8	7.4	3.1	0.0	0.0
Aurangabad	88.7	89.8	4.5	4.9	4.0	3.8	2.8	1.5
Banka	91.0	92.0	5.8	4.5	3.1	3.5	0.1	0.1
Bhabhua	94.1	94.1	4.1	4.3	1.4	1.6	0.4	0.1
Bhojpur	80.6	87.7	3.9	5.4	4.0	5.7	11.4	1.3
Buxar	92.1	90.1	4.0	4.7	3.8	5.0	0.1	0.2
W.Champaran	93.0	91.9	3.7	3.7	3.2	4.4	0.2	0.0
Gaya	88.9	90.7	6.9	6.8	3.5	2.1	0.7	0.4
Gopalganj	90.5	88.9	5.1	5.4	4.4	5.7	0.0	0.0
Jahanabad	87.4	90.4	5.6	6.6	6.5	2.7	0.5	0.3
Jamui	84.2	87.2	8.9	8.5	6.0	4.0	0.9	0.3
Kishanganj	93.4	90.9	3.7	4.2	2.9	4.9	0.0	0.0
Lakhisarai	85.7	88.4	6.8	6.0	6.1	4.8	1.5	0.9
Monghyr	81.6	83.9	8.0	7.1	9.9	8.0	0.4	1.0
Nalanda	88.9	89.3	5.1	6.3	5.5	4.4	0.5	0.1
Nawadha	85.6	89.7	6.6	7.0	7.4	3.1	0.4	0.2
Patna	85.8	86.0	6.2	7.9	3.9	4.9	4.1	1.2
Rohtas	91.0	90.8	4.1	4.0	2.2	2.1	2.7	3.2
Saran	87.0	82.6	7.1	7.4	5.8	10.0	0.0	0.0
Sheikhpura	89.6	89.9	5.8	6.0	4.3	3.8	0.3	0.3
Siwan	88.1	88.6	5.9	5.0	6.0	6.4	0.0	0.0
Vaishali	90.1	89.5	5.6	5.2	4.3	5.3	0.0	0.0

Source: Government of Bihar (2010)

Table A5.9 : Total Agriculture, Forestry and fishing Share in total Gross district Domestic Product At
Constant Price (%)

Bihar	2001TE	2006TE
<i>KBR</i>		
Araria	53.3	50.9
Begusarai	23.7	22.7
Bhagalpur	25.8	24.1
Katihar	46.7	43.9
Khagaria	44.8	47.4
Madhupura	64.0	61.3
Purnea	49.0	45.2
Saharsa	48.6	43.8
Samastipur	46.9	41.8
Supaul	60.3	58.9
Darbhangha	38.7	29.5
Muzafarpur	33.7	30.0
Sheohar	59.4	53.2
Sitamarhi	44.6	37.5
East Champaran	52.7	49.5
Madhubani	55.3	50.6
<i>NKBR</i>		
Araval	42.8	36.7
Aurangabad	48.4	45.3
Banka	51.1	54.6
Bhojpur	39.2	35.6
Buxar	50.5	44.4
Gaya	30.4	28.3
Gopalganj	45.1	42.5
Jahanabad	40.1	34.7
Jamui	35.0	31.1
Kaimur	62.1	54.4
Kishanganj	54.6	51.9
Lakhisarai	31.8	31.8
Monghyr	15.8	15.8
Nalanda	38.9	32.7
Nawadha	42.3	36.4
Patna	6.1	4.1
Rohtas	43.6	40.8
Saran	33.2	30.9
Sheikhpura	39.3	37.4
Siwan	42.2	43.3
Vaishali	39.8	38.9
West Champaran	55.0	51.5

Source: Government of Bihar (2010)

Table A5.10: Total Agriculture, Forestry and fishing % Share of Districts in Total State
Agriculture, Forestry and fishing (% Share)

Bihar	2001TE	2006TE
<i>KBR</i>		
Araria	2.6	2.7
Begusarai	2.3	2.5
Bhagalpur	2.2	2.6
Katihar	3.4	3.4
Khagaria	1.5	2.0
Madhupura	2.5	2.5
Purnea	3.5	3.4
Saharsa	2.3	2.2
Samastipur	4.8	4.4
Supaul	2.6	2.8
Darbhanga	3.7	2.9
Muzafarpur	4.4	4.4
Sheohar	0.6	0.5
Sitamarhi	2.8	2.5
East Champaran	5.3	5.4
Madhubani	5.7	5.4
<i>NKBR</i>		
Araval	0.6	0.5
Aurangabad	2.7	2.6
Banka	1.9	2.4
Bhojpur	2.9	2.6
Buxar	2.1	1.9
Gaya	3.1	3.2
Gopalganj	2.5	2.6
Jahanabad	1.0	0.9
Jamui	1.2	1.1
Kaimur	2.5	2.1
Kishanganj	1.9	1.9
Lakhisarai	0.7	0.8
Monghyr	0.8	0.9
Nalanda	2.7	2.4
Nawadha	1.8	1.7
Patna	3.6	3.2
Rohtas	3.7	3.8
Saran	2.7	2.8
Sheikhpura	0.5	0.5
Siwan	2.7	3.5
Vaishali	2.9	3.4
West Champaran	5.3	5.3

Source: Source: Government of Bihar (2010)

TableA5.11: Sources of Irrigation (%) (2010TE)

Districts	Canals		Tank	Wells		Other sources
	Govt.	Private		Tube Wells	Other wells	
KBR						
Araria	10.39	0.00	0.00	89.19	0.00	0.42
Begusarai	0.00	0.00	0.82	99.12	0.00	0.06
Bhagalpur	5.10	0.00	1.47	93.19	0.00	0.24
Katihar	1.09	0.00	0.00	97.71	0.02	1.18
Khagaria	0.00	0.00	0.00	99.94	0.00	0.06
Madhupura	8.03	0.00	0.01	91.60	0.00	0.37
Purnea	7.92	0.00	0.00	91.62	0.00	0.46
Saharsa	5.62	0.00	0.02	82.68	0.00	11.68
Samastipur	0.00	0.00	0.00	99.78	0.00	0.22
Supaul	15.44	0.00	0.00	83.86	0.00	0.70
Darbhanga	0.00	0.00	51.81	33.84	0.00	14.36
Muzafarpur	14.86	0.00	4.71	79.13	0.00	1.29
Sheohar	0.00	0.00	0.00	100.00	0.00	0.00
Sitamarhi	1.55	0.00	2.57	94.43	0.00	1.44
E.champaran	11.79	0.00	0.00	87.47	0.00	0.75
Madhubani	30.24	0.00	0.15	69.48	0.00	0.13
NKBR						
Araval	62.14	0.00	0.00	36.53	0.00	1.34
Aurangabad	46.13	0.00	0.06	33.56	1.37	18.88
Banka	44.73	0.00	1.17	51.32	1.01	1.77
Bhabhua	49.36	0.00	3.21	43.09	0.00	4.34
Bhojpur	66.00	0.00	0.00	30.94	0.17	2.90
Buxar	37.82	0.00	2.23	46.56	0.04	13.35
W.Champaran	54.72	0.00	0.00	44.85	0.00	0.43
Gaya	7.78	0.00	0.06	91.40	0.00	0.77
Gopalganj	42.79	0.00	0.00	57.09	0.00	0.12
Jahanabad	16.05	0.00	0.00	80.80	0.02	3.13
Jamui	3.58	0.00	1.95	70.36	1.02	23.08
Kishanganj	0.00	0.00	0.00	99.79	0.03	0.18
Lakhisarai	30.11	0.00	0.00	69.80	0.00	0.08
Monghyr	49.35	0.00	0.29	50.08	0.00	0.27
Nalanda	7.99	0.00	1.80	85.35	0.27	4.59
Nawadha	13.78	0.00	0.00	81.99	0.00	4.23
Patna	17.79	0.00	0.00	79.18	2.88	0.15
Rohtas	82.66	0.00	0.94	9.59	1.74	5.07
Saran	14.14	0.00	1.51	83.93	0.00	0.41
Sheikhpura	7.81	0.00	19.66	71.65	0.26	0.62
Siwan	16.84	0.00	2.59	79.35	0.00	1.22
Vaishali	0.00	0.00	0.09	87.97	0.00	11.94

Source: MoA (Website)

Table A5.12: Number & Area of Operational Holdings by size groups (2010)

	Marginal			Small			Semi-Medium			Medium			Large			All Size		
	No	Area	FSZ	No	Area	FSZ	No	Area	FSZ	No	Area	FSZ	No	Area	FSZ	No	Area	FSZ
	%		Hect.	%		Hect.	%		Hect.	%		Hect.	%		Hect.	'000	'000 Hect	Hect
<i>KBR</i>																		
Araria	85	48	0.31	10	23	1.29	5	21	2.55	1	8	5.37	0	0	16.81	389	216	0.56
Baghalpur	94	65	0.23	4	13	1.18	2	14	2.63	0	7	5.30	0	2	13.84	490	160	0.33
Begusarai	94	61	0.22	4	15	1.21	2	15	2.62	1	9	5.15	0	1	14.71	377	125	0.33
Katihar	90	58	0.32	7	19	1.40	3	17	2.71	1	6	5.31	0	1	12.60	420	206	0.49
Khagaria	87	49	0.25	8	20	1.17	4	22	2.38	1	9	4.65	0	0	12.93	219	99	0.45
Madhepura	91	58	0.24	6	20	1.29	2	17	2.70	0	6	4.89	0	0	13.28	363	139	0.38
Purnia	83	46	0.34	11	23	1.30	6	24	2.57	1	8	5.14	0	0	0.00	389	240	0.62
Samastipur	95	70	0.21	4	15	1.13	1	8	2.59	0	5	5.59	0	2	16.24	686	190	0.28
Sharsa	85	45	0.29	10	20	1.21	4	18	2.85	1	13	5.29	0	4	14.37	217	122	0.56
Supaul	89	54	0.27	7	18	1.23	3	19	2.55	1	8	5.00	0	1	12.54	376	169	0.45
Darbangha	93	62	0.20	5	19	1.26	2	15	2.57	0	4	4.92	0	0	22.88	546	168	0.31
Muzaffarpur	93	66	0.26	5	16	1.22	2	14	2.49	0	3	4.96	0	0	12.31	599	220	0.37
Sheohar	92	56	0.22	5	17	1.27	3	19	2.53	1	8	5.12	0	1	16.70	88	32	0.37
Sitamarhi	96	70	0.17	3	16	1.22	1	12	2.45	0	2	5.01	0	0	12.38	652	155	0.24
E.champaran	92	58	0.23	5	16	1.19	2	16	2.65	1	9	5.10	0	1	12.12	935	342	0.37
Madhubani	91	58	0.24	7	21	1.16	2	15	2.43	1	6	4.91	0	0	11.58	635	236	0.37
<i>NKBR</i>																		
Arwal	91	59	0.27	6	19	1.22	3	17	2.62	0	4	5.05	0	0	0.00	110	46	0.41
Aurangabad	82	45	0.33	11	24	1.28	6	25	2.64	1	6	4.94	0	0	13.84	404	242	0.60
Banka	93	67	0.33	5	17	1.47	2	12	2.85	0	4	5.33	0	0	21.23	333	151	0.45
Bhabhua	86	42	0.27	8	16	1.15	5	22	2.56	2	17	5.14	0	4	12.55	281	156	0.56
Bhojpur	90	56	0.29	7	17	1.21	3	16	2.59	1	11	5.02	0	1	11.46	392	184	0.47
Buxer	79	38	0.37	11	20	1.40	8	26	2.67	2	13	5.29	0	2	13.35	178	137	0.77
Gaya	89	53	0.28	7	20	1.26	3	19	2.61	1	8	5.14	0	0	12.42	746	347	0.46
Gopalganj	95	68	0.19	4	19	1.25	1	10	2.42	0	3	4.98	0	0	16.35	573	150	0.26
Jahanabad	86	48	0.25	10	26	1.21	4	21	2.51	1	5	4.66	0	0	13.08	168	77	0.46
Jamui	90	55	0.29	6	16	1.37	3	18	2.64	1	11	5.13	0	1	14.28	408	193	0.47
Kisanganj	91	63	0.30	5	16	1.28	3	20	2.62	0	1	4.94	0	0	12.00	311	136	0.44
Lakhisarai	94	78	0.35	5	15	1.20	1	6	2.67	0	0	5.19	0	0	11.00	147	62	0.42
Munger	94	64	0.20	4	17	1.17	2	14	2.36	0	6	5.04	0	0	0.00	156	46	0.30
Nalanda	87	52	0.32	9	23	1.37	4	19	2.71	1	5	4.85	0	1	100.78	346	182	0.52
Nawadaa	91	60	0.25	6	17	1.19	3	18	2.52	0	5	4.99	0	0	14.48	378	146	0.39
Patna	91	59	0.23	7	26	1.26	2	13	2.54	0	1	4.84	0	0	15.00	678	239	0.35
Rohtas	90	53	0.26	6	17	1.29	3	19	2.67	1	10	4.92	0	1	19.19	546	246	0.45
Saran	95	75	0.24	4	16	1.36	1	9	2.59	0	0	4.98	0	0	0.00	540	163	0.30
Shekhpura	94	71	0.27	4	13	1.22	2	13	2.64	0	3	5.36	0	1	14.05	157	56	0.36
Siwan	93	63	0.22	5	18	1.24	2	16	2.55	0	3	5.12	0	0	13.33	437	143	0.33
Vaishali	96	66	0.13	3	15	1.13	1	12	2.48	1	12	4.83	0	1	11.30	712	137	0.19
W.champaran	91	58	0.26	6	17	1.18	3	17	2.48	1	7	5.21	0	2	19.33	810	331	0.41

Note: No-Number; FSZ-Farm Size. Source: Agriculture Census (Website)

Table A 5.13: Trend in Average water table (In meter)

Bihar	YEAR				
	2007	2008	2009	2010	2011
<i>KBR</i>					
Araria	3.07	3.06	3.32	2.62	2.80
Baghalpur	5.07	3.84	4.31	5.42	5.02
Begusarai	5.12	3.56	5.39	6.14	5.57
Katihar	3.45	3.55	3.99	3.95	3.69
Khagaria	5.15	4.56	5.36	5.57	5.03
Madhepura	2.71	3.67	2.80	3.82	3.37
Purnia	3.18	3.09	3.13	3.30	3.31
Samastipur	4.47	3.23	4.62	6.09	6.41
Sharsa	2.68	2.93	2.75	3.24	3.24
Supaul	2.35	3.03	2.64	2.27	2.40
Darbangha	3.61	2.57	2.58	3.29	2.82
Muzaffarpur	3.56	3.14	3.52	4.13	3.55
Sheohar	0.00	0.00	1.77	0.92	1.65
Sitamarhi	2.82	2.19	2.18	3.17	2.58
E.champaran	3.47	2.81	2.58	2.81	3.30
Madhubani	2.43	2.93	2.61	2.84	2.85
<i>NKBR</i>					
Aurangabad	3.52	4.00	3.68	6.29	5.12
Banka	4.00	3.86	4.20	4.26	4.19
Kaimur(Bhabhua)	5.32	5.45	5.66	6.58	6.43
Bhojpur	4.83	4.02	4.29	5.44	4.45
Buxer	5.12	4.22	3.78	5.22	4.67
W.champaran	3.20	3.48	2.80	3.26	2.63
Gaya	4.24	3.97	5.87	7.47	7.30
Gopalganj	1.92	2.20	2.80	3.48	2.81
Jehnnabad	3.25	3.17	4.90	5.30	5.32
Jamui	7.08	6.93	6.98	8.02	7.98
Kishangang	3.98	4.35	3.71	4.14	3.68
lakhisarai	5.44	4.09	4.39	6.43	6.41
munger	4.19	3.75	4.19	5.24	5.04
Nalanda	1.92	2.56	3.95	4.48	3.69
Nawada	3.26	3.00	5.14	5.18	4.80
Patna	3.94	4.19	4.85	5.29	4.40
Rohtas	5.25	4.73	4.27	5.25	5.39
Saran	3.50	3.05	4.33	5.34	4.20
Sheikhpura	0.00	0.00	6.15	0.00	7.15
Siwan	3.75	3.51	3.86	4.83	3.86
Vaishali	5.09	4.33	4.93	6.24	5.36

Source: CGBW (Website)

Table A5.14: Seasonal variation in water table for the period 2007-2011 (in Meter) 2011

Disticts	JAN	MAY	AUG	NOV	AVERAGE
<i>KBR</i>					
Araria	3.32	3.37	1.97	2.55	2.80
Baghalpur	6.14	6.94	2.55	4.17	4.95
Begusarai	6.64	7.68	3.75	4.37	5.61
Katihar	4.36	5.25	2.83	3.24	3.92
Khagaria	6.55	7.55	2.55	3.87	5.13
Madhepura	3.26	4.10	2.86	3.11	3.33
Purnia	4.02	4.31	2.44	2.66	3.36
Samastipur	6.70	8.53	5.31	5.33	6.47
Sharsa	3.36	4.07	2.63	2.57	3.16
Supaul	2.64	2.60	2.01	2.27	2.38
Darbangha	3.96	4.15	1.72	2.16	3.00
Muzaffarpur	4.27	5.30	2.82	2.51	3.72
Sheohar	1.55	2.20		1.20	1.65
Sitamarhi	2.58	3.02	1.56	1.79	2.24
E.champaran	4.03	4.51	1.39	1.93	2.97
Madhubani	3.40	3.11	2.26	2.50	2.82
<i>NKBR</i>					
Aurangabad	6.47	7.62	3.28	3.15	5.13
Banka	5.50	5.84	2.36	3.15	4.21
Kaimur(Bhabhua)	7.14	8.26	4.55	4.19	6.03
Bhojpur	5.82	6.19	2.42	2.47	4.23
Buxer	5.41	6.52	3.26	2.82	4.50
W.champaran	2.87	5.52	1.59	2.77	3.19
Gaya	7.98	8.83	3.32	3.60	5.93
Gopalganj	2.66	4.31	1.08	2.45	2.63
Jehnabad	5.77	6.84	3.98	3.81	5.10
Jamui	8.56	10.31	6.81	5.99	7.92
Kishangang	2.63	5.61	3.55	4.39	4.05
lakhisarai	7.60	8.60	5.47	3.68	6.34
munger	5.30	7.12	4.02	3.72	5.04
Nalanda	4.19	5.40	2.60	2.56	3.69
Nawada	5.28	7.98	3.43	3.58	5.07
Patna	5.47	6.83	2.34	3.18	4.46
Rohtas	5.89	8.30	3.69	3.56	5.36
Saran	4.28	6.49	3.31	3.15	4.31
Sheikhpura	0.00	7.15	0.00	0.00	7.15
Siwan	4.23	6.07	2.22	3.35	3.97
Vaishali	6.52	8.12	4.30	4.28	5.81

Source: CGBW (Website)

Table A5.15: 18 Normal Rainfall (mm) of Bihar districts

Districts	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
<i>KBR</i>												
Araria	12.9	12.0	13.7	26.8	87.6	227.3	375.0	294.0	243.7	81.5	7.1	2.2
Brgusarai	14.3	15.3	12.3	10.5	34.9	149.1	294.7	270.4	211.2	76.9	5.4	2.1
Baghalpur	14.4	13.8	13.7	17.5	52.4	187.1	301.1	278.7	228.5	82.7	5.9	2.2
Katihar	12.3	12.3	12.2	25.2	72.2	210.2	322.6	281.8	231.4	84.7	6.5	2.4
Khagaria	14.0	13.7	12.4	12.9	42.2	165.6	300.0	269.1	219.7	78.4	6.2	1.9
madhepura	15.0	12.5	12.3	15.9	55.8	168.7	294.0	257.3	211.1	74.7	6.0	2.1
samastipur	15.2	14.9	11.2	9.8	34.1	141.3	306.8	275.0	209.8	79.3	5.8	2.2
Purnia	13.7	12.4	12.2	26.4	78.1	219.6	342.9	290.1	235.9	82.2	6.0	2.4
sharsa	14.4	13.0	12.1	14.4	47.4	163.8	311.8	266.5	219.6	76.9	6.2	2.1
Supaul	14.3	12.2	12.6	19.7	68.3	179.1	327.4	267.1	207.6	76.5	6.6	2.1
Darbangha	13.1	13.8	11.6	11.9	44.9	145.6	322.9	284.9	209.1	80.4	6.3	2.2
Muzaffurpur	13.1	16.8	10.5	9.9	38.3	135.7	334.3	297.0	203.4	78.3	7.1	2.2
Seohar	13.3	16.9	10.7	11.5	39.4	142.5	343.1	308.9	203.0	80.4	7.8	2.4
Sitamarhi	11.5	13.2	11.1	14.4	50.7	159.4	358.2	315.1	217.3	79.5	6.2	2.5
Champanan E	13.0	13.1	9.0	11.8	42.2	154.9	350.0	325.2	206.9	68.7	5.4	2.4
Madhubani	12.9	12.4	11.9	15.7	57.5	156.8	333.2	288.5	207.8	79.4	6.3	2.2
<i>NKBR</i>												
Aurangabad	16.9	20.3	15.2	11.1	22.5	142.7	319.7	326.2	196.0	57.9	8.6	2.9
Banka	12.4	19.3	15.8	16.5	41.1	196.7	310.1	309.3	231.5	81.8	5.1	2.1
Bhabhua	18.7	19.1	13.1	6.2	10.9	111.8	283.6	315.3	175.3	42.5	10.6	3.3
Bhojpur	15.8	19.4	11.5	7.3	20.2	123.8	290.7	303.0	182.4	57.7	8.0	2.4
Buxer	16.5	17.6	11.2	5.2	12.3	111.0	285.3	303.7	172.7	46.6	8.9	2.8
Gaya	16.6	22.2	15.7	11.8	20.8	148.4	301.9	320.2	196.9	62.2	7.3	2.8
Gopalganj	13.0	13.2	8.2	7.5	22.5	125.4	303.3	300.6	183.8	52.1	4.8	2.3
Jamui	13.3	20.0	14.6	13.7	32.8	168.4	281.2	297.4	211.6	74.7	4.9	3.1
jeenabad	15.5	19.4	11.7	8.2	15.0	124.8	282.0	298.2	181.8	59.1	7.1	1.8
Kishanganj	11.3	11.2	14.1	36.0	109.7	268.9	425.9	320.2	273.7	86.9	7.6	2.8
Lakhisarai	14.3	17.4	13.7	11.8	34.8	162.5	294.3	287.4	219.9	77.9	5.7	2.3
Munger	13.0	15.2	13.1	12.5	37.0	166.9	301.5	279.4	227.1	80.2	5.7	1.8
Nalanda	14.2	19.1	11.5	9.9	19.0	139.6	283.4	292.3	199.3	65.5	5.9	1.7
Nawada	15.0	22.3	15.4	12.8	27.1	163.3	297.8	324.6	212.2	72.4	6.3	3.1
Patna	15.2	20.9	12.1	9.4	22.7	139.9	304.0	300.5	198.6	65.2	6.4	1.8
Rohats	18.5	19.9	13.7	8.8	14.5	121.0	293.4	320.1	180.3	49.4	9.5	3.3
Saran	13.9	19.1	8.9	6.5	21.2	122.3	298.3	287.4	179.7	59.9	6.4	2.3
Shekhpura	15.1	19.3	13.1	10.8	24.1	152.7	299.1	297.5	216.2	73.7	5.5	1.9
siwan	13.1	12.8	7.8	5.6	19.2	115.3	303.1	301.8	185.8	49.7	5.4	2.1
Vaishali	13.2	22.2	10.8	8.1	25.4	132.9	299.6	281.4	194.0	75.1	5.6	1.8
W.champanan	12.9	13.4	10.0	16.8	42.4	187.8	384.7	355.6	229.9	63.8	4.6	3.9

Note: Average of 1901-2012 year. Source: IWP (Website) & IMD (Website)

Appendix Table 6

Table A6.1: Cropping Pattern (2010TE)

Districts	% Share							
	MC	CC	PLS	FR	VG	OLS	SC	OTH CROPS
<i>KBR</i>								
Araria	64.6	7.5	5.2	1.4	5.2	2.0	0.0	14.2
Begusarai	48.2	34.9	2.7	1.9	5.6	3.9	1.6	1.1
Bhagalpur	50.5	24.7	11.3	2.0	3.8	4.2	2.3	1.1
Katihar	53.8	19.3	4.3	2.0	5.0	2.4	0.1	13.1
Khagaria	44.0	43.0	5.7	1.4	2.5	2.8	0.1	0.5
Madhupura	53.8	22.2	11.6	1.6	3.5	3.7	0.9	2.6
Purnea	56.9	15.8	4.3	1.3	6.6	3.4	0.1	11.7
Saharsa	67.4	20.8	9.2	0.6	1.0	0.8	0.0	0.2
Samastipur	55.1	20.1	5.6	2.9	9.9	2.2	1.2	2.9
Supaul	65.8	6.2	12.7	1.4	4.3	3.3	0.0	6.3
Darbhanga	82.0	7.8	5.4	2.0	1.7	0.8	0.3	0.1
Muzafarpur	74.1	8.0	8.7	2.1	3.9	1.5	0.7	1.0
Sheohar	81.0	2.2	5.0	1.7	2.4	1.2	6.3	0.2
Sitamarhi	79.4	2.7	6.7	1.4	4.6	0.9	4.2	0.1
E.Champaran	81.1	5.0	3.4	2.7	4.2	0.7	2.7	0.2
Madhubani	84.4	1.3	5.6	4.1	3.5	0.6	0.1	0.3
<i>NKBR</i>								
Araval	79.0	1.9	12.9	0.1	4.5	1.4	0.1	0.2
Aurangabad	80.9	0.6	14.9	0.0	1.1	2.3	0.2	0.0
Banka	80.2	8.2	5.3	1.1	2.8	0.8	1.0	0.7
Bhabhua	87.2	0.5	9.1	0.0	1.2	1.6	0.2	0.1
Bhojpur	78.1	3.6	11.0	0.7	4.3	1.2	0.1	1.0
Buxar	82.6	3.2	6.1	0.6	5.4	1.2	0.2	0.7
Champaran(West)	60.3	3.9	6.0	1.2	4.8	5.1	18.3	0.5
Gaya	78.6	3.7	8.1	0.1	6.6	2.3	0.3	0.3
Gopalganj	74.4	7.0	1.3	1.5	2.2	1.2	12.3	0.1
Jahanabad	76.1	1.2	20.3	0.1	1.7	0.4	0.3	0.1
Jamui	86.0	6.4	3.5	0.0	1.9	0.2	1.9	0.1
Kishanganj	64.2	1.9	6.2	1.3	5.6	2.8	0.0	18.0
Lakhisarai	74.6	4.9	18.6	0.0	0.1	1.8	0.0	0.0
Monghyr	78.7	10.0	6.0	0.2	3.6	1.3	0.2	0.1
Nalanda	75.8	3.5	13.7	0.1	4.8	0.9	0.1	1.1
Nawadha	83.5	2.1	7.1	0.1	5.2	1.5	0.2	0.3
Patna	61.0	3.5	24.3	0.1	7.4	1.7	0.4	1.6
Rohtas	92.7	0.3	5.0	0.0	1.0	0.9	0.1	0.0
Saran	75.9	12.9	2.0	1.0	6.1	1.4	0.6	0.1
Sheikhpura	77.6	0.8	15.4	0.0	3.0	2.1	0.5	0.5
Siwan	82.0	8.0	1.7	1.3	4.2	0.9	1.5	0.3
Vaishali	53.0	18.3	6.3	4.8	10.2	2.0	0.1	5.4

Source: MoA (Website)

Table A6.2.a: Cropping pattern in *Rabi* season (2010TE)

Districts	RC	JW	WH	BL	MZ	OTH MLT	R &M	LS	SF	GRM	OTH PLS
<i>KBR</i>											
Araria	14.2	0.0	61.2	0.0	7.2	0.0	2.6	1.9	0.0	0.3	12.6
Begusarai	1.2	0.0	71.6	0.0	15.2	0.2	8.6	0.0	0.0	0.0	3.2
Bhagalpur	0.5	0.0	64.3	2.4	14.0	0.1	2.7	1.1	0.0	3.8	11.1
Katihar	33.0	0.0	35.6	0.1	20.1	0.2	4.2	0.5	0.0	0.1	6.1
Khagaria	0.1	0.0	52.1	0.2	37.0	0.0	4.1	0.0	0.0	0.4	6.1
Madhupura	1.1	0.0	47.9	0.0	21.7	0.0	1.8	0.3	0.0	0.0	27.2
Purnea	15.1	0.0	51.9	0.0	16.5	0.0	4.9	0.7	0.0	0.1	10.8
Saharsa	5.2	0.0	53.4	0.0	17.9	0.0	1.3	0.4	0.0	0.0	21.8
Samastipur	0.8	0.0	57.6	0.1	22.7	0.3	5.1	0.1	0.0	0.0	13.3
Supaul	6.0	0.0	56.6	0.0	5.2	0.0	1.3	1.2	0.0	0.0	29.7
Darbhangha	2.0	0.0	76.3	0.0	8.9	0.0	1.6	0.0	0.0	0.1	11.1
Muzafarpur	0.2	0.0	69.2	0.2	8.5	0.0	3.2	0.1	0.0	0.0	18.5
Sheohar	0.0	0.0	80.2	0.0	3.1	0.0	2.1	0.9	0.0	0.0	13.7
Sitamarhi	0.0	0.0	78.3	0.2	2.5	0.0	1.8	0.5	0.0	0.0	16.8
E.Champaran	0.1	0.0	85.1	0.3	4.2	0.0	1.4	0.6	0.0	0.1	8.1
Madhubani	1.8	0.0	81.5	0.0	0.6	0.0	1.1	0.4	0.0	0.0	14.6
<i>NKBR</i>											
Araval	0.0	0.0	61.0	0.8	1.2	0.0	2.0	0.9	0.0	5.0	29.0
Aurangabad	0.0	0.0	58.1	1.0	0.0	0.0	1.0	4.6	0.0	6.6	28.6
Banka	0.0	0.0	75.0	1.6	5.1	0.0	1.5	1.6	0.0	7.2	8.0
Bhabhua	0.0	0.0	76.7	1.1	0.0	0.0	1.7	1.8	0.0	8.1	10.5
Bhojpur	0.0	0.0	73.4	1.3	0.3	0.0	2.0	0.3	0.0	5.1	17.6
Buxar	0.0	0.0	85.2	1.6	0.0	0.0	1.4	0.8	0.0	2.6	8.3
W.Champaran	5.2	0.0	62.0	0.6	4.6	0.1	14.0	0.5	0.0	0.0	12.9
Gaya	0.0	0.0	78.3	0.9	0.7	0.1	1.7	2.2	0.0	5.6	10.6
Gopalganj	0.0	0.0	90.6	0.1	4.4	0.1	2.9	0.0	0.0	0.0	1.9
Jahanabad	0.0	0.0	67.0	0.2	0.2	0.0	0.4	0.2	0.0	4.0	27.9
Jamui	0.0	0.0	84.6	1.9	0.8	0.0	0.4	0.8	0.0	6.6	4.9
Kishanganj	27.5	0.0	45.0	0.0	2.2	0.0	3.3	5.2	0.0	0.5	16.2
Lakhisarai	0.0	0.0	68.0	0.0	0.3	0.0	2.7	0.0	0.0	7.2	21.7
Monghyr	0.3	0.0	78.7	0.2	5.6	0.0	2.4	1.0	0.0	4.8	6.9
Nalanda	0.0	0.0	70.7	0.2	1.4	0.0	1.6	0.0	0.0	4.3	21.8
Nawadha	0.0	0.0	84.8	0.2	0.6	0.0	1.6	0.9	0.0	2.5	9.5
Patna	0.0	0.0	54.2	0.2	0.6	0.0	1.8	0.6	0.0	6.1	36.6
Rohtas	0.0	0.0	87.4	0.6	0.0	0.0	0.9	0.9	0.0	2.5	7.7
Saran	0.2	0.0	87.8	0.1	5.5	0.1	2.9	0.0	0.0	0.1	3.2
Sheikhpura	0.0	0.0	70.8	0.1	0.2	0.0	2.2	1.3	0.0	4.2	21.1
Siwan	0.1	0.0	91.0	0.1	4.4	0.1	1.8	0.1	0.0	0.1	2.3
Vaishali	0.6	0.0	69.2	0.1	11.1	0.4	3.8	0.1	0.1	0.1	14.4

Source: MoA (Website)

Table A6.2.b: Cropping pattern in *Kharif* season (2010TE)

Disticts	RC	JW	BJ	MZ	RG	OTH MLT	GN	CS	SM	SF	TUR	Other PLS
<i>KBR</i>												
Araria	92.3	0.0	0.0	3.7	0.3	0.0	0.0	0.0	0.0	1.3	0.2	2.2
Begusarai	38.7	0.1	0.0	57.9	0.0	0.0	0.0	0.0	0.0	0.0	1.1	2.0
Bhagalpur	55.5	0.0	0.0	31.7	0.0	0.1	0.0	0.0	0.1	0.6	1.1	10.9
Katihar	80.6	0.0	0.0	12.2	0.0	0.7	0.0	0.0	0.0	1.0	0.0	5.4
Khagaria	52.8	0.0	0.0	35.5	0.1	1.9	0.0	0.0	0.0	2.2	0.4	7.0
Madhupura	83.3	0.0	0.0	7.7	0.3	0.0	0.0	0.0	0.1	7.0	0.0	1.6
Purnea	86.3	0.0	0.0	9.5	0.0	0.0	0.0	0.0	0.0	2.4	0.1	1.7
Saharsa	89.6	0.0	0.0	8.5	0.9	0.0	0.0	0.0	0.0	0.2	0.0	0.8
Samastipur	86.1	0.1	0.0	11.5	0.1	0.0	0.0	0.0	0.5	0.0	0.7	1.0
Supaul	89.3	0.0	0.0	1.4	1.8	0.0	0.0	0.0	0.2	4.8	0.0	2.4
Darbhanga	96.5	0.0	0.0	2.1	0.8	0.0	0.0	0.0	0.0	0.0	0.2	0.2
Muzafarpur	96.1	0.0	0.0	2.7	0.2	0.0	0.0	0.0	0.0	0.0	0.9	0.1
Sheohar	98.7	0.0	0.0	0.3	0.0	0.1	0.0	0.0	0.2	0.1	0.4	0.2
Sitamarhi	98.1	0.0	0.0	0.7	0.7	0.0	0.0	0.0	0.0	0.0	0.3	0.2
E.Champaran	95.3	0.0	0.0	3.5	0.0	0.0	0.0	0.0	0.0	0.0	0.7	0.5
Madhubani	83.3	0.0	0.0	7.7	0.3	0.0	0.0	0.0	0.1	7.0	0.0	1.6
<i>NKBR</i>												
Araval	96.5	0.0	0.0	1.7	0.1	0.0	0.0	0.0	0.2	0.1	1.1	0.3
Aurangabad	99.2	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.6	0.0
Banka	90.3	0.0	0.0	7.4	0.0	0.0	0.0	0.0	0.0	0.0	0.6	1.6
Bhabhua	98.1	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	1.7	0.0
Bhojpur	92.8	0.7	1.1	3.5	0.0	0.1	0.0	0.0	0.1	0.0	1.4	0.3
Buxar	93.4	1.2	3.0	1.7	0.0	0.0	0.0	0.0	0.0	0.0	0.6	0.1
W.Champaran	94.0	0.0	0.0	1.7	0.0	0.0	0.1	0.0	0.2	0.1	3.7	0.3
Gaya	90.2	0.0	0.0	4.6	0.2	0.3	0.9	0.0	0.2	0.0	1.6	1.9
Gopalganj	90.2	0.0	0.1	8.3	0.1	0.1	0.0	0.0	0.0	0.0	1.1	0.1
Jahanabad	97.2	0.0	0.0	1.9	0.2	0.0	0.0	0.0	0.0	0.0	0.5	0.1
Jamui	89.7	0.0	0.0	8.5	0.0	0.0	0.0	0.0	0.0	0.0	0.4	1.5
Kishanganj	93.5	0.0	0.0	3.4	0.2	0.0	0.0	0.0	0.4	0.0	0.1	2.5
Lakhisarai	86.8	0.0	0.0	12.3	0.0	0.0	0.0	0.0	0.0	0.0	0.8	0.1
Monghyr	86.6	0.0	0.0	10.5	0.0	0.0	0.0	0.0	0.0	0.0	0.5	2.4
Nalanda	94.2	0.0	0.0	4.9	0.0	0.0	0.0	0.0	0.0	0.2	0.4	0.3
Nawadha	94.2	0.3	0.0	1.9	0.4	0.0	0.2	0.0	0.1	0.0	1.7	1.2
Patna	91.4	0.0	0.0	7.6	0.0	0.0	0.0	0.0	0.0	0.1	0.9	0.0
Rohtas	99.4	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.4	0.2
Saran	81.4	0.0	0.0	17.0	0.3	0.1	0.0	0.0	0.0	0.0	0.9	0.2
Sheikhpura	97.9	0.1	0.0	1.1	0.0	0.0	0.0	0.0	0.0	0.0	1.0	0.0
Siwan	89.0	0.0	0.0	8.6	0.6	0.3	0.0	0.0	0.0	0.0	1.3	0.1
Vaishali	74.0	0.0	0.0	23.5	0.2	0.0	0.0	0.0	0.0	0.0	1.2	0.9

Source: MoA (Website)

Table A6.3: Season wise Maize crops share.

Districts	Maize- kharif (2000)	Maize- kharif (2010)	Maize- Rabi (2000)	Maize- Rabi (2010)	Maize- summer (2000)	Maize- summer (2010)
<i>KBR</i>						
Araria	37	22.9	21.7	31.9	41.2	45.2
Begusarai	67.8	66.8	26.8	19.7	5.4	13.5
Bhagalpur	62.5	56.7	17.6	24.6	19.9	18.7
Katihar	26.8	22.1	37.8	41.4	35.5	36.5
Khagaria	28.3	28.7	53.8	42.0	17.9	29.3
Madhupura	19.5	14.2	32	39.3	48.5	46.5
Purnea	25.3	25.5	29.8	35.5	44.9	39.0
Saharsa	24.7	20.5	28.6	35.6	46.6	44.0
Samastipur	34.2	23.0	53.9	43.8	11.9	33.2
Supaul	6.4	11.6	36.9	36.3	56.6	52.1
Darbhanga	10.9	13.7	65.1	56.7	24	29.6
Muzafarpur	23.7	15.3	42.8	48.5	33.5	36.2
Sheohar	9.1	8.1	55.2	51.6	35.6	40.3
Sitamarhi	20.2	13.6	38.1	38.5	41.7	47.9
E.Champaran	31.9	38.2	39.1	28.4	29	33.4
Madhubani	48.2	18.8	14.8	44.3	37.1	37.0
<i>NKBR</i>						
Araval	50.5	56.0	25.3	24.7	24.2	19.3
Aurangabad	94.4	96.2	2.8	1.5	2.8	2.4
Banka	69.6	62.8	11.9	15.2	18.5	22.0
Bhabhua	93.6	100.0	0.1	0.0	6.3	0.0
Bhojpur	90.9	80.6	1	6.9	8.2	12.5
Buxar	98.2	92.0	1	3.5	0.8	4.5
W.Champaran	13.3	18.4	40.7	42.6	46	39.1
Gaya	72	79.9	3.3	8.6	24.6	11.4
Gopalganj	52.8	53.5	35.9	26.3	11.3	20.3
Jahanabad	51.9	69.7	20	12.0	28.1	18.3
Jamui	90.3	94.5	3.5	1.9	6.2	3.6
Kishanganj	40.8	50.9	40.4	22.3	18.8	26.9
Lakhisarai	95.9	92.0	1	4.0	3.1	4.0
Monghyr	70	59.0	17.6	20.7	12.4	20.2
Nalanda	51	55.8	13	19.1	36	25.1
Nawadha	80	52.4	4.3	16.8	15.7	30.8
Patna	60.1	75.2	12	9.6	27.8	15.2
Rohtas	95.1	94.4	0.8	2.4	4.2	3.2
Saran	63.3	57.8	17	20.3	19.7	21.9
Sheikhpura	88.2	57.1	1.9	21.4	9.9	21.6
Siwan	58.4	53.5	24.3	25.2	17.3	21.4
Vaishali	47.4	46.7	17.9	23.4	34.8	30.0

Source: MoA (Website)

Table A6.4: Kharif and Rabi share in gross cropped area

% Share				% Share		
Districts	Year	Kharif	Rabi	Year	Kharif	Rabi
KBR						
Araria	TE2000	51.19	30.29	2010TE	49.92	29.32
Begusarai	TE2000	53.31	36.54	2010TE	51.59	38.12
Bhagalpur	TE2000	59.35	33.30	2010TE	52.60	35.79
Katihar	TE2000	30.16	29.63	2010TE	48.42	32.85
Khagaria	TE2000	69.25	41.67	2010TE	65.09	31.04
Madhepura	TE2000	39.33	34.91	2010TE	61.10	33.65
Purnea	TE2000	38.72	33.17	2010TE	55.28	28.14
Saharsa	TE2000	65.13	50.75	2010TE	60.95	33.87
Samastipur	TE2000	53.49	28.64	2010TE	50.95	28.12
Supaul	TE2000	47.26	37.42	2010TE	54.79	36.54
Darbhanga	TE2000	53.27	41.23	2010TE	53.18	42.80
Muzafarpur	TE2000	54.82	36.54	2010TE	54.57	41.01
Sheohar	TE2000	43.40	43.30	2010TE	52.94	32.03
Sitamarhi	TE2000	37.27	24.74	2010TE	44.38	37.62
East Champaran	TE2000	61.09	31.37	2010TE	57.98	32.27
Madhubani	TE2000	53.27	25.75	2010TE	56.45	34.72
NKBR						
Aurangabad	TE2000	62.69	36.01	2010TE	57.37	41.33
Banka	TE2000	73.91	22.18	2010TE	71.27	23.14
Kaimur(bhabhua)	TE2000	55.18	43.26	2010TE	54.63	43.79
Bhojpur	TE2000	52.30	42.78	2010TE	49.23	44.50
Buxar	TE2000	54.35	42.20	2010TE	39.85	53.28
West Champaran	TE2000	37.25	25.68	2010TE	47.71	33.11
Gaya	TE2000	67.34	36.31	2010TE	72.50	41.04
Gopalganj	TE2000	40.76	34.49	2010TE	44.89	37.52
Jahanabad	TE2000	49.28	30.64	2010TE	52.97	58.35
Jamui	TE2000	84.51	56.09	2010TE	85.14	22.21
Kishanganj	TE2000	49.38	24.64	2010TE	46.25	30.31
Lakhisarai	TE2000	60.01	52.74	2010TE	51.78	50.18
Monghyr	TE2000	123.67	96.37	2010TE	64.58	36.47
Nalanda	TE2000	36.47	36.21	2010TE	46.88	50.38
Nawadha	TE2000	55.60	46.80	2010TE	53.33	45.61
Patna	TE2000	34.79	42.17	2010TE	41.78	55.60
Rohtas	TE2000	49.35	34.36	2010TE	52.17	45.36
Saran	TE2000	52.39	38.15	2010TE	49.68	46.51
Sheikhpura	TE2000	93.24	89.77	2010TE	52.08	61.91
Siwan	TE2000	49.46	37.06	2010TE	54.44	42.16
Vaishali	TE2000	50.21	35.06	2010TE	49.22	34.20

Source: MoA (Website)

Table A6.5.a: Consumption of Fertilizer across agriculture seasons (In Kg/Hect) (TE2010)

Districts	Kharif				Rabi				Total			
	N	P	K	NPK	N	P	K	NPK	N	P	K	NPK
<i>KBR</i>												
Araria	48	16	12	76	113	42	54	209	72	26	28	126
Begusarai	127	64	34	225	305	101	87	493	203	79	57	338
Bhagalpur	182	48	28	258	352	74	72	498	251	59	46	356
Katihar	96	30	23	150	216	71	73	359	144	47	43	234
Khagaria	100	52	36	188	594	144	197	935	259	82	88	429
Madhupura	59	11	8	79	145	42	53	240	89	23	24	136
Purnea	139	33	27	200	349	95	123	567	210	54	60	323
Saharsa	44	12	6	62	151	42	53	246	83	23	23	128
Samastipur	99	39	24	162	301	87	66	454	171	56	38	266
Supaul	26	7	3	36	55	18	21	94	37	11	10	59
Darbhanga	79	28	14	122	148	51	28	228	110	39	20	169
Muzafarpur	121	38	22	180	209	59	44	312	159	47	31	237
Sheohar	54	20	18	92	115	40	22	178	76	27	19	122
Sitamarhi	116	28	13	158	165	45	26	237	137	36	19	191
E.Champaran	123	29	13	166	213	57	42	312	155	39	24	218
Madhubani	31	11	4	46	73	27	15	116	47	17	8	72
<i>NKBR</i>												
Jehnabad	293	119	29	441	195	72	19	286	203	73	22	299
Aurangabad	100	25	8	132	133	33	11	177	114	28	9	151
Banka	47	11	6	65	201	36	40	277	84	17	14	116
Bhabhua	84	28	7	120	117	29	10	155	99	28	9	136
Bhojpur	171	37	20	228	196	53	15	264	183	45	18	245
Buxar	140	37	11	188	114	33	9	156	125	34	10	170
Champaran(West)	170	36	18	224	185	61	39	284	176	47	27	250
Gaya	147	42	15	204	269	67	32	368	191	50	22	263
Gopalganj	74	16	5	95	100	34	13	146	86	24	8	118
Jamui	129	34	13	176	692	139	124	954	234	54	33	320
Kishanganj	35	11	9	55	80	22	23	125	53	16	14	83
Lakhisarai	64	30	9	103	114	42	22	178	88	36	16	140
Monghyr	61	26	11	98	154	47	32	233	95	33	19	147
Nalanda	169	50	25	244	201	52	27	281	184	49	25	258
Nawadha	82	19	6	108	110	20	9	139	95	20	7	122
Patna	167	34	4	205	122	24	5	151	145	29	4	178
Rohtas	117	32	12	161	150	35	12	197	132	33	12	178
Saran	96	28	14	139	144	44	24	212	119	36	19	174
Sheikhpura	145	28	11	185	135	29	12	175	136	27	11	175
Siwan	66	19	6	91	123	43	18	184	91	29	11	132
Vaishali	159	65	39	263	374	138	78	590	248	95	55	397

Source : The Fertilizer Association of India

Table A6.5.b: Fertilizer Ratio across agriculture season (2010TE)

Table 10.3.3: Fertilizer Ratio across agriculture season (2010-12)									
Districts	Kharif			Rabi			Total		
KBR									
Araria	4	1	1	2	1	1	3	1	1
Begusarai	4	2	1	4	1	1	4	1	1
Bhagalpur	6	2	1	5	1	1	5	1	1
Katihar	4	1	1	3	1	1	3	1	1
Khagaria	3	1	1	3	1	1	3	1	1
Madhupura	7	1	1	3	1	1	4	1	1
Purnea	5	1	1	3	1	1	4	1	1
Saharsa	8	2	1	3	1	1	4	1	1
Samastipur	4	2	1	5	1	1	4	1	1
Supaul	8	2	1	3	1	1	4	1	1
Darbhanga	6	2	1	5	2	1	5	2	1
Muzafarpur	6	2	1	5	1	1	5	1	1
Sheohar	3	1	1	5	2	1	4	1	1
Sitamarhi	9	2	1	6	2	1	7	2	1
E.Champaran	10	2	1	5	1	1	7	2	1
Madhubani	8	3	1	5	2	1	6	2	1
NKBR									
Jehnnabad	10	4	1	10	4	1	9	3	1
Aurangabad	13	3	1	12	3	1	12	3	1
Banka	7	2	1	5	1	1	6	1	1
Bhabhua	11	4	1	12	3	1	11	3	1
Bhojpur	9	2	1	13	3	1	10	3	1
Buxar	13	3	1	13	4	1	13	3	1
Champaran(West)	9	2	1	5	2	1	6	2	1
Gaya	10	3	1	9	2	1	9	2	1
Gopalganj	16	3	1	8	3	1	10	3	1
Jamui	10	3	1	6	1	1	7	2	1
Kishanganj	4	1	1	4	1	1	4	1	1
Lakhisarai	7	3	1	5	2	1	6	2	1
Monghyr	6	2	1	5	1	1	5	2	1
Nalanda	7	2	1	7	2	1	7	2	1
Nawadha	14	3	1	13	2	1	13	3	1
Patna	46	9	1	26	5	1	35	7	1
Rohtas	10	3	1	13	3	1	11	3	1
Saran	7	2	1	6	2	1	6	2	1
Sheikhpura	13	3	1	11	2	1	12	2	1
Siwan	12	3	1	7	2	1	8	3	1
Vaishali	4	2	1	5	2	1	5	2	1

Source: The Fertilizer Association of India

Table A6.6: District Wise Procurement Summary (2015) (PACS and Vyapar Mandal)

District	Societies	Farmers	Paddy from Farmers	Paddy to Rice Mill	CMR from Rice Mill	SFC to CMR
	No.	000 No.		000 Tonnes		
LOW						
Araria	189	3.6	27.4	22.2	14.2	14.4
Begusarai	179	3.5	15.5	9.9	6.1	5.5
Bhagalpur	99	4.6	23.4	15.9	9.2	9.2
Katihar	191	3.4	27.0	25.6	19.7	17.0
Khagaria	136	2.3	13.7	10.8	7.1	12.8
Madhepura	172	5.0	27.1	20.0	13.5	13.5
Purnia	235	5.4	35.1	22.1	16.2	16.8
Saharsa	150	4.6	30.2	14.7	9.3	9.6
Samastipur	347	6.5	40.2	35.6	31.9	17.8
Supaul	142	8.1	51.7	49.6	33.3	33.6
MID						
Darbhangha	310	4.7	27.5	19.6	12.0	12.0
Muzaffarpur	371	5.7	31.6	16.3	9.3	13.1
Sheohar	56	2.7	17.1	13.1	9.5	9.6
Sitamarhi	243	3.5	14.8	7.0	4.8	5.0
UPPER						
E Champaran	405	10.7	59.5	53.3	35.1	35.7
Madhubani	392	7.1	40.7	38.5	13.8	13.7
NKBR						
Arwal	68	5.1	37.4	37.3	24.3	24.4
Aurangabad	202	15.4	118.8	96.9	52.6	58.5
Banka	176	9.0	56.6	26.1	18.5	22.3
Bhojpur	202	8.0	55.6	36.3	21.8	20.1
Buxar	122	3.8	30.7	28.8	56.8	18.6
Gaya	340	18.5	126.0	105.9	68.1	76.6
Gopalganj	190	3.7	25.8	22.4	16.0	13.3
Jamui	162	7.1	45.8	40.9	26.7	27.3
Jehanabad	96	5.4	31.7	29.3	18.0	17.8
Kaimur	149	13.3	124.5	104.1	70.8	71.5
Kishanganj	126	3.8	21.9	19.8	17.4	17.6
Lakhisarai	50	3.8	27.9	17.7	11.0	11.6
Munger	66	4.1	20.6	15.9	9.2	9.8
Nalanda	282	25.2	160.5	169.0	79.8	74.1
Nawada	197	15.9	125.3	125.1	83.8	83.4
Patna	340	16.5	93.8	59.8	33.6	36.4
Rohtas	230	13.0	105.4	83.2	56.6	66.5
Saran	246	4.1	27.5	27.4	17.3	17.9
Sheikhpura	52	4.2	32.1	28.9	17.6	17.7
Siwan	301	4.1	19.1	13.6	10.4	10.0
Vaishali	201	1.7	8.0	7.2	4.7	4.6
W Champaran	303	8.3	51.5	46.7	32.1	32.6

Source; GoB-CD (Website)

Table A6.7: Milk Production scenario in Koshi Basin and Bihar

Region	Production	Production	Cross- bred	Indigenous	Buffalo	Others	
	2005-06	2013-14		2013-14			
	'000 Tonnes			Share (%)			
	LOW						
Araria	35.8	142.5	2.2	52.9	39.3	5.6	
Begusarai	244.9	276.8	82.4	2.6	13.9	1.1	
Bhagalpur	211.2	251.5	23.6	40.5	32.6	3.3	
Katihar	21.3	163.8	2.8	75.3	17.8	4.1	
Khagaria	24.7	186.5	49.5	18.5	30.3	1.7	
Madhepura	55.2	143.5	6.4	41.3	47.4	4.9	
Purnia	51.2	154.9	2.9	62.5	29.7	4.9	
Saharsa	91.7	178.8	8.5	43.5	44.2	3.9	
Samastipur	281.0	342.1	56.3	13.5	28.9	1.3	
Supaul	109.5	203.2	0.9	49.0	46.6	3.5	
MID							
Darbhanga	182.9	218.4	8.5	37.1	52.8	1.6	
Muzaffarpur	232.4	250.2	27.8	24.7	44.8	2.8	
Sheohar	5.2	28.6	5.3	30.6	58.8	5.2	
Sitamarhi	61.4	125.7	9.2	14.7	70.2	5.9	
UPPER							
E Champaran	72.5	210.2	4.0	42.6	48.9	4.5	
Madhubani	198.4	213.2	4.4	36.1	57.0	2.4	
NKBR							
Arwal	4.8	53.2	12.3	27.5	58.4	1.8	
Aurangabad	39.5	215.8	5.9	48.1	43.2	2.8	
Banka	250.0	184.1	6.2	59.9	30.6	3.3	
Bhojpur	330.4	307.2	29.4	21.9	47.5	1.2	
Buxar	238.8	205.8	24.1	23.4	51.9	0.6	
Gaya	51.0	369.7	7.1	48.2	41.6	3.1	
Gopalganj	27.5	150.5	22.7	26.0	49.1	2.2	
Jamui	154.5	210.7	9.5	62.5	24.8	3.2	
Jehanabad	10.9	77.3	10.2	23.3	65.0	1.5	
Kaimur	184.5	158.0	1.6	34.1	63.7	0.7	
Kishanganj	17.9	117.4	2.0	75.7	15.7	6.6	
Lakhisarai	72.4	76.4	34.7	27.2	36.5	1.5	
Munger	86.6	118.3	34.8	36.2	25.6	3.5	
Nalanda	90.8	203.4	15.0	19.8	63.2	2.0	
Nawada	41.3	163.9	4.4	50.1	41.6	3.9	
Patna	518.4	397.4	37.3	13.4	48.0	1.3	
Rohtas	259.5	259.9	13.0	26.3	58.6	2.1	
Saran	214.5	250.3	34.7	23.5	39.8	2.0	
Shekhpura	40.6	40.7	13.3	36.5	47.9	2.3	
Siwan	176.7	155.1	13.0	40.8	45.0	1.2	
Vaishali	101.9	210.0	58.4	9.7	29.1	2.8	
W. Champaran	56.1	182.0	6.2	44.5	43.3	6.0	

Source:

Table A6.8: Animal scenario in Koshi Basin and Bihar

LOW						
	Cattle	Buffalo 000 No	Total	Cattle	Buffalo Female (%)	Total
Araria	444.7	150.3	595.0	56.2	60.3	57.3
Begusarai	365.9	17.1	382.9	81.6	93.2	82.1
Bhagalpur	102.4	19.6	122.0	88.0	96.8	89.4
Katihar	989.8	119.6	1109.4	63.1	84.9	65.4
Khagaria	263.0	85.4	348.3	86.0	87.5	86.4
Madhepura	363.4	104.6	468.0	69.9	80.3	72.2
Purnia	460.8	176.8	637.6	72.8	76.4	73.8
Saharsa	173.2	96.3	269.5	65.5	81.5	71.2
Samastipur	476.0	184.4	660.4	84.4	85.0	84.5
Supaul	202.0	97.7	299.7	85.5	87.5	86.2
MID						
Darbhanga	342.9	128.3	471.2	85.8	87.3	86.2
Muzaffarpur	392.2	141.2	533.4	90.3	85.4	89.0
Sheohar	48.3	21.2	69.5	84.6	64.7	78.5
Sitamarhi	402.5	599.2	1001.7	64.9	59.2	61.5
UPPER						
E Champaran	606.4	573.0	1179.3	69.1	61.2	65.3
Madhubani	453.9	105.1	559.0	48.0	54.4	49.2
NKBR						
Arwal	90.6	31.6	122.2	55.9	60.9	57.2
Aurangabad	476.6	298.1	774.7	83.7	77.4	81.3
Banka	690.1	219.6	909.7	52.4	67.9	56.1
Bhojpur	189.7	150.7	340.4	89.8	91.8	90.7
Buxar	219.9	83.1	303.0	85.9	84.5	85.5
Gaya	739.2	405.3	1144.5	51.9	72.1	59.1
Gopalganj	324.6	86.0	410.6	81.1	91.4	83.3
Jahanabad	105.7	80.9	186.6	89.8	83.2	87.0
Jamui	772.8	549.1	1321.9	62.8	51.1	58.0
Kaimur	445.7	309.2	754.9	71.4	73.6	72.3
Kishanganj	500.6	154.6	655.1	68.5	78.3	70.8
Lakhisarai	159.9	25.5	185.4	79.9	79.7	79.9
Munger	122.4	25.3	147.7	73.7	76.4	74.1
Nalanda	232.9	106.4	339.2	73.7	72.9	73.4
Nawadah	337.2	102.0	439.2	62.6	82.6	67.2
Patna	390.4	592.4	982.7	83.9	60.4	69.7
Rohtas	627.2	374.9	1002.1	78.0	77.2	77.7
Saran	294.5	97.7	392.2	92.4	87.5	91.2
Shekhpura	138.5	28.5	167.0	78.6	71.4	77.4
Siwan	320.4	87.8	408.2	69.8	95.4	75.3
Vaishali	692.8	174.6	867.4	86.5	89.6	87.1
West. Champaran	403.7	212.1	615.9	57.9	86.9	67.9

Source: Agriculture Census (Website)

Appendix 7

Table A7.1 : Selected Model Estimates

Kharif Rice			Kharif maize			Rabi wheat			Rabi Maize		
Variable	Coeffi	Z-stat	Variable	Coeffi	Z-stat	Variable	Coeffi	Z-stat	Variable	Coeffi	Z-stat
Rc_Kh_rev	0.1	58.4	Mz_Kh_RevXRoad	0.3	1.8	Wh_Rb_Rev	0.1	61.3	Mz_Rb_Rev	0.1	31.8
Mz_Kh_Price	-182	-5.6	Mz_Kh_RevXElec	0.0	7.3	Rc_Rb_Price	-126	-5.3	Wh_Rb_Price	-68.2	-2.1
Urad_Kh_Price	-4.0	-2.1	Rc_Kh_RevXElec	0.0	-3.5	Mz_Rb_Rev	0.0	2.7	Mz_Rb_Rev*Elec	0.0	-1.2
Rf_Jul	-0.2	-2.0	RF_Sep	-0.7	-2.2	Rf_Sep	-0.2	-2.2	Rf_Nov	-16.5	-4.0
Rf_Oct	0.6	3.1	Temp_Jul	456.5	2.9	Rf_Dec	-13.8	-1.4	RF_Feb	6.0	2.1
Rf_OctXRf_Oct	0.0	-3.3				Rf_Jan	-2.3	-2.1	Temp_Oct	-228.3	-1.7
Temp_Jun	-58.0	-3.3				Rf_Mar	2.9	2.1	Temp_Nov	317.1	2.6
Temp_Jul	73.4	3.3				Temp_Oct	291.2	5.7	Temp_Mar	301.6	1.8
Temp_Aug	-236	-2.8				Temp_Apr	59.1	1.1			
Temp_Sep	106.4	1.7	Irr*RF_Aug	-0.6	-1.9	Irr*Rf_Aug	0.2	1.8			
IrrXRF_Jul	0.2	1.7									
IrrXRF_Aug	-1.6	-2.7									
Spatial interlinkages											
Rf_Jun_Jul_US	-2.3	-7.3	Rf_Extr_HT	-2.8	-1.7	Rf_Jul_MS	-0.5	-2.7	Rf_Jun_US	1.1	5.4
Rf_Jul_MS	4.2	6.8	Irr*RF_June_US	3.6	3.8	Irr*Rf_Jun_US	0.5	2.4	Irr*Rf_Aug_US	0.3	2.1
Rf_Aug_MS	-6.3	-1.6	Irr*RF_Jul_MS	-2.0	-3.6	Irr*Rf_Aug_MS	-0.4	-1.8	IRR*Rf_Sep_US	0.9	2.3
Rf_Extr_HT	2.3	4.8	Irr*RF_Aug_MS	1.7	2.2				IRR*Rf_Jul_MS	-0.8	-2.1
IrrXRF_Apr_US	14.8	5.3							IRR*Rf_Sep_MS	-0.5	-1.2
IrrXRF_Jul_US	0.1	1.4							IRR*Rf_Oct_MS	-1.7	-2.5
IrrXRF_Mar_MS	-10.9	-2.0									
IrrXRf_Oct_MS	-0.2	-1.3									
IrrXRf_Extr_HT	-0.7	-2.1									
R ² (Within)	0.73		R ² (Within)	0.97		R ² (Within)	0.91		R ² (Within)	0.97	
Harris-Tzavalis	-12.9	0.0		-8.8			-10.9			-13.9	

Table A7.2: t-statistics of non-spatial models of crop yield

Maize Rabi	OLS	RDD	PFE	Wheat Rabi	OLS	RDD	PFE	Rice Kharif	OLS	RDD	PFE	Maize Kharif	OLS	RDD	PFE
Mz_Rev	34.3	22.3	29.0	Wh_Rev	28.6	27.7	26.9	Rc_Rev	40.0	38.1	41.7	Mz_Rev_Road	1.7	3.4	1.9
Wh_Rev	-3.0	-1.9	-2.1	Rc_Price	-9.8	-11.6	-11.2	Mz_Price	-5.7	-5.0	-11.3	Mz_RevXElec	6.5	7.6	5.6
Rc_Price	-1.5	-2.0	-2.0	Mz_Rev	-1.0	0.9	1.4	Urad_Price	-2.5	-2.8	-6.6	Rc_RevXElec	-3.9	-4.4	-4.1
Mz_RevXElec	-2.3	0.0	0.0	Rf_Aug	-1.7	-1.3	-1.3	Rf_Jul	-3.0	-3.4	-4.4	Rf_Jun	2.4	1.6	1.7
Mz_RevXRoad	-2.0	-3.4	-2.0	Rf_Oct	4.4	2.7	2.3	Rf_Sep	2.2	2.0	2.5	Rf_Jul	-1.4	-1.7	-1.7
Wh_RevXRoad	3.3	2.5	2.9	Rf_Dec	-3.2	-4.2	-6.4	Temp_Jun	-8.2	-5.6	-10.1	Rf_Aug	-2.6	0.7	0.8
Rf_Jun	2.9	2.1	2.0	Rf_Jan	-1.2	0.2	0.3	Temp_Jul	6.2	6.4	8.0	Rf_Sep	1.7	0.8	0.9
Rf_Sep	-4.5	-4.5	-3.8	Rf_Mar	1.9	2.8	2.6	IrrXRf_Jul	2.9	3.0	3.3	Rf_SepXRf_Sep	-2.6	-1.9	-2.0
Rf_Nov	-3.1	-2.4	-2.5	Temp_Oct	6.8	3.5	3.5	IrrXRf_Aug	-2.1	-2.2	-2.6	Temp_Jun	-5.0	-2.4	-3.3
Rf_Feb	-5.8	-5.9	-6.6	Temp_Apr	5.0	6.9	8.0					Temp_Jul	3.3	4.5	5.9
Rf_Mar	3.2	2.8	3.5	IrrXRf_Jun	3.0	1.8	1.4					IrrXRf_Aug	3.4	-0.9	-0.9
Temp_Oct	-3.9	-3.4	-3.2	IrrXRf_Sep	1.6	-1.3	-2.0								
Temp_Nov	5.5	4.7	4.4												
Temp_Feb	5.0	5.0	4.5												
Temp_Apr	3.2	2.7	2.7												
IrrXRf_Aug	2.5	2.1	2.4												
Const	-7.1	-4.2	-4.1		-13.3		-16.1		-0.1	0.0	0.0		0.8		-2.0
Max_ Min_ R^2		Araria Bhagal. 0.98				Araria Samast. 0.94	0.92				East Champ. Samast. 0.94			Madhe. Sheoh. 0.66	
			0.98	R^2	0.9				0.93		0.91	R^2			0.72

Table A7.3: Moran coefficients (MCG) for autocorrelation

	Year	Actual Yield		Residual Yield (FE)		Residual Yield (Spatial)	
		Moran -G	P-Value	Moran I	P-Value	Moran I	P-Value
Kharif Rice	2007-08	0.5781	0.01	0.3313	0.16	0.2411	0.28
	2005-16	0.4276	0.02	0.2688	0.24	0.3641	0.07
	2000-01	0.086	0.27	0.0815	0.19	0.0456	0.20
Wheat	2007-08	0.4028	0.03	0.5985	0.01	0.4523	0.02
	2005-16	0.5123	0.02	0.5475	0.01	0.3065	0.11
	2000-01	0.0618	0.22	0.5423	0.02	0.2340	0.30
Kharif Maize	2007-08	0.5933	0.01	0.5691	0.01	0.3687	0.08
	2005-16	0.4074	0.02	0.5484	0.01	0.3603	0.09
	2000-01	0.1964	0.46	0.3971	0.05	0.1457	0.47
Rabi Maize	2007-08	0.1757	0.40	0.2912	0.18	0.6012	0.01
	2005-16	0.1446	0.39	0.2435	0.21	0.3065	0.09
	2000-01	0.1913	0.46	0.1576	0.45	0.1983	0.41

Notes: The P-value tests the hypothesis of no spatial autocorrelation.

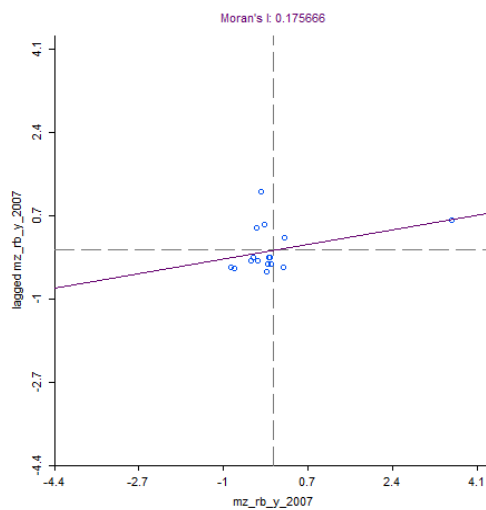
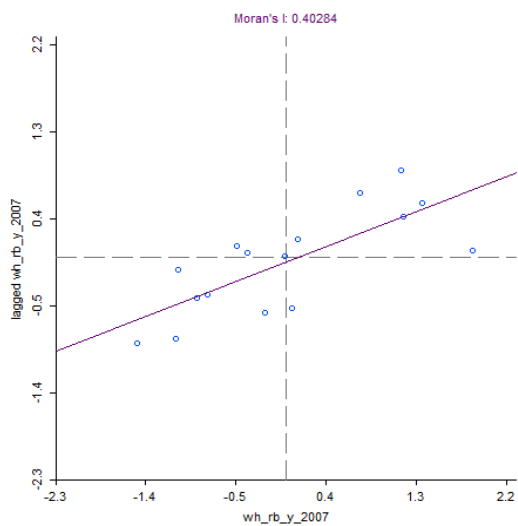
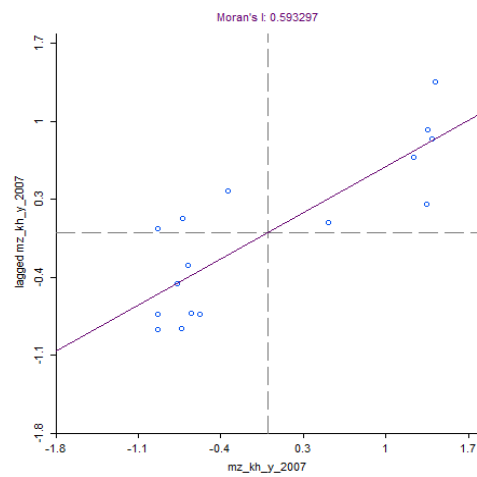
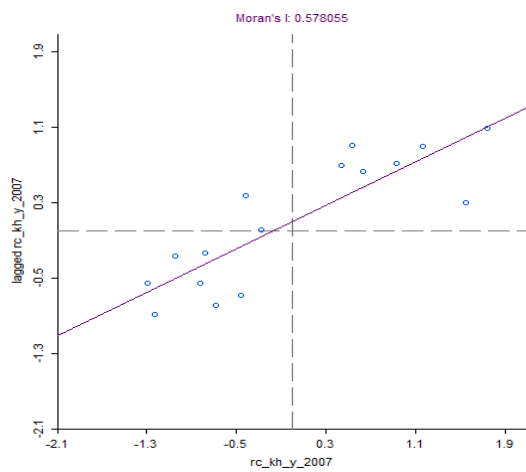


Figure A7.1: Scatter diagram for yield

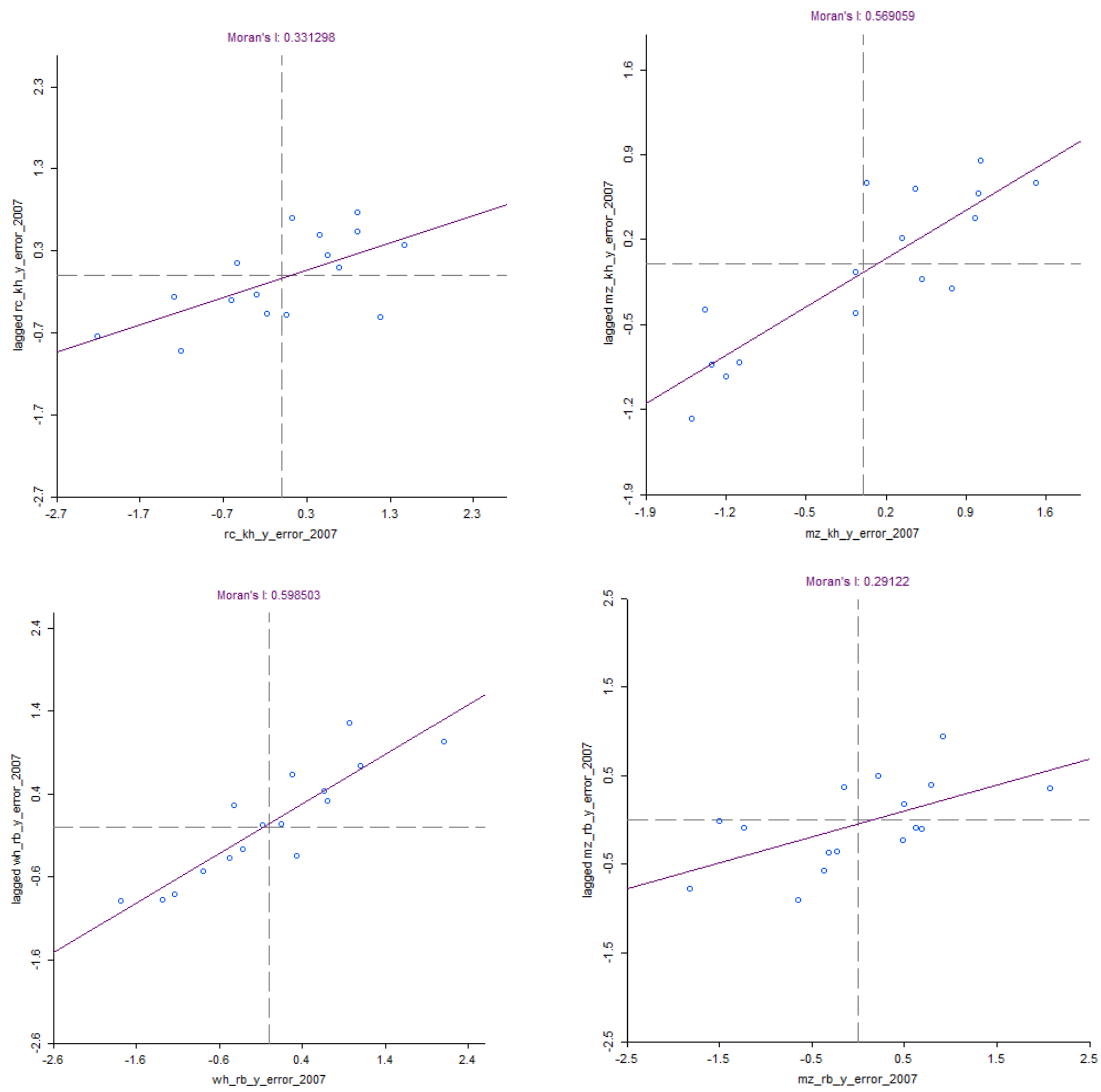


Figure A7.2: Scatter diagram for residuals of yield

Appendix 8

Appendix A8.1 Tables

Table A8.1: Sample averages of Area, Yield and Price of major crops grown only by Crop growing farmer households by size class

Farm class		Marginal	Small	Medium	Large	Total
Sample 1 (Khagaria)						
Rice (Paddy)*	Area (Hectare)	0.46	1.16	1.47	2.35	1.21
	Yield (Tonnes/hectare)	2.48	2.61	2.68	2.01	2.51
	Price (Rs./Kg)	10.75	11.15	10.91	10.56	10.85
Wheat	Area (Hectare)	0.32	0.62	1.04	1.80	0.86
	Yield (Tonnes/hectare)	2.66	2.45	2.41	2.33	2.47
	Producer price (Rs./Kg)	13.41	13.18	12.05	12.78	12.80
Maize	Area (Hectare)	0.54	1.04	1.54	3.05	1.34
	Yield (Tonnes/hectare)	7.24	5.95	6.36	8.24	6.81
	Producer price (Rs./Kg)	10.37	10.18	9.95	9.92	10.12
Sample 2 (Saharsa)						
Rice	Area (Hectare)	0.50	0.86	1.97	1.50	0.78
	Yield (Tonnes/hectare)	2.27	1.45	1.59	2.21	2.04
	Producer price (Rs./Kg)	10.27	9.52	9.21	9.39	9.97
Wheat	Area (Hectare)	0.38	0.51	0.82	1.00	0.49
	Yield ((Tonnes/hectare)	2.01	1.69	1.91	1.96	1.95
	Producer price (Rs./Kg)	12.11	11.89	11.91	11.90	12.04
Maize	Area (Hect.)	0.26	0.45	1.19	2.23	0.55
	Yield (Tonnes/hectare)	5.77	5.26	5.31	5.61	5.61
	Producer price (Rs./Kg)	9.56	9.30	9.48	8.89	9.46

Note: From secondary data of Bihar the average yield in Tonnes per hectare in triennium 2014-15 of Rice, Wheat and Maize are 1.996, 2.212 and 3.269 respectively. Conversion: 1.5Kg Paddy equivalent to 1Kg Rice. Source: Computed from sample data, MoA (Website).

Table A8.2(a): Fertilizer consumption by farm-class

	Rice	Wheat	Maize	(Gross)	(Net)
	KG/hectare				
Marginal	123	140	216	176	267
Small	172	193	280	231	335
Medium	263	243	314	281	340
Large	331	232	366	319	331
Khagaria	211	202	284	244	295
Marginal	97	126	118	111	239
Small	111	179	161	141	341
Medium	156	216	248	193	332
Large	110	157	224	178	176
Saharsa	107	148	150	130	238
Bihar (secondary)*	99	145	138	-	-

Note:* denotes taken from Cost of Cultivation data 2013-14 figure Source: Survey data and MoA (Website)

Table A8.2 (b) Labour hiring for cultivation by Sample crop growing households

	Rice	Wheat	Maize	Rice	Wheat	Maize
	% Hired in total farm labourers			Wage Rate in Rs/person days		
Khagaria						
Marginal	49.0	50.4	46.9	176	204	203
Small	61.5	56.4	56.0	184	192	188
Medium	72.4	59.1	63.2	181	188	188
Large	75.3	71.2	74.8	184	191	191
Sample 1	65.2	57.8	58.6	180	194	193
Saharsa						
Marginal	79.5	42.8	74.9	149	154	142
Small	52.0	47.5	50.8	154	150	156
Medium	58.5	55.4	60.9	179	179	179
Large	60.3	37.6	60.5	150	155	155
Sample 2	73.7	45.2	69.2	154	156	150
Bihar (Secondary)*	-	-	-	156	162	159

Note:* denotes taken from Cost of Cultivation data 2013-14 figure. Source: Compute from Survey data, MoA (Website)

Table A8.2(c) Cost of cultivation of Sample households and Labour share

	Rice Hired Labour Cost in Rs. '000/hectare	Wheat Cost in Rs. '000/hectare	Maize Cost in Rs. '000/hectare	Rice Other Cost in Rs. '000/hectare	Wheat Other Cost in Rs. '000/hectare	Maize Other Cost in Rs. '000/hectare	Rice Total Cost in Rs. '000/Hectare	Wheat Total Cost in Rs. '000/Hectare	Maize Total Cost in Rs. '000/Hectare	Rice Labour share in Total cost (%)	Wheat Labour share in Total cost (%)	Maize Labour share in Total cost (%)
Khagaria												
Marginal	5.8	9.2	8.6	10.6	25.3	21.9	16.4	34.5	30.4	36	27	28
Small	10.9	8.7	9.4	14.2	26.4	32.3	25.1	35.1	41.7	43	25	23
Medium	13.4	7.3	10.9	13.2	23.9	30.9	26.6	31.2	41.8	50	23	26
Large	14.6	11.3	14.7	18.4	26.0	35.3	33.1	37.4	50.0	44	30	29
Sample 1	10.7	8.8	10.4	13.3	25.2	29.1	24.0	34.0	39.5	44	26	26
Saharsa												
Marginal	28.8	5.6	20.8	11.2	18.0	21.0	40.0	23.6	41.7	72	24	50
Small	12.3	7.9	10.9	10.8	29.8	19.1	23.1	37.7	29.9	53	21	36
Medium	18.6	12.2	16.3	12.6	16.7	26.0	31.2	28.9	42.3	60	42	39
Large	17.5	8.3	14.4	7.0	19.3	23.2	24.4	27.6	37.5	72	30	38
Sample 2	24.5	6.9	18.3	11.1	19.9	21.4	35.6	26.8	39.7	69	26	46
Bihar (Secondary)*												
	8.2	3.5	5.8	25.2	28.8	25.9	33.4	32.3	31.7	25	11	18

Note: * denotes taken from Cost of Cultivation data 2013-14 figure Source: Compute from survey data

Table A8.2 (d): Per hectare Revenue, Cost and Income from Crops of Crop growing farmers and the weights of crops in the Total Sample of cultivating Households

Crops		Sample 1 (Khagaria)				Sample 2 (Saharsa)			
		Marginal	Small	Medium	Large	Marginal	Small	Medium	Large
Rice	Cost (Rs.000/Hectare)	16.40	25.06	26.64	33.07	40.02	23.08	31.25	24.40
	Income (Rs.000/Hectare)	10.26	4.05	2.59	-11.8	-16.7	-9.28	-16.6	-3.65
	Weight% (Hectare)*	23 (0.20)	22 (0.44)	24 (0.79)	18 (1.08)	48 (0.46)	50 (0.83)	52 (1.80)	27 (1.20)
Wheat	Cost (Rs.000/Hectare)	34.50	35.09	31.20	37.35	23.63	37.66	28.92	27.57
	Income (Rs.000/Hectare)	1.17	-2.80	-2.16	-7.57	0.71	17.56	-6.17	-4.25
	Weight% (Hectare)*	25 (0.22)	29 (0.59)	30 (0.97)	30 (1.80)	34 (0.33)	29 (0.48)	22 (0.75)	23 (1.00)
Maize	Cost (Rs.000/Hectare)	30.43	41.74	41.78	50.02	41.72	29.91	42.29	37.51
	Income (Rs.000/Hectare)	44.65	18.83	21.50	31.72	13.44	19.01	8.05	12.36
	Weight% (Hectare)*	52 (0.46)	49 (0.99)	46 (1.49)	51 (3.05)	18 (0.17)	22 (0.36)	26 (0.89)	50 (2.23)

Note: *The parenthesis figure are Average acreages of total sample of landed household. Source: Survey Data

Table A8.5: Access to food grain from market Tonnes per hectare under Maize

Table A6.5. Access to food grain from market founies per hectare under maize								
Farm	Sample 1 (Khagaria)				Sample 2 (Saharsa)			
Class	Physical in Tonnes	Relative To Yield	Physical in Tonnes	Relative To Yield	Physical in Tonnes	Relative To Yield	Physica l in Tonnes	Relative To Yield
	Access to Rice		Access to Wheat		Access to Rice		Access to Wheat	
At Public Distribution Price								
Marginal	14.88	6.00	22.32	8.39	4.48	1.97	6.72	3.34
Small	6.28	2.40	9.41	3.84	6.34	4.37	9.50	5.62
Medium	7.17	2.67	10.75	4.46	2.68	1.69	4.02	2.11
Large	10.57	5.26	15.86	6.81	4.12	1.86	6.18	3.15
Landed	9.79	3.90	14.69	5.95	4.46	2.19	6.69	3.43
At Retail Price (Annual High)								
Marginal	1.49	0.60	1.72	0.60	0.45	0.20	0.52	0.26
Small	0.63	0.24	0.72	0.24	0.63	0.44	0.73	0.43
Medium	0.72	0.27	0.83	0.27	0.27	0.17	0.31	0.16
Large	1.06	0.53	1.22	0.53	0.41	0.19	0.48	0.24
Landed	0.98	0.39	1.13	0.39	0.45	0.22	0.51	0.26

Source: Computed from survey data and MoA (Website)

Appendix A8.2: Heckman model

Let us assume that the farmer growing crop maximizes her utility (U) which is a function of her household food security (f), the cash returns (E) from sale to buy non-food commodities they can buy on cultivation and other relevant factors (compositely Z). Considering only food the utility is U_f

$$U_f = a_0 + a_1 f + a_2 (f)^2 + cZ \dots$$

And $f = F/FT$, F is the available food and FT is the total food requirement of the household based on taste and nutritional standards of the case. Food insecurity is denoted by $f < 1$ but it may be conjectured based on intuition that while utility increases with greater food security, as F approaches the total requirement F, or in other words as f increases towards one the gain in utility from each additional unit reduces. At the level where $f=1$ and food availed is equal to food required, utility maximizes and stagnates. U is a_0 which is

negative at f is zero and $a_1 > 0$ and $a_2 < 0$. Setting the derivative with respect to f to zero we get

$$f^* = -(a_1 / 2a_2) = 1$$

where f^* is utility maximizing level (optimum) F/FT . If we assume that F depends on the access to food from the market given at the per hectare level

$$F = \alpha A$$

Where α is access to food through market route

$$\alpha = (PY - C) / P_f \dots\dots 9.5$$

$$\text{optimum } A = A^* = -(a_1 / 2a_2) \cdot (FT / \alpha) = (FT / \alpha)$$

Which makes area (A) positively related to total household food needs but negatively too access per hectare (α).

Let us assume that the farmer growing a crop derives utility U_c also from cash returns (E) from sale of crops to buy non-food commodities (or save).

$$U_c = b_1 E + b_2 E^2$$

where

$$E = \alpha \cdot P_f \cdot A \dots\dots 9.9$$

In which E would be the food purchased expressed in monetary terms using either retail or the PDS price as relevant for the farmer. Coefficient $b_1 > 0$ signifying the value attached to cash earnings but the sign of b_2 is indeterminate and will be sensitive to the marginal utility of money as against the effort implied although it may be more reasonable to assume $b_2 > 0$ (as affluence improves additional earnings will be more valuable).

Maximizing total utility (U) as a average of U_f and U_c with equal weights (or unequal) optimum area (A^*) as

$$A^* = ((F_T)/\alpha) \cdot ((W_1/W_2) - (b_1 P_F F_T / 2a_2)) / (((W_1/W_2) + (2b_2 P_F F_T^2 / 2a_2))$$

Since W is non negative numerator is positive.

Denominator is positive is $b_2 < 0$ diminishing MU of cash.

Denominator is negative if b_2 positive and large and/or $b_2 > 0$ and $((W_1/W_2))$ is small relative to b_2/a_2 that is the weight attached to cash is much higher than weight attached to food security.

Appendix 9.1

Appendix Table

Table A9.1: Marketing channels adopted by primary producers (% Sellers)

	Type	Sample 1(Khagaria)			Sample 2(Saharsa)		
		Rice	Wheat	Maize Hybrid	Rice	Wheat	Maize Hybrid
Marginal	Landlord, Sahukar, Mahajan	0.0	0.0	8.3	4.4	0.0	3.7
	Village Trader	50.0	83.3	66.7	87.0	16.0	88.9
	Mandi	0.0	0.0	20.8	4.4	0.0	7.4
	Producer Company	0.0	0.0	0.0	4.4	0.0	0.0
	State Trader and PACS	0.0	0.0	0.0	0.0	0.0	0.0
	Others	50.0	16.7	4.2	0.0	0.0	0.0
Small	Landlord, Sahukar, Mahajan	0.0	25.0	20.0	0.0	0.0	0.0
	Village Trader	100	75.0	60.0	75.0	66.7	80.0
	Mandi	0.0	0.0	15.0	12.5	0.0	10.0
	Producer Company	0.0	0.0	5.0	12.5	0.0	0.0
	State Trader and PACS	0.0	0.0	0.0	0.0	0.0	0.0
	Others	0.0	0.0	0.0	0.0	33.3	10.0
Medium	Landlord, Sahukar, Mahajan	25.0	22.2	22.2	0.0	0.0	0.0
	Village Trader	62.5	61.1	44.4	44.4	50.0	66.7
	Mandi	12.5	0.0	33.3	44.4	0.0	33.3
	Producer Company	0.0	0.0	0.0	11.1	0.0	0.0
	State Trader and PACS	0.0	0.0	0.0	0.0	0.0	0.0
	Others	0.0	16.7	0.0	0.0	50.0	0.0
Large	Landlord, Sahukar, Mahajan	66.7	40.0	30.8	0.0 100.	0.0	0.0
	Village Trader	0.0	30.0	23.1	0	50.0	75.0
	Mandi	33.3	20.0	38.5	0.0	0.0	25.0
	Producer Company	0.0	0.0	0.0	0.0	0.0	0.0
	State Trader and PACS	0.0	0.0	0.0	0.0	0.0	0.0
	Others	0.0	10.0	7.7	0.0	50.0	0.0
All	Landlord, Sahukar, Mahajan	22.2	23.8	19.1	2.4	3.3	2.0
	Village Trader	61.1	59.5	51.2	76.2	76.7	82.0
	Mandi	11.1	4.8	26.2	14.3	16.7	14.0
	Producer Company	0.0	0.0	1.2	4.8	0.0	0.0
	State Trader and PACS	0.0	0.0	0.0	0.0	0.0	0.0
	Others	5.6	11.9	2.4	2.4	3.3	2.0

Source: Computed from sample data

Table A9.2 (a): Marketed surplus and Sources of food all total sample 1 (Khagaria)

	Household	Landless	Marginal	Small	Medium	Large	Sample 1
Rice	Production (Kg/Year)	0	337	778	1467	1313	735
	Marketed (%)	-	21.0	51.0	37.9	43.4	39.3
	Consumption (Kg/Year)	814	629	704	760	890	740
	Farm Produced (%)	0.0	34.0	50.8	47.4	37.3	33.2
	Work for Food (%)	14.2	0.0	0.0	0.0	0.0	3.1
	Public Distribution (%)	23.9	21.9	15.7	14.2	5.1	17.2
	Market (%)	62.1	44.1	33.7	38.1	57.5	46.5
	Surplus (%)	-	-86.8	9.5	48.2	32.2	-0.8
Wheat	Production (Kg/Year)	0	594	1340	2381	3912	1412
	Marketed (%)	-	20.4	22.8	46.7	58.0	43.1
	Consumption (Kg/Year)	844	698	845	872	1075	838
	Farm Produced (%)	0.0	52.9	81.1	86.9	98.8	62.4
	Work for Food (%)	14.8	0.0	0.0	0.0	0.0	3.0
	Public Distribution (%)	16.7	12.7	8.5	10.2	1.3	10.5
	Market (%)	68.5	34.4	10.3	3.0	0.0	24.0
	Surplus (%)	-	-17.6	37.0	63.4	72.5	40.7
Pulses	Production (Kg/Year)	0.0	0.0	0.0	0.0	0.0	0.0
	Marketed (%)	0.0	0.0	0.0	0.0	0.0	0.0
	Consumption (Kg/Year)	74.4	86.4	120	126	160.8	106.8
	Farm Produced (%)	0.0	0.0	7.0	0.0	0.0	1.4
	Market (%)	100	100	92.0	96.2	100	98.3
Per Capital Consumption							
Rice and Wheat (Kg/Year)		215.9	188.3	206.2	225.6	207.4	209.2
Pulses (Kg/Year)		9.7	12.3	16.0	17.4	17.0	14.2

Source: Computed from sample data

Table A9.2 (b): Marketed surplus and Sources of food all total sample 2 (Saharsa)

	Household	Landless	Marginal	Small	Medium	Large	Sample 2
Rice	Production (Kg/Year)	0	681	798	1817	1279	665
	Marketed (%)	-	33.3	39.5	5.9	28.2	40.4
	Consumption (Kg/Year)	545	618	635	876	812	635
	Farm Produced (%)	0.0	57.2	58.9	63.9	60.7	46.8
	Work for Food (%)	11.0	0.0	0.0	0.0	0.0	2.3
	Public Distribution (%)	31.1	24.2	19.6	9.7	21.6	23.0
	Market (%)	57.9	18.6	21.4	26.4	17.7	27.9
	Surplus (%)	-	9.3	20.4	51.8	36.5	4.5
Wheat	Production (Kg/Year)	0	608	816	1352	2014	615
	Marketed (%)	-	20.1	14.7	35.1	5.3	16.7
	Consumption (Kg)	991	635	642	826	845	747
	Farm Produced (%)	0.0	62.0	73.9	89.7	83.8	47.5
	Work for Food (%)	14.0	0.0	0.0	0.0	0.0	4.4
	Public Distribution (%)	11.5	15.3	13.0	7.5	13.4	13.0
	Market (%)	74.4	22.7	13.1	2.8	2.8	35.1
	Surplus (%)	-	-4.4	21.4	38.9	58.1	-21.5
Pulses	Production (Kg/Year)	0	14.1	15.7	45.6	55.8	20.4
	Marketed (%)	-	0.0	0.0	0.0	0.0	0.0
	Consumption (Kg)	96.7	107.3	138.4	188.0	189.6	119.4
	Farm Produced (%)	0.0	25.5	37.6	52.6	55.7	28.2
	Market (%)	100	73.0	60.1	46.3	44.3	70.7
Per Capital Consumption							
Rice and Wheat (Kg/Year)		223.5	194.5	175.6	212.7	165.7	199.4
Pulses (Kg/Year)		14.1	16.7	19.0	23.5	19.0	17.2

Source: Computed from sample data

Table A9.3 (c): Households holding Ration Card (%) for food in PDS by types of eligibility cards (%) in total Samples

Farm Class	Sample 1 (Khagaria)					Sample 2 (Saharsa)				
	Ration Card	APL	BPL	Types of Card Anatyo daya	Others	Ration Card	APL	BPL	Types of Card Anatyo daya	Others
Landless	88	24	71	5	0	83	20	4	0	76
Marginal	94	50	43	7	0	84	15	15	2	69
Small	76	63	38	0	0	73	9	0	0	91
Medium	77	78	22	0	0	75	33	22	0	44
Large	85	100	0	0	0	60	0	0	0	100
Sample	84	58	39	3	0	81	17	11	1	72

Note: See Chapter 4 for specifications of eligible categories. Source: Calculated from survey data.

Source: Computed from sample data

Table A 9.4: Animal husbandry and Milk production in sample households

	Animal				Milk				
	Possessed %	No	Value (Rs.000)	Cost Rs.000/ Month	Yield	Production Litre/ Day	Price	Income Rs.000 /Year	Home Consumption %
Khagaria									
Landless	37.50	1.30	28.89	2.50	4.4	6.0	28	6.29	58.5
Marginal	59.40	1.60	34.53	2.72	5.7	8.2	30	20.44	41.2
Small	76.20	2.60	73.25	2.30	4.7	11.6	30	47.80	39.1
Medium	73.30	3.00	70.18	2.15	4.2	11.0	30	45.23	37.5
Large	92.30	2.20	48.28	2.24	4.9	10.5	31	43.21	46.0
Sample 1	65.00	2.20	53.99	2.37	4.8	9.9	30	35.91	42.0
Saharsa									
Landless	30.00	1.60	14.82	2.09	1.3	1.7	28	-14.71	100.0
Marginal	78.10	1.80	26.63	2.20	2.6	4.0	28	-2.38	57.5
Small	93.30	2.60	37.86	2.30	2.9	6.2	30	12.69	47.9
Medium	83.30	2.60	39.59	2.43	2.6	6.2	28	8.00	52.2
Large	80.00	3.00	43.25	2.08	3.6	8.8	31	33.69	41.5
Sample 2	69.00	2.10	29.47	2.23	2.6	4.6	28	0.93	57.5

Note: Assumed milk production for 27 days in a month and 8 months in a year. Feed cost covers fodder and medicine including artificial insemination. Source: Computed from Survey data

Table 9.5 (a): Time (Hours) spent on different chores by Women with overlaps

	Cooking, Children Caring	Fetching Water, Making dung Cake	Agriculture work, Caring Pets	Work on Other Farm	Non-Agriculture Work
Khagaria					
Landless	6.00	2.70	8.80	5.10	3.00
Marginal	5.70	2.80	7.00	5.20	4.00
Small	5.40	3.00	7.40	3.50	3.40
Medium	6.10	2.90	6.20	4.80	3.80
Large	6.40	2.50	7.60		3.30
Sample 1	5.90	2.80	7.20	4.90	3.60
Saharsa					
Landless	6.60	2.70	8.80	4.50	3.60
Marginal	5.90	2.90	7.00	4.50	2.90
Small	5.50	2.40	7.40	3.50	3.10
Medium	5.70	1.90	6.20		3.80
Large	5.60	2.00	7.60	5.00	2.70
Sample 2	5.90	2.80	7.20	4.50	3.10

Source: Computed from sample data

Table 9.5 (b): Decision making of Women in farm households as Full or Partial Participation in various matters (%)

Farm Class	Landless	Marginal	Small	Medium	Large	Sample	Full Participation
Khagaria							
Variety (crops, animals)	8.3	25.0	23.8	23.3	15.4	20.0	1.7
Machine Purchase	0.0	3.1	0.0	6.7	0.0	2.5	0.0
Technology Adoption	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Hiring Farm Labour	20.8	18.8	28.6	30.0	31.0	25.0	3.3
Selling Product	0.0	21.9	42.9	53.3	30.8	30.0	2.5
Livestock Feed	58.3	53.1	57.1	73.3	53.8	60.0	17.5
Land Leased	0.0	15.6	15.0	6.7	0.0	8.4	1.7
Household Expenditure	87.5	71.8	80.0	96.7	92.3	84.9	29.4
Saharsa							
Variety	23.3	7.9	0.0	41.7	0.0	26.4	12.8
Machine Purchase	6.7	7.9	0.0	41.7	0.0	26.4	0.8
Technology Adoption	23.3	7.9	0.0	41.7	0.0	26.4	3.2
Hiring Farm Labour	33.3	7.9	0.0	41.7	0.0	26.4	19.8
Selling Product	20.0	7.9	0.0	41.7	0.0	26.4	13.5
Livestock Feed	50.0	7.9	0.0	41.7	0.0	26.4	54.8
Land Leased	23.3	7.9	0.0	41.7	0.0	26.4	3.2
Household Expenditure	93.3	7.9	0.0	41.7	0.0	26.4	88.9

Source: Computed from sample data

Appendix 9.2

Village Profile

Village*	Saidpur District: Khagaria Area: 486 Hactare.
Household Number	Total: 1580 and Cultivator: 720.
Nonfarm employment inside the village	Mason, Agriculture labour, Fishery, MANREGA, Shop, Vegetable Sale.
Employment outside the village	Labour, Rickshaw puller, Loading, Tempo
Migration Targets	Delhi, Rajasthan, Gujarat, UP
Minority	10 %
Nearest town	Khagaria which is 15 km away from village and mode of transportation is Auto and Bus.
Name of district Head quarter	Khagaria which is 15 km away from village.
Accessible by road	Yes by Paved Road but not usable during Monsoon and this Paved Road is 0.5 km away from village.
Does this village have electricity	Yes, 90% Household , 12 hours a day
Does village have phone service	Yes about 90% Household.
Source of drinking water	Hand pump but Bad quality
Source of cooking fuels	Gas, Crop residue
Facilities in village	PDS shop, Input Market, Grocery shop,Bank,cooperative,Anganvadi, Post office, Primary school, Middle school, Higher Secondary School, Madarsa, Girls School, Community Health Center, Untrained doctor clinic, Self Help Group, Storage (private use).
Water logged area	0.82 % area
Major crops grown in the village	Maize, Wheat, Flower, Vegetable, Paddy
Floods in last five years	2011,2012,2013,2014 (Due to Kosi Erosion) 50-80% Area affected. Impacts: Crop, Cattle, House damage and Income loss.
Droughts in last five year	2013, 2014(September to October). Impacts: High cost of Rice cultivation and Low Yield.
River near the Village	Kosi, Ganga are near the Village. Village is located between the River kosi and Ganga.
Embankments near the village	Yes
Early Flood warning	
Visit Extension Agent	

Village*	NayaGaon	District: Khagaria	Area: 600 Hactare.
Household Number	Total: 1080 and Cultivator: 756.		
Nonfarm employment inside the village	Carpenter, Own shop, Vegetable hawker, Tractor service, Groundman, Auto driver.		
Employment outside the village	Own shop, Service, Tuition Center, Helper, and Trading.		
Migration targets	Punjab, Haryana, Delhi, Bombay, Assam.		
Minority			
Nearest town	Khagaria which is 40 km away from village and mode of transportation is Bus.		
Name of district Head quarter	Khagaria which is 40 km away from village.		
Accessible by road	Yes by 50 % KatchaRoad and 50% Paved road but not usable during Monsoon.		
Does this village have electricity	Yes, 80% Household , 6 hours a day		
Does village have phone service	Yes about 85% Household.		
Source of drinking water	Private Hand pump withBad quality of water and one government hand pump with good quality of water.		
Source of cooking fuels	Gas, Crop residue, Dung cake		
Facilities in village	PDS shop, Output market, Input Market, Grocery shop, cooperative, Anganvadi, Primary school, Middle school, Higher Secondary School, Untrained doctor clinic, Self Help Group, Storage (private use).		
Water logged area	27%		
Major crops grown in the village	Maize, Wheat, oil seed.		
Floods in last five years	2011,2012,2013,2014, (Due to heavy Rainfall and River movement) 30-50% Area affected in 2011, 2012 and 80-100% area affected in 2013 and 50-80% area affected in 2014. Impacts: Damage of Road and House, Fodder loss. Got support from government.		
Droughts in last five year	None		
River near the Village	Kosi, Ganga is near the Village and Village is located Between the River Kosi and Ganga.		
Embankments near the village	Yes, Distance from embankments is 150 meter		
Early Flood warning	Yes, Mode of communication is Radio, Paper.		
Visit Extension Agent	Yes.		

Village*	Govindpur	District: Khagaria	Area: 880 Hectare.
Household Number	Total: 2483 and Cultivator: 869		
Nonfarm employment inside the village	Mason, MNREGA, Dairy, Private tutor, Trading.		
Employment outside the village	Shop assistant, Hawker, Agent, Coaching center, Stitching, Driver, Small scale industry.		
Migration targets	Punjab, Haryana, Delhi, Bombay, Manipur, Gujarat.		
Minority	None.		
Nearest town	Khagaria which is 50 km away from village and mode of transportation is Private and Own vehicle.		
Name of district Head quarter	Khagaria which is 50 km away from village.		
Accessible by road	Yes by Paved Road and this Paved Road is usable during Monsoon. Paved Road is 2 KM away from Village.		
Does this village have electricity	Yes, 60% Household, 6 hours a day.		
Does village have phone service	Yes about 60% Household.		
Source of drinking water	Private Hand Pump and the water quality of hand pump is good and one Government Hand Pump and the Water quality is good.		
Source of cooking fuels	Gas, Crop residue, Dung cake		
Facilities in village	PDS shop, Grocery shop, Anganvadi, Post office, Primary school, Middle school, Higher Secondary School, Primary health center, Community health center, Untrained doctor clinic, Self Help Group.		
Water logged area	10%		
Major crops grown in the village	Maize, Wheat, Banana.		
Floods in last five years	2012, 2013, 2014, (Due to Rainfall and Low land) 50-80% Area affected. Impacts: Transport problems, Fodder and Yield loss.		
Droughts in last five year	2011, 2012 (March-June). Impacts: Low crop production, Delayed sowing of next crop.		
River near the Village	Ganga is near the Village and its 2 Km away from Village.		
Embankments near the Village	Yes, Distance from embankments is 2 Km. and Village is located in between River and Embankment.		
Early Flood warning	Yes, and Mode of communication is DM, DDO		
Visit Extension Agent	Yes but Rarely in Kharif Season. Advisory service Is also Useful.		

Village*	East Borne	District: Khagaria	Area: 1400 Hectare.
Household Number	Total: 1518 and Cultivator: 534.		
Nonfarm employment inside the village	Own shop, Dairy.		
Employment outside the village	Shop assistant, Service, Labour.		
Migration targets	Punjab, Haryana, Delhi, Gujarat.		
Minority	4 %		
Nearest town	Khagaria which is 35 km away from village and mode of transportation is Bus accessible at 1.5 km. away from Village.		
Name of district Head quarter	Khagaria which is 35 km away from village.		
Accessible by road	No. but Paved Road is 1.5 km. away from Village.		
Does this village have electricity	Yes, 70% Household, 7 hours a day.		
Does village have phone service	Yes about 72% Household.		
Source of drinking water	Hand pump and Quality of water is good.		
Source of cooking fuels	Gas, Crop residue, Dung cake		
Facilities in village	PDS shop, Grocery shop, Bank, cooperative, Anganvadi, Post office, Primary school, Middle school, Primary health center, Community health center, Private clinic(Trained doctor), Untrained doctor clinic, Police station, Self Help Group.		
Water logged area	34%		
Major crops grown in the village	Maize, Wheat, Peas, Paddy, Raincha.		
Floods in last five years	2010, 2011, 2012, 2013, 2014, (Due to rainfall in Nepal) 30% Area affected in 2010-2013, 40 % Area affected in 2013 and 50 % Area affected in 2014. Impacts: Crop, Cattle, Soil degradation.		
Droughts in last five year	2013 (June-July). Impacts: Crop and Fodder loss.		
River near the Village	Kosi and Bagmati are near the Village and are 90 meter away from Village.		
Embankments near the Village	Yes, Distance from embankments is 1.5 Km. and Village is located in North-South (Middle) from Embankments.		
Early Flood warning	Yes.		
Visit Extension Agent	No, but There is Advisory service for Flood and Drought.		

Village*	Pachaut	District: Khagaria	Area: 680 Hectare.
Household Number	Total: 1132 and Cultivator: 326.		
Nonfarm employment inside the village	Own shop, Hawker.		
Employment outside the village	Mason, Rickshaw puller, Driver.		
Migration Targets	Punjab, Kolkata, Delhi, Gujarat, Andhra Pradesh.		
Minority	4 %		
Nearest town	Khagaria which is 48 km away from village and mode of transportation is Bus from Beldaur next Village.		
Name of district Head quarter	Khagaria which is 48 km away from village.		
Accessible by road	No. but there is road about 60 % Katcha and Paved road is 4 km away from Village.		
Does this village have electricity	None		
Does village have phone service	Yes (Mobile Phone) about 60% Household.		
Source of drinking water	Hand pump and Quality of water is Bad.		
Source of cooking fuels	Crop residue, Dung cake		
Facilities in village	PDS shop, Grocery shop, Anganvadi, Primary school, Middle school, Higher secondary school, Primary health center, Untrained doctor.		
Water logged area	3.5%		
Major crops grown in the village	Maize, Wheat, Paddy.		
Floods in last five years	2012, 2014, (Due to Kosi Movement) 30- 50% Area affected in 2012, 50-80 % Area affected in 2014. Impacts: Crop and Cattle loss, House damage and No support from government.		
Droughts in last five year	None.		
River near the Village	Kosi and Bagmati are near the Village and 10 Km. away from Village.		
Embankments near the Village	Yes, Distance from embankments is 10 Km. and Village is located Outside from Embankments.		
Early Flood warning	Yes. mode of communication of extension agent is Verbal		
Visit Extension Agent	Yes and the agent has visited in December month and. There is Advisory service for Flood and Drought and it's useful.		

Village*	Nawada	District: Khagaria	Area: 1803 Hectare
Household Number	Total: 1087 and Cultivator: 510.		
Nonfarm employment inside the village	Dairy, Own shop.		
Employment outside the village	Labour, Rickshaw puller, Auto driver, Shop assistant, Tutor.		
Migration Targets	Delhi, Punjab, Haryana, Gujarat, Andhra Pradesh.		
Minority	None.		
Nearest town	Khagaria which is 30 km away from village and mode of transportation is Auto taxi and Bus.		
Name of district Head quarter	Khagaria which is 30 km away from village.		
Accessible by road	Yes by Paved Road but not usable during Monsoon and this Paved Road is 1 km away from village.		
Does this village have electricity	Yes, 50% Household, 10 hours a day		
Does village have phone service	Yes about 80% Household.		
Source of drinking water	Hand pumpbut Bad quality, Public hand pump and quality of water is good.		
Source of cooking fuels	Gas, Crop residue, wood.		
Facilities in village	PDS shop, Input Market, Grocery shop, Anganvadi, Post office, Primary school, Middle school, Primary health center, Untrained doctor clinic, Self Help Group.		
Water logged area	11.1%		
Major crops grown in the village	Maize, Wheat, Moong.		
Floods in last five years	2010, 2011,2012,2013,2014 (Due to kosi and Bagmati movements) 30-50 % Area affected in 2011-2013 and 50-80 % Area is affected in 2014. Impacts: Crop and Fodder loss, House damage and Transportation problems. No support from government.		
Droughts in last five year	2011, 2012, 2013, 2014 (July-September). Impacts: Yield and Fodder loss.		
River near the Village	Bagmati is near the Village and Village is located Between the kosi and Ganga River.		
Embankments near the Village	Yes, and village is located in middle from Embankments.		
Early Flood warning	Yes, from District magistrate.		
Visit extension agent	Yes, last year		

Village*	Balha District: Khagaria Area: 581.7 Hectare.
Household Number	Total: 1696 and Cultivator: 425.
Nonfarm employment inside the village	Labour, Own shop, Driver, Coaching center.
Employment outside the village	Labour, Rickshaw puller, Commission agent, Shop assistant.
Migration Targets	Delhi, Punjab, Uttar Pradesh, Banglore, Guhati, Kolkatta.
Minority	15 %
Nearest town	Khagaria which is 16 km away from village and mode of transportation is Jeep, Train and Bus.
Name of district Head quarter	Khagaria which is 16 km away from village.
Accessible by road	Yes by Paved Road but not usable during Monsoon and this Paved Road is 0 km away from village.
Does this village have electricity	Yes, 85% Household , 18 hours a day
Does village have phone service	Yes about 60% Household.
Source of drinking water	Hand pump but Bad quality, Supply water with good quality of water and Community hand pump with good quality of water.
Source of cooking fuels	Gas, Crop residue, Dung cake.
Facilities in village	PDS shop, Output market, Input Market, Grocery shop, Bank, Post office, Madarsa, Primary health center, Private clinic (Trained doctor), Untrained doctor clinic, Self Help Group.
Water logged area	1.7%
Major crops grown in the village	Maize, Wheat, Paddy, Raicha.
Floods in last five years	None.
Droughts in last five year	2011, 2012, 2013, 2014 (August-October). Impacts: Yield loss.
River near the Village	Kosi is near the Village and Village is 3 Km away from the River.
Embankments near the Village	Yes, and village is 2 Km away from embankments and located Outside from Embankments.
Early Flood warning	Yes, and mode of communication is Radio and B.D.O.
Visit extension agent	Yes.

Village*	Beldaur	District: Khagaria	Area: 880 Hectare.
Household Number	Total: 3773 and Cultivator: 1509.		
Nonfarm employment inside the village	Labour, driver, Own shop, Dairy.		
Employment outside the village	Mason, Beldari, Driver, Milkman, Shop keeper, Own shop.		
Migration Targets	Delhi, Punjab, Rajasthan, Maharashtra.		
Minority	2 %		
Nearest town	Khagaria which is 50 km away from village and mode of transportation is Auto taxi and Bus.		
Name of district Head quarter	Khagaria which is 50 km away from village.		
Accessible by road	Yes by Paved Road but not usable during Monsoon and this Paved Road is 3 km away from village.		
Does this village have electricity	Yes, 35% Household , 6 hours a day		
Does village have phone service	Yes about 80% Household.		
Source of drinking water	Hand pump but Bad quality, Supply water is good quality.		
Source of cooking fuels	Gas, Crop residue.		
Facilities in village	PDS shop, Output market, Input Market, Grocery shop, Bank, Anganvadi, Post office, Primary school, Middle school, Higher Secondary School, Madarsa, Girls School, Primary health center, Community Health Center, maternity center, Private clinic (Trained doctor), Untrained doctor clinic, Police station, Self Help Group.		
Water logged area	2.27 %		
Major crops grown in the village	Maize, Wheat, Paddy.		
Floods in last five years	2011,2012,2013,2014 (Due to incomplete of Dam) 50-80 % Area affected. Impacts: Crop and Fodder loss, House damage and Transportation problems, No support from government.		
Droughts in last five year	None.		
River near the Village	Kosi and Bagmati is near the Village and Village is 12 Km. away from the Rivers..		
Embankments near the Village	Yes, but not complete and Village is 11 km away from the Embankments and Village is located Outside from Embankments.		
Early Flood warning	Yes.		
Visit extension agent	None.		

Village*	Dharampur	District: Sharsa	Area: 634 Hectare
Household Number	Total: 800 and Cultivator: 480.		
Nonfarm employment inside the village	Dairy, Trading.		
Employment outside the village	None.		
Migration	Delhi, Punjab, Haryana.		
Minority	None.		
Nearest town	Khagaria by boat which is 22 km away from village and mode of transportation is Boat.		
Name of district Head quarter	Sharsa which is 29 km away from village.		
Accessible by road	None.		
Does this village have electricity	None.		
Does village have phone service	Yes about 60% Household.		
Source of drinking water	Hand pump but Bad quality.		
Source of cooking fuels	Gas, Crop residue, Dung cake.		
Facilities in village	PDS shop, Grocery shop, Anganvadi, Primary school, Middle school.		
Water logged area	30 %		
Major crops grown in the village	Maize, Wheat, Paddy, Mustard, Pulses.		
Floods in last five years	2011,2012,2013,2014 (Due to kosi Movement) 50-80 % Area affected.		
Droughts in last five year	2011 (September-October). Impacts: wilting of crops.		
River near the Village	Kosi is near the Village and Village is 3 Km. away from the River.		
Embankments near the Village	Yes, and Village is 4 km away from the Embankments and Village is located middle of River from Embankments.		
Early Flood warning	Yes.		
Visit extension agent	None.		

Village*	Hansari	District: Sharsa	Area: 370.28 Hectare
Household Number	Total: 700 and Cultivator: 266.		
Nonfarm employment inside the village	None.		
Employment outside the village	Construction work, Labour.		
Migration Targets	Delhi, Punjab, Uttrakhand.		
Minority	None.		
Nearest town	Simiri Bahktiyarpur which is 12 km away from village and mode of transportation is tempo, Train, Owns vehicle.		
Name of district Head quarter	Sharsa which is 36 km away from village.		
Accessible by road	Yes, Katcha road approx 3 Km and not usable during monsoon. Paved Road is 2 Km away from Village.		
Does this village have electricity	Yes, 100% Household, 12 hours a day.		
Does village have phone service	Yes about 90% Household.		
Source of drinking water	Hand pump good quality water.		
Source of cooking fuels	Crop residue, Dung cake.		
Facilities in village	PDS shop, Grocery shop, Anganvadi, Post office, Primary school, Middle school, Untrained doctor clinic, Self help group.		
Water logged area	54%		
Major crops grown in the village	Maize, Wheat, Paddy, Moong, Potato.		
Floods in last five years	2011,2012,2013,2014, 2015 (Due to Nepal rainfall) 50-80 % Area affected in 2011 and 30-50% Area affected in 2012-2015. Impacts: cattle loss, Transportation problems, Starvation, Yield loss. 5400 Rs per Household support by Government.		
Droughts in last five year	None.		
River near the Village	Kosi is near the Village and Village is 4 Km. away from the Rivers.		
Embankments near the Village	Yes, and Village is 3 km away from the Embankments and Village is located Outside from Embankments.		
Early Flood warning	Yes. Mode of communication is personal advisor.		
Visit extension agent	Yes, and agent last Visited on October 2015.		

Village*	Raipura	District: Sharsa	Area: 199.91 Hectare
Household Number	Total: 617 and Cultivator: 277.		
Nonfarm employment inside the village	Driver, Own shop, Fishery, MNREGA.		
Employment outside the village	Labour, Teacher, Mason, Hawker.		
Migration Targets	Delhi, Punjab, Rajasthan, Himachal Pradesh.		
Minority	10 %.		
Nearest town	Simiri Bahktiyarpur which is 4 km away from village and mode of transportation is tempo, Bus.		
Name of district Head quarter	Sharsa which is 17 km away from village.		
Accessible by road	Yes, Paved Road is 2 Km away from Village.		
Does this village have electricity	Yes, 80% Household, 14 hours a day.		
Does village have phone service	Yes about 90% Household.		
Source of drinking water	Hand pump good quality water and Supply water.		
Source of cooking fuels	Crop residue, Dung cake, Gas.		
Facilities in village	PDS shop, Grocery shop, Bank, Anganvadi, Post office, Primary school, Middle school, Madrsa, Primary health center, Untrained doctor clinic, Self-help group, Storage (Private use).		
Water logged area	10%		
Major crops grown in the village	Maize, Wheat, Paddy, Moong, Potato, Kobi.		
Floods in last five years	2012, 30 % Area affected. Impacts: Crop and fodder loss, delayed for next crop, support by Government.		
Droughts in last five year	2013 (Mach-August). Impact: Crop loss.		
River near the Village	Kosi is near the Village and Village is 12 Km. away from the Rivers.		
Embankments near the Village	Yes, and Village is 10 km away from the Embankments.		
Early Flood warning	Yes, from News.		
Visit extension agent	Yes, and agent frequently 4-5 visit in a month, mode of communication is personal.		

Village*	Mahishi	District: Sharsa	Area: 1259.36
Household Number	Total: 1330 and Cultivator: 465.		
Nonfarm employment inside the village	Own shop, Hawker.		
Employment outside the village	None.		
Migration Targets	Delhi, Punjab, Haryana.		
Minority	14.7 %.		
Nearest town	Sharsa which is 25 km away from village and mode of transportation is Own vehicle, auto rickshaw, Bus.		
Name of district Head quarter	Sharsa which is 25 km away from village.		
Accessible by road	Yes, 0-50%, Paved Road is 0 Km away from Village.		
Does this village have electricity	Yes, 30% Household, 8 hours a day.		
Does village have phone service	Yes about 80% Household.		
Source of drinking water	Hand pump good quality water and Sealed water.		
Source of cooking fuels	Crop residue, Gas.		
Facilities in village	PDS shop, Output market, Input market, Grocery shop, Bank, Cooperative, Anganvadi, Post office, Primary school, Middle school, Higher secondary school, Primary health center, Untrained doctor clinic, Self help group, Storage(Government).		
Water logged area	80%		
Major crops grown in the village	Garma Rice, Aghani Rice, Wheat, Pulses, Oilseeds, Maize.		
Floods in last five years	None.		
Droughts in last five year	2015 (Mach-August). Impacts: Migration of labour, Crop loss, Crop wilting. No government support.		
River near the Village	Kosi is near the Village and Village is 1 Km. away from the Rivers.		
Embankments near the Village	Yes, and Village is .20 km away from the Embankments. Village is located Outside from the embankment.		
Early Flood warning	Yes, mode of communication by publicity.		
Visit extension agent	Yes and agent last visit in a year ago,		

Village*	Kodillaptti	District: Sharsa	Area: 498.16
Household Number	Total: 724 and Cultivator: 362.		
Nonfarm employment inside the village	Dairy, Labour		
Employment outside the village	None.		
Migration Targets	Delhi, Punjab, Haryana.		
Minority	13 %.		
Nearest town	Sharsa which is 25 km away from village and mode of transportation is auto rickshaw, Bus.		
Name of district Head quarter	Sharsa which is 24 km away from village.		
Accessible by road	Yes, 90%, Katcha Road and the road is not usable during monsoon.		
Does this village have electricity	Yes, 30% Household, 14 hours a day.		
Does village have phone service	Yes about 80% Household.		
Source of drinking water	Hand pump good quality water.		
Source of cooking fuels	Crop residue, Gas, Manechar.		
Facilities in village	PDS shop, Input market, Grocery shop, Anganvadi, Post office, Primary school, Middle school, Higher secondary school, Untrained doctor clinic, Self-help group, Public toilet.		
Water logged area	75%		
Major crops grown in the village	Paddy, Manechar, Wheat, Maize.		
Floods in last five years	2010, 2011, 2012, 2013, 2014(Due to soil erosion in Nepal), 80-100% Area affected. Impacts: Crop and Cattle loss, Delayed for next crop.		
Droughts in last five year	2013-2014 (June-July). Impacts: Migration of labour.		
River near the Village	Kosi is near the Village and Village is .50 Km. away from the Rivers.		
Embankments near the Village	Yes, and Village is 1 km away from the Embankments. Village is located in Middle from the embankment.		
Early Flood warning	Yes.		
Visit extension agent	Yes and agent last visit in June month.		

Village*	Hati	District: Sharsa	Area: 319.70
Household Number	Total: 615 and Cultivator: 31.		
Nonfarm employment inside the village	Dairy.		
Employment outside the village	None.		
Migration Targets	Delhi, Punjab, Haryana, Jammu & Kashmir, Saudi arab.		
Minority	None.		
Nearest town	Supaul which is 15 km away from village and mode of transportation is Cycle, auto rickshaw, Three whealers.		
Name of district Head quarter	Sharsa which is 35 km away from village.		
Accessible by road	No, but 80% katcha and not usable during monsoon and Paved road 3 Km away from village.		
Does this village have electricity	None.		
Does village have phone service	Yes about 40% Household.		
Source of drinking water	Hand pump but bad quality.		
Source of cooking fuels	Crop residue, Dung cake.		
Facilities in village	PDS shop, Grocery shop, Anganvadi, Post office, Middle school.		
Water logged area	100%		
Major crops grown in the village	Rice, Oilseed, Paddy, Manechar, Wheat, Maize.		
Floods in last five years	2011, 2012, 2013, 2014, 2015(Due to Kosi movement), 80-100% Area affected. Impacts: Crop and Cattle loss, House damage, Transportation problems, Migration of labour, Toilet problems, Fodder loss. Support by government.		
Droughts in last five year	None.		
River near the Village	Kosi is near the Village and Village is 1 Km. away from the Rivers.		
Embankments near the Village	Yes, and Village is 2.5 km away from the Embankments. Village is located in Middle from the embankment.		
Early Flood warning	None.		
Visit extension agent	Yes and agent last visit in 2015.		

Village*	Bankatti	District: Sharsa	Area: 490.07 Hectare
Household Number	Total: 975 and Cultivator: 425.		
Nonfarm employment inside the village	None.		
Employment outside the village	None.		
Migration Targets	Delhi, Punjab..		
Minority	None.		
Nearest town	Sharsa which is 32 km away from village and mode of transportation is Train.		
Name of district Head quarter	Sharsa which is 32 km away from village.		
Accessible by road	Yes, 50% Paved road and this Paved road is not usable during monsoon and Paved road 7 Km away from village.		
Does this village have electricity	Yes, 70% Household.		
Does village have phone service	Yes about 80% Household.		
Source of drinking water	Hand pump but bad quality, Supply water with good quality.		
Source of cooking fuels	Crop residue, Dung cake, gas.		
Facilities in village	PDS shop, Grocery shop, Anganvadi, Post office, Primary school, Middle school, Self help group, Subject matter specialist.		
Water logged area	83%		
Major crops grown in the village	Wheat, Maize, Paddy.		
Floods in last five years	2011, 2012, 2013, 2014, 2015(Due to Nepal rainfall), 30% Area affected in2011, 2013, 2014, 2015, 30-50% Area affected in 2012. Impacts: Cattle loss. Support by government per household (5400 /-).		
Droughts in last five year	None.		
River near the Village	Kosi is near the Village and Village is 2 Km. away from the Rivers.		
Embankments near the Village	No, but Village is located Outside from the embankment.		
Early Flood warning	Yes. Mode of communication is personal.		
Visit extension agent	Yes and agent visit in every season.		

Village*	Balahapur	District: Sharsa	Area: 263.85 Hectare.
Household Number	Total: 600 and Cultivator: 328.		
Nonfarm employment inside the village	Shop owner, Shop keeper, MNREGA, Vegetable hawker.		
Employment outside the village	Vegetable hawker, Own shop, Mason, Driver.		
Migration Targets	Delhi, Punjab.Haryana, Himachal Pradesh.		
Minority	200		
Nearest town	Simri bakhtyarpur which is 5 km away from village and mode of transportation is Auto rickshaw, Taxi.		
Name of district Head quarter	Sharsa which is 17 km away from village.		
Accessible by road	Yes, Paved road and this Paved road is usable during monsoon and Paved road 0 Km away from village.		
Does this village have electricity	Yes, 90% Household, 16 hours a day.		
Does village have phone service	Yes about 80% Household.		
Source of drinking water	Hand pump but bad quality, Sealed water.		
Source of cooking fuels	Crop residue, Dung cake, gas.		
Facilities in village	PDS shop, Grocery shop, Bank, Anganvadi, Post office, Primary school, Middle school, Madarsa, Primary health center, Untrained doctor clinic, Self help group, Storage (Private use).		
Water logged area	10%		
Major crops grown in the village	Wheat, Maize, Paddy, Potato, Moong, Vegetable.		
Floods in last five years	None.		
Droughts in last five year	2013, 2015. Impacts: Yield loss, Migration of labour. 30% household got the support by government.		
River near the Village	Kosi is near the Village and Village is 13 Km. away from the Rivers.		
Embankments near the Village	Yes, and Village is 17 Km away from embankment.		
Early Flood warning	Yes, mode of communication is personal contacts.		
Visit extension agent	Yes.		

Appendix 9.3

(Notes)

A9.1. Institutions for Food, Agriculture and Water management

Food, agriculture and water have dimensions of public concern, goods and property. Their management is not entirely left to the private sector or individual domain where there arises severe competition for limited and valuable natural resources in which the rich and the powerful are likely to win and the weaker sections suffer deprivation from basic needs. Conventionally the state had a major role in the distribution and managed intervention in these sectors through rules, regulations, restrictions and devolution of public budget for social ends.

Under India's Constitution food, water and water are State subjects but the Centre usually guides the states in choosing the direction of legislation in tune with the spirit of the time and policy of the ruling government. Over time as in other sectors, the state's domination came under pressure both due to the paradigm change accompanying economic reforms and the financial unsustainability as also the failure of governments to generate a fair and representative system free of corruption. However, given that food and water are basic human needs for survival and that food, water and agriculture draw from shared natural resources, there is a need for sound institutions that can prevent deprivation even in a market based system. The institutions in India have been witnessing dynamics in the last decades and are still undergoing transitions in a formative stage towards a new paradigm which evolve through dialogues, conflicts, criticisms and corrections. This before outlining the systems in place currently the next section provides a glimpse at the transitions.

A9.1.1. *Transitions in public support systems*

India's systems of food and water management worked on both sides of production and consumption and were financed largely by public subsidies and investments on infrastructure. Contradiction in particular surrounded subsidies. They are blamed for being budgetary burden for purposes that the market could accomplish, for creating opportunities for leakage and corruption and for distorting the market in favour of products and systems that are not suitable for India. Many have argued that subsidies, drawing from limited budget restrain public investment and that public investments are superior to subsidies for development purposes.

On the other hand, there are strong defenses in favour of state's support to farmers and consumers of food. Public investment is also not beyond questioning especially for its effect of crowding out private investment (Kumar, 1992, Kirit Parikh paper**). Private investment on irrigation (tube wells, diesel pumps) financed by banks and informal sources has served well when public assets languished for want of resources, managerial proficiencies and designing defects. Similarly, control over agricultural produce marketing through rigid regulations have restricted human enterprise and choices. Recently there is a move to change the subsidies in ways that are more flexible and non-distorting and marketing to be more private initiative driven.

In the period 2014-16 drawing from the technological developments in the field of communication, a landmark development was instituted for the population of India, namely the Unique Identification Authority of India UIDAI issuing an Aadhar card for individual identity which, combined with the mobile telephony and banking outreach (Jandhan), created the so called JAM for revolutionizing delivery of government's services, particularly the subsidies (Economic Survey 2016). India may now be shifting towards Direct Benefit Transfer (DBT) from the

age old subsidy system with its complex ways of implementation although the outlook is still uncertain.

The chief economic adviser (CEA) to the finance ministry, Arvind Subramanian recently described DBT as a “game changer” for India. The DBT holds a promise of reducing malpractices, exclusions and pilferage while economizing on public budgetary outflow. Applied for delivering cooking gas (LPG) subsidy it has resulted in a 24% reduction in the sale of subsidized LPG by excluding “ghost beneficiaries” and savings public resources. According to the Finance Ministry, instead of distributing food grains, fertiliser and kerosene through the PDS shops, the intended beneficiaries need to be given money through cash transfers and be allowed to buy commodities from wherever they want to. As the Survey for 2014-2015 pointed out: “Recent experimental evidence documents that unconditional cash transfers – if targeted well – can boost household consumption and asset ownership and reduce food security problems for the ultra poor.”

A9.1.2. *Public distribution of food*

Public distribution system (PDS) is the leading support system which provides rice and wheat at very low prices to poor households across the country. Having its roots in colonial times of the World War II the system of rationing food to the needy as a quota was continued, extended and modified several times over the years in independent India. Food procured from farmers with surplus are distributed to a large number beneficiary through a large government run machinery and implemented by the Food Corporation of India (FCI) and a nation-wide network of fair price shops at a price lower than the cost of procurement, the difference being compensated by the government as a subsidy. The PDS has been charged for failure to reach the actually needy and for misuse of public resource.

With this understanding and synchronized with economic reforms of the 1990s the system gave way to a targeted PDS (or TPDS) in which only the people identified to be below the poverty line (BPL) were entitled to public subsidy while others holding ration cards (APL) could avail food at the cost price. Even this system was a victim of leakage, corruption, political patronage and more pertinently culpable for excluding needy sections even as the measurement of the poverty line which determined the beneficiary set began to be questioned. Recently, while the merits of the PDS were facing attack, Government of India made a significant change in the policy which in fact increases its coverage of PDS by legislating a National Food Security Act (NFSA), 2013 (also Right to Food Act) that converts the access to food into a legal entitlement. It includes the Midday Meal Scheme, Integrated Child Development Services (ICDS) scheme, maternity entitlements and the TPDS reaching out about two-thirds of the population, 75% in rural areas and 50% in urban areas who are entitled to 5 kilograms of food per person per month of cereals at low prices (Rice: Rs 3.00/Kg, Wheat: Rs2/Kg and Coarse cereals: Rs 1 /Kg).

The 'biggest ever experiment in the world for distributing highly subsidized food by any government through a rights based approach' in a country marked by mal-nutrition, the bill has been severely critiqued for reasons both political and financial. The implementation would demand huge amounts of public procurement and large budgetary burden. Moreover, the enactment could make more demands on growing food grains as against cash crops, impose pressure on other crops like pulses and create price rise in the market. The implementation deadline was 4 October 2014 but by then only 11 states had either implemented the Act or declared readiness to do so and in November 2014 the union government announced allocation of foodgrains to 11 States/Union Territories (UTs) including Bihar. The remaining 25 States/UTs had not completed the preparatory measures required for implementation of the Act.

A further milestone came in the way with the formulation of the Aadhar and the JAM and with the release of the Shanta Kumar Committee report. The committee recommends a gradual move to cash transfers. Indications are clear that the government is considering moving to cash transfers and replacing the public distribution system (PDS) targeting Aadhaar-linked bank accounts of beneficiaries in Puducherry, Chandigarh and Dadra and Nagar Haveli. In Daman and Diu, PDS benefits will be provided on the basis of Aadhaar-linked biometric verification. The DBT in public food distribution has raised debates on whether the shift from the older system is justified given that it was already showing signs of performing well with reduced leakages and how best food can be distributed. The fear that DBT will encourage the poor to use their coupons or cash for something else is dismissed on the ground that the same risk is present in the older system (the poor can sell the grain or the kerosene for buying other things with that money).

A9.1.3. Public procurement

Government's food policy is implemented by the FCI which has to procure grains from farmers at minimum support prices (MSP) or procurement prices (earlier these prices were different) and stock grains in order to achieve inter-temporal and spatial equity of distribution to bridge surplus and deficit times and regions. The procurement policy of the government of India and the FCI have also been a targets of criticism for years. While budgetary burden, administrative efficiency, corruption and in the current paradigm, the distortionary implications are serious problems of the system not only with national but also international dimensions. Like the PDS, the support price serves as a safety net for farmer's welfare and an incentive to farmers. With the launch of economic reforms procurement was gradually being decentralized.

The high-level committee (Shanta Kumar Committee), suggested restructuring the state-owned FCI, which created at a time when food being distributed came

largely from foreign aid, has become anachronistic. The Committee's recommendation reinforced decentralization as ways to reduce cost of production and food policy operations.

A9.1.4. *Fertilizer Policy*

Fertilizer is an important component of the package technology for the green revolution and a contributor for India's breakthrough in food production. While investment in the fertilizer sector by the state created substantial domestic production capacity, India is deficit in fertilizer and mostly the inputs required for manufacturing chemical fertilizers. To make fertilizer affordable subsidies were given for its use but the subsidy system was complex starting with the retention price scheme later modified and made more nutrient-based as imbalances appeared in the use of nitrogenous, phosphatic and potassic fertilizers and the value of micro-nutrients were ignored. Use of urea often reaches excess levels. Recent policy emphasizes soil testing to decide fertilizer use and promotes neem-coated urea to stop mal-practices and enrich soil. In general organic fertilizers are including bio-fertilizer are gaining significance for sustainable agriculture. Cash transfer is also explored for fertilizers. Farmers often buy urea in the black market at above the stipulated minimum retail price (MRP) as substantial portion of the subsidized urea is suspected to be smuggled across the border or illegally appropriated by the industries forcing farmers to buy back fertilizers in the black market.

A9.1.5. *Institution of food management in Bihar*

Growing outreach of PDS in Bihar from 2 per cent of households in 2004-05 to 43 per cent in 2011-12, shown by NSS data and other evidences cited are used for criticism. The Government of Bihar government initiated a system of coupons in year 2007 to administer the public distribution system (PDS). The system is supposed to operate as follows. Every beneficiary household is provided with 12

coupons annually for wheat, rice and kerosene, which specify their entitlements and the price they have to pay for each commodity. Coupons are distributed through village camps, organised by the local gram panchayat under the monitoring and surveillance of officials from the block development office (the process of coupon distribution is video-recorded). Every month, beneficiary households redeem one coupon each against each of the specified PDS commodities at a local fair price shop (FPS) (beneficiaries can ostensibly choose the FPS they use). Then, the FPS owner forwards coupons to the block/district level authorities in order to get next month's supplies. Apart from the above, Government of Bihar government initiated a system of coupons in year 2007 to administer the public distribution system (PDS). The system is supposed to operate as follows. Every beneficiary household is provided with 12 coupons annually for wheat, rice and kerosene, which specify their entitlements and the price they have to pay for each commodity. Coupons are distributed through village camps, organised by the local gram panchayat under the monitoring and surveillance of officials from the block development office (the process of coupon distribution is video-recorded). Every month, beneficiary households redeem one coupon each against each of the specified PDS commodities at a local fair price shop (FPS) (beneficiaries can ostensibly choose the FPS they use). Then, the FPS owner forwards coupons to the block/district level authorities in order to get next month's supplies. Procurement is made through agencies of FCI, BISCOAUN, NAFED, PACS/cooperatives and Rice Millers. Minimum support prices are fixed by government of India and the FCI sets up purchasing centre for the marketing season.

A9.1.6. *Water Management*

Institutions involved in the field of hydrology and water resources are divided into eight categories; (1) Ministries, (2) Central Government Organization, (3) River Basin Organizations, (4) State Government Organizations, (5) Academic

Organizations, (6) Professional Societies, (7) Non-Government Organizations, and (8) International Organizations.

A9.1.7. Flood management

Flood is the major challenge facing by Bihar state of India. There are 28 districts known as flood prone district. Flood management is domain subject of Water Resources Department in Bihar. Notwithstanding, several state and centre level departments coordinate with the Water Resources Department for flood management. These departments are mentioned below.

State department	Centre department
Home	Ganges Flood Control Commission
Disaster management	Centre Water Commission
Public work	Indian Meteorology Department
Rural development	
Directorate of Economics and Statistics	
Information and public relation	
Environment and forest	

Flood management works in Bihar focus on construction and maintenance of embankments, mainly. Total length of embankments constructed in Bihar, considering all together, is about 3732 km, which protects about 36.33 lakh ha of area.

To face the above challenge, the Government of Bihar prepared a guideline consisting of standard operating procedure for flood management in 2013. Following standard procedures are suggested in this guideline.

1. Inspection of embankments by engineers and breach control committee
2. Finish all maintenance work before 15th May
3. Regular monitoring of level of water in the river, rainfall and embankments.

4. Provide information related to floods and early warning through all possible mediums of communication, for example newspaper, radio, television etc. Chief engineers at local level and public relation officer at head office level are responsible for dissemination of all necessary information to citizens.

A9.1.8. Disaster Management

The disaster risks are very high in Bihar. The state faces both natural as well manmade disasters. In Bihar, the major disasters are flood, drought, earthquake, cyclones, heat waves, cold waves, fire or attacks by extremists. The government has taken many important steps to manage these disaster. These steps are presented below.

5. Establishment of Bihar State Disaster Management Authority in Patna.
6. The state government has provided 9 battalions of NDRF (National Disaster Response Force) at Bihar, Patna.
7. The state government has provided 75 acres of land for NDRF battalions. The NDRF battalion consists of specialized teams, trained to respond in various types of natural and manmade disasters. There are a total 1149 personnel and 18 teams. Each team comprises 45 personnel, well equipped and trained for rendering effective response in any disaster situation. For effective search and rescue operations in North Bihar, a process has started to establish a regular company of NDRF in Supaul district.
8. There is provision for Agriculture Input Subsidy, House Damage Subsidy, and Subsidy for cattle loss, in addition to Gratuitous Relief and cash dole for those who are affected by natural disasters.
9. A Standard Operating Procedure for flood, drought and drinking Water Crisis has been formulated.
10. There is provision for fund to be given as compensation to acquire the land for the rehabilitation of families displaced due to erosion caused by rivers.

11. For those who die in man-made disaster, there is a provision for ex-gratia payment and arrangement of free treatment of injured in hospitals.
12. The state government has organised training and awareness programme to enhance the capacity of government officials and community for disaster management. These include — training of divers for search and rescue during flood, training of volunteers, training for quick medical response, and training for motor boat driving. In addition, engineers, contractors, architects and masons are being trained to make Earthquake Resistant Building.
13. The state government has formulated 'Bihar Scheme for Assistance to Farmers in Farm Distress, 2015' to provide assistance to farmers and take measures to prevent farm distress. Under this scheme, there is a provision of an ex-gratia payment of Rs. 4.00 lakh to be deposited in the name of the next-of-kin of the farmer who has died. This creates a fund with a Nationalized Bank which provides monthly interest to the account holder as a financial relief.
14. The state government has provided 40 Fibre Reinforced Plastic Boats and 310 inflatable motor boats to 28 flood-prone districts.
15. An amount Rs. 149.58 lakh has been provided to Beltron for the procurement of 85 satellite phones. Till date, 27 satellite phones have been installed.
16. The state government has identified 9 flood-prone districts as the nodal districts to procure tents for mega relief camp. Till date, 4729 tents have been procured.
17. The State Disaster Response Force has been equipped with modern equipment like inflatable motor boat, life jacket, deep diving suits and latest equipment of search and rescue during flood.
18. In collaboration with All India Institute of Medical Sciences (AIIMS), Patna, two phases of training for Quick Medical Response Teams have been completed. Under this, 3 officials of Health Department and 2 Police

personnel from each district were trained. A total of 328 personnel have been trained to operate during multihazard disasters.

19. A total of 830 GPS sets were made available to all Divisional Commissioners, District Magistrates, Police Superintendents, Sub-Divisional Officers, Sub-Divisional Police Officers and Block Development Officers.

BOX A9.1

Ministries	Ministry of Water Resources, Ministry of Agriculture, Ministry of Power, Ministry of Environment & Forests, Ministry of Surface Transport, Ministry of Urban Affairs, Ministry of Rural Development, Planning Commission, Ministry of Science & Technology, and Department of Space
Central Government Organization	Central Water Commission, Central Ground Water Board, National Institute of Hydrology, Central Water and Power Research Station, Central Soil and Material Research Station, National Water Development Agency (NWDA), Water and Power Consultancy Services (India) Limited, Narmada Control Authority, Upper Yamuna River Board, National Projects Construction Corporation Ltd, Sardar Sarovar Construction Advisory Committee, Brahmaputra Board, India Meteorological Department (IMD), Indian Council of Agriculture Research (ICAR), Water Technology Center for Eastern Region, Bhubaneswar Bureau of Indian Standards (BIS), Central Pollution Control Board, National Environment Engineering Research Institute, Central Arid Zone Research Institute (CAZRI), Indian Institute of Tropical Meteorology (IITM), National Center for Medium Range Weather Forecasting Central Electricity Authority, G . B. Pant Institute for Himalayan Environment & Development, National Hydroelectric Power Corporation (NHPC), National Thermal Power Corporation (NTPC)
River Basin Organizations	Damodar Valley Corporation, Tungabhadra Board, Bhakra-Beas Management Board, Mahi Control Board, Chambal River Board, Betwa River Board, Ganga Flood Control Commission, Brahmaputra Board, Sone River Commission, Narmada Control Authority, Farakka Barrage Project, Upper Yamuna River Board, and Bansagar Control Board

State Government Organizations	Water and Land Management Institutes, and State Water Resource Departments
Academic Organizations	IITs, NITs
Professional Societies	Indian Association of Hydrologists (IAH), Indian Water Resources Society (IWRS), Indian Society for Hydraulics Institution of Engineers, Association of Hydrologists of India (IAHS)
Non-Government Organizations	India Water Portal Center for Science and Environment (CSE), The Energy and Resources Institute (TERI), Tarun Bharat Sangh (TBS), Narmada Bachao Andolan (NBA), The Sathya Sai Trust
International Organizations	BIS, FAO, International Commission on Irrigation and Drainage, International Crop Research Institute for Semi-Arid Tropics (ICRISAT), ISO, IWMI, UNESCO, UNDP, UNEP, USGS, and WMO

Water is the state subject mainly in India. Indian Constitution provides power to states to legislate these subjects related to water – water supplies, irrigation and canals, drainage and embankments, water storage and water power, subject of provisions of Entry 56 of List 1. However, Union government is enabled to legislate regulation and development of inter-state river related issues. Union list of the Constitution, Entry 56, states that “Regulation and development of inter-State rivers and river valleys, to the extent to which such regulation and development under the control of the Union is declared by Parliament by law to be expedient in the public interest.”

Water resources in Bihar are indeed abundant, as it receives an average of 1009 mm of rainfall each year, around 85 percent of it from the south-west monsoon. The annual rainfall is reasonably adequate for the state’s agricultural operations. However, it is the year-to-year variation which tends to create flood or draught-like situations in some years. During the period 2001 to 2013, the annual rainfall has varied from being 678 mm in 2010 to 1506 mm in 2007. To manage the water resources and provide access to safe drinking water to all population, both center and state government offer several programmes and provide many institutional framework to facilitate their offered programmes and policies. Water supply and sanitation has been given top priority by the government since

assuming the post of Chief Minister by Mr. Nitish Kumar. It reflects from the fact that expenditure on water supply and sanitation has an increasing trend. In year 2010 -11, this figure was Rs. 1698 crore which increased to Rs. 3639 crore in 2014 – 15. Besides providing drinking water, irrigation has also been top priority of the state government. Irrigation potential created in the state has increasing trend, however created potential is not being fully utilized which reflects from increasing gap between potential created and irrigated area (see Figure 1 and Figure 2).

In Bihar, there are two major government departments related to water in Bihar state of India. These two departments are Water Resources and Minor Water Resources. The major focuses of Department of Water resources are major irrigation schemes and flood control and management, while minor irrigation schemes, tube well schemes, and groundwater schemes are the major functional areas of Department of Minor Water Resources.

Figure A9.1: Irrigation Potential Created and Irrigated Area in Bihar in Since 1987

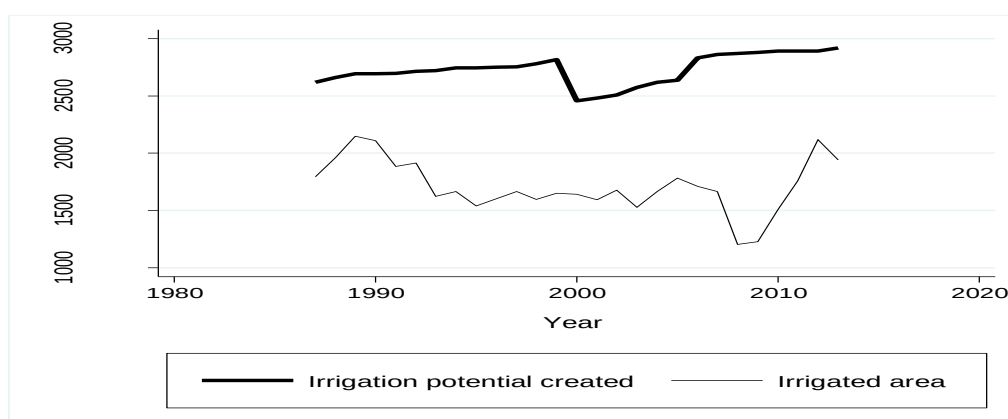
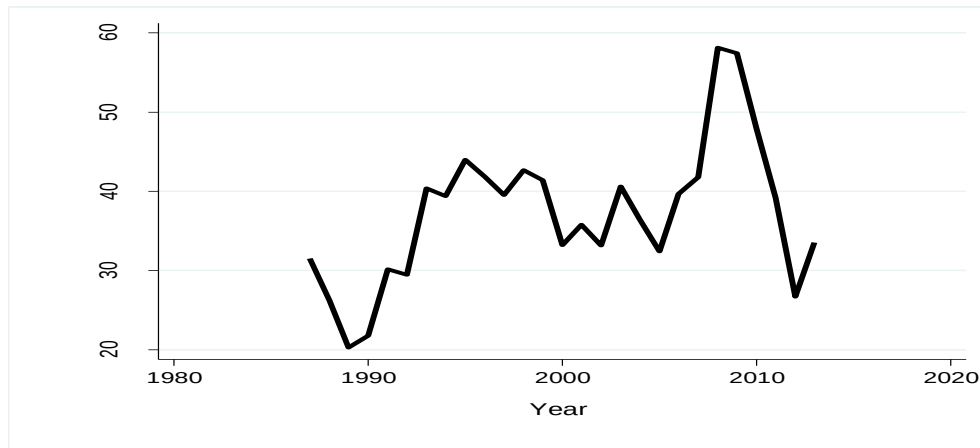


Figure A9.2: Gap between Irrigation Potential Created and Irrigated Area in Bihar



A9.1.9. Inter-linking River Programme

There are large disparities in the availability of water between different river Basins in India – few of them are water abundant and few of them are water deficit. For example, Per-capita utilizable water availability-based on 1991 population varies from 182 M³ in Sabarmati Basin to 2,500 M³ in Mahanadi, and 3082 M³ in Narmada Basin. Utilizable water availability per ha of culturable area varies from 1244 M³ in Sabarmati Basin to 8320 M³ in Mahanadi and 7669 M³ in Narmada Basin. Taking the above fact into account the National Water Policy has underlined the need for making water available to water short areas by transfer from water surplus areas including transfers from one river Basin to another based on a national perspective, after taking into account the requirements, of the areas/basins. The Ministry of Water Resources has prepared in 1980 a National Perspective Plan for development of country's water resources, disregarding political boundaries of States. This National Plan is in two parts namely, (i) development of Peninsular Rivers which is entirely within the control of the central and state Governments and (ii) Himalayan Rivers Development which calls for cooperation of India, Nepal, Bangladesh and Bhutan. The Himalayan and Peninsular River Plan would add to the presently

estimated ultimate irrigation potential besides providing large quantum of hydropower, flood control and in some reaches, navigation.

In order to implement the above plan, the Government of India has set up the National Water Development Agency. This agency aims at to carry out necessary surveys and prepare inter-basin transfer of water has completed pre-feasibility studies of 16 links in Peninsular component and 14 links in Himalayan component. Out of 14 Himalayan components, 6 river-rink canals are related to Bihar.

These are:

1. Brahmaputra – Ganga Alternative
2. Koshi – Mechi
3. Koshi – Ghaghara
4. Gandak – Ganga
5. Chunar – Sone Barrage
6. Sone Dam – Southern Tributaries of Ganga

The above schemes envisage construction of three dams across Kosi, Gandak and Sone River besides the concerned link canals. The Gandak - Ganga link canal is partly related to Bihar although it will not pass through Bihar as it will have significant impact on water utilization and flood moderation in Bihar. Besides the above six link canals of the Himalayan Components the following link canals will also have impact on Bihar.

1. Ghaghara – Yamuna
2. Sharda – Yamuna
3. Yamuna – Rajasthan
4. Rajasthan – Sabarmati

Though these rivers and these link canals do not pass through Bihar, they are going to have significant impact on water resources and issues related to Bihar as they are located on the tributaries of Ganga.

Beside the above inter-state linking schemes, fifteen intra-linking schemes were identified in the state for preparing details project report in different stages. In First Phase, preparation of DPR of following 6 intra linking schemes was allotted to different consultants upon their expression of interest.

1. Construction of 2nd barrage on river Gandak near Areraj and stabilising water availability in Gandak Canal System by transferring water from Burhi Gandak and Baya rivers
2. Construction of barrage at Dheng on river Bagmati
3. Construction of barrages on river Sakri (at Baksoti) and Nata to link both the rivers.
4. Link between Dhanrjay and Fulwaria reservoirs
5. Drainage and water management in Mokama Tal
6. Construction of 2nd barrage near Alwar on river Sone

In Second Phase, preparation of DPR of following 6 schemes was allotted to National Water Development Agency (NWDA):

1. Transfer of water from Kosibasin to Mahananda basin through Kosi-Mechi Link Canal
2. Transferring Ganga water to South Bihar through Pump Canal Scheme near Barh.

Flood Sector

1. Kohra – Chandrawat Link
2. Bagmati(Belwadhar) – Burhi Gandak Link
3. Burhi Gandak – None – Baya – Ganga Link
4. Kosi – Bagmati – Ganga Link

In Third Phase, preparation of DPR of following 3 schemes was allotted to NWDA. Pre-Feasibility Report (PPR), of these schemes were prepared by Water and Land Management Institute (WALMI).

1. Kosi – Adhwara – Bagmati Link
2. Transferring water to South Bihar through pump canal scheme at Buxar
3. Development of Badua – Chandan basin

Based on pre-feasibility reports of above 15 schemes, following 8 schemes have been identified as intra state linking scheme, while remaining 7 are conventional irrigation schemes.

1. Transferring Ganga water to South Bihar through Pump Canal Scheme near Barh.
2. Construction of barrages on river Sakri (at Baksoti) and Nata to link both the rivers.
3. Link between Dhanrjay and Fulwaria reservoirs.
4. Kohra – Chandrawat Link
5. Bagmati(Belwadhar) – Burhi Gandak Link
6. Burhi Gandak – None – Baya – Ganga Link
7. Kosi – Bagmati – Ganga Link
8. Kosi – Adhwara – Bagmati Link

The present status of detailed project report preparation work on these intra-linking schemes are presented in Table 1.

A9.1.10. Land Leasing

Land leasing laws relating to rural agricultural land in Indian states were enacted during decades immediately following the independence. At the time, the abolition of Zamindari and redistribution of land to the tiller were the highest

policy priorities. Top leadership of the day saw tenancy and sub-tenancy as integral to the feudal land arrangements that India had inherited from the British who instituted three different systems of land tenure namely, Zamindari (mostly in east and some parts of south having many layers of intermediaries between the landlord and the cultivator making up rack-renting and exploitation), rayatwari (with no intermediary between the State and the tenant cultivator and the Mahalwari (community based in northern India). Tenancy reform laws that various states adopted not only were meant to transfer ownership rights to the tenant but also either prohibited or heavily discouraged leasing and sub-leasing of land. In recent times there is an outcry in favour of legalizing and relaxing leasing of land so that land can be farmed by those who are interested and needy while landed persons more interested in other jobs can move out of cultivation with assurance of property rights. Leasing by this argument would be efficiency enhancing. Moreover, land reforms in reality were hardly effective as politically influential landowners successfully subverted the reform (P.S. Appu, 1996) and till as late as 1992, ownership rights were transferred to the cultivator on just 4% of the operated land and just seven states, Assam, Gujarat, Himachal Pradesh, Karnataka, Kerala, Maharashtra, and West Bengal, accounted for some 97% of this transfer. Leasing too in reality continued albeit in unrecorded forms making both owners and tenants highly insecure leading to underinvestment and underutilization of land. Land use was also prescribed by law and lately the preponderance of agricultural land possessed by large numbers of impoverished peasants is hindering industrialization that arguably provide alternative and more viable earning for the same people who are also seen to lose interest on low low income generating farming. The covert for of land leasing is also coming in the way of delivering benefits like credit, insurance and support to cultivators many of whom have no authentication and the benefits reach the persons who are not deserving of public support.

Bihar along with states Telengana, Karnataka, Madhya Pradesh and Uttar Pradesh ban land leasing with exceptions granted to landowners among widows,

minors, disabled and defence personnel. Tenancy is banned in Kerala for long but the state has only recently permitted only self-help groups to lease land. States like Punjab, Haryana, Gujarat, Maharashtra and Assam do not ban leasing but the tenant acquires a right to purchase the leased land from the owner after a specified period of tenancy. Despite restriction and the differentiated histories of states, leasing continues. According to 70 round of National sample Survey for the year 2012-13, the proportion of area under leasing is as high as 33.7 percent in Andhra Pradesh, where zamindari system prevailed, and 24.6 percent in Punjab where mahalwari or ryotwari system prevailed. But it is as low as 0.1 percent in Jammu & Kashmir and 8.1 percent in Kerala where there is complete ban on land leasing. The other states with highly restrictive land leasing laws, namely Gujarat (5.6 percent), Maharashtra (3.7 percent), Tripura (4.7 percent), and Madhya Pradesh (5.1 percent) have lower incidence of leasing. The terms of lease also vary from region to region. The system of fixed cash rent predominantly exists in Andhra Pradesh, Haryana, Punjab, while share cropping is the dominant in Assam, Bihar, Chhattisgarh, Jharkhand, Karnataka, Madhya Pradesh, Maharashtra, Manipur, Mizoram, Orissa, Rajasthan, Sikkim, Tripura and Uttar Pradesh. Some field level studies further show that irrigated areas in each region has higher incidence of fixed cash tenancy, while farmers in rainfed areas prefer share cropping arrangement. Considering the country as a whole, share cropping accounts for about 39.6 percent of the total leased in land, followed by fixed cash (31.0 percent), fixed produce (15.3 percent) and other terms (14.1 percent) (Govt. of India: 2006) . Recently attempts to change land laws for land use, tenancy and ownership transfer motivated more for efficient use, industrialization and housing are facing huge political challenges. The land acquisition bill passed by the central government till date has made little impact while further modification was stalled.

Tenancy in Bihar is guided by Land reforms Act of 1961 which prohibits leasing except by disabled ryots i.e. minor, widow, or unmarried, divorced or separated

woman, or person with physical or mental disability or a person in the armed forces or a public servant in receipt of salary not exceeding a minimum share cropping is the dominant in Bihar. A tenant (under raiyat) in Bihar is entitled to purchase ownership right if he is in continuous occupation of land for 12 years, on payment of 24 times the land revenue. In spite of restrictions 21.04% of land is leased in Bihar against all India average of 10.10% (NSSO **) . The Bihar Agriculture Land (Conversion for Non Agriculture Purposes) Rules, 2011 is under consideration.

A9.1.11. Participatory Irrigation Management Programme (PIM)

This program was started in 1997 as per the guidelines of Government of India. Under the program, operation and maintenance of different distribution systems of the completed irrigation schemes are to be handed over, in phases, to the Societies of the beneficiaries formed for the purpose. The Societies besides being responsible for operation and maintenance would also collect water charges and would pay 30 percent of such collections to the department and spend the remaining 70 percent on the operation and maintenance works of the canal system under their jurisdiction. The Canal Head Works and water supplies would be under the control of the department and it would also be responsible for major repairs and for providing any technical assistance required.

The Water Resources Department, Bihar has introduced the Bihar PIM model in 2005 (GoB, 2005). This was based on the guidelines prepared for implementing the participatory irrigation management by the Water Resources Department (WRD) in the year 2000. The Model basically suggested to made organizational changes both at the state and lower level to implement PIM in the state. It has envisaged setting up of a) state organizations and b) farmer organizations. At the state level, a steering committee headed by the Chief Secretary/Development Commissioner to guide and co-ordinate the activities with the concerned departments. At WRD level, establishment of PIM cell headed by a

superintending engineer. The cell is supposed to make annual as well as a perspective programme for PIM in the state and need to ensure effective implementation. In addition, there are a) PIM advisory committee to advise PIM cell, b) WALMI, to provide capacity building of stakeholders, c) PIM Unit at the chief engineer level, and d) Field implementation team (FIT), at the executive engineer level. But till December 2005, owing to severe shortage of staff and resources, these units/cells are handicapped to perform effectively. At users level, the model had envisaged farmer's organizations (FO). Each FO comprises: a) village irrigation committee (VIC) wherein, every water user of the village will be a member. The executive committee of the VIC comprises 5 to 11 members through election; this executive committee is responsible for proper management of irrigation through all the outlets within the village, in accordance to the decisions taken by the SLC. b) System level committee (SLC). Here system refers to distributory, sub-distributory or minor, where the management transfers of the system takes place. SLC are registered bodies under the Societies Registration Act, 1860. As of December 2005, there were 12 active SLCs, popularly known as Water Users Associations (WUAs) in Bihar, spreading both in Sone and Gandak command areas. Societies in a phased manner, under the Participatory Irrigation Management Programme. Out of 79 registered societies in the state which have already given their consent, the management of irrigation system has been transferred to 53 societies and the process of transfer is on-going for 26 remaining societies.

A9.1.12. Command Area Development and Water Management Programme

The Government of India launched a Centrally Sponsored Command Area Development Programme (CADP) in December 1974 across the country to improve irrigation potential utilisation and optimise agricultural production from irrigated land through integrated and coordinated approach of efficient water management. In tune with these objectives of the programme, a number of components such as construction of field channels and field drains, enforcement

of warabandi, land levelling and shaping, realignment of field boundaries/consolidation of holdings, introduction of suitable cropping patterns, strengthening of extension services etc. were included in the programme. Subsequently, in view of emergent needs a few more components like farmers' participation and reclamation of waterlogged areas were included in the programme with effect from 1st April, 1996 to make the programme more beneficial to the farmers.

A9.1.13. Accelerated Irrigation Benefits Programme (AIBP)

A large number of major and medium irrigation projects in the country are languishing due to various reasons, the most important of them being inadequate provision of funds by the concerned State Governments. As a result, large amount of funds spent on these projects are locked up and the benefits envisaged at the time of formulation of project reports could not be achieved. This is a cause for concern to the nation and initiative is required at the national level to remedy the situation. Since the irrigation projects are capital intensive, and states with limited resources at their disposal find themselves unable to meet the desired fund demands of all the projects, the implementation of these projects get delayed.

Keeping the above in view, Central Govt., during 1996-97, launched an Accelerated Irrigation Benefits Programme (AIBP) to provide Central Loan Assistance (CLA) to major/medium irrigation projects in the country, with the objective to accelerate the implementation of those projects which were beyond resource capability of the states or were in advanced stage of completion. While selecting the projects, special emphasis was to be given to Pre-fifth and Fifth Plan projects. Priorities were also given to those projects which were benefiting Tribal and Drought Prone Areas. However, under the revised AIBP Guidelines from the year 1999-2000 onwards, Central Loan Assistance under AIBP could also be extended to minor surface irrigation projects of special category states

(N.E. States & Hilly States of H. P., Sikkim, J&K, Uttarakhand and projects benefiting KBK districts of Orissa). 9 projects have been assigned under AIBP programme in Bihar. These projects and their status are presented in Table 2.

A9.1.14. National Project for Repair, Renovation & Restoration (RRR) of Water Bodies

The Government of India sanctioned a Pilot Scheme for “National Project for Repair, Renovation & Restoration (RRR) of Water Bodies directly linked to Agriculture” in January, 2005 with an estimated cost of Rs.300 crore to be shared by Centre and State in the ratio of 3:1 proposed to be implemented during the X Plan period. The objectives of the Scheme were to restore and augment storage capacities of water bodies, and also to recover and extend their lost irrigation potential. The Scheme has been approved for 26 district projects in 15 States, viz., Andhra Pradesh, Bihar, Chhattisgarh, Jharkhand, Karnataka, Madhya Pradesh, Orissa, Rajasthan, Tamil Nadu, West Bengal, Himachal Pradesh, J&K, Gujarat, Kerala and Maharashtra at an estimated cost of Rs.299.92 crore and Central Share of Rs.197.30 crore has been released to the states so far. These projects cover 1098 water bodies with total original culturable command area of 1.72 lakh hectares. After RRR works, an additional irrigation potential of 0.78 lakh hectares is likely to be generated. The physical work for restoration has been completed in 1054 water bodies in 15 states so far. The spillover activities are being continued during the current financial year i.e. 2009-10 in the state of Maharashtra. The irrigation potential created subsequent to pilot scheme is 1.73 lakh ha. Approximately.

Government of India approved two schemes on repair, renovation and restoration of water bodies (i) one with external assistance with an outlay of Rs. 1500 crore and (ii) the other with domestic support with an outlay of Rs. 1250 crore for implementation during XI Plan Period. Under the scheme covered by external assistance, the GoI provides assistance to the extent of 25% and borrows necessary funds as loan from World Bank, 75% State share is to be borrowed

from the World Bank by Concerned State. Under the scheme with domestic support funding is also in the ratio of 25:75 (Centre: State) for non-special category States and in the ratio of 90:10 for special category States (North Eastern States including Sikkim, HP, Uttarakhand, J&K and undivided Koraput, Bolangir and Kalahandi (KBK) districts of Orissa), drought prone/naxal affected/tribal areas. These schemes were approved during the end of financial year 2008-09. Public water bodies are covered under the scheme. The status of this scheme in Bihar is presented in Table 3.

A9.1.15. *Minor irrigation scheme*

Minor irrigation scheme falls under the Department of Minor Water Resources. The department is functioning with the objective of providing irrigation by executing and maintaining small schemes having command area up to 2000 hectares by utilizing available ground water & surface water. Minor Irrigation Schemes implement in Bihar could broadly be divided into three categories - 1. Surface Water Flow Irrigation Schemes, 2. Surface Water Lift Irrigation Schemes, and 3. Ground Water Irrigation Schemes.

A9.1.16. *Surface Water Flow Irrigation Schemes*

In this scheme, surface water under gravity flow is utilized for irrigation purpose. Surface water flow irrigation projects comprising storage and diversion works occupy a conspicuous place in the complex hierarchy of irrigated agriculture in the country particularly in the undulating areas and the hilly regions. These are the only means of irrigation available in major part of the chronically drought affected areas. Such schemes are labour intensive and offer extensive opportunities for rural employment. These are also of considerable help in recharging the resource of ground water in the hard rock areas. Construction period of such schemes is rather long in comparison to other categories of Minor Irrigation schemes. Construction cost is also more in comparison to other Minor

Irrigation schemes, but there is no recurring cost involved in providing irrigation once scheme is complete. Surface water Flow Irrigation Schemes could further be classified into two categories.

A9.1.17. Surface Water Storage Irrigation Schemes

In this scheme, water brought in by seasonal rivers, rivulets, drains etc. during rainy season is stored by constructing small earthen dams at appropriate place. In lean season such stored water is provided for irrigation through a network of distribution channels. Ponds, Check dams, traditional Ahar & Pynes etc. are examples of this type of scheme.

A9.1.18. Surface Water Diversion Schemes

Under this scheme, an "obstacle" or "barrier" is proposed to construct across the small seasonal river, rivulet, drain etc. at an appropriate place. This results in increasing the water level of the water body at that place and flow of water is diverted into irrigation channels. Depending upon requirement, cost involved, terrain & topography of the site the "obstacle" or "barrier" constructed could be of "permanent" or "temporary" nature. Weir is an example of such type of scheme.

A9.1.9. Surface Water Lift Irrigation Schemes

These schemes are playing a very useful role on sites where available surface water cannot be used for irrigation through construction of flow irrigation schemes due to topographical limitations. In such type of schemes surface water available in flowing water body is lifted by means of pumps and is made available for irrigation. The pumps are either electricity or diesel driven. Investment on these schemes is comparatively less and these can be completed very quickly but recurring cost of providing irrigation is quite high due to high energy costs.

Besides this, as the structures like Intake well, Pump house are constructed at any one bank of flowing water body, fields on only that side can be irrigated. Sometimes due to shifting or change in water course of river, intake well becomes defunct. Very often "Intake Well" gets filled with sand & silt. To overcome these problems of Lift Irrigation Scheme an innovative lift irrigation scheme by the name "Barge" has been launched. In this scheme, a diesel pump is fitted on a floating platform made up of large hollow iron cylinders (Pontoon). Due to floating platform this scheme achieves the much required "mobility" rendering it very useful. As the Barge can be towed it can be moved to either of the two banks of river and fields on both sides of river can be irrigated. There is no risk of scheme becoming defunct due to siltation or shifting of river course.

A9.1.20. *Ground Water Irrigation Schemes*

In this scheme, the groundwater is used for irrigation by lifting from underground areas through pumps. These pumps are energized by either electricity or diesels. Ground water exploitation forms the major part of the minor irrigation programme and includes construction of dug wells, dug-cum-bore wells, filter points, private shallow tube wells and deep public or state tube wells. It is essentially a people's programme implemented primarily through individual and cooperative efforts with finance obtained mainly from institutional sources. The programme thus, imposes very little burden on the public exchequer.

The programme of ground water exploitation has become a vital factor in promoting modern agriculture through high yielding crops. Ground water is abundantly available in most part of the state and hence provides instant and assured sources of irrigation in the hands of the cultivators. It provides critical help in improving the status of irrigation supply and in controlling water logging and salinity in the canal commands. The programme also provides abundant opportunities for rural employment.

A9.1.21. Other Minor Irrigation Schemes

Under NABARD (Phase XI), 31,200 hectares irrigation potential has been created through 775 tubewells. A total of 10,242 state tubewells which were constructed under the NABARD (Phase-III), NABARD (Phase-VIII) and NABARD (Phase-XI), are to be soon energized. Besides, 2871 state tubewells have been energized by the end of 2014-15. Again, Rs. 85.39 crore was sanctioned for renovation of 2397 state tubewells which were energized during NABARD (Phase-III) and NABARD (Phase-XI).

A Water Regulatory Authority has been established under Bihar State Water Policy, 2015 by Department of Water Resources. A proposal to keep the Ground Water Authority as a subsidiary of the Water Regulatory Authority has been submitted to Department of Water Resources.

Under the Integrated Action Plan, a total of 184 projects for Irrigation Channels, Ahars, Tanks, Water Harvesting Structure, etc. have been completed at a total expenditure of Rs. 20.24 crore.

To encourage the installation of private tubewells by farmers, the state government has two ambitious schemes — Million Shallow Tube Wells Programme (MSTP) and Bihar Ground Water Irrigation Scheme (BGWIS). An irrigation potential of 2.78 lakh hectare has been created through installation of 99,570 private tubewells under BIGWIS during 2009-10 to 2012-13.

A new scheme named ‘Bihar Shatabdi Niji Nalkup Yojana’ has been launched by the state government, which is a subsidy-based scheme. Under this scheme, two types of tubewells will be installed — shallow tubewells (70m depth) and medium depth tubewell (70m to 100m depth). For the installation of shallow tubewells, the maximum subsidy is Rs. 15,000 and, for the medium tubewells, it is Rs.

35,000. There is also a programme of giving a maximum subsidy of Rs. 10,000 for pump sets or half the market rate price, whichever is less.

A9.1.22. National Rural Drinking Water Programme

National Rural Drinking Water Programme (NRDWP) aims at to provide every rural person with adequate safe water for drinking, cooking and other basic domestic needs on a sustainable basis, with a minimum water quality standard, which should be conveniently accessible at all times and in all situations. In this programme, the center government provides financial and technical assistance to the state government to supplement its efforts to provide adequate safe drinking water to the rural population. Under this programme, 24.3 thousand handpumps have been installed till 2014 – 15 in the state.

A9.2.1. Recent initiatives taken by the Government of Bihar for improving Drinking water facility and sanitation

Lohiya Swachhatta Yojana: A scheme called Lohiya Swachhatta Yojana with a budget of Rs. 1431.12 crore was sanctioned to cover APL families excluded from Swachh Bharat Mission (Gramin) for construction of individual household toilets in rural areas.

Pranjal :A state-level Training-cum-Research centre 'Pranjal' has been established under Bihar State Water and Sanitation Mission of the Department of Public Health Engineering.

Mukhyamantri Chapakal Yojana: Under Mukhyamantri Chapakal Yojana, a target of 5 hand pumps per Gram Panchayats in rural areas, 3 hand pumps per ward in Nagar Nigam, 2 hand pumps per ward in Nagar Parishad , 1 hand pump per ward in Nagar Panchayat of urban areas and 100 hand pumps per Hon'ble

Member of Bihar Legislative Council was set to strengthen the drinking water supply in all districts by 2015-16.

Mukhya Mantri Kshetra Vikas Yojna: Under Mukhya Mantri Kshetra Vikas Yojna, 382 Ghats and public ponds were constructed. Presently, another 301 schemes with a cost of Rs. 18.80 crore are under progress.

Piped Water Scheme : Implementation of multi-villages piped water supply scheme based on surface water (Ganga River) with 5 years operation and maintenance was started in 2015, to provide potable drinking water in 111 arsenic affected villages. These affected villages are — Matihani, Barauni and Begusarai blocks of Begusarai districts. In 45 villages in Biddupur block and adjoining areas of Vaishali district, 130 villages in Simari block and adjoining areas of Buxar district and 25 villages of Maner block of Patna district, a similar scheme is under progress.

Table A9.1: Intra-linking Schemes in Bihar and Thier Status

Name of Scheme	Total Length of Channel or Link	Consultant	Project Cost (in crore)	NORTH OR SOUTH BIHAR	PURPOSE	Benefited Districts	Type of Scheme	Present Status as on 23 rd April 2014
2	3	4	5	6	7	8	9	10
Kohra-Chandravat Link 80 cumec water is to be transferred (Burhi gandak - Gandak) Line canal- 17 Km Lalbegi river - 9 Km	26 Km	National Water Development Agency (NWDA)	168.86	NORTH	Flood Mitigation	West and East Champaran	Inter linking of rivers	PPR preparation work is under process under Patna Division.
Bagmati-Burhi Gandak through Belwadhar 500 Cumec water is to be transferred Lined canal- 1 Km Belwadhar-59 Km	60 Km	National Water Development Agency (NWDA)	125.96	NORTH	Flood Mitigation	Sheohar, Sitamarhi and Muzaffarpur	Inter linking of rivers	PPR preparation work is under process. under Lucknow Division.
Burhi Gandak- Noon-Baya-Ganga Link 492 Cumec water is to be transferred CCA- 1,26,000 Ha Lined canal-28.95 Km	71 Km	National Water Development Agency (NWDA)	4213.8	NORTH	Flood Mitigation	Samastipur, Begusarai and Khagaria	Inter linking of rivers	DPR submitted to CWC in january 2014 .by NWDA.
Kosi-Ganga Link (Bagmati) Lined canal-1 Km Mushradhar-3.5 Km Kasraiya River-4.5 Km	10 Km	National Water Development Agency (NWDA)	88.93	NORTH	Flood Mitigation	Khagaria and Bhagalpur	Inter linking of rivers	PPR preparation work is under process under Patna Division.

Transfer of Kosi Basin Water to Mahananda Basin through Kosi-Mechi Link Canal. Diversion of 1814 million cumec surplus water of Kosi river. Length of canal-117.5 km. CCA- 2,10,516 Ha	120.15 Km	National Water Development Agency (NWDA)	2903.03	NORTH	Irrigation	Supaul, Saharsa, Araria, Kishanganj and Purnea	Inter linking of rivers	DPR submitted to CWC.
Development of Bagmati Irrigation & Drainage Project-Phase-II (Barrage near Kataunjha in Muzaffarpur district) and Adhwara Multipurposed Project with Kosi-Adhwara-Bagmati Link		National Water Development Agency (NWDA)		NORTH	Irrigation and Flood Mitigation	East Champaran, Muzaffarpur, Samstipur, Begusarai, Khagaria, Darbhanga, Madhubani and Sitamarhi	Inter linking of rivers	Detailed Proposal for preparation of Detailed Project Report submitted to NWDA in December 2011.
Baksoti Barrage Scheme on river Sakri and conversion of Nata Weir to a Barrage on river Nata in Nawada district with suitable link from Sakri river to Nata river from Baksoti barrage site. Paura Weir CCA-31370 Ha Nata-Kaurihari		M/S Integrated Hydro Development Forum, Patna.	572.38	SOUTH	Irrigation	Nawadah, Nalanda and Sheikhpura	Inter linking of rivers	DPR submitted to CWC as per their observation on PPR.

CCA- 17438 Ha Baksoti barrage CCA- 20000 Ha								
Dhanarjay Reservoir Scheme and link to Phulwaria Reservoir Scheme. Link Canal 6.55 Km Submergence - 490Ha (232Bh, 258JH) Forest - 255Ha (133 Bh,102JH)	6.55 Km	M/s Algorithm Consultancy Services Pvt. Ltd., Patna.	272.9	SOUTH	Energy	Nawada and Gaya	Inter linking of rivers	As per the CWC comments on PPR of project Bihar government is trying to make consensus with Jharkhand government on various issue of project.
Transfer of Ganga- Water to South Bihar through Pump Canal Scheme at Barh with Multipurpose Development of these basins. 7 Nos. Barrage, 1 no. Reservoir (Masathu) Lined canal- 83.5 Km. CCA- 7.0 Lakh Ha.		National Water Development Agency (NWDA)	12298	SOUTH	Multi Purpose	Patna, Nalanda, Nawadah, Gaya, Aurangabad, Jamui	Conventional Irrigation Schemes	PPR to be ready by December 2013 under Lucknow Division.
Transfer of Ganga- water to South Bihar through Pump Canal Schemes at Buxar.		National Water Development Agency (NWDA)		SOUTH	Irrigation		Conventional Irrigation Schemes	Detailed Proposal for praparation of DPR submitted to NWDA in December 2011.
Development of- Badua-Chandan Basin.		National Water Development Agency (NWDA)		SOUTH	Irrigation		Conventional Irrigation Schemes	Detailed Proposal for praparation of DPR submitted to NWDA in December 2011.

Augmentation of water availability in the existing Gandak Canal System through construction of a second barrage near Areraj on it under Gandak Project Phase-II. Link canal- 16+21 = 37 Km CCA- 3,73,000 Ha.	-	WAPCOS, Haryana	2000	NORTH	Irrigation	West and East Champaran, Muzaffarpur, Khagaria, Chapra, Siwan, Gopalganj and Begusarai	Conventional Irrigation Schemes	PPR submitted to CWC for clearance.
Development of Bagmati Irrigation & Drainage Project-Phase-I (Barrage near Nepal Border at Dheng). Link canal- 62.71 Km CCA- 1,03,000 Ha.	-	WAPCOS, Haryana	1275	NORTH	Irrigation	Muzaffarpur, Sheohar and East Champaran	Conventional Irrigation Schemes	PPR submitted to CWC for clearance.
Drainage improvement & best economic use of water in Mokama Tal. Tal area- 1.06 Lakh Ha.	-	M/s Sigma Resources Development Consultants Pvt. Ltd. Patna	691	SOUTH	Multi Purpose	Patna, Nalanda and Gaya	Conventional Irrigation Schemes	DPR is under process. Consultants are directed to revise the Draft DPR as per the observation given by Task Force Team. Presentation of PPT before Honourable Chief Minister is pending.
Provision of second barrage at Arwal at river Sone to augment water availability in Sone Canal System to reduce its dependence on Indrapuri Barrage	-	WAPCOS, Haryana		SOUTH	Irrigation		Conventional Irrigation Schemes	The preparation of DPR for second barrage over river Sone has been stopped vide dept. letter no. 1185 dated 14 November 2008. In July 2009, a meeting was held between officials of WAPCOS and WRD

and utilize water thus saved at Indrapuri Barrage for transfer to Punpun-Harohar-Kiul Basin.								Bihar to review of the preparation of DPR, which has been stopped by GOB. In the light of the minutes of the meeting further study of availability of water claimed by the WAPCOS is being examined by REC. Another reminder has been recently sent to REC.
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Table A9.2: Name and Status of Projects Benifited from AIBP in Bihar

Sl.N o.	Name of Major Medium Irrigation Projects	AIBP Project Name	State	Year of inclusion under AIBP	Status (Completed /Ongoing)	Year of completi on (Rs. Crore)	Target Potenti al under AIBP (Th ha)	Latest Estimated Cost of AIBP Componen ts (Rs. Crore)	Total CLA/Gra nt relesaed upto (3/12) (Rs. Crore)	Total expenditu re incurred upto (3/2011) (Rs. Crore)	Total potential created upto(3/201 2) (Th ha)
1	Sone Barrage Remodelling And Link Canal	Sone Modernisati on	Bihar	1998-99	Completed	2008-09	314.6	664.3	168.4	635.5	290.6
2	Bansagar Dam Major Irrigation Project_Bihar	Bansagar	Bihar	1997-98	Ongoing		94.0	366.8	83.5	334.9	0.0
3	Batane Major Irrigation Project_Bihar	Batane	Bihar	2000-01	Ongoing		4.8	77.0	3.3	40.7	0.0
4	Bilasi Medium Irrigation Project	Bilasi Reservoir	Bihar	1997-98	Completed	2000-01	4.0	16.8	3.4	16.8	4.0
5	Durgavati Major Irrigation Project	Durgawati	Bihar	1996-97	Ongoing		20.3	896.7	65.1	399.5	3.3
6	Orni Medium Irrigation Project	Orni Reservoir	Bihar	1997-98	Completed	2006-07	9.6	28.9	11.4	29.1	9.5
7	Punpun Barrage Major Irrigation Project	Punpun	Bihar	2007-08	Ongoing		13.7	619.1	43.9	196.2	0.0

8	Upper Kiul Major Irrigation Project	Upper Kiul	Bihar	1996-97	Completed	2006-07	11.7	94.5	22.6	83.5	10.2
9	Western Kosi Canal Major Irrigation Project	Western Kosi	Bihar	1996-97	Ongoing		212.1	1069.3	252.1	986.0	145.7

Table A9.3: Status of RRR Schemes of XI Plan

S.No.	Name of State	No. of water bodies	Total project cost in crore	CCA (ha)	Potential to be restored (ha)	Committed Central Share	Total funds released (Rs. in crore)	Committed centre share to be released	Number of water bodies completed	No. of water bodies in progress	Potential restored (ha)	Fund Utilised (Rs in Cr)
1	Orissa	1321	451	64979	64979	406	244	162	1128	193	48643	242
2	Karnataka	427	233	8182		209	199	10	424	3	21789	221
3	Andhra Pradesh*	345	90	36674	12585	81	75	6	251	91		73
4	Telangana#	684	227		21415	205	114	90	326	324	10232	
5	Bihar	15	64	15718	15446	55	53	3	14	1	15598	58
6	U.P. (Bundelkhand)	28	46	29697	17695	42	39	2	28	0	17695	44
7	M.P. (Bundelkhand)	78	42	25254	15560	10	10	1	72	6		11
8	Meghalaya Umiam Lake(cost related to irrigation only)	1	3	405		3	2	0	1	0		3
9	Maharashtra**	258	135	1352		119	81	39	105	150	19128	57
10	Gujarat	34	17	6574	2842	16	11	5	23	11	1230	12
11	Chhattisgarh	131	123	89951	10703	111	73	38	111	20	10026	77
12	Rajasthan	16	11	5749		7	7	0	15	1	1470	12
13	Haryana	3	40	24936		10	10	1	3	0		11
	Total	3341	1484	309471	161225	1275	917	357	2501	800	145810	820

Table A9.3: Status of RRR Schemes of XII Plan

Sl No	State	No. of Water Bodies identified by the State Govt during XII Plan	Estimated Cost (Rs in crore)	No. of water Bodies approved	Total cost Rs in crore	Funds released in crore
1	Odisha	1355	529	760	362	53
2	Maharashtra	362	168			
3	Madhya Pradesh	134	198	134	198	38
4	Meghalaya	9	11	9	11	3
5	Rajasthan	36	90	32	90	
6	Uttarakhand	5	13	5	13	
7	Manipur	4	85	4	65	10
8	Assam	9	98			
9	Bihar	39	127			
10	Karnataka	128	81			
11	Nagaland	19	15			
12	Chhattisgarh	122	213			
13	West Bengal	2	74			
14	Uttar Pradesh	63	80	8	37	
15	Himachal Pradesh	1	4			
16	Kerala	80	27			
17	Tamil nadu	214	101	105	55	
18	Telangana	335	293			
19	Andhra Pradesh	49	19			
Total		2966	2226	1057	831	103

A9.3. Agriculture Sector

A9.3.1. Seed sector

Development of quality seeds is the major focus of the Government of Bihar, as is reflected from Agriculture Roadmap prepared by the government. Following initiatives regarding seed development have been taken in Bihar. As a result of these initiatives, the seed replacement rate of paddy and wheat has increased to 38 per cent and 35 per cent, respectively.

- Bihar State Seed Corporation was revived in year 2006. The corporation again started producing seed through farmers. It also procures seeds from other government organizations and agencies and supplies certified seeds to farmers. The corporation produced about 2 lakh quintal seeds in year 2010-11.
- The seed corporation has built 31 warehouses of 3.10 lakh MT capacity.
- Three new seed processing plants were set up at Hajipur, Bhagalpur, and Begusarai.
- Seven mobile seed processing plants were set up.
- Chief Minister's Crash Seed Programme was launched in 2008 – 09. The features of this programme are as: Distribution of foundation seed of selected crops in small packets, Two farmers selected from each village for each selected crops, Seed given to farmers at half the cost, and Certified seed produced by farmers spread through traditional method of exchange i.e.-*Dedha* (1.5 times), *Sawaiya* (1.25 times).
- The seed village programme has been getting implemented since 2007 – 08. In this programme, four villages are chosen from each block and intending farmers of these select villages are given foundation seeds at half rate. Besides, farmers are given training in seed production and seed bins at subsidized rate. Paddy, Wheat, Lentil and Gram are included in this programme.
- Certified seeds are distributed to farmers at subsidized rate. In year 2011-12, about 6 lakh quintal seeds were distributed.

A9.3.2. *Mechanization in agriculture*

To increase mechanization in Bihar agriculture, the government provides financial assistance to farmers through subsidy in purchasing of agriculture machines. Amount of subsidy is jointly shared by both the central and state government. Subsidies is provide almost all equipment related agriculture. Except for few items such as tractor, power tiller etc., the major chunk of subsidy come from state government. Subsidy structure in Bihar seems to be promoted to resource conservation technologies such as laser land leveling etc. in the state.

A9.3.3. *Focus of High value crops*

To promote horticulture development, the National Horticulture Mission was launched by the Government of India in 2005 – 06. The mission has been started in 23 district of Bihar. In the similar line, the Government of Bihar introduced Chief Minister Horticulture Mission (CMHM) in year 2006 -07. The CMHM has been implemented in those district where the National Horticulture Mission (NHM) is not working. Such provision was made to avoid overlap between CMHM and NHM. Besides, the state government has attempted identify a specific horticulture crop in each districts which is being developed through Agriculture Technology Management Agency (ATMA).

ATMA is an agency which is constructed in each district for the purpose of technology dissemination. It has linkage with all the line departments, research organizations, non-governmental organizations and agencies associated with agricultural development in the district. Research and Extension units within the project districts such as ZRS or substations, KVKs and the key line Departments of Agriculture, Animal Husbandry, Horticulture and Fisheries etc. are supposed to be constituent members or Key stake holders of ATMA. This Scheme was approved on 29th March, 2005. The Scheme has made extension system farmer driven and farmer accountable.

Apart from the above, a mission based programme was also launched by the Government of India to encourage farmers to cultivate medicinal plant. It is called National Mission on Medicinal Plant and is being implemented in each district of Bihar. Under this programme, the government provides subsidy for all aspects related to medicinal plant such as cultivation, marketing, processing etc. This programme provides value chain and helps to develop Agriculture Processing Industry in the state.

Bihar Horticulture Development Society at the state level and horticulture committee at the district level are responsible for implementation of horticulture schemes.

A9.3.4. Agriculture Marketing

In Bihar, Agricultural Produce Market Committee Act was repealed in 2006. The market yards of dissolved marketing board and committee was converted into rent free markets. Agriculture produce can now be directly purchased from farmers. As a result of these changes, the market development could involve public, cooperative, private and joint sector participation. Farmers could be organized for marketing of the perishable produce like fruits and vegetables.

A9.3.5. Drip and Sprinkle Irrigation

Several irrigation projects and programmes of both the center and state government have been discussed in the previous section. Here, it would be interesting to mention sprinkle and drip irrigation systems because these systems contribute to water conservation. Both the center and state government provide subsidy to those farmers who adopt drip and sprinkler irrigation systems. The above scheme is sponsored by the central government while it is being implemented the state government. In Bihar, both irrigation

schemes launched in 2011. In these schemes, 90% subsidy has been provided to small and marginal farmers and 80% subsidy is provided to other farmers, 40% subsidy comes from the state government in both the cases.

A9.3.6. National Food Security Mission in Bihar

National Food Security Mission' (NFSM), has been launched from 2007-08 to increase the production of rice by 10 million tons, wheat by 8 million tons and pulses by 2 million tons by the end of the Eleventh Plan (2011-12). National Food Security Mission has been implemented in those districts of the state where productivity is lower than state average productivity. Following are the various intervention under Rice, Wheat and pulses of NFSM.

A9.3.7. Demonstration

The objective of the demonstration is to convince the farmers regarding new technology and crop production practices applied particularly in Rice and Wheat. The demonstrations of improved package of practices of Rice and Wheat, System of Rice Intensification (SRI) and Hybrid Rice will be conducted at farmer's field to create the awareness about the new technology. The area of demonstration is 0.4 ha and for every 100 ha area of Rice and 50 ha area of Wheat one demonstration is conducted. The fund to meet the critical inputs and other expenses for Rice demonstration Rs 2500/per demonstration on IPP, Rs 3000/per demonstration on SRI and Hybrid Rice and Rs 2000/ per demonstration for wheat on IPP are provided.

A9.3.8. Seed Production

Seed is one of the most critical inputs for enhancing the productivity of crops. The Mission provides financial assistance for production of seed of hybrid Rice and breeder, foundation and certified seed of improved variety of pulses.

A9.3.9. Subsidy on Seed

To enhance the production and productivity of food grain varietal replacement of Rice, pulses and Wheat is an essential component. To bring the Seed Replacement Rate up to 33% in identified district by the end of 11th five year plan and to increase in the productivity an assistance of subsidy on purchase of certified seed released during last 10 years is provided on rice and wheat @ 50% of the cost or Rs 700/ per quintal or whichever is less and on pulses @ 50% of the cost or Rs 2000/ per quintal or whichever is less. The assistance will be limited to 5 hectare only to each introducing farmers.

A9.3.10. Seed minikit programme of rice and wheat

Minikit seeds are meant for introduction and popularization of latest released/pre-released variety /hybrid among farmers. The size of the minikit of HYV Rice is of 5 kg and Hybrid Rice is of 6 kg whereas, size of wheat minikit 10 kg each. This is supplied on full subsidy to the farmers.

N A9.3.11. nutrient management/soil ameliorates

To promote the application of lime/liming material and micronutrients an incentive up-to @ Rs 500/ Ha and micronutrient @ Rs 500/ Ha whichever is less in rice is given to improve the soil health.

Application of micronutrient and gypsum in wheat a package assistance for micronutrient and gypsum together will be 50% of the cost limited to Rs 1000/ per ha (@ Rs 500/ Ha for gypsum and micronutrient @ Rs 500/ Ha) is given.

The assistance of gypsum will be provided to those farmers whose soil is saline and alkaline PH.

An assistance for use of micronutrient /lime/gypsum will be provided @ 50% of the cost limited to Rs 1250 /ha to the farmers. The assistance will be limited to Rs 750/ha for lime/gypsum and Rs 500/ha for micronutrients.

This assistance will be admissible to farmers for maximum 5 ha only.

A9.3.12. Mechanization

Mechanization will help in timely completion of field operations in Rice, Wheat and Pulses. Inputs identified are required to be comply the ISI standard or tested at FMTTI/ CIAE Bhopal or RAU. Following interventions have been identified under NFSM-Rice, wheat & Pulses.

BOX A9.2

NFSM Rice	
Incentive for Cono-weeders & Other Implement	Rs. 3000/-per unit Machine
Incentive on Knap Sack Sprayers.	Rs. 3000/-per unit Machine
Incentive on Zero Till Seed Drills.	Rs. 15000/-per unit Machine
Incentive on Seed Drills.	Rs. 15000/-per unit Machine
Incentive on multi-crop planter.	Rs. 15000/-per unit Machine
Incentive on Rotavator	Rs. 30000/-per unit Machine
Incentives for Pump Sets up-to 10 HP	Rs. 10000/-per unit Machine
NFSM -Pulses	
Incentive on Sprinkler Sets.	Rs. 7,500/-per Ha
Incentive on Zero Till Seed Drills.	Rs. 15,000/-per machine
Incentive on Seed Drills.	Rs. 15000/-per unit Machine
Incentive on multi-crop planter.	Rs. 15000/-per unit Machine
Incentive on Rotavators.	Rs. 30,000/-per machine
Incentives for Pump Sets up-to 10 HP	Rs. 10,000/-per machine
Incentive on Knap Sack Sprayers.	Rs. 3,000/-per machine
NFSM Wheat	
Incentive on Knap Sack Sprayers.	Rs. 3000/-per unit Machine
Incentive on Zero Till Seed Drills.	Rs. 15000/-per unit Machine
Incentive on Seed Drills.	Rs. 15000/-per unit Machine
Incentive on multi-crop planter.	Rs. 15000/-per unit Machine
Incentive on Rotavator	Rs. 30000/-per unit Machine
Incentives for Pump Sets up-to 10 HP	Rs. 10000/-per unit Machine
Incentive on Sprinkler Sets.	Rs. 7,500/-per Ha

A9.3.13. Integrated pest management

Financial assistance of Rs 500/per ha in case of Rice and Rs 750/per ha in case ofPulses or 50% of the cost whichever is less will be provided for use of inputs to safe guard rice and pulses crops. A plan of prophylactic and

contingent measures conceptualizing the idea of integrated pest management the bio-pesticides and plant protection chemicals has been incorporated in the package. Assistance will be limited to a maximum of 5 ha per farmer. IPM in Pulses.

A9.3.14. Fisheries

The state government has recently taken steps to enhance the skill of the fishermen through on-site training, and exposure visits to various places in India. Such skill includes both managerial and participatory skills. The fishermen of the state have been sent to Andhra Pradesh and West Bengal for upgradation of their skills for modern fishing techniques. During 2012-13, around 1552 fishermen were trained in the state; in 2013-14, another about 2000 fishermen were inducted for training.

In 2011-12, a new insurance scheme for fish farmers was introduced. One scheme is being implemented by the Oriental Insurance Company, a public sector enterprise. Under this scheme, a premium of Rs. 3200 is fixed for each hectare of fishing area, which is shared by the beneficiary and the state government on 50:50 basis.

The schemes undertaken by the state government, like distribution of fingerlings on subsidies, distribution of easy loans to farmers for maintenance and renovation of private ponds, and free houses for poor fishermen have immensely benefited the fish farmers of the state. The development of ponds and 'chaur' have increased the water area under fishing. In 2013-14, 4812.85 lakh fish seeds were produced in the state.

Recently, several new schemes have been undertaken by the state government for the upliftment of the poor fishermen in the state. A scheme of stocking of reservoirs with fingerlings have been newly introduced by the state government. The National Fisheries Development Board (NFDB) is assisting the process of stocking. The required credit is provided to the fishermen for

the creation of hatcheries which have the capacity to breed 8-10 million fries. To maximize carp production, hatchery is being developed on Public-Private Partnership (PPP) basis. The unit cost of hatcheries is Rs. 15.00 lakh, for which 50 percent subsidy is provided. There is a target to construct 40 such hatcheries. Any beneficiary may construct a carp hatchery on his fish farm and can avail 50 percent of subsidy.

There is a proposal to create a Fish Research Centre in Mithapur (Patna) at a cost of Rs. 129.37 lakh. The state government also proposes to establish a Sonaru fish seed farm in Fatuha at a cost of Rs. 99.68 lakh. The state government also plans to provide installation of tubewells and pumpsets to produce quality fish seed in water logged areas of the state to increase fish production.

A9.3.5. Organic Farming

- Following initiatives were taken to promote organic farming in Bihar.
- Now soil testing laboratories were set up in 16 districts of the state.
- Pesticide testing laboratory was started in Patna.
- Fertilizer testing laboratory has been equipped with latest equipment.
- Bio control laboratory has been set up in Patna.
- Fertilizer, Pesticide, and Bio control laboratory is set up at Muzaffarpur, Bhagalpur, and Saharsa.
- Subsidy is started to provide on micronutrient.
- Subsidy is provided to farmers to produce vermin compost. Besides, farmers get subsidy for establishment of biogas system.
- A programme for cultivation of Dhaincha as green manure crop was implemented for maintaining the soil fertility.

A9.3.16. Livestock

The state government has taken several important steps for its livestock development. These include breed upgradation, health and nutrition, insurance scheme for milch animals, and marketing of the products of the

sectors. Further, training programmes have been arranged for rural workers to enhance their skill in animal husbandry.

The state government has several schemes like medical treatment of animals, sterilization, artificial insemination, immunization and free distribution of fodder seeds for the growth of the animal husbandry sector.

The state government has also started providing ambulatory vans (Mobile Veterinary Clinic) since 2013-14, which are equipped with latest technology and modern medicines and tools. At present, 50 such vans are providing veterinary services to remote rural areas. The state government plans to provide more such vans in the coming annual plans.

Further, the state government has established silos in Patna, Munger, Gaya, Saharsa, Purnea, Muzaffarpur, Darbhanga, Chhapra and Banka for the storage of liquid nitrogen for artificial insemination. The state government has also taken steps to renovate the artificial insemination centres in all the districts. In the new programme of artificial insemination, the non-governmental organisations (NGOs) would also be associated, where the expected number of new calves is rather high. The state government also proposes to create a modern slaughter house in each of the 38 districts, at a cost of Rs. 1 crore per unit, for the production of clean and hygienic meat.

A9.3.17. Procurement and Primary Agricultural Credit Society

Bihar government is engaged in paddy procurement from its farmers through three agencies - Food Corporation of India (FCI), Bihar State Food Corporation (BSFC) and Primary Agriculture Credit Societies (PACS). The BSFC has been procuring rice and wheat under Decentralized Procurement (DCP) scheme of Govt. of India in the marketing Season involving direct purchase of paddy/wheat from farmers & milling it into rice for distribution in various schemes under Public Distribution System (PDS). Decentralized procurement of paddy was introduced to to purchase paddy/wheat from farmers directly on payment of Minimum Support Price (MSP), to mill paddy

into rice and utilize under Public Distribution System (PDS) and to avoid distress sale of paddy /wheat in the State.

Primary Agricultural Credit Society (PACS) is a basic unit and smallest co-operative credit institutions in India which works on the grassroots level (gram panchayat and village level). In Bihar PACS worked in 16 districts, with 15 Civil Society Organisation (CSO) partners and 18 network partners. PACS projects covered 116 blocks, 4012 villages and reached 2,254,454 people. In times of decentralized procurement and targeted distribution the role of PACS increased. In 2009 the governments initiated to hand over the management of PDS to PACS. According to then Chief Minister “Since all works of villages and farmers will be under the purview of PACS for implementation and surveillance, it will be better to give them the additional responsibility to avoid irregularities,” More effective was the government’s decision to ensure purchase of paddy and wheat through PACs in the selected districts under a pilot project in the next Kharif season.

The plan was to ensure that every PAC has a storage capacity of 200 tonnes of foodgrains, to equip 8500 PACs with computers and link them with headquarters for early survey of crops for Kharif purchase and have elections regularly. This was a revival of the cooperative structure even in the remote village of the state. A similar system of procurement of paddy and PDS in another state Chhattisgarh was an inspiration. All these steps will be taken to wipe out black marketing of the foodgrains, The state government had also decided not to allot the work of PDS to an individual but to a PAC or self-help groups. Food Corporation of India (FCI) was criticized for its alleged apathy towards purchase of foodgrains and their storage.

However the PCAS is also blamed with complaints of behaving in a high-handed manner in rice-procurement in Bihar and collusion with the local traders to fleece the poor farmers. It is alleged that along with District cooperative bank the PACS contrives to delay the process of procurement for a month or two thereby forcing the hand-to-mouth farmers to sell their grain

to the local traders at a much lower rate. Centre-state cooperation is also coming in the way. The rice procurement and payment of arrears to cooperative banks and primary cooperative societies, for example, have suffered on account of the circumstances. PACS is under the department of cooperative but the absence of a proper minister is also responsible for the non-receipt of fund from the Central Government. Crop Insurance money and various other payments are missed out or pending, because there is no whole time minister looking after the Cooperative Department.

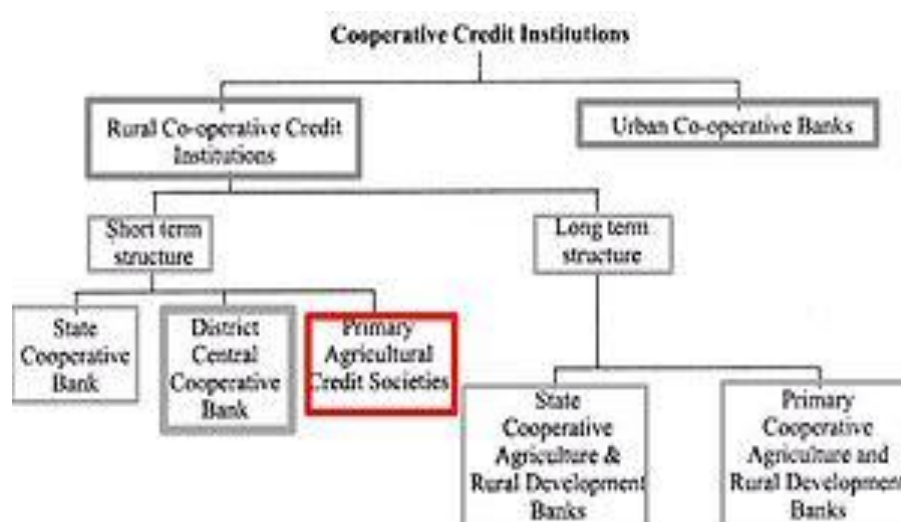


Diagram A9.1: PACS structure

A9.3.18. Other occupations

Among non-crop economic activities, livestock rearing or dairy is most dominant occupation in Khagaria. Members of the households also reported earning from non-farm activities like tailoring, Carpentry, shop ownership, Cycle repair, trading in grain and other small food (restaurants) business. They reported being skilled as masons, drives, security guards and tractor drivers. Only 8 households out of the 120 reported members being employed

in MNREGA job in the reference period. In Saharsa on the other hand, 26 out of the 126 households earned from MNREGA. Other occupations are similar to the Khagaria sample but members also reported working as contractors, furniture makers and even tutors.

A9.3.19. Dairy: Cooperative movement

Cooperatives did exist prior to India's independence¹, but they were of minor significance. Exploitation of farmers by traders, manifested in an unrest in the Polson factory, led to the creation of the Kaira cooperative which inspired the cooperative revolution to follow. Complemented by innovative ideas of statesmen, the failure of the market to function led to the institution of the celebrated Gujarat Co-operative Milk Marketing Federation (GCMMF) of Gujarat with its AMUL brand and the 3-tiered 'Anand pattern' of cooperative structure made up of primary Societies, the Unions and the apex Federation. Created in 1965 to replicate the Anand pattern in all states, the National Dairy Development Board (NDDB) not only brought large numbers of farmers including small holders and poor farmers together to sell their milk cooperatively as a team but also implemented the Operation Flood (OF) that gradually enhanced milk production. India became world's largest milk producer in the course of this white revolution.

Two major instruments of the OF became contemporary drivers of India's dairy. They were (i) Genetic transformation of indigenous animals by cross-breeding them with more productive foreign breeds. (ii) Improving the feeding practices to realize the potentials of the high yielding animals. Despite the success, it is believed that bureaucratic control and political interference inhibited success in many parts of the country (Raj, 1969) but Post-OF efforts to spread the cooperative movement to areas where it is dysfunctional, thin or non-existent still continue. The National Dairy Plan (NDP) and Intensive

¹ A Cooperative Act was passed in 1904

Dairy Development Plan (IDDP) are more recent such initiatives but reforms have already set a new direction.

The compounded pressure on land coming from the food security compulsion, urbanization, industrial growth and environmental conservation makes meeting the requirements of improved feed increasingly difficult leading to straying of animals. Not just indigenous, but also exotic animals are vulnerable to such indifference because the poor farmers can barely afford their greater feed demands adequately. The incessant spree of artificial insemination (AI) has diluted the local breeds that are more adapted to Indian conditions of weather and feed availability and are resistant to many deadly diseases. Market reforms have brought competition from importers along with the potential for reaching out to the export markets. The cost of milk production now needs to be reduced even while several contradictions arise in a relatively protected market.

Thus cooperation, though seen to be an 'ethical' practice (Abbott 2014, Chahal and Gupta 1986, Ibitoye 2012) is under challenge today, especially with the enactment of Milk and Milk Product Order (MMPO) in 1992 which allowed private dairy plants to start business with flexibility. Foreign dairy majors are interested in doing business in India and countries efficient at producing milk are trying to enter Indian market under WTO compliant conditions. Moreover, a philosophical change in attitude towards sustainable farming becoming popular worldwide, is shared by India. Organic farming, the farming system approach, community participation, sustainable animal breeding, animal welfare and ethics and food safety are notions that are entering into the domains of policy making and extension. In the wake of globalization, animal health issues have become serious concerns.

A9.3.20. Private sector Dairy Firms

In Bihar a private dairy called Ganga Dairy Limited (GDL or Ganga) competes with the Sudha brand cooperative for procuring milk from farmers. GDL

specializes in the manufacture and storage of milk products. Situated 1020 km from Patna on the railway route from Delhi to Guwahati on NH31, GDL was created in 1997 with a capacity of 10,000 litre per day of liquid milk. It also entered milk powder business in 2006 and gradually began to produce many milk products selling to neighbouring states and to bulk consuming institutions. GDL maintains relationship with farmers, builds cold chain infrastructure and controls quality with latest technology. Over time its plant capacity has been expanded and upgraded.

Appendix 10



Figure A10.1 Silt deposited landscape which was very fertile agriculture Farm land before Kusaha Breach in 2008

Note: The photographs are taken by Amarnath Tripathi.



Figure 10.2 Indian part (Bihar) of Koshi basin

Note: The photographs are taken by Amarnath Tripathi.



Figure A**10.3**: East Canal.

Note: The photographs are taken by Amarnath Tripathi.



Figure A10.4: East Embankment of Koshi river, river side



Figure A10.5: Koshi barrage.

Note: The photographs are taken by Amarnath Tripathi.



Figure A10.6: Farm Women

Note: The photographs are taken by Nialbja Ghosh



Figure A10.7: Anganwadi Worker *Note:* The photographs are taken by Nialbja Ghosh



FigureA 10.8: Irrigation
Note: The photographs are taken by Nialbja Ghosh



Figure A10.9: Market in town
Note: The photographs are taken by Nialbja Ghosh



Figure A10.10: Maize field *Note:* The photographs are taken by Nialbja Ghosh



Figure A10.11: Carrying ration by Boat *Note:* The photographs are taken by Nilabja Ghosh



Figure A10.12: Knowledge Forum in Patna



Figure A10.13: Knowledge Forum in Delhi *Note: The photographs are taken by Nilabja Ghosh*